

SCFT & LST bases

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1 Description

LST with P-type endpoint bases with $T \leq 193$

2 List of P-type LSTs

2.1 0 node(502)

2.1.1 non-instanton Side links + Noble Molecules with extra curve)(119)

² 131,	¹ 131,	³ 12321
¹ 12321,	¹ 123151321,	⁵ 12231513221
¹ 12231513221,	³ 1315131,	² 1315131
¹ 1315131,	² 1321,	¹ 1321
³ 1321,	² 13221,	¹ 13221
³ 13151321,	² 13151321,	¹ 13151321
³ 131513221,	² 131513221,	¹ 131513221
⁵ 131513221,	¹ 1231513221,	⁵ 1231513221
³ ¹ 231,	¹ 231,	²³ 1231
¹² 321,	²² 231,	¹ 2313221
² 2313221,	¹ 22313221,	¹ 313221
⁵ 313221,	¹ 231321,	² 231321
⁵ 231321,	¹ 31321,	² 23131
⁵ 223131,	¹ 3131,	³ 3131
²¹ 15131,	¹ 15131,	¹ ² 15131
¹ ³ 15131,	²¹ 151321,	¹ 151321
²¹ 1513221,	¹ 1513221,	¹ ⁵ 1513221
³ ¹ 315131,	³ ¹ 3151321,	¹ ⁵ 3151321
³ ¹ 31513221,	¹ ⁵ 2315131,	² 1512321
¹ 1512321,	¹ 321513221,	¹ 2321513221
¹ 32151321,	⁵ 32151321,	¹ 232151321
¹ 31512321,	⁵ 31512321,	¹ 3215131
⁵ 23215131,	² 151231,	¹ 151231
² 151231,	³ 151231,	¹ 512231
² 215131,	³ 215131,	⁵ 21513221

² 15131513221,	¹ 15131513221,	² 1513151321
¹ 1513151321,	1513151321, ⁵	² 151315131
¹ 151315131,	¹ 51231513221,	¹ 2315131513221
¹ 22315131513221,	¹ 315131513221,	¹ 5123151321
5123151321, ⁵	¹ 231513151321,	¹ 31513151321
¹ 512315131,	¹ 3151315131,	3151315131 ⁵
³ 2132,	² 1511,	¹ 1511
² 1512,	¹ 51322,	³ 31512
¹ 31512,	¹ 31513,	⁵ 13215132
22215, ¹	231315, ¹	2215132 ¹
³ 215123,	22151322, ¹	³ 221513
221513, ¹	21513215, ¹	31512315 ¹
32151315, ¹	³ 1513151,	¹ 1513151 ¹
¹ 1513151, ²	¹ 15132151,	21513151 ²
1513151315, ¹	¹ 51232151	

2.1.2 A,D series like LSTs(383)

$$\overbrace{1222 \cdots 2}^n 1 \quad 1 \leq n \leq 192$$

$$\overbrace{222 \cdots 2}^n 1 \quad 2 \leq n \leq 192$$

2.2 1 node(947)

2.2.1 Sg(43)

³ 1321s0,	² 1321s0,	⁵ 12315131s0
¹ 3221s0,	² 221s0,	² 3131s0
¹ 2231s0,	¹ 512321s0,	¹ 23151321s0
¹ 223151321s0,	¹ 3151321s0,	315131s0 ¹
³ 151231s0,	³ 215131s0,	¹ 51315131s0
² 151321s0,	¹ 151321s0	

$\overset{5}{1}231513221\mathfrak{e}_6,$
 $\overset{2}{2}2221\mathfrak{e}_6,$
 $\overset{1}{2}3151321\mathfrak{e}_6,$
 $\overset{3}{2}1513221\mathfrak{e}_6,$
 $\overset{3}{1}513151321\mathfrak{e}_6$

$\overset{2}{3}13221\mathfrak{e}_6,$
 $\overset{1}{3}1513221\mathfrak{e}_6,$
 $\overset{2}{3}1512321\mathfrak{e}_6,$
 $\overset{1}{5}131513221\mathfrak{e}_6,$

$\overset{2}{2}31321\mathfrak{e}_6$
 $\overset{2}{3}2151321\mathfrak{e}_6$
 $\overset{1}{2}2315131\mathfrak{e}_6$
 $\overset{2}{3}151315131\mathfrak{e}_6$

$\overset{2}{2}313221\mathfrak{e}'_7,$
 $\overset{1}{2}31513221\mathfrak{e}'_7,$
 $\overset{2}{3}1513151321\mathfrak{e}'_7,$

$\overset{2}{2}2221\mathfrak{e}'_7,$
 $\overset{1}{2}23151321\mathfrak{e}'_7,$
 $\overset{3}{1}5131513221\mathfrak{e}'_7$

$\overset{2}{3}21513221\mathfrak{e}'_7$
 $\overset{2}{2}31512321\mathfrak{e}'_7$

$\overset{2}{2}22221\mathfrak{e}_7,$
 $\overset{2}{2}31513151321\mathfrak{e}_7$

$\overset{1}{2}231513221\mathfrak{e}_7,$

$\overset{2}{3}15131513221\mathfrak{e}_7$

$\overset{2}{2}222222221\mathfrak{e}_8$

2.2.2 SgS(388)

$\overset{1}{2}2313221\mathfrak{e}_7,$
 $\overset{1}{2}231512321\mathfrak{e}_7,$
 $\overset{1}{2}31513221\mathfrak{e}_7,$
 $\overset{2,3}{4,5} \otimes \mathfrak{e}_7,$
 $\overset{2,3}{2}31321\mathfrak{e}_7,$
 $\overset{2,3}{3}1512321\mathfrak{e}_7,$
 $\overset{2,3}{1}3151321\mathfrak{e}_7,$
 $\overset{2,4}{2,4} \otimes \mathfrak{e}_7,$
 $\overset{2,4}{5,4} \otimes \mathfrak{e}_7,$
 $\overset{2,4}{2}151321\mathfrak{e}_7,$

$I^{\oplus 7} \mathfrak{e}_7,$
 $\overset{1}{3}15131513221\mathfrak{e}_7,$
 $\overset{1}{2}23151321\mathfrak{e}_7,$
 $\overset{2,3}{5,5} \otimes \mathfrak{e}_7,$
 $I^{\oplus 5} \mathfrak{e}_7,$
 $\overset{2,3}{2}1513221\mathfrak{e}_7,$
 $\overset{2,3}{1}513151321\mathfrak{e}_7,$
 $\overset{2,4}{3,4} \otimes \mathfrak{e}_7,$
 $I^{\oplus 4} \mathfrak{e}_7,$
 $\overset{2,4}{1}2321\mathfrak{e}_7,$

$\overset{1}{2}2321513221\mathfrak{e}_7$
 $\overset{1}{2}31513151321\mathfrak{e}_7$
 $\overset{2,3}{3,5} \otimes \mathfrak{e}_7$
 $\overset{2,3}{3}13221\mathfrak{e}_7$
 $\overset{2,3}{3}2151321\mathfrak{e}_7$
 $\overset{2,3}{5}123151321\mathfrak{e}_7$
 $\overset{2,3}{1}1513221\mathfrak{e}_7$
 $\overset{2,4}{4,4} \otimes \mathfrak{e}_7$
 $\overset{2,4}{3}1321\mathfrak{e}_7$
 $\overset{2,4}{1}512321\mathfrak{e}_7$

$$\begin{aligned}
& 151321 \overset{1}{\otimes} \overset{2,4}{\mathfrak{e}_7}, \\
& 5,5 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& I^{\oplus 5} \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 21513221 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 1513151321 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 4,4 \overset{3,4}{\otimes} \mathfrak{e}_7, \\
& 31321 \overset{3,4}{\otimes} \mathfrak{e}_7, \\
& 1512321 \overset{3,4}{\otimes} \mathfrak{e}_7, \\
& 4,4 \overset{4,4}{\otimes} \mathfrak{e}_7, \\
& 31321 \overset{4,4}{\otimes} \mathfrak{e}_7, \\
& 1512321 \overset{4,4}{\otimes} \mathfrak{e}_7, \\
& 5,4 \overset{5,4}{\otimes} \mathfrak{e}_7, \\
& 2151321 \overset{5,4}{\otimes} \mathfrak{e}_7, \\
& 151321 \overset{5,4}{\otimes} \mathfrak{e}_7, \\
& 1513221 \overset{3221}{\mathfrak{e}_7}, \\
& 5131513221 \overset{321}{\mathfrak{e}_7}, \\
& 2231321 \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 231512321 \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 31513221 \overset{1}{\mathfrak{e}_7} \overset{I^{\oplus 2}}, \\
& I^{\oplus 5} \overset{I^{\oplus 3}}{\mathfrak{e}_7}, \\
& 21513221 \overset{I^{\oplus 3}}{\mathfrak{e}_7}, \\
& 1513151321 \overset{I^{\oplus 3}}{\mathfrak{e}_7}, \\
& 31321 \overset{I^{\oplus 4}}{\mathfrak{e}_7}, \\
& 1512321 \overset{I^{\oplus 4}}{\mathfrak{e}_7}, \\
& 2151321 \overset{31321}{\mathfrak{e}_7}, \\
& 151321 \overset{1}{\mathfrak{e}_7} \overset{31321}, \\
& 513151321 \overset{512321}{\mathfrak{e}_7},
\end{aligned}$$

$$\begin{aligned}
& 3,5 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 313221 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 32151321 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 5123151321 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 1513221 \overset{1}{\mathfrak{e}_7} \overset{3,3}, \\
& 5,4 \overset{3,4}{\otimes} \mathfrak{e}_7, \\
& 2151321 \overset{3,4}{\otimes} \mathfrak{e}_7, \\
& 151321 \overset{1}{\mathfrak{e}_7} \overset{3,4}, \\
& 5,4 \overset{4,4}{\otimes} \mathfrak{e}_7, \\
& 2151321 \overset{4,4}{\otimes} \mathfrak{e}_7, \\
& 151321 \overset{1}{\mathfrak{e}_7} \overset{4,4}, \\
& I^{\oplus 4} \overset{5,4}{\otimes} \mathfrak{e}_7, \\
& 2321 \overset{1}{\mathfrak{e}_7} \overset{5,4}, \\
& I^{\oplus 3} \overset{5,5}{\otimes} \mathfrak{e}_7, \\
& I^{\oplus 2} \overset{2313221}{\mathfrak{e}_7}, \\
& I^{\oplus 3} \overset{231321}{\mathfrak{e}_7}, \\
& 321513221 \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 51231513221 \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 23151321 \overset{1}{\mathfrak{e}_7} \overset{I^{\oplus 2}}, \\
& 32151321 \overset{I^{\oplus 3}}{\mathfrak{e}_7}, \\
& 5123151321 \overset{I^{\oplus 3}}{\mathfrak{e}_7}, \\
& 1513221 \overset{1}{\mathfrak{e}_7} \overset{I^{\oplus 3}}, \\
& 2151321 \overset{I^{\oplus 4}}{\mathfrak{e}_7}, \\
& 151321 \overset{1}{\mathfrak{e}_7} \overset{I^{\oplus 4}}, \\
& 2321 \overset{1}{\mathfrak{e}_7} \overset{31321}, \\
& 3151321 \overset{513221}{\mathfrak{e}_7}, \\
& 1513221 \overset{512321}{\mathfrak{e}_7},
\end{aligned}$$

$$\begin{aligned}
& 4,5 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 231321 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 31512321 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
& 3151321 \overset{1}{\mathfrak{e}_7} \overset{3,3}, \\
& 3,4 \overset{3,4}{\otimes} \mathfrak{e}_7, \\
& I^{\oplus 4} \overset{3,4}{\otimes} \mathfrak{e}_7, \\
& 2321 \overset{1}{\mathfrak{e}_7} \overset{3,4}, \\
& I^{\oplus 3} \overset{3,5}{\otimes} \mathfrak{e}_7, \\
& I^{\oplus 4} \overset{4,4}{\otimes} \mathfrak{e}_7, \\
& 2321 \overset{1}{\mathfrak{e}_7} \overset{4,4}, \\
& I^{\oplus 3} \overset{4,5}{\otimes} \mathfrak{e}_7, \\
& 31321 \overset{5,4}{\otimes} \mathfrak{e}_7, \\
& 1512321 \overset{5,4}{\otimes} \mathfrak{e}_7, \\
& 513151321 \overset{3221}{\mathfrak{e}_7}, \\
& I^{\oplus 3} \overset{313221}{\mathfrak{e}_7}, \\
& I^{\oplus 6} \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 232151321 \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 31513151321 \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 15131513221 \overset{I^{\oplus 2}}{\mathfrak{e}_7}, \\
& 31512321 \overset{I^{\oplus 3}}{\mathfrak{e}_7}, \\
& 3151321 \overset{1}{\mathfrak{e}_7} \overset{I^{\oplus 3}}, \\
& I^{\oplus 4} \overset{I^{\oplus 4}}{\mathfrak{e}_7}, \\
& 2321 \overset{1}{\mathfrak{e}_7} \overset{I^{\oplus 4}}, \\
& 31321 \overset{31321}{\mathfrak{e}_7}, \\
& 1512321 \overset{31321}{\mathfrak{e}_7}, \\
& 51321 \overset{31513221}{\mathfrak{e}_7}, \\
& 2151321 \overset{2151321}{\mathfrak{e}_7}
\end{aligned}$$

$$2321 \overset{1}{\mathfrak{e}_7} \overset{2151321}{},$$

$$151321 \overset{513151321}{\mathfrak{e}_7},$$

$$151321 \overset{1}{\mathfrak{e}_7} \overset{2321}{},$$

$$151321 \overset{1}{\mathfrak{e}_7} \overset{1512321}{},$$

$$1512321 \overset{2151321}{\mathfrak{e}_7},$$

$$2321 \overset{1}{\mathfrak{e}_7} \overset{2321}{},$$

$$151321 \overset{1513221}{\mathfrak{e}_7},$$

$$151321 \overset{1}{\mathfrak{e}_7} \overset{151321}{},$$

$$151321 \overset{1}{\mathfrak{e}_7} \overset{2151321}{},$$

$$1512321 \overset{1}{\mathfrak{e}_7} \overset{2321}{},$$

$$1512321 \overset{1512321}{\mathfrak{e}_7},$$

$$\overset{3,5}{\otimes} \overset{1}{\mathfrak{e}_6},$$

$$313221 \overset{1}{\mathfrak{e}_6},$$

$$223131 \overset{1}{\mathfrak{e}_6},$$

$$23215131 \overset{1}{\mathfrak{e}_6},$$

$$3151315131 \overset{1}{\mathfrak{e}_6},$$

$$1513151321 \overset{1}{\mathfrak{e}_6},$$

$$\overset{3,4}{\otimes} \overset{2,2}{\mathfrak{e}_6},$$

$$I^{\oplus 4} \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$3215131 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$\overset{1}{\mathfrak{e}_6} \overset{2,2}{\otimes},$$

$$151315131 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$\overset{3,3}{\otimes} \overset{2,3}{\mathfrak{e}_6},$$

$$\overset{3,3}{\otimes} \overset{2,3}{\mathfrak{e}_6},$$

$$215131 \overset{2,3}{\otimes} \mathfrak{e}_6,$$

$$\overset{3,3}{\otimes} \overset{3,3}{\mathfrak{e}_6},$$

$$\overset{3,3}{\otimes} \overset{3,3}{\mathfrak{e}_6},$$

$$215131 \overset{3,3}{\otimes} \mathfrak{e}_6,$$

$$\overset{4,3}{\otimes} \overset{4,3}{\mathfrak{e}_6},$$

$$I^{\oplus 3} \overset{4,3}{\otimes} \mathfrak{e}_6,$$

$$\overset{1}{\mathfrak{e}_6} \overset{4,3}{\otimes},$$

$$\overset{3,3}{\otimes} \overset{5,3}{\mathfrak{e}_6},$$

$$\overset{4,5}{\otimes} \overset{1}{\mathfrak{e}_6},$$

$$231321 \overset{1}{\mathfrak{e}_6},$$

$$32151321 \overset{1}{\mathfrak{e}_6},$$

$$21513221 \overset{1}{\mathfrak{e}_6},$$

$$3151321 \overset{1}{\mathfrak{e}_6},$$

$$1513221 \overset{1}{\mathfrak{e}_6},$$

$$\overset{4,4}{\otimes} \overset{2,2}{\mathfrak{e}_6},$$

$$31321 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$2151321 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$\overset{1}{\mathfrak{e}_6} \overset{2,2}{\otimes},$$

$$151321 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$\overset{4,3}{\otimes} \overset{2,3}{\mathfrak{e}_6},$$

$$I^{\oplus 3} \overset{2,3}{\otimes} \mathfrak{e}_6,$$

$$\overset{1}{\mathfrak{e}_6} \overset{2,3}{\otimes},$$

$$\overset{4,3}{\otimes} \overset{3,3}{\mathfrak{e}_6},$$

$$I^{\oplus 3} \overset{3,3}{\otimes} \mathfrak{e}_6,$$

$$\overset{1}{\mathfrak{e}_6} \overset{3,3}{\otimes},$$

$$\overset{5,3}{\otimes} \overset{4,3}{\mathfrak{e}_6},$$

$$3131 \overset{4,3}{\otimes} \mathfrak{e}_6,$$

$$I^{\oplus 2} \overset{4,4}{\otimes} \mathfrak{e}_6,$$

$$I^{\oplus 3} \overset{5,3}{\otimes} \mathfrak{e}_6,$$

$$\overset{5,5}{\otimes} \overset{1}{\mathfrak{e}_6},$$

$$I^{\oplus 5} \overset{1}{\mathfrak{e}_6},$$

$$31512321 \overset{1}{\mathfrak{e}_6},$$

$$5123151321 \overset{1}{\mathfrak{e}_6},$$

$$2315131 \overset{1}{\mathfrak{e}_6},$$

$$\overset{2,4}{\otimes} \overset{2,2}{\mathfrak{e}_6},$$

$$\overset{5,4}{\otimes} \overset{2,2}{\mathfrak{e}_6},$$

$$23131 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$512315131 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$1512321 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$\overset{2,3}{\otimes} \overset{2,3}{\mathfrak{e}_6},$$

$$\overset{5,3}{\otimes} \overset{2,3}{\mathfrak{e}_6},$$

$$3131 \overset{2,3}{\otimes} \mathfrak{e}_6,$$

$$I^{\oplus 2} \overset{2,4}{\otimes} \mathfrak{e}_6,$$

$$\overset{5,3}{\otimes} \overset{3,3}{\mathfrak{e}_6},$$

$$3131 \overset{3,3}{\otimes} \mathfrak{e}_6,$$

$$I^{\oplus 2} \overset{3,4}{\otimes} \mathfrak{e}_6,$$

$$\overset{3,3}{\otimes} \overset{4,3}{\mathfrak{e}_6},$$

$$215131 \overset{4,3}{\otimes} \mathfrak{e}_6,$$

$$\overset{5,3}{\otimes} \overset{5,3}{\mathfrak{e}_6},$$

$$3131 \overset{5,3}{\otimes} \mathfrak{e}_6,$$

$$\begin{aligned}
& 215131 \overset{5,3}{\otimes} \mathfrak{e}_6, \\
& 321 \overset{3221}{\mathfrak{e}_6}, \\
& 51315131 \overset{321}{\mathfrak{e}_6}, \\
& \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& 215131 \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& 31321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 2151321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{1}{315131} \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{1}{151321} \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 215131 \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& 1513221 \overset{31}{\mathfrak{e}_6}, \\
& \overset{1}{15131} \overset{3131}{\mathfrak{e}_6}, \\
& 5131 \overset{3151321}{\mathfrak{e}_6}, \\
& 15131 \overset{51315131}{\mathfrak{e}_6},
\end{aligned}$$

$$\begin{aligned}
& \overset{1}{15131} \overset{5,3}{\otimes} \mathfrak{e}_6, \\
& 15131 \overset{3221}{\mathfrak{e}_6}, \\
& 151321 \overset{321}{\mathfrak{e}_6}, \\
& \overset{3,3}{I^{\oplus 3}} \mathfrak{e}_6, \\
& \overset{1}{15131} \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& 23131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 512315131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 1512321 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& \overset{1}{15131} \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& 3131 \overset{3131}{\mathfrak{e}_6}, \\
& 315131 \overset{51321}{\mathfrak{e}_6}, \\
& 215131 \overset{215131}{\mathfrak{e}_6}, \\
& 151321 \overset{15131}{\mathfrak{e}_6},
\end{aligned}$$

$$\begin{aligned}
& \overset{5,4}{I^{\oplus 2}} \overset{\otimes}{\mathfrak{e}_6}, \\
& 512321 \overset{321}{\mathfrak{e}_6}, \\
& 51321 \overset{2321}{\mathfrak{e}_6}, \\
& 3131 \overset{3,3}{\bigcirc} \mathfrak{e}_6, \\
& \overset{I^{\oplus 4}}{\mathfrak{e}_6}, \\
& 3215131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& \overset{1}{2321} \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 151315131 \overset{I^{\oplus 2}}{\mathfrak{e}_6}, \\
& 3131 \overset{I^{\oplus 3}}{\mathfrak{e}_6}, \\
& 513151321 \overset{31}{\mathfrak{e}_6}, \\
& 215131 \overset{3131}{\mathfrak{e}_6}, \\
& 15131 \overset{512321}{\mathfrak{e}_6}, \\
& \overset{1}{15131} \overset{215131}{\mathfrak{e}_6}, \\
& \overset{1}{15131} \overset{15131}{\mathfrak{e}_6},
\end{aligned}$$

$$\begin{aligned}
& \overset{I^{\oplus 11}}{\mathfrak{e}_8}, \\
& \overset{2,4}{I^{\oplus 8}} \overset{\otimes}{\mathfrak{e}_8}, \\
& 2321513221 \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& 22313221 \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& 315131513221 \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& \overset{5,5}{I^{\oplus 7}} \overset{\otimes}{\mathfrak{e}_8}, \\
& 231513221 \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& 321513221 \overset{2313221}{\mathfrak{e}_8}, \\
& 15131513221 \overset{2313221}{\mathfrak{e}_8}, \\
& 21513221 \overset{22313221}{\mathfrak{e}_8}, \\
& 2321513221 \overset{313221}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 10}}{\mathfrak{e}_8}, \\
& 2315131513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8},
\end{aligned}$$

$$\begin{aligned}
& 2315131513221 \overset{2,4}{\otimes} \mathfrak{e}_8, \\
& 22313221 \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& 315131513221 \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& \overset{4,5}{I^{\oplus 7}} \mathfrak{e}_8, \\
& 231513221 \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& 2321513221 \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& 2313221 \overset{2313221}{\mathfrak{e}_8}, \\
& 51231513221 \overset{2313221}{\mathfrak{e}_8}, \\
& 313221 \overset{22313221}{\mathfrak{e}_8}, \\
& \overset{1}{1513221} \overset{22313221}{\mathfrak{e}_8}, \\
& 315131513221 \overset{313221}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 9}}{\mathfrak{e}_8}, \\
& 2231513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8},
\end{aligned}$$

$$\begin{aligned}
& 2231513221 \overset{2,4}{\otimes} \mathfrak{e}_8, \\
& \overset{3,5}{I^{\oplus 7}} \overset{\otimes}{\mathfrak{e}_8}, \\
& 231513221 \overset{3,5}{\otimes} \mathfrak{e}_8, \\
& 2321513221 \overset{4,5}{\otimes} \mathfrak{e}_8, \\
& 22313221 \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& 315131513221 \overset{5,5}{\otimes} \mathfrak{e}_8, \\
& \overset{I^{\oplus 6}}{\mathfrak{e}_8}, \\
& 31513221 \overset{2313221}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 5}}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 7}}{\mathfrak{e}_8}, \\
& 231513221 \overset{313221}{\mathfrak{e}_8}, \\
& 22315131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8}, \\
& \overset{I^{\oplus 8}}{\mathfrak{e}_8},
\end{aligned}$$

$$\begin{array}{lll}
I^{\oplus 7} \mathfrak{e}_8, & 2321513221 \mathfrak{e}_8^{I^{\oplus 5}}, & 315131513221 \mathfrak{e}_8^{I^{\oplus 5}} \\
231513221 \mathfrak{e}_8^{I^{\oplus 5}}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 6}}, & 321513221 \mathfrak{e}_8^{I^{\oplus 6}} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 6}}, & 31513221 \mathfrak{e}_8^{I^{\oplus 6}}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 6}} \\
21513221 \mathfrak{e}_8^{I^{\oplus 7}}, & 1513221 \mathfrak{e}_8^{I^{\oplus 7}}, & 321513221 \mathfrak{e}_8^{321513221} \\
51231513221 \mathfrak{e}_8^{321513221}, & 31513221 \mathfrak{e}_8^{321513221}, & 15131513221 \mathfrak{e}_8^{321513221} \\
21513221 \mathfrak{e}_8^{2321513221}, & 1513221 \mathfrak{e}_8^{2321513221}, & 315131513221 \mathfrak{e}_8^{21513221} \\
231513221 \mathfrak{e}_8^{21513221}, & 51231513221 \mathfrak{e}_8^{51231513221}, & 31513221 \mathfrak{e}_8^{51231513221} \\
15131513221 \mathfrak{e}_8^{51231513221}, & 1513221 \mathfrak{e}_8^{315131513221}, & 31513221 \mathfrak{e}_8^{31513221} \\
15131513221 \mathfrak{e}_8^{31513221}, & 1513221 \mathfrak{e}_8^{231513221}, & 15131513221 \mathfrak{e}_8^{15131513221}
\end{array}$$

$$\begin{array}{lll}
2313221 \mathfrak{e}'_7, & I^{\oplus 6} \mathfrak{e}'_7, & 2231321 \mathfrak{e}'_7 \\
321513221 \mathfrak{e}'_7, & 232151321 \mathfrak{e}'_7, & 231512321 \mathfrak{e}'_7 \\
51231513221 \mathfrak{e}'_7, & 31513151321 \mathfrak{e}'_7, & 31513221 \mathfrak{e}'_7 \\
23151321 \mathfrak{e}'_7, & 15131513221 \mathfrak{e}'_7, & \begin{array}{c} 2,3 \\ 2,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array} \\
\begin{array}{c} 2,3 \\ 3,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 2,3 \\ 4,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 2,3 \\ 5,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array} \\
I^{\oplus 4} \mathfrak{e}'_7, & \begin{array}{c} 2,3 \\ \otimes \end{array}, & \begin{array}{c} 2,3 \\ \otimes \end{array} \\
\begin{array}{c} 2,3 \\ \otimes \end{array}, & \begin{array}{c} 2,3 \\ \otimes \end{array}, & \begin{array}{c} 2,3 \\ \otimes \end{array} \\
2321 \mathfrak{e}'_7, & 1512321 \mathfrak{e}'_7, & 151321 \mathfrak{e}'_7 \\
\begin{array}{c} 2,4 \\ 3,3 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 2,4 \\ I^{\oplus 3} \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 3,3 \\ 3,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array} \\
\begin{array}{c} 3,3 \\ 4,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 3,3 \\ 5,4 \otimes \\ \otimes \mathfrak{e}'_7 \end{array}, & \begin{array}{c} 3,3 \\ I^{\oplus 4} \mathfrak{e}'_7 \end{array} \\
\begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,3 \\ \otimes \end{array} \\
31321 \mathfrak{e}'_7, & 2151321 \mathfrak{e}'_7, & 2321 \mathfrak{e}'_7 \\
\begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,3 \\ \otimes \end{array}, & \begin{array}{c} 3,4 \\ \otimes \end{array} \\
1512321 \mathfrak{e}'_7, & 151321 \mathfrak{e}'_7, & I^{\oplus 3} \mathfrak{e}'_7 \\
\begin{array}{c} 3,5 \\ \otimes \end{array}, & \begin{array}{c} 4,4 \\ \otimes \end{array}, & \begin{array}{c} 4,5 \\ \otimes \end{array} \\
I^{\oplus 2} \mathfrak{e}'_7, & I^{\oplus 3} \mathfrak{e}'_7, & I^{\oplus 2} \mathfrak{e}'_7 \\
\begin{array}{c} 5,4 \\ \otimes \end{array}, & \begin{array}{c} 5,5 \\ \otimes \end{array}, & \\
I^{\oplus 3} \mathfrak{e}'_7, & I^{\oplus 2} \mathfrak{e}'_7, & \begin{array}{c} 3221 \\ 3221 \mathfrak{e}'_7 \end{array} \\
512321 \mathfrak{e}'_7, & 151321 \mathfrak{e}'_7, & I^{\oplus 2} \mathfrak{e}'_7 \\
513151321 \mathfrak{e}'_7, & 1513221 \mathfrak{e}'_7, & 513221 \mathfrak{e}'_7 \\
I^{\oplus 2} \mathfrak{e}'_7, & I^{\oplus 5} \mathfrak{e}'_7, & 32151321 \mathfrak{e}'_7
\end{array}$$

$$\begin{aligned}
& 31512321 \overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
& 3151321 \overset{1}{\overset{I^{\oplus 2}}{\mathfrak{e}'_7}}, \\
& I^{\oplus 4} \overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
& 2321 \overset{1}{\overset{I^{\oplus 3}}{\mathfrak{e}'_7}}, \\
& 3151321 \overset{51321}{\mathfrak{e}'_7}, \\
& 151321 \overset{151321}{\mathfrak{e}'_7}
\end{aligned}$$

$$\begin{aligned}
& 21513221 \overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
& 1513151321 \overset{I^{\oplus 2}}{\mathfrak{e}'_7}, \\
& 31321 \overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
& 1512321 \overset{I^{\oplus 3}}{\mathfrak{e}'_7}, \\
& 512321 \overset{512321}{\mathfrak{e}'_7},
\end{aligned}$$

$$\begin{aligned}
& 5123151321 \overset{I^{\oplus 2}}{\mathfrak{e}'_7} \\
& 1513221 \overset{1}{\overset{I^{\oplus 2}}{\mathfrak{e}'_7}} \\
& 2151321 \overset{I^{\oplus 3}}{\mathfrak{e}'_7} \\
& 151321 \overset{1}{\overset{I^{\oplus 3}}{\mathfrak{e}'_7}} \\
& 151321 \overset{512321}{\mathfrak{e}'_7}
\end{aligned}$$

$$\begin{aligned}
& \overset{2,3}{\otimes} \overset{1}{\mathfrak{so}}, \\
& \overset{5,3}{\otimes} \overset{1}{\mathfrak{so}}, \\
& 3131 \overset{1}{\mathfrak{so}}, \\
& 215131 \overset{1}{\mathfrak{so}}, \\
& 15131 \overset{1}{\mathfrak{so}}, \\
& \overset{4,2}{\otimes} \overset{2,2}{\mathfrak{so}}, \\
& \overset{3,2}{\otimes} \overset{3,2}{\mathfrak{so}}, \\
& I^{\oplus 2} \overset{3,2}{\otimes} \mathfrak{so}, \\
& I^{\oplus 2} \overset{4,2}{\otimes} \mathfrak{so}, \\
& I^{\oplus 2} \overset{I^{\oplus 2}}{\mathfrak{so}}, \\
& 5131 \overset{231}{\mathfrak{so}}
\end{aligned}$$

$$\begin{aligned}
& \overset{3,3}{\otimes} \overset{1}{\mathfrak{so}}, \\
& \overset{3,3}{\bigcirc} \overset{1}{\mathfrak{so}}, \\
& 151231 \overset{1}{\mathfrak{so}}, \\
& 231 \overset{1}{\overset{1}{\mathfrak{so}}}, \\
& \overset{2,2}{\otimes} \overset{2,2}{\mathfrak{so}}, \\
& 231 \overset{2}{\overset{2,2}{\otimes} \mathfrak{so}}, \\
& \overset{4,2}{\otimes} \overset{3,2}{\mathfrak{so}}, \\
& \overset{4,2}{\otimes} \overset{4,2}{\mathfrak{so}}, \\
& I^{\oplus 2} \overset{2}{\overset{231}{\mathfrak{so}}}, \\
& 51231 \overset{31}{\mathfrak{so}},
\end{aligned}$$

$$\begin{aligned}
& \overset{4,3}{\otimes} \overset{1}{\mathfrak{so}} \\
& I^{\oplus 3} \overset{1}{\mathfrak{so}} \\
& 512231 \overset{1}{\mathfrak{so}} \\
& 1231 \overset{2}{\overset{1}{\mathfrak{so}}} \\
& \overset{3,2}{\otimes} \overset{2,2}{\mathfrak{so}} \\
& I^{\oplus 2} \overset{2,2}{\otimes} \mathfrak{so} \\
& 231 \overset{2}{\overset{3,2}{\otimes} \mathfrak{so}} \\
& 231 \overset{2}{\overset{4,2}{\otimes} \mathfrak{so}} \\
& 31 \overset{321}{\mathfrak{so}} \\
& 15131 \overset{31}{\mathfrak{so}}
\end{aligned}$$

2.2.3 SSgS(333)

$$I^{\oplus 10} \overset{1}{\mathfrak{e}_8} 1,$$

$$2321513221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$51231513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$51231513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$2313221 \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$22313221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$315131513221 \overset{2,4}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$31513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$31513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 6} \overset{5,5}{\otimes} \mathfrak{e}_8 1,$$

$$I^{\oplus 7} \overset{2,4}{\otimes} \mathfrak{e}_8 1$$

$$231513221 \overset{1}{\overset{2,4}{\otimes} \mathfrak{e}_8} 1$$

$$321513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1$$

$$15131513221 \overset{3,5}{\otimes} \mathfrak{e}_8 1$$

$$321513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1$$

$$15131513221 \overset{4,5}{\otimes} \mathfrak{e}_8 1$$

$$321513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1$$

$$\begin{array}{lll}
51231513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1, & 31513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1, & 15131513221 \overset{5,5}{\otimes} \mathfrak{e}_8 1 \\
313221 \overset{2313221}{\mathfrak{e}_8} 1, & I^{\oplus 5} \overset{2313221}{\mathfrak{e}_8} 1, & 21513221 \overset{2313221}{\mathfrak{e}_8} 1 \\
1513221 \overset{1}{\mathfrak{e}_8} \overset{2313221}{1}, & I^{\oplus 4} \overset{22313221}{\mathfrak{e}_8} 1, & I^{\oplus 6} \overset{313221}{\mathfrak{e}_8} 1 \\
321513221 \overset{313221}{\mathfrak{e}_8} 1, & 51231513221 \overset{313221}{\mathfrak{e}_8} 1, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{313221}{1} \\
15131513221 \overset{313221}{\mathfrak{e}_8} 1, & I^{\oplus 9} \overset{I^{\oplus 2}}{\mathfrak{e}_8} 1, & 22315131513221 \overset{I^{\oplus 2}}{\mathfrak{e}_8} 1 \\
2315131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1, & 2231513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 3}}{1}, & I^{\oplus 8} \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1 \\
I^{\oplus 7} \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1, & 2321513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1, & 315131513221 \overset{I^{\oplus 4}}{\mathfrak{e}_8} 1 \\
231513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 4}}{1}, & I^{\oplus 6} \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1, & 321513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1 \\
51231513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 5}}{1}, & 15131513221 \overset{I^{\oplus 5}}{\mathfrak{e}_8} 1 \\
21513221 \overset{I^{\oplus 6}}{\mathfrak{e}_8} 1, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 6}}{1}, & 21513221 \overset{321513221}{\mathfrak{e}_8} 1 \\
1513221 \overset{1}{\mathfrak{e}_8} \overset{321513221}{1}, & 51231513221 \overset{21513221}{\mathfrak{e}_8} 1, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{21513221}{1} \\
15131513221 \overset{21513221}{\mathfrak{e}_8} 1, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{51231513221}{1}, & 5131513221 \overset{5131513221}{\mathfrak{e}_8} 1 \\
1513221 \overset{1}{\mathfrak{e}_8} \overset{31513221}{1}, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{15131513221}{1}, & \overset{2,4}{\otimes} \overset{2,4}{\mathfrak{e}_8} \overset{4,2}{\otimes} \\
I^{\oplus 4} \overset{2,4}{\mathfrak{e}_8} \overset{4,2}{\otimes}, & I^{\oplus 3} \overset{3,5}{\mathfrak{e}_8} \overset{4,2}{\otimes}, & I^{\oplus 3} \overset{4,5}{\mathfrak{e}_8} \overset{4,2}{\otimes} \\
I^{\oplus 3} \overset{5,5}{\mathfrak{e}_8} \overset{4,2}{\otimes}, & 1513221 \overset{32214,2}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{23132214,2}{\mathfrak{e}_8} \otimes \\
I^{\oplus 3} \overset{3132214,2}{\mathfrak{e}_8} \otimes, & I^{\oplus 6} \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes, & 321513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes \\
51231513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes, & 31513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 24,2}}{\otimes}, & 15131513221 \overset{I^{\oplus 24,2}}{\mathfrak{e}_8} \otimes \\
I^{\oplus 5} \overset{I^{\oplus 34,2}}{\mathfrak{e}_8} \otimes, & 21513221 \overset{I^{\oplus 34,2}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 34,2}}{\otimes} \\
I^{\oplus 4} \overset{I^{\oplus 44,2}}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3,5}{\mathfrak{e}_8} \overset{5,3}{\otimes}, & I^{\oplus 2} \overset{4,5}{\mathfrak{e}_8} \overset{5,3}{\otimes} \\
I^{\oplus 2} \overset{5,5}{\mathfrak{e}_8} \overset{5,3}{\otimes}, & 3221 \overset{32215,3}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3132215,3}{\mathfrak{e}_8} \otimes \\
I^{\oplus 5} \overset{I^{\oplus 25,3}}{\mathfrak{e}_8} \otimes, & 21513221 \overset{I^{\oplus 25,3}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 25,3}}{\otimes} \\
I^{\oplus 4} \overset{I^{\oplus 35,3}}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{4,5}{\mathfrak{e}_8} \overset{5,4}{\otimes}, & I^{\oplus 2} \overset{5,5}{\mathfrak{e}_8} \overset{5,4}{\otimes} \\
3221 \overset{32215,4}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3132215,4}{\mathfrak{e}_8} \otimes, & I^{\oplus 5} \overset{I^{\oplus 25,4}}{\mathfrak{e}_8} \otimes \\
21513221 \overset{I^{\oplus 25,4}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 25,4}}{\otimes}, & I^{\oplus 4} \overset{I^{\oplus 35,4}}{\mathfrak{e}_8} \otimes \\
I^{\oplus 2} \overset{5,5}{\mathfrak{e}_8} \overset{5,5}{\otimes}, & 3221 \overset{32215,5}{\mathfrak{e}_8} \otimes, & I^{\oplus 2} \overset{3132215,5}{\mathfrak{e}_8} \otimes \\
I^{\oplus 5} \overset{I^{\oplus 25,5}}{\mathfrak{e}_8} \otimes, & 21513221 \overset{I^{\oplus 25,5}}{\mathfrak{e}_8} \otimes, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{I^{\oplus 25,5}}{\otimes} \\
I^{\oplus 4} \overset{I^{\oplus 35,5}}{\mathfrak{e}_8} \otimes, & 313221 \overset{3221}{\mathfrak{e}_8} 1223, & I^{\oplus 5} \overset{3221}{\mathfrak{e}_8} 1223 \\
21513221 \overset{3221}{\mathfrak{e}_8} 1223, & 1513221 \overset{1}{\mathfrak{e}_8} \overset{3221}{1223}, & 5131513221 \overset{I^{\oplus 3}}{\mathfrak{e}_8} 1223
\end{array}$$

$$\begin{array}{lll}
1513221 \mathfrak{e}_8^{I^{\oplus 4}} 1223, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} 1223132, & I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} 1223132 \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} 12231322, & I^{\oplus 2} \mathfrak{e}_8^{313221} 122313, & I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2}} 122313 \\
21513221 \mathfrak{e}_8^{I^{\oplus 2}} 122313, & \mathfrak{e}_8^{11513221} I^{\oplus 2} 122313, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}} 122313 \\
2315131513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}, & 2231513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}, & I^{\oplus 8} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2} \\
I^{\oplus 7} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & 2321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & 315131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2} \\
231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2}, & 321513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2}, & \mathfrak{e}_8^{131513221} I^{\oplus 4} I^{\oplus 2}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 2} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 5}} I^{\oplus 2}, & 21513221 \mathfrak{e}_8^{I^{\oplus 5}} I^{\oplus 2}, & \mathfrak{e}_8^{11513221} I^{\oplus 5} I^{\oplus 2} \\
21513221 \mathfrak{e}_8^{21513221} I^{\oplus 2}, & \mathfrak{e}_8^{11513221} I^{\oplus 2} I^{\oplus 2}, & 1513221 \mathfrak{e}_8^{5131513221} I^{\oplus 2} \\
\mathfrak{e}_8^{11513221} I^{\oplus 2} I^{\oplus 2}, & I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3}, & 321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3} \\
51231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3}, & \mathfrak{e}_8^{131513221} I^{\oplus 3} I^{\oplus 3}, & 15131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3} \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 3}, & 21513221 \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 3}, & \mathfrak{e}_8^{11513221} I^{\oplus 4} I^{\oplus 3} \\
31513221 \mathfrak{e}_8^{513221} I^{\oplus 3}, & 1513221 \mathfrak{e}_8^{1513221} I^{\oplus 3}, & I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 4}} I^{\oplus 4}
\end{array}$$

$$\begin{array}{lll}
2313221 \mathfrak{e}_7^1 1, & I^{\oplus 6} \mathfrak{e}_7^1 1, & 2231321 \mathfrak{e}_7^1 1 \\
321513221 \mathfrak{e}_7^1 1, & 232151321 \mathfrak{e}_7^1 1, & 231512321 \mathfrak{e}_7^1 1 \\
51231513221 \mathfrak{e}_7^1 1, & 31513151321 \mathfrak{e}_7^1 1, & \mathfrak{e}_7^{131513221} 1 \\
23151321 \mathfrak{e}_7^1 1, & 15131513221 \mathfrak{e}_7^1 1, & \mathfrak{e}_7^{2,4 \otimes 2,3} 1 \\
\mathfrak{e}_7^{3,4 \otimes 2,3} 1, & \mathfrak{e}_7^{4,4 \otimes 2,3} 1, & \mathfrak{e}_7^{5,4 \otimes 2,3} 1 \\
I^{\oplus 4} \mathfrak{e}_7^{2,3 \otimes} 1, & 31321 \mathfrak{e}_7^{2,3 \otimes} 1, & 2151321 \mathfrak{e}_7^{2,3 \otimes} 1 \\
\mathfrak{e}_7^{12321 \otimes 2,3} 1, & 1512321 \mathfrak{e}_7^{2,3 \otimes} 1, & \mathfrak{e}_7^{1151321 \otimes 2,3} 1 \\
\mathfrak{e}_7^{3,3 \otimes 2,4} 1, & I^{\oplus 3} \mathfrak{e}_7^{2,4 \otimes} 1, & \mathfrak{e}_7^{3,4 \otimes 3,3} 1 \\
\mathfrak{e}_7^{4,4 \otimes 3,3} 1, & \mathfrak{e}_7^{5,4 \otimes 3,3} 1, & I^{\oplus 4} \mathfrak{e}_7^{3,3 \otimes} 1 \\
31321 \mathfrak{e}_7^{3,3 \otimes} 1, & 2151321 \mathfrak{e}_7^{3,3 \otimes} 1, & \mathfrak{e}_7^{12321 \otimes 3,3} 1 \\
1512321 \mathfrak{e}_7^{3,3 \otimes} 1, & \mathfrak{e}_7^{1151321 \otimes 3,3} 1, & I^{\oplus 3} \mathfrak{e}_7^{3,4 \otimes} 1 \\
I^{\oplus 2} \mathfrak{e}_7^{3,5 \otimes} 1, & I^{\oplus 3} \mathfrak{e}_7^{4,4 \otimes} 1, & I^{\oplus 2} \mathfrak{e}_7^{4,5 \otimes} 1 \\
I^{\oplus 3} \mathfrak{e}_7^{5,4 \otimes} 1, & I^{\oplus 2} \mathfrak{e}_7^{5,5 \otimes} 1, & 3221 \mathfrak{e}_7^{3221} 1 \\
512321 \mathfrak{e}_7^{3221} 1, & 151321 \mathfrak{e}_7^{3221} 1, & I^{\oplus 2} \mathfrak{e}_7^{313221} 1
\end{array}$$

$513151321 \mathfrak{e}_7^{321} 1,$	$1513221 \mathfrak{e}_7^{321} 1,$	$513221 \mathfrak{e}_7^{2321} 1$
$I^{\oplus 2} \mathfrak{e}_7^{231321} 1,$	$I^{\oplus 5} \mathfrak{e}_7^{I^{\oplus 2}},$	$32151321 \mathfrak{e}_7^{I^{\oplus 2}} 1$
$31512321 \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$21513221 \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$5123151321 \mathfrak{e}_7^{I^{\oplus 2}} 1$
$\overset{1}{3151321} \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$1513151321 \mathfrak{e}_7^{I^{\oplus 2}} 1,$	$\overset{1}{1513221} \mathfrak{e}_7^{I^{\oplus 2}} 1$
$I^{\oplus 4} \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$31321 \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$2151321 \mathfrak{e}_7^{I^{\oplus 3}} 1$
$\overset{1}{2321} \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$1512321 \mathfrak{e}_7^{I^{\oplus 3}} 1,$	$\overset{1}{151321} \mathfrak{e}_7^{I^{\oplus 3}} 1$
$3151321 \mathfrak{e}_7^{51321} 1,$	$512321 \mathfrak{e}_7^{512321} 1,$	$151321 \mathfrak{e}_7^{512321} 1$
$151321 \mathfrak{e}_7^{151321} 1,$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 2,3,2},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 3,3,2}$
$321 \mathfrak{e}_7^{\otimes 321,2},$	$I^{\oplus 3} \mathfrak{e}_7^{\otimes 23,2},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,2}$
$I^{\oplus 2} \mathfrak{e}_7^{\otimes 3,3,3},$	$321 \mathfrak{e}_7^{\otimes 321,3,3},$	$I^{\oplus 3} \mathfrak{e}_7^{\otimes 23,3}$
$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,3},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,4},$	$I^{\oplus 2} \mathfrak{e}_7^{\otimes 24,5}$
$I^{\oplus 2} \mathfrak{e}_7^{321} 1223,$	$I^{\oplus 3} \mathfrak{e}_7^{321} 123,$	$512321 \mathfrak{e}_7^{I^{\oplus 2}} 123$
$151321 \mathfrak{e}_7^{I^{\oplus 2}} 123,$	$51321 \mathfrak{e}_7^{I^{\oplus 2}} 1232,$	$I^{\oplus 4} \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}$
$31321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$2151321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$\overset{1}{2321} \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2}$
$1512321 \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$\overset{1}{151321} \mathfrak{e}_7^{I^{\oplus 2}} I^{\oplus 2},$	$I^{\oplus 3} \mathfrak{e}_7^{I^{\oplus 3}} I^{\oplus 2}$
$\overset{2,4}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{3,4}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{4,4}{\otimes} \mathfrak{e}_6^1 1$
$\overset{5,4}{\otimes} \mathfrak{e}_6^1 1,$	$I^{\oplus 4} \mathfrak{e}_6^1 1,$	$31321 \mathfrak{e}_6^1 1$
$23131 \mathfrak{e}_6^1 1,$	$3215131 \mathfrak{e}_6^1 1,$	$2151321 \mathfrak{e}_6^1 1$
$512315131 \mathfrak{e}_6^1 1,$	$\overset{1}{2321} \mathfrak{e}_6^1 1,$	$\overset{1}{315131} \mathfrak{e}_6^1 1$
$1512321 \mathfrak{e}_6^1 1,$	$151315131 \mathfrak{e}_6^1 1,$	$\overset{1}{151321} \mathfrak{e}_6^1 1$
$\overset{2,3}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{3,3}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{4,3}{\otimes} \mathfrak{e}_6^1 1$
$\overset{5,3}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{3,3}{\otimes} \mathfrak{e}_6^1 1,$	$I^{\oplus 3} \mathfrak{e}_6^1 1$
$\overset{2,2}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{2,2}{\otimes} \mathfrak{e}_6^1 1,$	$\overset{1}{15131} \mathfrak{e}_6^1 1$
$I^{\oplus 2} \mathfrak{e}_6^1 1,$	$I^{\oplus 2} \mathfrak{e}_6^1 1,$	$I^{\oplus 2} \mathfrak{e}_6^1 1$
$I^{\oplus 2} \mathfrak{e}_6^1 1,$	$31 \mathfrak{e}_6^1 1,$	$321 \mathfrak{e}_6^1 1$
$15131 \mathfrak{e}_6^1 1,$	$5131 \mathfrak{e}_6^1 1,$	$I^{\oplus 2} \mathfrak{e}_6^1 1$
$I^{\oplus 3} \mathfrak{e}_6^1 1,$	$3131 \mathfrak{e}_6^1 1,$	$215131 \mathfrak{e}_6^1 1$

$15131 \overset{1}{\mathfrak{e}_6} I^{\oplus 2},$	$512321 \overset{31}{\mathfrak{e}_6} 1,$	$51315131 \overset{31}{\mathfrak{e}_6} 1$
$151321 \overset{31}{\mathfrak{e}_6} 1,$	$315131 \overset{5131}{\mathfrak{e}_6} 1,$	$15131 \overset{15131}{\mathfrak{e}_6} 1$
$\overset{2,2}{2,2} \otimes \overset{2,2}{2,2} \otimes \mathfrak{e}_6 \otimes,$	$I^{\oplus 2} \overset{2,2}{2,2} \otimes \mathfrak{e}_6 \otimes,$	$31 \overset{321^{2,2}}{\mathfrak{e}_6} \otimes$
$I^{\oplus 2} \overset{I^{\oplus 2,2}}{\mathfrak{e}_6} \otimes,$	$15131 \overset{31^{2,2}}{\mathfrak{e}_6} \otimes,$	$31 \overset{31^{3,2}}{\mathfrak{e}_6} \otimes$
$31 \overset{31^{3,3}}{\mathfrak{e}_6} \otimes,$	$31 \overset{31^{3,4}}{\mathfrak{e}_6} \otimes,$	$31 \overset{31^{3,5}}{\mathfrak{e}_6} \otimes$
$31 \overset{I^{\oplus 2}}{\mathfrak{e}_6} 123,$	$31 \overset{31^{3,3}}{\mathfrak{e}_6} \bigcirc,$	$I^{\oplus 2} \overset{I^{\oplus 2}}{\mathfrak{e}_6} I^{\oplus 2}$
$15131 \overset{31}{\mathfrak{e}_6} I^{\oplus 2},$	$31 \overset{31}{\mathfrak{e}_6} I^{\oplus 3},$	$3131 \overset{31}{\mathfrak{e}_6} 13$
$215131 \overset{31}{\mathfrak{e}_6} 13,$	$15131 \overset{31}{\mathfrak{e}_6} 13$	
$\overset{3,5}{3,5} \overset{1}{\mathfrak{e}_7'} 1,$	$\overset{4,5}{4,5} \overset{1}{\mathfrak{e}_7'} 1,$	$\overset{5,5}{5,5} \overset{1}{\mathfrak{e}_7'} 1$
$313221 \overset{1}{\mathfrak{e}_7'} 1,$	$231321 \overset{1}{\mathfrak{e}_7'} 1,$	$I^{\oplus 5} \overset{1}{\mathfrak{e}_7'} 1$
$32151321 \overset{1}{\mathfrak{e}_7'} 1,$	$31512321 \overset{1}{\mathfrak{e}_7'} 1,$	$21513221 \overset{1}{\mathfrak{e}_7'} 1$
$5123151321 \overset{1}{\mathfrak{e}_7'} 1,$	$3151321 \overset{1}{\mathfrak{e}_7'} 1,$	$1513151321 \overset{1}{\mathfrak{e}_7'} 1$
$1513221 \overset{1}{\mathfrak{e}_7'} 1,$	$\overset{2,3}{2,3} \otimes \overset{2,3}{2,3} \otimes \mathfrak{e}_7' 1,$	$\overset{2,3}{3,3} \otimes \overset{2,3}{3,3} \otimes \mathfrak{e}_7' 1$
$I^{\oplus 3} \overset{2,3}{\mathfrak{e}_7'} 1,$	$I^{\oplus 2} \overset{2,4}{\mathfrak{e}_7'} 1,$	$\overset{3,3}{3,3} \otimes \overset{3,3}{3,3} \otimes \mathfrak{e}_7' 1$
$I^{\oplus 3} \overset{3,3}{\mathfrak{e}_7'} 1,$	$I^{\oplus 2} \overset{3,4}{\mathfrak{e}_7'} 1,$	$I^{\oplus 2} \overset{4,4}{\mathfrak{e}_7'} 1$
$I^{\oplus 2} \overset{5,4}{\mathfrak{e}_7'} 1,$	$321 \overset{3221}{\mathfrak{e}_7'} 1,$	$512321 \overset{321}{\mathfrak{e}_7'} 1$
$151321 \overset{321}{\mathfrak{e}_7'} 1,$	$51321 \overset{2321}{\mathfrak{e}_7'} 1,$	$I^{\oplus 4} \overset{I^{\oplus 2}}{\mathfrak{e}_7'} 1$
$31321 \overset{I^{\oplus 2}}{\mathfrak{e}_7'} 1,$	$2151321 \overset{I^{\oplus 2}}{\mathfrak{e}_7'} 1,$	$\overset{1}{2321} \overset{I^{\oplus 2}}{\mathfrak{e}_7'} 1$
$1512321 \overset{I^{\oplus 2}}{\mathfrak{e}_7'} 1,$	$151321 \overset{1}{\mathfrak{e}_7'} I^{\oplus 2},$	$I^{\oplus 3} \overset{I^{\oplus 3}}{\mathfrak{e}_7'} 1$
$I^{\oplus 2} \overset{I^{\oplus 2,3,2}}{\mathfrak{e}_7'} \otimes,$	$I^{\oplus 2} \overset{I^{\oplus 2,3,3}}{\mathfrak{e}_7'} \otimes,$	$I^{\oplus 2} \overset{321}{\mathfrak{e}_7'} 123$
$I^{\oplus 3} \overset{I^{\oplus 2}}{\mathfrak{e}_7'} I^{\oplus 2}$		
$\overset{2,2}{2,2} \overset{1}{\mathfrak{s}\mathfrak{o}} 1,$	$\overset{3,2}{3,2} \overset{1}{\mathfrak{s}\mathfrak{o}} 1,$	$\overset{4,2}{4,2} \overset{1}{\mathfrak{s}\mathfrak{o}} 1$
$\overset{2}{231} \overset{1}{\mathfrak{s}\mathfrak{o}} 1,$	$I^{\oplus 2} \overset{1}{\mathfrak{s}\mathfrak{o}} 1,$	$31 \overset{31}{\mathfrak{s}\mathfrak{o}} 1$

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$I^{\oplus 9} \overset{1}{\mathfrak{e}_8} 1,$	$22315131513221 \overset{1}{\mathfrak{e}_8} 1,$	$2313221 \overset{1}{\mathfrak{e}_8} \overset{4,2}{\otimes} 1$
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$$\begin{array}{l}
I^{\oplus 6} \mathfrak{e}_8^{1\ 4,2} \otimes, \\
31513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{4,5\ 1\ 5,3} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 5,3} \otimes, \\
\otimes \mathfrak{e}_8^{4,5\ 1\ 5,4} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 5,4} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ 1\ 5,5} \otimes, \\
21513221 \mathfrak{e}_8^{1\ 5,5} \otimes, \\
22313221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
21513221 \mathfrak{e}_8^{1\ 122313}, \\
2231513221 \mathfrak{e}_8^{I^{\oplus 2}} 1, \\
2321513221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
I^{\oplus 6} \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
31513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
21513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 5}, \\
1513221 \mathfrak{e}_8^{1\ 1} 12231512, \\
\otimes \mathfrak{e}_8^{2,4\ I^{\oplus 3} 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3} 4,2} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,5} \otimes, \\
1513221 \mathfrak{e}_8^{I^{\oplus 3}} 1223, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} 122313,
\end{array}
\begin{array}{l}
321513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
15131513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ 1\ 5,3} \otimes, \\
21513221 \mathfrak{e}_8^{1\ 5,3} \otimes, \\
\otimes \mathfrak{e}_8^{5,5\ 1\ 5,4} \otimes, \\
21513221 \mathfrak{e}_8^{1\ 5,4} \otimes, \\
313221 \mathfrak{e}_8^{1\ 5,5} \otimes, \\
1513221 \mathfrak{e}_8^{1\ 5,5} \otimes, \\
313221 \mathfrak{e}_8^{1\ 122313}, \\
1513221 \mathfrak{e}_8^{1\ 1} 122313, \\
I^{\oplus 8} \mathfrak{e}_8^{I^{\oplus 2}} 1, \\
315131513221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
321513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
15131513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
1513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 5}, \\
1513221 \mathfrak{e}_8^{1\ 1} 1223151315, \\
\otimes \mathfrak{e}_8^{3,5\ I^{\oplus 2} 4,2} \otimes, \\
3221 \mathfrak{e}_8^{1\ 1} 1223, \\
\otimes \mathfrak{e}_8^{2,4\ I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,3} \otimes, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{1\ 1} 1223, \\
I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} 1223132, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} 122313,
\end{array}
\begin{array}{l}
51231513221 \mathfrak{e}_8^{1\ 4,2} \otimes, \\
\otimes \mathfrak{e}_8^{3,5\ 1\ 5,3} \otimes, \\
313221 \mathfrak{e}_8^{1\ 5,3} \otimes, \\
1513221 \mathfrak{e}_8^{1\ 5,3} \otimes, \\
313221 \mathfrak{e}_8^{1\ 5,4} \otimes, \\
1513221 \mathfrak{e}_8^{1\ 5,4} \otimes, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 5,5} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{1\ 1} 1223132, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 1} 122313, \\
2315131513221 \mathfrak{e}_8^{I^{\oplus 2}} 1, \\
I^{\oplus 7} \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
231513221 \mathfrak{e}_8^{I^{\oplus 3}} 1, \\
51231513221 \mathfrak{e}_8^{1\ 1} I^{\oplus 4}, \\
I^{\oplus 5} \mathfrak{e}_8^{1\ 1} I^{\oplus 5}, \\
21513221 \mathfrak{e}_8^{1\ 1} 12231512, \\
1513221 \mathfrak{e}_8^{1\ 1} 1223151, \\
\otimes \mathfrak{e}_8^{4,5\ I^{\oplus 2} 4,2} \otimes, \\
313221 \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes, \\
1513221 \mathfrak{e}_8^{I^{\oplus 2} 4,2} \otimes, \\
I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,3} \otimes, \\
I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,5} \otimes, \\
5131513221 \mathfrak{e}_8^{I^{\oplus 2}} 1223, \\
I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}} 12231322, \\
I^{\oplus 7} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}
\end{array}$$

$$2321513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$21513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2}} 1223151,$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 6} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$21513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$315131513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$321513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$15131513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 4},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$321513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$15131513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$231513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$51231513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2}} 122315,$$

$$21513221 \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 3},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 24,2}} \otimes$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} 1223$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$51231513221 \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2},$$

$$I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 4},$$

$$\otimes \mathfrak{e}_7^{1,3,5} 1,$$

$$313221 \mathfrak{e}_7^{1,1},$$

$$32151321 \mathfrak{e}_7^{1,1},$$

$$5123151321 \mathfrak{e}_7^{1,1},$$

$$1513221 \mathfrak{e}_7^{1,1},$$

$$\otimes \mathfrak{e}_7^{1,2,3} 1,$$

$$\otimes \mathfrak{e}_7^{1,3,3} 1,$$

$$\otimes \mathfrak{e}_7^{1,5,4} 1,$$

$$151321 \mathfrak{e}_7^{1,1} 123,$$

$$31321 \mathfrak{e}_7^{1,1} 1,$$

$$\otimes \mathfrak{e}_7^{1,4,5} 1,$$

$$231321 \mathfrak{e}_7^{1,1},$$

$$31512321 \mathfrak{e}_7^{1,1},$$

$$3151321 \mathfrak{e}_7^{1,1},$$

$$\otimes \mathfrak{e}_7^{1,2,3} 1,$$

$$\otimes \mathfrak{e}_7^{1,2,4} 1,$$

$$\otimes \mathfrak{e}_7^{1,2,4} 1,$$

$$321 \mathfrak{e}_7^{1,1} 1223,$$

$$51321 \mathfrak{e}_7^{1,1} 1232,$$

$$2151321 \mathfrak{e}_7^{1,1} 1,$$

$$\otimes \mathfrak{e}_7^{1,5,5} 1,$$

$$I^{\oplus 5} \mathfrak{e}_7^{1,1},$$

$$21513221 \mathfrak{e}_7^{1,1},$$

$$1513151321 \mathfrak{e}_7^{1,1},$$

$$\otimes \mathfrak{e}_7^{1,3,3} 1,$$

$$\otimes \mathfrak{e}_7^{1,3,3} 1,$$

$$\otimes \mathfrak{e}_7^{1,4,4} 1,$$

$$512321 \mathfrak{e}_7^{1,1} 123$$

$$I^{\oplus 4} \mathfrak{e}_7^{1,1} 1$$

$$2321 \mathfrak{e}_7^{1,1} 1$$

$$\begin{aligned}
&1512321 \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}_7 \\ 1 \end{smallmatrix} 1, \\
&I^{\oplus 2} \begin{smallmatrix} I^{\oplus 23,2} \\ \mathfrak{e}_7 \\ 1 \end{smallmatrix} \otimes, \\
&I^{\oplus 2} \begin{smallmatrix} I^{\oplus 3} \\ \mathfrak{e}_7 \\ 1 \end{smallmatrix} I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
&151321 \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}_7 \\ 1 \end{smallmatrix} 1, \\
&I^{\oplus 2} \begin{smallmatrix} I^{\oplus 23,3} \\ \mathfrak{e}_7 \\ 1 \end{smallmatrix} \otimes, \\
&I^{\oplus 2} \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}_7 \\ I^{\oplus 2} \end{smallmatrix} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
&I^{\oplus 3} \begin{smallmatrix} I^{\oplus 3} \\ \mathfrak{e}_7 \\ 1 \end{smallmatrix} 1 \\
&321 \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}_7 \\ 1 \end{smallmatrix} 123
\end{aligned}$$

$$\begin{smallmatrix} 2,3 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} 3,3 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} 4,3 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1$$

$$\begin{smallmatrix} 5,3 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} 3,3 \\ \bigcirc \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} I^{\oplus 3} \\ 1 \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1$$

$$3131 \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$215131 \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$15131 \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1$$

$$\begin{smallmatrix} 2,2 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} \begin{smallmatrix} 2,2 \\ \otimes \\ 1 \end{smallmatrix},$$

$$\begin{smallmatrix} 2,2 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$31 \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 123$$

$$I^{\oplus 2} \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 1,$$

$$15131 \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 13,$$

$$31 \begin{smallmatrix} 1 \\ \mathfrak{e}_6 \\ 2,2 \\ \otimes \\ 1 \end{smallmatrix} 13$$

$$31 \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}_6 \\ 1 \end{smallmatrix} 13$$

$$\begin{smallmatrix} 2,4 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} 3,4 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} 4,4 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1$$

$$\begin{smallmatrix} 5,4 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$I^{\oplus 4} \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$31321 \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1$$

$$2151321 \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$2321 \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$1512321 \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1$$

$$151321 \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} 2,3 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$\begin{smallmatrix} 3,3 \\ \otimes \\ 1 \end{smallmatrix} \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1$$

$$321 \begin{smallmatrix} 1 \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 123,$$

$$I^{\oplus 2} \begin{smallmatrix} I^{\oplus 3} \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} 1,$$

$$I^{\oplus 2} \begin{smallmatrix} I^{\oplus 2} \\ \mathfrak{e}'_7 \\ 1 \end{smallmatrix} I^{\oplus 2}$$

$$1501 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix}$$

2.3 2 node(1504)

2.3.1 gLg(4)

$$\begin{smallmatrix} 3,3 \\ \otimes \\ \mathfrak{so} \end{smallmatrix} \otimes \mathfrak{so},$$

$$\begin{smallmatrix} 3,3 \\ \bigcirc \\ \mathfrak{so} \end{smallmatrix} \otimes \mathfrak{so}$$

$$\begin{smallmatrix} 3,5 \\ \otimes \\ \mathfrak{so} \end{smallmatrix} \otimes \mathfrak{e}_6$$

$$\begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_6 \end{smallmatrix} \otimes \mathfrak{e}_6$$

2.3.2 111gLg(3)

$$\begin{smallmatrix} 1 & 4,3 \\ 1\mathfrak{e}_7 & \otimes \mathfrak{so} \\ 1 \end{smallmatrix}$$

$$\begin{smallmatrix} 1 & 4,5 \\ 1\mathfrak{e}_7 & \otimes \mathfrak{e}_6 \\ 1 \end{smallmatrix}$$

$$\begin{smallmatrix} 1 & 3,3 \\ 1\mathfrak{e}'_7 & \otimes \mathfrak{so} \\ 1 \end{smallmatrix}$$

2.3.3 SgLg(115)

$$513151321\mathfrak{e}_7 \otimes \mathfrak{so},$$

$$2231\mathfrak{so} \otimes \mathfrak{e}_7,$$

$$\begin{smallmatrix} 2,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix}$$

$$\begin{smallmatrix} 5,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix}$$

$$2151321\mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\begin{smallmatrix} 1 & 3,3 \\ 151321\mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix}$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{so},$$

$$1513221\mathfrak{e}_7 \otimes \mathfrak{so},$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\begin{smallmatrix} 3,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix}$$

$$I^{\oplus 4}\mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\begin{smallmatrix} 1 & 3,3 \\ 2321\mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix}$$

$$\begin{smallmatrix} 2,3 & 4,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}, \end{smallmatrix}$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{so}$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{so}$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\begin{smallmatrix} 4,4 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so} \end{smallmatrix}$$

$$31321\mathfrak{e}_7 \otimes \mathfrak{so}$$

$$1512321\mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\begin{smallmatrix} 3,3 & 4,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{so} \end{smallmatrix}$$

$$3221\mathfrak{e}_6 \otimes \mathfrak{so},$$

$$51315131\mathfrak{e}_6 \otimes \mathfrak{so},$$

$$231\mathfrak{so} \otimes \mathfrak{e}_6,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6 \otimes \mathfrak{so},$$

$$2231\mathfrak{so} \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_6 \otimes \mathfrak{so},$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{so},$$

$$\begin{smallmatrix} 2,2 & 3,3 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{so}, \end{smallmatrix}$$

$$\begin{smallmatrix} 2,2 & 3,3 \\ \otimes \mathfrak{e}_6 & \bigcirc \mathfrak{so}, \end{smallmatrix}$$

$$512321\mathfrak{e}_6 \otimes \mathfrak{so}$$

$$321\mathfrak{e}_6 \otimes \mathfrak{so}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_6 \bigcirc \mathfrak{so}$$

$$31513221\mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$2315131\mathfrak{e}_6 \otimes \mathfrak{e}_7,$$

$$315131\mathfrak{e}_6 \otimes \mathfrak{e}_7,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$22315131\mathfrak{e}_6 \otimes \mathfrak{e}_7,$$

$$2321\mathfrak{e}_6 \otimes \mathfrak{e}_7,$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$\begin{smallmatrix} 2,3 & 4,5 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, \end{smallmatrix}$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$3151321\mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$2321\mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$\begin{smallmatrix} 3,3 & 4,5 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6 \end{smallmatrix}$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$3221\mathfrak{e}'_7 \otimes \mathfrak{so},$$

$$2231\mathfrak{so} \otimes \mathfrak{e}'_7,$$

$$512321\mathfrak{e}'_7 \otimes \mathfrak{so}$$

$$\begin{aligned}
&151321\mathfrak{e}'_7 \overset{3,2}{\otimes} \mathfrak{so}, \\
&\overset{2,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{so}, \\
&I^{\oplus 2}\mathfrak{e}'_7 \overset{4,3}{\otimes} \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
&321\mathfrak{e}'_7 \overset{4,2}{\otimes} \mathfrak{so}, \\
&\overset{3,3}{\otimes} \mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{so}, \\
&1\mathfrak{e}'_7 \overset{5,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
&231\mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}'_7 \\
&I^{\oplus 3}\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
&3151321\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}_6, \\
&2321\mathfrak{e}'_7 \overset{4,3}{\otimes} \mathfrak{e}_6, \\
&15131\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}'_7, \\
&31\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}'_7,
\end{aligned}$$

$$\begin{aligned}
&2315131\mathfrak{e}_6 \overset{3,3}{\otimes} \mathfrak{e}'_7, \\
&315131\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}'_7, \\
&321\mathfrak{e}'_7 \overset{4,4}{\otimes} \mathfrak{e}_6, \\
&1\mathfrak{e}'_7 \overset{5,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&2321\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}'_7 \\
&321\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}'_7 \\
&I^{\oplus 2}\mathfrak{e}'_7 \overset{4,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&23151321\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6, \\
&321\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}_6, \\
&I^{\oplus 2}\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_6, \\
&2321\mathfrak{e}_6 \overset{3,3}{\bigcirc} \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
&2321\mathfrak{e}_6 \overset{3,3}{\otimes} \mathfrak{e}_6, \\
&15131\mathfrak{e}_6 \overset{3,4}{\otimes} \mathfrak{e}_6, \\
&31\mathfrak{e}_6 \overset{4,4}{\otimes} \mathfrak{e}_6, \\
&315131\mathfrak{e}_6 \overset{3,3}{\bigcirc} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&315131\mathfrak{e}_6 \overset{3,3}{\otimes} \mathfrak{e}_6 \\
&\overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_6 \\
&1\mathfrak{e}_6 \overset{4,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&231513221\mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}'_7, \\
&3221\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}'_7, \\
&512321\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}'_7,
\end{aligned}$$

$$\begin{aligned}
&223151321\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
&2321\mathfrak{e}'_7 \overset{4,5}{\otimes} \mathfrak{e}_7, \\
&151321\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}'_7,
\end{aligned}$$

$$\begin{aligned}
&3151321\mathfrak{e}_7 \overset{4,4}{\otimes} \mathfrak{e}'_7 \\
&2321\mathfrak{e}_7 \overset{5,4}{\otimes} \mathfrak{e}'_7 \\
&321\mathfrak{e}_7 \overset{5,5}{\otimes} \mathfrak{e}'_7
\end{aligned}$$

$$\begin{aligned}
&2313221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6, \\
&51231513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
&I^{\oplus 6}\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6, \\
&\overset{1}{31513221}\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6,
\end{aligned}$$

$$\begin{aligned}
&321513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6 \\
&15131513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6
\end{aligned}$$

$$\begin{aligned}
&2313221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so}, \\
&51231513221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
&I^{\oplus 6}\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so}, \\
&\overset{1}{31513221}\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so},
\end{aligned}$$

$$\begin{aligned}
&321513221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so} \\
&15131513221\mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$\begin{aligned}
&2321\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}, \\
&1\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{so}
\end{aligned}$$

$$315131\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so},$$

$$31\mathfrak{so} \overset{2,2}{\otimes} \mathfrak{so}$$

$$\begin{aligned}
&2231513221\mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7, \\
&2321\mathfrak{e}_7 \overset{5,5}{\otimes} \mathfrak{e}_7
\end{aligned}$$

$$23151321\mathfrak{e}_7 \overset{4,4}{\otimes} \mathfrak{e}_7,$$

$$3151321\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_7$$

$$23151321\mathfrak{e}'_7 \overset{3,3}{\otimes} \mathfrak{e}'_7,$$

$$2321\mathfrak{e}'_7 \overset{4,4}{\otimes} \mathfrak{e}'_7,$$

$$321\mathfrak{e}'_7 \overset{4,5}{\otimes} \mathfrak{e}'_7$$

2.3.4 SgLgS(552)

$$1\epsilon_7 \otimes \epsilon_7^{3,3} 122315132,$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,2},$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5},$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \epsilon_7 1231512,$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \epsilon_7 123151,$$

$$\otimes \epsilon_7^{3,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,3},$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7^{4,3},$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7^{4,3},$$

$$\otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4},$$

$$31321\epsilon_7^{3,3} \otimes \epsilon_7^{4,4},$$

$$\otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,3} \epsilon_7 1232151,$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7^{4,5},$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7^{4,5},$$

$$3221\epsilon_7^{3,3} \otimes \epsilon_7 122315,$$

$$I^{\oplus 3} \epsilon_7^{3,3} \otimes \epsilon_7 1223151,$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7 1231512,$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} 123151,$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7 12313,$$

$$513221\epsilon_7^{3,3} \otimes \epsilon_7 123215,$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7 1231512,$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 1232,$$

$$1512321\epsilon_7^{3,3} \otimes \epsilon_7 1232151,$$

$$1\epsilon_7^{4,4} \otimes \epsilon_7 1231513,$$

$$I^{\oplus 3} \epsilon_7^{4,4} \otimes \epsilon_7^{3,2},$$

$$I^{\oplus 2} \epsilon_7^{4,4} \otimes \epsilon_7 1223,$$

$$I^{\oplus 2} \epsilon_7^{4,4} \otimes \epsilon_7 123151,$$

$$2321\epsilon_7^{5,4} \otimes \epsilon_7 1,$$

$$I^{\oplus 2} \epsilon_7^{5,4} \otimes \epsilon_7^{3,2},$$

$$I^{\oplus 2} \epsilon_7^{5,4} \otimes \epsilon_7 I^{\oplus 3},$$

$$\otimes \epsilon_7^{2,3} \otimes \epsilon_7^{3,3} 123151315,$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,3},$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7^{4,2},$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \epsilon_7 1232,$$

$$\otimes \epsilon_7^{3,3} \otimes \epsilon_7 123151315,$$

$$\otimes \epsilon_7^{3,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4},$$

$$31321\epsilon_7^{3,3} \otimes \epsilon_7^{4,3},$$

$$\otimes \epsilon_7^{3,4} \otimes \epsilon_7^{3,3} \epsilon_7 1232151,$$

$$\otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5},$$

$$\otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,3} \epsilon_7 1231512,$$

$$\otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,3} \epsilon_7 123151,$$

$$31321\epsilon_7^{3,3} \otimes \epsilon_7^{4,5},$$

$$5,4 \epsilon_7^{3,3} \otimes \epsilon_7 1232151,$$

$$I^{\oplus 2} \epsilon_7^{3,3} \otimes \epsilon_7 12231513,$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7 I^{\oplus 4},$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} 1232,$$

$$31321\epsilon_7^{3,3} \otimes \epsilon_7 12313,$$

$$31321\epsilon_7^{3,3} \otimes \epsilon_7 1232151,$$

$$151321\epsilon_7^{3,3} \otimes \epsilon_7 122315,$$

$$1512321\epsilon_7^{3,3} \otimes \epsilon_7 1231512,$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7 1232151,$$

$$151321\epsilon_7^{1,3,3} \otimes \epsilon_7 1232151,$$

$$\otimes \epsilon_7^{2,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,2},$$

$$\otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,3},$$

$$321\epsilon_7^{4,4} \otimes \epsilon_7 12315,$$

$$I^{\oplus 3} \epsilon_7^{4,4} \otimes \epsilon_7 I^{\oplus 3},$$

$$1\epsilon_7^{5,4} \otimes \epsilon_7 123215,$$

$$I^{\oplus 2} \epsilon_7^{5,4} \otimes \epsilon_7^{3,3},$$

$$1\epsilon_7^{5,5} \otimes \epsilon_7 123,$$

$$\otimes \epsilon_7^{2,3} \otimes \epsilon_7^{3,3} 1223151$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,4}$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \epsilon_7 12313$$

$$\otimes \epsilon_7^{2,4} \otimes \epsilon_7^{3,3} \epsilon_7 1232151$$

$$\otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \epsilon_7 1223151$$

$$\otimes \epsilon_7^{3,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5}$$

$$\otimes \epsilon_7^{3,4} \otimes \epsilon_7^{3,3} \epsilon_7 1231512$$

$$\otimes \epsilon_7^{3,4} \otimes \epsilon_7^{3,3} \epsilon_7^{1,3,3} 123151$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7^{4,4}$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7^{4,4}$$

$$\otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5}$$

$$\otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \epsilon_7 1231512$$

$$\otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \epsilon_7^{1,3,3} 123151$$

$$I^{\oplus 3} \epsilon_7^{3,3} \otimes \epsilon_7 123151315$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7 12313$$

$$I^{\oplus 4} \epsilon_7^{3,3} \otimes \epsilon_7 1232151$$

$$31321\epsilon_7^{3,3} \otimes \epsilon_7 1231512$$

$$31321\epsilon_7^{3,3} \otimes \epsilon_7^{1,3,3} 123151$$

$$2151321\epsilon_7^{3,3} \otimes \epsilon_7 1231512$$

$$151321\epsilon_7^{1,3,3} \otimes \epsilon_7 1231512$$

$$2321\epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 123151$$

$$151321\epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 123151$$

$$\otimes \epsilon_7^{2,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7^{3,3}$$

$$I^{\oplus 3} \epsilon_7^{4,4} \otimes \epsilon_7^{3,3}$$

$$I^{\oplus 2} \epsilon_7^{4,4} \otimes \epsilon_7 123215$$

$$1\epsilon_7^{5,4} \otimes \epsilon_7 1223$$

$$1\epsilon_7^{5,4} \otimes \epsilon_7 123151$$

$$321\epsilon_7^{5,4} \otimes \epsilon_7 I^{\oplus 2}$$

$$I^{\oplus 2} \epsilon_7^{5,5} \otimes \epsilon_7 I^{\oplus 2}$$

$$\begin{aligned}
& 151321\mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7{}^{3,2}, \\
& \otimes \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7{}^{4,2}, \\
& \otimes \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7{}^{4,4}, \\
& \otimes \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 I^{\oplus 4}, \\
& \otimes \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 1231512, \\
& \otimes \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 1232151, \\
& I^{\oplus 3} \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7{}^{4,4}, \\
& 51321\mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 1223, \\
& I^{\oplus 2} \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 123151315, \\
& I^{\oplus 3} \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 12313, \\
& I^{\oplus 3} \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 1232, \\
& I^{\oplus 3} \mathbf{e}'_7{}^{3,3} \otimes \mathbf{e}_7 123151, \\
& 1\mathbf{e}'_7{}^{4,4} \otimes \mathbf{e}_7 1223, \\
& 1\mathbf{e}'_7{}^{4,4} \otimes \mathbf{e}_7 123151, \\
& 321\mathbf{e}'_7{}^{4,4} \otimes \mathbf{e}_7 I^{\oplus 2}, \\
& 1\mathbf{e}'_7{}^{5,4} \otimes \mathbf{e}_7{}^{3,3}, \\
& 1\mathbf{e}'_7{}^{5,4} \otimes \mathbf{e}_7 I^{\oplus 3},
\end{aligned}$$

$$\begin{aligned}
& \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7 1232151, \\
& I^{\oplus 3} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7, \\
& \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7, \\
& \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7 12313, \\
& \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7 1232, \\
& \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7 123151, \\
& I^{\oplus 3} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7, \\
& 321 \mathbf{e}'_7 \otimes \mathbf{e}_7 122315, \\
& I^{\oplus 2} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7 1223151, \\
& 512321 \mathbf{e}'_7 \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& 151321 \mathbf{e}'_7 \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& 51321 \mathbf{e}'_7 \otimes \mathbf{e}_7 123215, \\
& 2321 \mathbf{e}'_7 \otimes \mathbf{e}_7 1, \\
& I^{\oplus 2} \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7, \\
& I^{\oplus 2} \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7 I^{\oplus 3}, \\
& 1 \mathbf{e}'_7 \otimes \mathbf{e}_7 123, \\
& I^{\oplus 2} \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \otimes \mathbf{e}_7 I^{\oplus 2}.
\end{aligned}$$

$$\begin{array}{l}
\begin{array}{c} 2,3 \\ \otimes \end{array} \begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 1 \\ \mathfrak{e}_7 \end{array} 123151 \\
\begin{array}{c} 3,3 \\ \otimes \end{array} \begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 4,3 \\ \mathfrak{e}_7 \end{array} \otimes \\
3221\begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 3,3 \\ \mathfrak{e}_7 \end{array} \otimes \\
512321\begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 3,3 \\ \mathfrak{e}_7 \end{array} \otimes \\
151321\begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 3,3 \\ \mathfrak{e}_7 \end{array} \otimes \\
I^{\oplus 3} \begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 4,3 \\ \mathfrak{e}_7 \end{array} \otimes \\
3221\begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 I^{\oplus 3} \\
3151321\begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 I^{\oplus 2} \\
I^{\oplus 3} \begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 I^{\oplus 4} \\
I^{\oplus 3} \begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 1231512 \\
I^{\oplus 3} \begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 1232151 \\
51321\begin{array}{c} 3,3 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 123151 \\
1\begin{array}{c} 4,4 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 123215 \\
I^{\oplus 2} \begin{array}{c} 4,4 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 3,3 \\ \mathfrak{e}_7 \end{array} \otimes \\
1\begin{array}{c} 5,4 \\ \mathfrak{e}_7' \end{array} \otimes \begin{array}{c} 3,2 \\ \mathfrak{e}_7 \end{array} \otimes \\
321\begin{array}{c} 4,5 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 1 \\
1\begin{array}{c} 5,5 \\ \mathfrak{e}_7' \end{array} \otimes \mathfrak{e}_7 I^{\oplus 2}
\end{array}$$

$$\begin{array}{lll}
321\mathbf{e}'_7 \otimes \mathbf{e}_8 I^{\oplus 5}, & 321\mathbf{e}'_7 \otimes \mathbf{e}_8 12231512, & 321\mathbf{e}'_7 \otimes \mathbf{e}_8 1223151^1 \\
2321\mathbf{e}'_7 \otimes \mathbf{e}_8 I^{\oplus 4}, & I^{\oplus 2}\mathbf{e}'_7 \otimes \mathbf{e}_8 I^{\oplus 6}, & I^{\oplus 2}\mathbf{e}'_7 \otimes \mathbf{e}_8 122315123 \\
I^{\oplus 2}\mathbf{e}'_7 \otimes \mathbf{e}_8 12231513215, & I^{\oplus 2}\mathbf{e}'_7 \otimes \mathbf{e}_8 12231513^1, & I^{\oplus 2}\mathbf{e}'_7 \otimes \mathbf{e}_8 12231513151 \\
1\mathbf{e}'_7 \otimes \mathbf{e}_8 1223132, & 1\mathbf{e}'_7 \otimes \mathbf{e}_8 I^{\oplus 6}, & 1\mathbf{e}'_7 \otimes \mathbf{e}_8 122315123 \\
1\mathbf{e}'_7 \otimes \mathbf{e}_8 12231513215, & 1\mathbf{e}'_7 \otimes \mathbf{e}_8 12231513^1, & 1\mathbf{e}'_7 \otimes \mathbf{e}_8 12231513151
\end{array}$$

$$\begin{array}{lll}
2313221\mathbf{e}_8 \otimes \mathbf{e}_8 1223132, & I^{\oplus 6}\mathbf{e}_8 \otimes \mathbf{e}_8 1223132, & 2313221\mathbf{e}_8 \otimes \mathbf{e}_8 122315123 \\
2313221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513215, & 2313221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513^1, & 2313221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513151 \\
I^{\oplus 6}\mathbf{e}_8 \otimes \mathbf{e}_8 I^{\oplus 6}, & I^{\oplus 6}\mathbf{e}_8 \otimes \mathbf{e}_8 122315123, & I^{\oplus 6}\mathbf{e}_8 \otimes \mathbf{e}_8 12231513215 \\
I^{\oplus 6}\mathbf{e}_8 \otimes \mathbf{e}_8 12231513^1, & I^{\oplus 6}\mathbf{e}_8 \otimes \mathbf{e}_8 12231513151, & 321513221\mathbf{e}_8 \otimes \mathbf{e}_8 122315123 \\
321513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513215, & 31513221\mathbf{e}_8 \otimes \mathbf{e}_8 122315123, & 321513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513151 \\
51231513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513215, & 31513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513215, & 15131513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513215 \\
31513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513^1, & 31513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513151, & 15131513221\mathbf{e}_8 \otimes \mathbf{e}_8 12231513151
\end{array}$$

$$\begin{array}{lll}
1\mathbf{so} \otimes \mathbf{e}_8 12231322, & 1\mathbf{so} \otimes \mathbf{e}_8 I^{\oplus 7}, & 1\mathbf{so} \otimes \mathbf{e}_8 1223151232 \\
1\mathbf{so} \otimes \mathbf{e}_8 122315131513, & 1\mathbf{so} \otimes \mathbf{e}_8 122315132^1, & 2231\mathbf{so} \otimes \mathbf{e}_8 \otimes \\
231\mathbf{so} \otimes \mathbf{e}_8 \otimes, & 231\mathbf{so} \otimes \mathbf{e}_8 \otimes, & 231\mathbf{so} \otimes \mathbf{e}_8 \otimes \\
31\mathbf{so} \otimes \mathbf{e}_8 1223132, & 231\mathbf{so} \otimes \mathbf{e}_8 122313, & 2231\mathbf{so} \otimes \mathbf{e}_8 I^{\oplus 4} \\
231\mathbf{so} \otimes \mathbf{e}_8 I^{\oplus 5}, & 31\mathbf{so} \otimes \mathbf{e}_8 I^{\oplus 6}, & 31\mathbf{so} \otimes \mathbf{e}_8 122315123 \\
31\mathbf{so} \otimes \mathbf{e}_8 12231513215, & 31\mathbf{so} \otimes \mathbf{e}_8 12231513^1, & 31\mathbf{so} \otimes \mathbf{e}_8 12231513151 \\
231\mathbf{so} \otimes \mathbf{e}_8 12231512, & 231\mathbf{so} \otimes \mathbf{e}_8 1223151^1, &
\end{array}$$

$$\begin{array}{lll}
1\mathbf{so} \otimes \mathbf{e}_7 \otimes, & 1\mathbf{so} \otimes \mathbf{e}_7 \otimes, & 1\mathbf{so} \otimes \mathbf{e}_7 \otimes \\
1\mathbf{so} \otimes \mathbf{e}_7 \otimes, & 1\mathbf{so} \otimes \mathbf{e}_7 I^{\oplus 4}, & 1\mathbf{so} \otimes \mathbf{e}_7 12313 \\
2231\mathbf{so} \otimes \mathbf{e}_7 1, & 1\mathbf{so} \otimes \mathbf{e}_7 1231512, & 1\mathbf{so} \otimes \mathbf{e}_7 1232^1 \\
1\mathbf{so} \otimes \mathbf{e}_7 1232151, & 1\mathbf{so} \otimes \mathbf{e}_7 123151^1, & 31\mathbf{so} \otimes \mathbf{e}_7 \otimes \\
31\mathbf{so} \otimes \mathbf{e}_7 \otimes, & 231\mathbf{so} \otimes \mathbf{e}_7 I^{\oplus 2}, & 31\mathbf{so} \otimes \mathbf{e}_7 I^{\oplus 3} \\
1\mathbf{so} \otimes \mathbf{e}_7 \otimes, & 1\mathbf{so} \otimes \mathbf{e}_7 \otimes, & 1\mathbf{so} \otimes \mathbf{e}_7 I^{\oplus 3} \\
231\mathbf{so} \otimes \mathbf{e}_7 1, & 31\mathbf{so} \otimes \mathbf{e}_7 I^{\oplus 2}
\end{array}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}123, \\
& \begin{smallmatrix} 2,2 & 1,1 & 2,2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& \begin{smallmatrix} 2,2 & 1,1 & 2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \mathbf{132}, \end{smallmatrix} \\
& \begin{smallmatrix} 3,2 & 1,1 & 2,4 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& \begin{smallmatrix} 4,2 & 1,1 & 2,4 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so}132, \\
& \mathbf{1so} \otimes \mathbf{so}1
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}13215, \\
& \begin{smallmatrix} 2,2 & 1,1 & 2,3 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} \otimes \mathbf{so}, \\
& \begin{smallmatrix} 3,2 & 1,1 & 2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \mathbf{132}, \end{smallmatrix} \\
& \begin{smallmatrix} 4,2 & 1,1 & 2 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \mathbf{132}, \end{smallmatrix} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{so}13151 \\
& \begin{smallmatrix} 2,2 & 1,1 & 2,4 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix} \\
& \begin{smallmatrix} 3,2 & 1,1 & 2,3 \\ \otimes \mathbf{so} & \otimes \mathbf{so} & \otimes \end{smallmatrix} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} \otimes \mathbf{so} \\
& I^{\oplus 2} \mathbf{so} \otimes \mathbf{so} \otimes \mathbf{so} \\
& \mathbf{31so} \otimes \mathbf{so}1315
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_61313, \\
& \mathbf{1so} \otimes \mathbf{e}_613151, \\
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_61
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_6 \bigcirc, \\
& 231\mathbf{so} \otimes \mathbf{e}_61, \\
& 31\mathbf{so} \otimes \mathbf{e}_6 \otimes, \\
& \mathbf{1so} \otimes \mathbf{e}_6 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}_6 \otimes \\
& \mathbf{1so} \otimes \mathbf{e}_6 I^{\oplus 3} \\
& \mathbf{1so} \otimes \mathbf{e}_6131512 \\
& 31\mathbf{so} \otimes \mathbf{e}_6 I^{\oplus 2} \\
& 31\mathbf{so} \otimes \mathbf{e}_61
\end{aligned}$$

$$\begin{aligned}
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_71231513, \\
& I^{\oplus 3} \mathbf{e}'_7 \otimes \mathbf{e}'_7 \otimes, \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_71223, \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_7123151, \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_7 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \begin{smallmatrix} 2,3 & 3,3 & 3,2 \\ \otimes \mathbf{e}'_7 & \otimes \mathbf{e}'_7 & \otimes \end{smallmatrix}, \\
& \begin{smallmatrix} 3,3 & 3,3 & 3,3 \\ \otimes \mathbf{e}'_7 & \otimes \mathbf{e}'_7 & \otimes \end{smallmatrix}, \\
& 321\mathbf{e}'_7 \otimes \mathbf{e}'_712315, \\
& I^{\oplus 3} \mathbf{e}'_7 \otimes \mathbf{e}'_7 I^{\oplus 3}, \\
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_7 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \begin{smallmatrix} 2,3 & 3,3 & 3,3 \\ \otimes \mathbf{e}'_7 & \otimes \mathbf{e}'_7 & \otimes \end{smallmatrix} \\
& I^{\oplus 3} \mathbf{e}'_7 \otimes \mathbf{e}'_7 \otimes \mathbf{e}'_7 \\
& I^{\oplus 2} \mathbf{e}'_7 \otimes \mathbf{e}'_7123215 \\
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_7123 \\
& 1\mathbf{e}'_7 \otimes \mathbf{e}'_71
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}'_7 \otimes, \\
& 231\mathbf{so} \otimes \mathbf{e}'_71, \\
& 31\mathbf{so} \otimes \mathbf{e}'_71
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}'_7 \otimes, \\
& 31\mathbf{so} \otimes \mathbf{e}'_7 I^{\oplus 2},
\end{aligned}$$

$$\begin{aligned}
& \mathbf{1so} \otimes \mathbf{e}'_7 I^{\oplus 3} \\
& \mathbf{1so} \otimes \mathbf{e}'_7 I^{\oplus 2}
\end{aligned}$$

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$$\begin{aligned}
& 3221\mathbf{e}_8 \otimes \mathbf{so}, \\
& 1513221\mathbf{e}_8 \otimes \mathbf{so}, \\
& \begin{smallmatrix} 5,5 & 1 & 5,3 \\ \otimes \mathbf{e}_8 & \otimes \mathbf{so} & \otimes \end{smallmatrix},
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 4} \mathbf{e}_8 \otimes \mathbf{so}, \\
& \begin{smallmatrix} 3,5 & 1 & 5,3 \\ \otimes \mathbf{e}_8 & \otimes \mathbf{so} & \otimes \end{smallmatrix}, \\
& 313221\mathbf{e}_8 \otimes \mathbf{so},
\end{aligned}$$

$$\begin{aligned}
& 5131513221\mathbf{e}_8 \otimes \mathbf{so} \\
& \begin{smallmatrix} 4,5 & 1 & 5,3 \\ \otimes \mathbf{e}_8 & \otimes \mathbf{so} & \otimes \end{smallmatrix} \\
& I^{\oplus 5} \mathbf{e}_8 \otimes \mathbf{so}
\end{aligned}$$

$$21513221 \mathfrak{e}_8^{1 \ 5,3} \otimes \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} \ 5,3} \otimes \mathfrak{so},$$

$$1513221 \mathfrak{e}_8^{1 \ 5,3} \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} \ 5,3} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4 \ 5,3} \otimes \mathfrak{so}$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,3} \otimes \mathfrak{e}_6,$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,4} \otimes \mathfrak{e}_6,$$

$$\mathfrak{e}_8^{5,5 \ 1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$5131513221 \mathfrak{e}_8^{1 \ 5,4} \otimes \mathfrak{e}_6,$$

$$\mathfrak{e}_8^{3,5 \ 1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$313221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 3} \mathfrak{e}_8^{3221 \ 5,4} \otimes \mathfrak{e}_6$$

$$\mathfrak{e}_8^{4,5 \ 1 \ 5,5} \otimes \mathfrak{e}_6$$

$$I^{\oplus 5} \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6$$

$$21513221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} \ 5,5} \otimes \mathfrak{e}_6,$$

$$1513221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} \ 5,5} \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4 \ 5,5} \otimes \mathfrak{e}_6$$

$$3221 \mathfrak{e}_7^{1 \ 3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7^{321 \ 3,2} \otimes \mathfrak{so},$$

$$\mathfrak{e}_7^{3,3 \ 1 \ 3,3} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \ 4,3} \otimes \mathfrak{so},$$

$$512321 \mathfrak{e}_7^{1 \ 3,2} \otimes \mathfrak{so},$$

$$321 \mathfrak{e}_7^{1 \ 4,2} \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_7^{1 \ 3,3} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7^{1 \ 5,3} \otimes \mathfrak{so}$$

$$151321 \mathfrak{e}_7^{1 \ 3,2} \otimes \mathfrak{so}$$

$$\mathfrak{e}_7^{2,3 \ 1 \ 3,3} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathfrak{so}$$

$$321 \mathfrak{e}_6^{1 \ 2,2} \otimes \mathfrak{so},$$

$$31 \mathfrak{e}_6^{I^{\oplus 2} \ 2,2} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_6^{1 \ 3,3} \bigcirc \mathfrak{so}$$

$$15131 \mathfrak{e}_6^{1 \ 2,2} \otimes \mathfrak{so},$$

$$31 \mathfrak{e}_6^{1 \ 3,2} \otimes \mathfrak{so},$$

$$31 \mathfrak{e}_6^{2,2 \ 2,2} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6^{1 \ 3,3} \otimes \mathfrak{so}$$

$$3151321 \mathfrak{e}_7^{1 \ 3,3} \otimes \mathfrak{e}_6,$$

$$321 \mathfrak{e}_7^{1 \ 4,4} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathfrak{e}_7^{2321 \ 3,3} \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \ 4,5} \otimes \mathfrak{e}_6,$$

$$2321 \mathfrak{e}_7^{1 \ 4,3} \otimes \mathfrak{e}_6$$

$$1 \mathfrak{e}_7^{1 \ 5,5} \otimes \mathfrak{e}_6$$

$$31513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,4} \otimes \mathfrak{e}_7',$$

$$1513221 \mathfrak{e}_8^{I^{\oplus 2} \ 5,5} \otimes \mathfrak{e}_7'$$

$$5131513221 \mathfrak{e}_8^{1 \ 5,5} \otimes \mathfrak{e}_7',$$

$$I^{\oplus 3} \mathfrak{e}_8^{3221 \ 5,5} \otimes \mathfrak{e}_7'$$

$$321 \mathfrak{e}_7'^{1 \ 3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 \ 3,3} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7'^{1 \ 4,3} \otimes \mathfrak{so}$$

$$2321 \mathfrak{e}_7'^{1 \ 3,3} \otimes \mathfrak{e}_6,$$

$$31 \mathfrak{e}_6^{1 \ 3,5} \otimes \mathfrak{e}_7',$$

$$1 \mathfrak{e}_7'^{1 \ 4,5} \otimes \mathfrak{e}_6$$

$$23151321 \mathfrak{e}_7^{1 \ 3,3} \otimes \mathfrak{e}_7',$$

$$2321 \mathfrak{e}_7^{1 \ 4,4} \otimes \mathfrak{e}_7',$$

$$321 \mathfrak{e}_7^{1 \ 4,5} \otimes \mathfrak{e}_7'$$

$$2315131\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2},$$

$$31\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{3,4},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{3,5}$$

$$231513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7,$$

$$31513221\mathfrak{e}_8^{I^{\oplus 2} 5,5} \otimes \mathfrak{e}_7$$

$$223151321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3},$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{4,5}$$

$$231\mathfrak{so}^1 \otimes \mathfrak{so}^{1,1}$$

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$$1\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 123151322,$$

$$\otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1 5,4 3,2},$$

$$21513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,2},$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5 4,2},$$

$$\otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4 3,3},$$

$$313221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,3},$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,3},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5 5,4},$$

$$\otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1 5,4} I^{\oplus 3},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 122313,$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 12231512,$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 6},$$

$$I^{\oplus 2} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 12315132,$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 12231513,$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 1231513,$$

$$I^{\oplus 4} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 123215,$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4 3,2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4 3,3},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes \mathfrak{e}_7^{3,3},$$

$$I^{\oplus 2} \mathfrak{e}_8^{3221 5,4} \otimes \mathfrak{e}_7^{12315},$$

$$\otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4 3,2},$$

$$313221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,2},$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,2},$$

$$\otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} 123215,$$

$$\otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{5,4 3,3},$$

$$I^{\oplus 5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,3},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5 5,3},$$

$$\otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{1 5,4} I^{\oplus 3},$$

$$I^{\oplus 4} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 1223,$$

$$313221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 3},$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 1223151,$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 122315123,$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 12231513215,$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 12231513151,$$

$$21513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 3},$$

$$I^{\oplus 4} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 123151,$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{3,2},$$

$$I^{\oplus 2} \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} I^{\oplus 3},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{1223},$$

$$513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{123},$$

$$\otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_7^{5,4 3,2}$$

$$I^{\oplus 5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,2}$$

$$\otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} 1223$$

$$\otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} 123151$$

$$\otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4 3,3}$$

$$21513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4 3,3}$$

$$\otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{1 5,4} I^{\oplus 3}$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5 5,5}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} 1223132$$

$$321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 5}$$

$$2321\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 4}$$

$$31513221\mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^1$$

$$5131513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2}$$

$$I^{\oplus 5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 3}$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 3}$$

$$1513221\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} 12315$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes \mathfrak{e}_7^{3,2}$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$I^{\oplus 3} \mathfrak{e}_8^{3221 5,4} \otimes \mathfrak{e}_7^{1223} I^{\oplus 2}$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{1231513}$$

$$\begin{aligned} & I^{\oplus 2} \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7 I^{\oplus 3}, \\ & I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2}, 3,3} \otimes \mathbf{e}_7 12313, \\ & I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2}, 3,3} \otimes \mathbf{e}_7 1232151, \\ & 2321 \mathbf{e}_7^1 \otimes \mathbf{e}_7 1, \\ & I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{4,4} \otimes \mathbf{e}_7^{3,2}, \\ & I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{4,4} I^{\oplus 3}, \\ & 1 \mathbf{e}_7^1 \otimes \mathbf{e}_7 123, \\ & I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{4,5} I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathbf{e}_7^{2321} \otimes \mathbf{e}_7 I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_7 1231512, \\
& I^{\oplus 2} \mathbf{e}_7^{I^{\oplus 2} \ 3,3} \otimes \mathbf{e}_7 123151, \\
& 1 \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7 123215, \\
& I^{\oplus 2} \mathbf{e}_7^{1 \ 4,4} \otimes \mathbf{e}_7^{3,3}, \\
& 1 \mathbf{e}_7^{1 \ 5,4} \otimes \mathbf{e}_7^{3,2}, \\
& 321 \mathbf{e}_7^{1 \ 4,5} \otimes \mathbf{e}_7 1, \\
& 1 \mathbf{e}_7^{1 \ 5,5} \otimes \mathbf{e}_7 I^{\oplus 2}
\end{aligned}$$

$$\begin{array}{l}
I^{\oplus 2} \epsilon_7^{\oplus 2, 3, 3} \otimes \epsilon_7 I^{\oplus 4} \\
I^{\oplus 2} \epsilon_7^{\oplus 2, 3, 3} \otimes \epsilon_7 1232^1 \\
1\epsilon_7^1 \otimes \epsilon_7 1223^{4, 4} \\
1\epsilon_7^1 \otimes \epsilon_7 123151^{4, 4} \\
321\epsilon_7^1 \otimes \epsilon_7 I^{\oplus 2, 4, 4} \\
1\epsilon_7^1 \otimes \epsilon_7^{\oplus 2, 3, 3} \\
1\epsilon_7^1 \otimes \epsilon_7 I^{\oplus 3, 5, 4}
\end{array}$$

$$\begin{aligned}
& 1^1_6 \otimes^3_5 \mathbf{e}_8 1223132, \\
& 1^1_8 \otimes^5_3 \mathbf{e}_6 13151322, \\
& 1^1_6 \otimes^3_5 \mathbf{e}_8 12231^1 513, \\
& 4,5^1_8 \otimes^5_3 \mathbf{e}_6^2_2 \otimes, \\
& I^{\oplus 5} \mathbf{e}_8^1 \otimes^5_3 \mathbf{e}_6^2 \otimes, \\
& 2,4^1_8 \otimes^5_3 \mathbf{e}_6 123, \\
& 31^1_6 \otimes^3_5 \mathbf{e}_8^5_3, \\
& 5,5^1_8 \otimes^5_3 \mathbf{e}_6 I^{\oplus 2}, \\
& 313221^1 \mathbf{e}_8 \otimes^5_3 \mathbf{e}_6 I^{\oplus 2}, \\
& I^{\oplus 3} \mathbf{e}_8^1 \otimes^5_3 \mathbf{e}_6 1232, \\
& 21513221^1 \mathbf{e}_8 \otimes^5_3 \mathbf{e}_6 I^{\oplus 2}, \\
& I^{\oplus 3} \mathbf{e}_8^1 \otimes^5_3 \mathbf{e}_6 131513, \\
& 31^1_6 \otimes^3_5 \mathbf{e}_8 12231512, \\
& I^{\oplus 2} \mathbf{e}_8^2 \otimes^5_3 \mathbf{e}_6^2 \otimes, \\
& I^{\oplus 2} \mathbf{e}_8^2 \otimes^5_3 \mathbf{e}_6 I^{\oplus 2}, \\
& I^{\oplus 2} \mathbf{e}_8^{\oplus 2} \otimes^5_3 \mathbf{e}_6 1232, \\
& I^{\oplus 3} \mathbf{e}_8^{\oplus 3} \otimes^5_3 \mathbf{e}_6 I^{\oplus 2}, \\
& 3,5^1_8 \otimes^5_4 \mathbf{e}_6 1, \\
& 313221^1 \mathbf{e}_8 \otimes^5_4 \mathbf{e}_6 1, \\
& 1^1 1513221^1 \mathbf{e}_8 \otimes^5_4 \mathbf{e}_6 1,
\end{aligned}$$

$$\begin{aligned}
& 1\epsilon_6^1 \epsilon_8^{3,5} \otimes \epsilon_8 I^{\oplus 6}, \\
& 1\epsilon_6^1 \epsilon_8^{3,5} \otimes \epsilon_8 12231513215, \\
& 1\epsilon_6^1 \epsilon_8^{3,5} \otimes \epsilon_8 12231513151, \\
& 5,5 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \epsilon_8 \otimes \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \epsilon_6 \otimes \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix}, \\
& 21513221 \epsilon_8^1 \otimes \epsilon_6^{5,3} \otimes \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix}, \\
& 2,4 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \epsilon_8 \otimes \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \epsilon_6 13151, \\
& 4,5 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \epsilon_8 \otimes \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \epsilon_6 I^{\oplus 2}, \\
& 31 \epsilon_6^1 \otimes \epsilon_8^{3,5} \otimes \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}, \\
& 31 \epsilon_6^1 \otimes \epsilon_8^{3,5} 122313, \\
& I^{\oplus 5} \epsilon_8^1 \otimes \epsilon_6^{5,3} I^{\oplus 2}, \\
& 1513221 \epsilon_8^{I^{\oplus 2} \ 5,3} \otimes \epsilon_6 1, \\
& I^{\oplus 4} \epsilon_8^1 \otimes \epsilon_6^{5,3} 13151, \\
& 31 \epsilon_6^1 \otimes \epsilon_8^{3,5} 1223151, \\
& I^{\oplus 4} \epsilon_8^{I^{\oplus 2} \ 5,3} \otimes \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \otimes \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix}, \\
& I^{\oplus 2} \epsilon_8^{3221 \ 5,3} \otimes \epsilon_6 1315, \\
& I^{\oplus 4} \epsilon_8^{I^{\oplus 2} \ 5,3} \otimes \epsilon_6 I^{\oplus 2}, \\
& I^{\oplus 3} \epsilon_8^{I^{\oplus 2} \ 5,3} \otimes \epsilon_6 13151, \\
& 4,5 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \epsilon_8 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \epsilon_6 1, \\
& I^{\oplus 5} \epsilon_8^1 \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \epsilon_6 1, \\
& I^{\oplus 2} \epsilon_8^{2,4 \ 5,4} \otimes \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \epsilon_6 1,
\end{aligned}$$

$$\begin{array}{l}
1^{3,5} \\
1\mathbf{c}_6 \otimes \mathbf{c}_8 122315123 \\
5131513221^1 5,3 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 1 \\
3,5^1 5,3^1 2,2 \\
\otimes \mathbf{c}_8 \otimes \mathbf{c}_6 \otimes \\
313221^1 5,3^1 2,2 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 \otimes \\
1^1 1513221^1 5,3^1 2,2 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 \otimes \\
3,5^1 5,3^1 \\
\otimes \mathbf{c}_8 \otimes \mathbf{c}_6 I^{\oplus 2} \\
31\mathbf{c}_6^1 3,5^1 5,4 \\
\otimes \mathbf{c}_8 \otimes \\
I^{\oplus 3} \mathbf{c}_8^{221} 5,3^1 \\
\otimes \mathbf{c}_6 1 \\
I^{\oplus 4} \mathbf{c}_8^1 5,3^1 \\
\otimes \mathbf{c}_6 123 \\
I^{\oplus 2} \mathbf{c}_8^1 5,3^1 \\
\otimes \mathbf{c}_6 1315132 \\
1^1 1513221^1 5,3^1 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 I^{\oplus 2} \\
31\mathbf{c}_6^1 3,5^1 \\
\otimes \mathbf{c}_8 I^{\oplus 5} \\
1513221^1 5,3^1 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 1315 \\
I^{\oplus 3} I^{\oplus 3} 5,3^1 2,2 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 \otimes \\
I^{\oplus 3} \mathbf{c}_8^1 I^{\oplus 2} 5,3^1 \\
\otimes \mathbf{c}_6 123 \\
I^{\oplus 2} I^{\oplus 2} 5,3^1 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 131513 \\
513221^1 I^{\oplus 2} 5,3^1 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 13 \\
5,5^1 5,4^1 \\
\otimes \mathbf{c}_8 \otimes \mathbf{c}_6 1 \\
21513221^1 5,4^1 \\
\mathbf{c}_8 \otimes \mathbf{c}_6 1 \\
2,4^1 5,4^1 \\
\otimes \mathbf{c}_8 \otimes \mathbf{c}_6 13
\end{array}$$

$$I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_6 1,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_6 13$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3} 5,4} \otimes \mathfrak{e}_6 1,$$

$$I^{\oplus 4} \mathfrak{e}_8^{1 5,4} \otimes \mathfrak{e}_6 13$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1223,$$

$$315131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1,$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,2} \otimes \mathfrak{e}_6 \otimes,$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,5} \otimes \mathfrak{e}_6 \otimes,$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 3},$$

$$15131 \mathfrak{e}_6^{1 2,2 2,2} \otimes \mathfrak{e}_6 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,3} \otimes \mathfrak{e}_6 \otimes,$$

$$321 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 3},$$

$$15131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$5131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 13151,$$

$$31 \mathfrak{e}_6^{2,2 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$31 \mathfrak{e}_6^{31 2,2 3,4} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{I^{\oplus 2} 2,2} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 131512,$$

$$1 \mathfrak{e}_6^{1 3,3} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$1 \mathfrak{e}_6^{1 3,3 2,2} \bigcirc \mathfrak{e}_6 \otimes,$$

$$1 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 1231513,$$

$$1 \mathfrak{e}_7'^{1 3,3} \otimes \mathfrak{e}_7 1223151,$$

$$\mathfrak{e}_7^{3,3 1 3,3 3,2} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes,$$

$$\mathfrak{e}_7^{2,3 1 3,3} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 3},$$

$$321 \mathfrak{e}_7'^{1 3,3 3,3} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,3} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 1223,$$

$$321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 12315,$$

$$2321 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1,$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 13151315,$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,3} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes,$$

$$321 \mathfrak{e}_6^{1 2,2 2,2} \otimes \mathfrak{e}_6 \otimes,$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 1313,$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 13151,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,4} \otimes \mathfrak{e}_6 \otimes,$$

$$5131 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 123,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 1313,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 13151,$$

$$31 \mathfrak{e}_6^{2,2 2,2 2,2} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{31 2,2 3,2} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{31 2,2 3,5} \otimes \mathfrak{e}_6 \otimes,$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 I^{\oplus 3},$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 13151,$$

$$31 \mathfrak{e}_6^{1 3,3} \otimes \mathfrak{e}_6 1,$$

$$1 \mathfrak{e}_6^{1 3,3} \bigcirc \mathfrak{e}_6 I^{\oplus 2},$$

$$3151321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 1,$$

$$\mathfrak{e}_7^{2,3 1 3,3 3,2} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes,$$

$$321 \mathfrak{e}_7'^{1 3,3 3,2} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,2} \otimes \mathfrak{e}_7 \otimes,$$

$$I^{\oplus 3} \mathfrak{e}_7^{1 3,3 3,3} \otimes \mathfrak{e}_7' \otimes,$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,4} \otimes \mathfrak{e}_7 \otimes,$$

$$3221 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$51321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' 123,$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 123215$$

$$1 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 123151$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,4} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes$$

$$\mathfrak{e}_6^{2,2 1 2,2 3,3} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc$$

$$\mathfrak{e}_6^{2,2 1 2,2} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 131512$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,2} \otimes \mathfrak{e}_6 \otimes$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,5} \otimes \mathfrak{e}_6 \otimes$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2 3,3} \otimes \mathfrak{e}_6 \bigcirc$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 131512$$

$$31 \mathfrak{e}_6^{1 2,2} \otimes \mathfrak{e}_6 12315$$

$$31 \mathfrak{e}_6^{I^{\oplus 2} 2,2 2,2} \otimes \mathfrak{e}_6 \otimes$$

$$31 \mathfrak{e}_6^{31 2,2 3,3} \otimes \mathfrak{e}_6 \otimes$$

$$31 \mathfrak{e}_6^{31 2,2 3,3} \otimes \mathfrak{e}_6 \bigcirc$$

$$31 \mathfrak{e}_6^{31 2,2} \otimes \mathfrak{e}_6 1313$$

$$1 \mathfrak{e}_6^{1 3,3 2,2} \otimes \mathfrak{e}_6 \otimes$$

$$1 \mathfrak{e}_6^{1 3,4} \otimes \mathfrak{e}_6 1$$

$$31 \mathfrak{e}_6^{1 3,3} \bigcirc \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_7'^{1 3,3} \otimes \mathfrak{e}_7 123151315$$

$$\mathfrak{e}_7^{2,3 1 3,3 3,3} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes$$

$$I^{\oplus 3} \mathfrak{e}_7^{1 3,3 3,2} \otimes \mathfrak{e}_7' \otimes$$

$$\mathfrak{e}_7^{3,3 1 3,3 3,3} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes$$

$$\mathfrak{e}_7^{3,3 1 3,3} \otimes \mathfrak{e}_7' I^{\oplus 3}$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 3,3 4,5} \otimes \mathfrak{e}_7 \otimes$$

$$321 \mathfrak{e}_7'^{1 3,3} \otimes \mathfrak{e}_7 I^{\oplus 3}$$

$$2321 \mathfrak{e}_7^{1 3,3} \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8^{1\ 4,5} 12231513151, \\
& \otimes \mathfrak{e}_8^{4,5\ 1\ 5,4} \mathfrak{e}'_7 I^{\oplus 2}, \\
& 313221\mathfrak{e}_8^{1\ 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 5} \mathfrak{e}_8^{1\ 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& 1513221\mathfrak{e}_8^{1\ 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}\ 5,4} \otimes \mathfrak{e}'_7 1232, \\
& \otimes \mathfrak{e}_8^{3,5\ 1\ 5,5} \mathfrak{e}'_7 1, \\
& 313221\mathfrak{e}_8^{1\ 5,5} \otimes \mathfrak{e}'_7 1, \\
& 1513221\mathfrak{e}_8^{1\ 5,5} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}\ 5,5} \otimes \mathfrak{e}'_7 1
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_8^{2,4\ 1\ 5,4} \mathfrak{e}'_7 123, \\
& \otimes \mathfrak{e}_8^{5,5\ 1\ 5,4} \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 4} \mathfrak{e}_8^{1\ 5,4} \otimes \mathfrak{e}'_7 123, \\
& 21513221\mathfrak{e}_8^{1\ 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 2} \mathfrak{e}_8^{2,4\ 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}\ 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& \otimes \mathfrak{e}_8^{4,5\ 1\ 5,5} \mathfrak{e}'_7 1, \\
& I^{\oplus 5} \mathfrak{e}_8^{1\ 5,5} \otimes \mathfrak{e}'_7 1, \\
& I^{\oplus 2} \mathfrak{e}_8^{2,4\ 5,5} \otimes \mathfrak{e}'_7 1,
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_8^{3,5\ 1\ 5,4} \mathfrak{e}'_7 I^{\oplus 2} \\
& I^{\oplus 3} \mathfrak{e}_8^{3221\ 5,4} \otimes \mathfrak{e}'_7 1 \\
& I^{\oplus 3} \mathfrak{e}_8^{1\ 5,4} \otimes \mathfrak{e}'_7 1232 \\
& 1513221\mathfrak{e}_8^{I^{\oplus 2}\ 5,4} \otimes \mathfrak{e}'_7 1 \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}\ 5,4} \otimes \mathfrak{e}'_7 123 \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}\ 5,4} \otimes \mathfrak{e}'_7 I^{\oplus 2} \\
& \otimes \mathfrak{e}_8^{5,5\ 1\ 5,5} \mathfrak{e}'_7 1 \\
& 21513221\mathfrak{e}_8^{1\ 5,5} \otimes \mathfrak{e}'_7 1 \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}\ 5,5} \otimes \mathfrak{e}'_7 1
\end{aligned}$$

$$\begin{aligned}
& 2313221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}1, \\
& 51231513221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 3} \mathfrak{e}_8^{2,4\ 4,2} \otimes \mathfrak{so}1, \\
& \otimes \mathfrak{e}_8^{3,5\ 1\ 4,2} \mathfrak{so}13, \\
& I^{\oplus 2} \mathfrak{e}_8^{5,5\ 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 2} \mathfrak{e}_8^{313221\ 4,2} \otimes \mathfrak{so}1, \\
& 21513221\mathfrak{e}_8^{I^{\oplus 2}\ 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 3} \mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}1322, \\
& 21513221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}13, \\
& I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}\ 4,2} \otimes \mathfrak{so}1322, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 3}\ 4,2} \otimes \mathfrak{so}13
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 6} \mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}1, \\
& 31513221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}1, \\
& \otimes \mathfrak{e}_8^{2,4\ 1\ 4,2} \mathfrak{so}132, \\
& I^{\oplus 2} \mathfrak{e}_8^{4,5\ 4,2} \otimes \mathfrak{so}1, \\
& \otimes \mathfrak{e}_8^{5,5\ 1\ 4,2} \mathfrak{so}13, \\
& 313221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}13, \\
& 1513221\mathfrak{e}_8^{I^{\oplus 2}\ 4,2} \otimes \mathfrak{so}1, \\
& I^{\oplus 4} \mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}132, \\
& 1513221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}13, \\
& I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}\ 4,2} \otimes \mathfrak{so}132,
\end{aligned}$$

$$\begin{aligned}
& 321513221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}1 \\
& 15131513221\mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}1 \\
& I^{\oplus 2} \mathfrak{e}_8^{3,5\ 4,2} \otimes \mathfrak{so}1 \\
& \otimes \mathfrak{e}_8^{4,5\ 1\ 4,2} \mathfrak{so}13 \\
& 3221\mathfrak{e}_8^{3221\ 4,2} \otimes \mathfrak{so}1 \\
& I^{\oplus 5} \mathfrak{e}_8^{I^{\oplus 2}\ 4,2} \otimes \mathfrak{so}1 \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 3}\ 4,2} \otimes \mathfrak{so}1 \\
& I^{\oplus 5} \mathfrak{e}_8^{1\ 4,2} \otimes \mathfrak{so}13 \\
& I^{\oplus 2} \mathfrak{e}_8^{2,4\ 4,2} \otimes \mathfrak{so}13 \\
& I^{\oplus 4} \mathfrak{e}_8^{I^{\oplus 2}\ 4,2} \otimes \mathfrak{so}13
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes, \\
& 321\mathfrak{e}'_7 \otimes \mathfrak{e}_6 1, \\
& I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_6^{1\ 3,3\ 2,2}, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6^{1\ 4,3\ 2,2}, \\
& 31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 I^{\oplus 3}, \\
& I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_6^{1\ 3,3} I^{\oplus 2}, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 I^{\oplus 2}, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 123 \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 13151 \\
& 31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 I^{\oplus 2} \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 I^{\oplus 2} \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_6 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}1223, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}123151, \\
& 321\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}I^{\oplus 2}, \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,4,5}1
\end{aligned}$$

$$\begin{aligned}
& 2321\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}1, \\
& I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}{}^{3,2}, \\
& I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}I^{\oplus 3},
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}123215 \\
& I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,3,3}{}^{3,3} \\
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7{}^{1,4,4}I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& {}^{2,3}1\mathfrak{e}_7 \otimes \mathfrak{so}1, \\
& 1\mathfrak{e}_7 \otimes \mathfrak{so}1{}^{3,2}32, \\
& I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{so}1{}^{1,4,2}
\end{aligned}$$

$$\begin{aligned}
& {}^{3,3}1\mathfrak{e}_7 \otimes \mathfrak{so}1, \\
& I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{so}1{}^{1,3,2}{}^{3,2}, \\
& 1\mathfrak{e}_7 \otimes \mathfrak{so}1{}^{1,4,2}3
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{so}1{}^{1,3,2} \\
& I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{so}1{}^{1,3,2}3
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so} \otimes \mathfrak{so}{}^{1,1}{}^{2,2}, \\
& 1\mathfrak{so} \otimes \mathfrak{so}1{}^{1,1}{}^232,
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so} \otimes \mathfrak{so}{}^{1,1}{}^{2,3}, \\
& 1\mathfrak{so} \otimes \mathfrak{so}I^{\oplus 2}{}^{1,1},
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so} \otimes \mathfrak{so}{}^{1,1}{}^{2,4} \\
& 31\mathfrak{so} \otimes \mathfrak{so}1{}^{1,1}
\end{aligned}$$

$$\begin{aligned}
& {}^{2,2}1\mathfrak{e}_6 \otimes \mathfrak{so}1, \\
& 31\mathfrak{e}_6 \otimes \mathfrak{so}1{}^{31,2,2}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{so}1{}^{1,2,2}, \\
& 1\mathfrak{e}_6 \otimes \mathfrak{so}1{}^{1,3,2}
\end{aligned}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{so}1{}^{1,2,2}3$$

$$I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{so}1{}^{1,3,2}$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{so}1{}^{1,3,2}3$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{so}1{}^{1,4,2}$$

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$$\begin{aligned}
& {}^{2,4}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}1223151315, \\
& {}^{3,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}15, \\
& {}^{3,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}12231512, \\
& {}^{4,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}15, \\
& {}^{4,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}12231512, \\
& 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}15, \\
& {}^{5,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}1223151, \\
& 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}12231512, \\
& I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}1223151315, \\
& I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}1223151,
\end{aligned}$$

$$\begin{aligned}
& {}^{3,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}15, \\
& 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}15, \\
& {}^{3,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}1223151, \\
& 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}15, \\
& {}^{4,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}1223151, \\
& I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}15, \\
& 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}122313, \\
& 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}1223151, \\
& I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}I^{\oplus 5}, \\
& 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}12231512,
\end{aligned}$$

$$\begin{aligned}
& {}^{3,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}15, \\
& I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}15, \\
& {}^{4,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}15, \\
& I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}15, \\
& {}^{5,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}15, \\
& {}^{5,5}1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{5,5}12231512, \\
& I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}122313, \\
& 1\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}I^{\oplus 2}1223151322, \\
& I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}12231512, \\
& 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8{}^{1,5,5}12231512
\end{aligned}$$

$$\begin{aligned}
& 1513221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 1223151, \\
& \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& I^{\oplus 5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& 1513221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& \otimes \mathbf{e}_8^{4,5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 2}, \\
& \otimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& 313221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& 5131513221 \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& I^{\oplus 4} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 1223151, \\
& I^{\oplus 5} \mathbf{e}_8^1 \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}, \\
& I^{\oplus 2} \mathbf{e}_8^{2,4} \otimes \mathbf{e}_8^{5,5} I^{\oplus 24,2}, \\
& I^{\oplus 2} 3221 \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 122315, \\
& I^{\oplus 3} \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} I^{\oplus 2} 1223151, \\
& I^{\oplus 3} \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1223151315, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1^5 4, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 122313, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1\epsilon_8 I^{\oplus 5}, \\
& I^{\oplus 2} 1\epsilon_7^1 \otimes \epsilon_8^{4,5} 1223151, \\
& 2321\epsilon_7^1 \otimes \epsilon_8^{4,5} I^{\oplus 2} I^{\oplus 2}, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 1^5 3, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 122313, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} 1223151, \\
& 1\epsilon_7^1 \otimes \epsilon_8^{5,5} I^{\oplus 3} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned} & 1\epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 1231513, \\ & I^{\oplus 3} \epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,2}, \\ & I^{\oplus 2} \epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 1223, \\ & I^{\oplus 2} \epsilon_7^{1,3,3} \otimes \epsilon_7^{1,3,3} 123215, \end{aligned}$$

[illegible]

$$\begin{aligned}
& 321 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,4,2}, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{1,5,5}, \\
& 321 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8 I^{\oplus 4}, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8 12231512, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes I^{\oplus 2,4,2}, \\
& I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} \otimes I^{\oplus 2} I^{\oplus 4}, \\
& 1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,5,4}, \\
& 1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8 I^{\oplus 5}, \\
& 1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes I^{\oplus 2,4,2}, \\
& 1 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8 \otimes,
\end{aligned}$$

$$\begin{aligned} & \begin{smallmatrix} 2,3 & 1 & 3,3 & 1 & 3,2 \\ \otimes & \mathfrak{e}_7 & \otimes & \mathfrak{e}_7 & \otimes \end{smallmatrix}, \\ & \begin{smallmatrix} 3,3 & 1 & 3,3 & 1 & 3,3 \\ \otimes & \mathfrak{e}_7 & \otimes & \mathfrak{e}_7 & \otimes \end{smallmatrix}, \\ & 321 \begin{smallmatrix} 1 & 3,3 & 1 \\ \otimes & \mathfrak{e}_7 & \otimes \mathfrak{e}_7 \end{smallmatrix} 12315, \\ & I^{\oplus 2} \begin{smallmatrix} 1 & 3,3 & 1 \\ \otimes & \mathfrak{e}_7 & \otimes \mathfrak{e}_7 \end{smallmatrix} 123151, \end{aligned}$$

$$\begin{array}{l}
\begin{array}{c} 4,5 \\ \otimes \end{array} \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 24,2} \\
313221 \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 24,2} \\
\begin{array}{c} 2,4 \\ \otimes \end{array} \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 2} \\
\begin{array}{c} 3,5 \\ \otimes \end{array} \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 3} \\
\begin{array}{c} 5,5 \\ \otimes \end{array} \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 2} \\
313221 \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 2} \\
I^{\oplus 3} \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 2} \\
21513221 \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 2} \\
1513221 \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 2} \\
1513221 \begin{array}{c} 1 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 3} \\
I^{\oplus 2} \begin{array}{c} 2,4 \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 3} \\
I^{\oplus 2} \begin{array}{c} I^{\oplus 2} \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 2} \\
I^{\oplus 4} \begin{array}{c} I^{\oplus 2} \\ \otimes \end{array} \begin{array}{c} 5,5 \\ \otimes \end{array} I^{\oplus 3}
\end{array}$$

$$\begin{array}{l}
I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{1,5,3} \\
1 \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} 1223 \\
2321 \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} 1 \\
1 \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 2}} 1223151 \\
321 \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} I^{\oplus 2} \\
I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{I^{\oplus 3}} I^{\oplus 3} \\
1 \mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8^{1,5,5} \\
1 \mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8^{1,5,5} 12231512 \\
1 \mathbf{e}_7^{1,5,5} \otimes \mathbf{e}_8^{I^{\oplus 2}} I^{\oplus 4}
\end{array}$$

$$\begin{array}{c} 2,3 \ 1 \ 3,3 \ 1 \ 3,3 \\ \otimes e_7 \ \otimes \ e_7 \ \otimes \\ I^{\oplus 3} e_7^1 \ \otimes \ e_7^{3,3} \ 1 \ 3,3 \\ 1 e_7^1 \ \otimes \ e_7^{3,3} I^{\oplus 2} \ 1232 \\ I^{\oplus 3} e_7^1 \ \otimes \ e_7^{3,3} I^{\oplus 3} \ 1 \end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^{2,3}{}_1 {}^{3,3} {}^{I^{\oplus 2}} \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 {}^{I^{\oplus 2}}, \\
I^{\oplus 3} {}^1{}_7 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^{I^{\oplus 2}}, \\
I^{\oplus 2} {}^1{}_7 \otimes {}^{4,4} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^1{}_7,
\end{array}
&
\begin{array}{l}
{}^{3,3}{}_1 {}^{3,3} {}^{I^{\oplus 2}} \\
\otimes \mathbf{e}_7 \otimes \mathbf{e}_7 {}^{I^{\oplus 2}}, \\
I^{\oplus 2} {}^{I^{\oplus 2}}{}_7 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^{I^{\oplus 2}}, \\
1 {}^1{}_7 \otimes {}^{5,4} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^1{}_7,
\end{array}
&
\begin{array}{l}
I^{\oplus 2} {}^1{}_7 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 \otimes \mathbf{e}_7 {}^{123} \\
1 {}^1{}_7 \otimes {}^{4,4} {}^1{}_7 \\
\mathbf{e}_7 {}^{123} \\
1 {}^1{}_7 \otimes {}^{5,5} {}^1{}_7 \\
\mathbf{e}_7 {}^1{}_7
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,5} {}^1{}_8 \otimes {}^{5,3} \\
\mathbf{e}_8 \otimes, \\
{}^1{}_6 \otimes {}^{3,5} {}^1{}_8 {}^{122313}, \\
{}^1{}_6 \otimes {}^{3,5} {}^1{}_8 {}^{1223151}, \\
{}^1{}_6 \otimes {}^{3,5} {}^{I^{\oplus 2}} \\
\mathbf{e}_8 {}^{I^{\oplus 4}}, \\
31 {}^1{}_6 \otimes {}^{3,5} {}^{I^{\oplus 3}} \\
\mathbf{e}_8 {}^{I^{\oplus 2}}
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,5} {}^1{}_8 \otimes {}^{5,4} \\
\mathbf{e}_8 \otimes, \\
{}^1{}_6 \otimes {}^{3,5} {}^1{}_8 {}^{I^{\oplus 5}}, \\
{}^1{}_6 \otimes {}^{3,5} {}^{I^{\oplus 24,2}} \\
\mathbf{e}_8 \otimes, \\
{}^1{}_6 \otimes {}^{3,5} {}^{I^{\oplus 3}} \\
\mathbf{e}_8 {}^{I^{\oplus 3}},
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,5} {}^1{}_8 \otimes {}^{5,5} \\
\mathbf{e}_8 \otimes \\
{}^1{}_6 \otimes {}^{3,5} {}^1{}_8 {}^{12231512} \\
31 {}^1{}_6 \otimes {}^{3,5} {}^1{}_8 \otimes {}^{4,2} \\
\mathbf{e}_8 \otimes \\
31 {}^1{}_6 \otimes {}^{3,5} {}^1{}_8 {}^{I^{\oplus 4}}
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_7 \otimes {}^{3,3} {}^1{}_7 {}^{1223}, \\
{}^1{}_7 \otimes {}^{3,3} {}^1{}_7 {}^{123151}, \\
{}^1{}_7 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^{123}, \\
I^{\oplus 2} {}^1{}_7 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^{I^{\oplus 2}},
\end{array}
&
\begin{array}{l}
2321 {}^1{}_7 \otimes {}^{3,3} {}^1{}_7, \\
I^{\oplus 2} {}^1{}_7 \otimes {}^{3,3} {}^1{}_7 \otimes {}^{3,2} \\
\mathbf{e}_7 \otimes, \\
321 {}^1{}_7 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^1{}_7, \\
1 {}^1{}_7 \otimes {}^{4,4} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^1{}_7,
\end{array}
&
\begin{array}{l}
{}^1{}_7 \otimes {}^{3,3} {}^1{}_7 {}^{123215} \\
I^{\oplus 2} {}^1{}_7 \otimes {}^{3,3} {}^1{}_7 \otimes {}^{3,3} \\
\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \\
I^{\oplus 2} {}^1{}_7 \otimes {}^{3,3} {}^{I^{\oplus 3}} \\
\mathbf{e}_7 {}^1{}_7 \\
1 {}^1{}_7 \otimes {}^{4,5} {}^1{}_7 \\
\mathbf{e}_7 {}^1{}_7
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_6 \otimes {}^{2,2} {}^1{}_6 {}^{123}, \\
I^{\oplus 2} {}^1{}_6 \otimes {}^{2,2} {}^1{}_6 \otimes {}^{2,2} \\
\mathbf{e}_6 \otimes, \\
{}^1{}_6 \otimes {}^{2,2} {}^{I^{\oplus 2}} \\
\mathbf{e}_6 {}^{13}, \\
31 {}^1{}_6 \otimes {}^{2,2} {}^{I^{\oplus 2}} \\
\mathbf{e}_6 {}^1{}_6,
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{2,2} {}^1{}_6 {}^{13151}, \\
31 {}^1{}_6 \otimes {}^{2,2} {}^1{}_6 \otimes {}^{2,2} {}^1{}_6, \\
31 {}^1{}_6 \otimes {}^{2,2} {}^1{}_6 {}^{1315}, \\
{}^1{}_6 \otimes {}^{3,3} {}^1{}_6,
\end{array}
&
\begin{array}{l}
{}^{2,2}{}_6 \otimes {}^{2,2} {}^1{}_6 \otimes {}^{2,2} \\
\mathbf{e}_6 \otimes \mathbf{e}_6 \otimes \\
I^{\oplus 2} {}^1{}_6 \otimes {}^{2,2} {}^{I^{\oplus 2}} \\
\mathbf{e}_6 {}^1{}_6 \\
31 {}^1{}_6 \otimes {}^{2,2} {}^1{}_6 \otimes {}^{2,2} \\
\mathbf{e}_6 \otimes \mathbf{e}_6 \otimes \\
{}^1{}_6 \otimes {}^{3,3} {}^1{}_6
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_7 \otimes {}^{4,5} {}^1{}_8 \otimes {}^{5,3} \\
\mathbf{e}_8 \otimes, \\
{}^1{}_7 \otimes {}^{4,5} {}^1{}_8 {}^{122313}, \\
{}^1{}_7 \otimes {}^{4,5} {}^1{}_8 {}^{1223151}, \\
{}^1{}_7 \otimes {}^{4,5} {}^{I^{\oplus 3}} \\
\mathbf{e}_8 {}^{I^{\oplus 3}}
\end{array}
&
\begin{array}{l}
{}^1{}_7 \otimes {}^{4,5} {}^1{}_8 \otimes {}^{5,4} \\
\mathbf{e}_8 \otimes, \\
{}^1{}_7 \otimes {}^{4,5} {}^1{}_8 {}^{I^{\oplus 5}}, \\
{}^1{}_7 \otimes {}^{4,5} {}^{I^{\oplus 24,2}} \\
\mathbf{e}_8 \otimes,
\end{array}
&
\begin{array}{l}
{}^1{}_7 \otimes {}^{4,5} {}^1{}_8 \otimes {}^{5,5} \\
\mathbf{e}_8 \otimes \\
{}^1{}_7 \otimes {}^{4,5} {}^1{}_8 {}^{12231512} \\
{}^1{}_7 \otimes {}^{4,5} {}^{I^{\oplus 2}} \\
\mathbf{e}_8 {}^{I^{\oplus 4}}
\end{array}
\end{array}$$

$$\begin{array}{lll}
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,3} {}^1{}_7 \otimes {}^{3,2} \\
\mathbf{e}_7 \otimes, \\
{}^1{}_6 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^{I^{\oplus 2}}, \\
31 {}^1{}_6 \otimes {}^{3,4} {}^1{}_7, \\
\mathbf{e}_7 {}^1{}_7,
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,3} {}^1{}_7 \otimes {}^{3,3} \\
\mathbf{e}_7 \otimes, \\
31 {}^1{}_6 \otimes {}^{3,3} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^1{}_7, \\
1 {}^1{}_6 \otimes {}^{3,5} {}^1{}_7 \\
\mathbf{e}_7 {}^1{}_7
\end{array}
&
\begin{array}{l}
{}^1{}_6 \otimes {}^{3,3} {}^{I^{\oplus 3}} \\
\mathbf{e}_7 {}^1{}_7 \\
{}^1{}_6 \otimes {}^{3,4} {}^{I^{\oplus 2}} \\
\mathbf{e}_7 {}^1{}_7
\end{array}
\end{array}$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7 123,$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}'_7 1,$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} 1$$

2.4 3 node(1907)

2.4.1 gLgLg(11)

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}'_7 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7$$

$$\mathfrak{so} \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

2.4.2 gLg(8)

$$\mathfrak{so} \otimes \overset{2,4}{\mathfrak{e}_8} \otimes \overset{I^{\oplus 3} 4,2}{\mathfrak{so}}$$

$$\mathfrak{so} \otimes \overset{2,4}{\mathfrak{e}_8} \otimes \overset{I^{\oplus 2} 5,5}{\mathfrak{e}_7'}$$

$$\mathfrak{so} \otimes \overset{2,4}{\mathfrak{e}_8} \otimes \overset{I^{\oplus 2} 5,4}{\mathfrak{e}_6}$$

$$\mathfrak{so} \otimes \overset{2,3}{\mathfrak{e}_7} \otimes \overset{1 \ 3,2}{\mathfrak{so}}$$

$$\mathfrak{so} \otimes \overset{2,2}{\mathfrak{e}_6} \otimes \overset{1 \ 2,2}{\mathfrak{so}}$$

$$\mathfrak{e}_7' \otimes \overset{5,5}{\mathfrak{e}_8} \otimes \overset{1 \ 5,5}{\mathfrak{e}_7'}$$

$$\mathfrak{e}_6 \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{1 \ 5,5}{\mathfrak{e}_7'}$$

$$\mathfrak{e}_6 \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{1 \ 5,4}{\mathfrak{e}_6}$$

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$$513221\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$\overset{2,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,2}{\mathfrak{so}},$$

$$\overset{2,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$\overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$2151321\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$\overset{1}{151321\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$\overset{3,5}{\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$313221\mathfrak{e}_8 \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$\overset{1}{1513221\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$I^{\oplus 3} \overset{4,5}{\mathfrak{e}_7} \otimes \overset{5,3}{\mathfrak{e}_8} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \overset{4,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{e}_7} \otimes \mathfrak{so},$$

$$\overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,2}{\mathfrak{so}},$$

$$\overset{3,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$I^{\oplus 4} \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \mathfrak{so},$$

$$\overset{1}{2321\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}},$$

$$\overset{4,5}{\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}},$$

$$I^{\oplus 5} \overset{5,4}{\mathfrak{e}_8} \otimes \overset{3,2}{\mathfrak{e}_7} \otimes \mathfrak{so},$$

$$\overset{2,3}{\mathfrak{e}_7} \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{5,3}{\mathfrak{so}},$$

$$I^{\oplus 2} \overset{5,5}{\mathfrak{e}_7} \otimes \overset{5,3}{\mathfrak{e}_8} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_7 \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}}$$

$$I^{\oplus 3} \overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,2}{\mathfrak{e}_7} \otimes \mathfrak{so}$$

$$\overset{4,4}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$31321\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$1512321\mathfrak{e}_7 \otimes \overset{3,3}{\mathfrak{e}_7} \otimes \overset{3,3}{\mathfrak{so}}$$

$$\overset{5,5}{\mathfrak{e}_8} \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}}$$

$$21513221\mathfrak{e}_8 \otimes \overset{5,4}{\mathfrak{e}_7} \otimes \overset{3,2}{\mathfrak{so}}$$

$$\overset{3,3}{\mathfrak{e}_7} \otimes \overset{4,5}{\mathfrak{e}_8} \otimes \overset{5,3}{\mathfrak{so}}$$

$$\begin{array}{ccccc} 5,5 & 5,3 & 2,2 & & \\ \otimes e_8 & \otimes e_6 & \otimes s_0 & & \\ 21513221e_8 & \otimes e_6 & \otimes s_0 & & \\ 2,2 & 3,5 & 5,3 & & \\ \otimes e_6 & \otimes e_8 & \otimes s_0 & & \end{array}$$

$$\begin{array}{c}
I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{so} \\
\begin{array}{ccc}
3,3 & & 2,2 \\
\mathfrak{e}_6 & \otimes & \mathfrak{e}_7 \otimes \mathfrak{so}
\end{array} \\
\begin{array}{ccc}
2,2 & 3,3 & 3,3 \\
\otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{so}
\end{array}
\end{array}$$

$$\begin{array}{l} 2321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,4} \otimes \mathfrak{e}_7 \\ 321\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_7 \\ 31\mathfrak{e}_6^{3,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{l} 2321\epsilon'_7 \otimes^{3,3} \epsilon_7 \otimes^{4,4} \epsilon_7 \\ 151321\epsilon_7 \otimes^{3,3} \epsilon_7 \otimes^{4,4} \epsilon'_7 \\ 321\epsilon'_7 \otimes^{3,3} \epsilon_7 \otimes^{4,5} \epsilon_7 \end{array}$$

$$\begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{\prime 3,2} \otimes \mathfrak{so} \\ I^{\oplus 2} \mathfrak{e}_7^{\prime 3,3} \otimes \mathfrak{e}_7^{4,2} \otimes \mathfrak{so} \\ I^{\oplus 3} \mathfrak{e}_7^{\prime 3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{c} 151321e_7 \otimes^{3,3} e_7 \otimes^{4,3} e_6 \\ 321e_7 \otimes^{4,4} e_7 \otimes^{3,3} e_6 \end{array}$$

$$315131\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7,$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7,$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}'_7$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6,$$

$$321\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}'_7 \otimes \mathfrak{e}_6$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$2231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7$$

$$5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$51321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 3}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_6 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_6,$$

$$231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_6,$$

$$31\mathfrak{e}_6 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$15131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc \mathfrak{e}_6$$

$$2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$51231513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$2313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$51231513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 6}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$321513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$15131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$51321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6 \bigcirc \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$\begin{array}{c} 5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ 1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{c} 2,2 \quad 3,5 \quad 5,5 \\ \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\begin{array}{c} 3151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7, \\ 151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \end{array}$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7,$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7' \otimes \mathfrak{so}$$

$$5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_7' \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$231\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7',$$

$$31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_7'$$

$$3151321\mathfrak{e}_7'(\otimes \mathfrak{e}_7')^2,$$

$$321\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7' \otimes \mathfrak{so}$$

$$51321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7'$$

$$321\mathfrak{e}_7' \otimes \mathfrak{e}_7' \otimes \mathfrak{e}_6$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$2321\mathfrak{e}_7'(\otimes \mathfrak{e}_7')^2$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}$$

[illegible]

$$\begin{array}{ccccc}
15131\mathbf{e}_6 & \overset{2,2}{\otimes} \mathbf{e}_6 & \overset{3,4}{\otimes} \mathbf{e}_7 & 1 \\
\overset{2,2}{\otimes} \mathbf{e}_6 & \overset{2,2}{\otimes} \mathbf{e}_6 & \overset{3,5}{\otimes} \mathbf{e}_7 & 1 \\
\overset{2,2}{\otimes} \mathbf{e}_6 & \overset{3,3}{\otimes} \mathbf{e}_7 & \overset{3,3}{\otimes} \mathbf{e}_6 & \overset{2,2}{\otimes} \\
1\mathbf{e}_6 & \overset{3,3}{\otimes} \mathbf{e}_7 & \overset{4,3}{\otimes} \mathbf{e}_6 & 1
\end{array}$$

$$\begin{array}{l}
{}^{2,2} \otimes \mathfrak{e}_6 \quad {}^{3,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 122315123 \\
{}^{2,2} \otimes \mathfrak{e}_6 \quad {}^{3,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 12231513151 \\
5131\mathfrak{e}_6 \quad {}^{3,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 \otimes \\
I^{\oplus 2} \mathfrak{e}_6 \quad {}^{3,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 I^{\oplus 6} \\
I^{\oplus 2} \mathfrak{e}_6 \quad {}^{3,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 12231513 \\
5131\mathfrak{e}_6 \quad {}^{3,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 12231512 \\
1\mathfrak{e}_6 \quad {}^{4,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 I^{\oplus 6} \\
1\mathfrak{e}_6 \quad {}^{4,5} \otimes \mathfrak{e}_8 \quad {}^{5,5} \otimes \mathfrak{e}_8 12231513
\end{array}$$

$$\begin{array}{l}
1\epsilon_6^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \\
1\epsilon_6^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{12231512} \\
31\epsilon_6^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8 I^{\oplus 4} \\
\otimes \epsilon_6^{2,2} \otimes \epsilon_8^{3,5} \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\
I^{\oplus 2} \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7^{3,3} \\
5131\epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_7^1 \\
1\epsilon_6^{4,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7^{3,2} \\
1\epsilon_6^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_7 I^{\oplus 2}
\end{array}$$

$$\begin{array}{l}
2313221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 122315123 \\
2313221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513151 \\
I^{\oplus 6} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513215 \\
321513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 122315123 \\
321513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513151 \\
15131513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513215
\end{array}$$

$$31513221\mathbf{e}_8 \overset{1}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 12231513151, \quad 15131513221\mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 12231513151$$

$$\begin{array}{l}
I^{\oplus 2} \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,3} \\
I^{\oplus 2} \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{122313} \\
I^{\oplus 2} \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8 I^{\oplus 5} \\
1 \epsilon_7^{\prime 4,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7^{12315} \\
I^{\oplus 2} \epsilon_7^{\prime 4,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 I^{\oplus 3} \\
1 \epsilon_7^{\prime 5,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 I^{\oplus 3}
\end{array}$$

$$\begin{array}{l}
150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 122315123 \\
150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 12231513151 \\
3150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 \otimes \\
223150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 I^{\oplus 3} \\
3150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 12231512 \\
3150 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} 5013
\end{array}$$

$$\begin{array}{c}
1\epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7 123215 \\
\otimes \epsilon_7^{\prime 2,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \\
\otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \\
321\epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7 I^{\oplus 2} \\
I^{\oplus 3} \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} \otimes \epsilon_7^{\prime 3,3} I^{\oplus 3}
\end{array}$$

$$\begin{array}{cccc}
2,2 & 1,1 & 1,1 & 2,3 \\
\otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \\
I^{\oplus 2} \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} \\
3,2 & 1,1 & 1,1 & 2 \\
\otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \mathfrak{so} 132 \\
4,2 & 1,1 & 1,1 & 2 \\
\otimes \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \mathfrak{so} 132 \\
I^{\oplus 2} \mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{so} & \mathfrak{so} I^{\oplus 2}
\end{array}$$

$$\begin{array}{c}
1\epsilon_6^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} \otimes \epsilon_7 I^{\oplus 3} \\
\otimes \epsilon_6^{\textcolor{teal}{2},2} \otimes \epsilon_7^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} \\
I^{\oplus 2} \epsilon_6^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} \\
5131 \epsilon_6^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{3},3} I^{\oplus 2} \\
1\epsilon_6^{\textcolor{teal}{3},3} \otimes \epsilon_7^{\textcolor{teal}{4},4} \otimes \epsilon_7^{\textcolor{teal}{3},3} 1
\end{array}$$

$$\begin{array}{l} I^{\oplus 2} \mathfrak{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 122315123 \\ I^{\oplus 2} \mathfrak{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 12231513151 \\ 1\mathfrak{e}'_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 122315123 \\ 1\mathfrak{e}'_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 12231513151 \end{array}$$

$$\begin{array}{l} 150 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{4,4}{\otimes} \\ 150 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 12313 \\ 150 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 1232151 \end{array}$$

$$\begin{array}{l} 1\mathfrak{so}^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,4} \\ 1\mathfrak{so}^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6 I^{\oplus 3} \\ 1\mathfrak{so}^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{13151} \end{array}$$

$$\begin{array}{c} 2,4 \quad 5,4 \\ 150 \otimes e_8 \otimes e_7 123151 \\ 3150 \otimes e_8 \otimes e_7 I^{\oplus 3} \end{array}$$

$$\begin{array}{c} 1\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,3} \oplus \mathfrak{e}_7 I^{\oplus 3} \\ 1\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \oplus \mathfrak{e}_7 I^{\oplus 2} \end{array}$$

$$15131\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7 1$$

$$\begin{array}{c} \begin{array}{ccc} \begin{array}{c} 2,2 \\ \otimes \end{array} \mathbf{e}_6 & \begin{array}{c} 2,2 \\ \otimes \end{array} \mathbf{e}_6 & \begin{array}{c} 3,3 \\ \otimes \end{array} \mathbf{e}'_7 I^{\oplus 2}, \\ I^{\oplus 2} \mathbf{e}_6 & \begin{array}{c} 2,2 \\ \otimes \end{array} \mathbf{e}_6 & \begin{array}{c} 3,4 \\ \otimes \end{array} \mathbf{e}'_7 1, \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$1 \mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{ccc} 2,2 & 2,2 & 3,4 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}'_7 1 \end{array}$$

$$\begin{array}{c} 3,5 \quad 5,4 \\ 5131e_6 \otimes e_8 \otimes e'_7 1, \\ 2,2 \quad 3,5 \quad 5,5 \\ \otimes e_6 \otimes e_8 \otimes e'_7 1, \\ 4,5 \quad 5,5 \\ 1e_6 \otimes e_8 \otimes e'_7 1 \end{array}$$

$$\begin{array}{c} \begin{array}{ccc} 2,2 & 3,5 & 5,4 \\ \otimes e_6 & \otimes e_8 & \otimes e'_7 I^{\oplus 2}, \end{array} \\ I^{\oplus 2} e_6 \begin{array}{ccc} 3,5 & 5,5 \\ \otimes e_8 & \otimes e'_7 1, \end{array} \end{array}$$

$$\begin{array}{ccccc} I^{\oplus 2} \mathbf{e}_6 & \overset{3,5}{\otimes} & \mathbf{e}_8 & \overset{5,4}{\otimes} & \mathbf{e}'_7 I^{\oplus 2} \\ 1 \mathbf{e}_6 & \overset{4,5}{\otimes} & \mathbf{e}_8 & \overset{5,4}{\otimes} & \mathbf{e}'_7 I^{\oplus 2} \end{array}$$

$$\begin{array}{l} 3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 1223132, \\ 3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 12231513215, \end{array}$$

$$3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 I^{\oplus 6},$$

$$3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 12231513, \quad 1$$

$$\begin{array}{l} 3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 122315123 \\ 3150 \otimes^{2,4} \epsilon_8 \otimes^{5,5} \epsilon_8 12231513151 \end{array}$$

$$\begin{array}{l} 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6 123, \\ 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6 I^{\oplus 2}, \end{array}$$

$$\begin{array}{l} 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6 13151, \\ 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_6 13, \end{array}$$

$$\begin{array}{c} 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \epsilon_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \epsilon_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \\ 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \epsilon_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \epsilon_6 1 \end{array}$$

$$\begin{array}{l} 1_{50} \otimes^{1,1} 5_0 \otimes^{2,2} e_6 \otimes^{2,2}, \\ 1_{50} \otimes^{1,1} 5_0 \otimes^{2,3} e_6 1, \end{array}$$

$$\begin{array}{cc} 1,1 & 2,2 \\ 1s_0 \otimes s_0 & \otimes e_6 I^{\oplus 2}, \\ 2,2 & 2,2 \\ 1s_0 \otimes e_6 & \otimes s_0 1 \end{array}$$

$$31\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} \oplus \mathfrak{e}_6 1$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1e'_7 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e'_7 1$$

$$150 \otimes^{2,4} e_8 \otimes^{5,4} e'_7 123,$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e'_7 I^{\oplus 2},$$

$$3150 \otimes^{2,4} e_8 \otimes^{5,5} e'_7 1$$

$$150 \otimes^{2,3} e_7 \otimes^{3,3} e'_7 \otimes^{3,2},$$

$$150 \otimes^{2,3} e_7 \otimes^{3,3} e'_7 \otimes^{3,3},$$

$$150 \otimes^{2,3} e_7 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,3}{\otimes} e'_7 I^{\oplus 2},$$

$$3150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e'_7 1,$$

$$1\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \oplus \mathfrak{e}'_7 1$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix},$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_6 I^{\oplus 2}$$

$$1e'_7 \overset{3,3}{\otimes} e'_7 \overset{3,3}{\otimes} e'_7 123,$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}'_7 I^{\oplus 2}$$

$$1\mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$31e_6 \overset{3,3}{\otimes} e'_7 \overset{3,3}{\otimes} e'_7 1$$

2.4.5 SgLIgIg(65)

$$\begin{aligned} & 5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \\ & 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \\ & I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{aligned}$$

$$\begin{aligned} & 5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\ & I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \\ & 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \\ & 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \\ & I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \\ & 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{aligned}$$

$$\begin{aligned} & 321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ & 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{aligned}$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$51321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$3221\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$512321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7,$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$5131\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$2321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7,$$

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$2321\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$315131\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$5131513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7$$

$$321\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$31\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$3151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$2321\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7$$

$$23151321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

2.4.6 SgLIgLgS(128)

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7 123151,$$

$$321\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 2},$$

$$I^{\oplus 2}\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$I^{\oplus 2}\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 3}$$

$$I^{\oplus 2}\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8,$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 122313,$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 1223151$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8,$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 I^{\oplus 5},$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8 12231512$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 1223132,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 12231513215,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 I^{\oplus 6},$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 12231513,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 122315123$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_8 12231513151$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 123,$$

$$I^{\oplus 2}\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6,$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 13151,$$

$$I^{\oplus 2}\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 I^{\oplus 2},$$

$$\otimes \epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6$$

$$31\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_6 1315$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 3}$$

$$1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 123,$$

$$1\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 1$$

$$I^{\oplus 2}\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 2},$$

$$I^{\oplus 2}\epsilon_7 \otimes \epsilon_8 \otimes \epsilon_7 1$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$31\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 2}$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 \otimes \epsilon_7,$$

$$1\epsilon_6 \otimes \epsilon_7 \otimes \epsilon_7 I^{\oplus 3}$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_6 \otimes \epsilon_6,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_6 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_6 1$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_7 I^{\oplus 2},$$

$$31\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_7 1,$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_7 1$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \mathfrak{so} 1,$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7' I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \epsilon_8 \otimes \epsilon_7' 1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 I^{\oplus 2},$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 123,$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1$$

2.4.7 SSgLg(80)

$$513221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$313221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$21513221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$1513221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$\otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$313221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$21513221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$1513221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so},$$

$$I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$51321 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\begin{array}{c} 2,3 \ 1 \ 3,3 \quad 3,3 \\ \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \\ I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{c} 3,3 \ 1 \ 3,3 \quad 3,3 \\ \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \end{array}$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\begin{array}{c} 2,4 \ 1 \ 5,4 \quad 3,2 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \\ I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}, \end{array}$$

$$\begin{array}{c} I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}, \\ 1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{array}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$\begin{array}{c} 2,4 \ 1 \ 5,3 \quad 2,2 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}, \\ 1 \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{array}$$

$$I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$\begin{array}{c} 1513221 \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6, \\ 1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_6 \otimes \mathfrak{e}_7 \otimes \mathfrak{so}$$

$$5131513221 \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$1513221 \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$31 \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$31 \mathfrak{e}_6 \otimes \mathfrak{e}_6 \bigcirc \mathfrak{e}_6$$

$$5131 \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so}$$

$$321 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$321 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$321 \mathfrak{e}_7 \otimes \mathfrak{e}_7' \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6$$

$$31 \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_7',$$

$$1 \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_7'$$

$$31513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_6^{5,5}$$

$$2321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4},$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5}$$

$$2321 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2,$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5}$$

$$2321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$31 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,5}$$

$$513221 I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,5}$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{so}^{3,2} \otimes \mathfrak{so}^{1,1}$$

$$31 \mathfrak{e}_6^1 \otimes \mathfrak{so}^{2,2} \otimes \mathfrak{so}^{1,1}$$

$$31513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7'^{3,3}$$

$$231513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_6^{5,5}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{so}^{5,3}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{so}^{3,2}$$

$$1 \mathfrak{e}_7'^1 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{so}^{2,2}$$

2.4.8 SSgLgLgS(548)

$$\begin{array}{l}
I^{\oplus 2} \epsilon_7^{r^{\oplus 2} \, 3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{1 \, 1232} \\
1 \epsilon_7^{1 \, 4,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2} \\
321 \epsilon_7^{1 \, 3,3} \otimes \epsilon_7^{4,4} \otimes \epsilon_7^1 \\
I^{\oplus 2} \epsilon_7^{1 \, 3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_7^1
\end{array}$$

$$\begin{array}{l}
1 \quad 3,3 \quad 4,5 \\
1\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 1223151315 \\
2,4 \quad 1 \quad 5,4 \quad 3,3 \quad 3,3 \\
\otimes \mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \\
I^{\oplus 4} 1 \quad 5,4 \quad 3,3 \quad 3,3 \\
\mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \\
I^{\oplus 2} 1 \quad 3,3 \quad 4,5 \quad 5,5 \\
\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \\
I^{\oplus 2} 1 \quad 3,3 \quad 4,5 \\
\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 122313 \\
2321\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 I^{\oplus 3} \\
I^{\oplus 2} 1 \quad 3,3 \quad 4,5 \\
\mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \mathbf{e}_8 12231512 \\
I^{\oplus 4} 1 \quad 5,4 \quad 3,3 \\
\mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 I^{\oplus 3} \\
I^{\oplus 3} I^{\oplus 2} 5,4 \quad 3,3 \quad 3,2 \\
\mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 \otimes \\
I^{\oplus 2} 3221 \quad 5,4 \quad 3,3 \\
\mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 I^{\oplus 2} \\
I^{\oplus 3} I^{\oplus 2} 5,4 \quad 3,3 \\
\mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7 I^{\oplus 3} \\
I^{\oplus 2} 1 \quad 4,5 \quad 5,4 \quad 3,3 \\
\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \\
1 \quad 5,5 \quad 5,4 \quad 3,3 \\
1\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_7 \otimes \\
1 \quad 5,5 \quad 5,5 \\
1\mathbf{e}_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_7 I^{\oplus 2}
\end{array}$$

$$\begin{array}{l}
1^{3,5} \quad 5,5 \\
1\epsilon_6^1 \otimes \epsilon_8 \otimes \epsilon_8 122315123 \\
\\
1^{3,5} \quad 5,5 \\
1\epsilon_6^1 \otimes \epsilon_8 \otimes \epsilon_8 12231513151 \\
\\
5,5 \quad 1 \quad 5,5 \quad 5,3 \quad 2,2 \\
\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6 \otimes \\
\\
21513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{2,2} \\
\\
3,5 \quad 1 \quad 5,5 \quad 5,3 \\
\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_6 I^{\oplus 2} \\
\\
313221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 I^{\oplus 2} \\
\\
21513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 I^{\oplus 2} \\
\\
I^{\oplus 2} \otimes \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{2,2} \\
\\
I^{\oplus 2} \otimes \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 I^{\oplus 2} \\
\\
I^{\oplus 3} \otimes \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6 1315
\end{array}$$

$$I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_7' 1$$

$$\begin{array}{l}
1_{\mathbf{e}_6}^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{4,4} \\
1_{\mathbf{e}_6}^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{12313} \\
1_{\mathbf{e}_6}^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1232} \\
\otimes \mathbf{e}_7^{2,3,1} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{2,2} \\
\otimes \mathbf{e}_7^{2,3,1} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{I^{\oplus 2}} \\
I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{1315} \\
I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{4,3} \otimes \mathbf{e}_6^1
\end{array}$$

$$\begin{array}{l} \begin{array}{l} 5,5 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1 \end{array} \\ 21513221 \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1 \\ \begin{array}{l} 2,4 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}13 \end{array} \\ I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1 \\ I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}132 \end{array}$$

$$\begin{array}{l}
I^{\oplus 3} \epsilon_8^{+5,4} \otimes \epsilon_7^{3,3} \otimes \epsilon_6^1 \\
1 \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7^{3,2} \\
1 \epsilon_7^{4,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{1315} \\
1 \epsilon_7^{5,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{2,2} \\
I^{\oplus 2} \epsilon_7^{4,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_6^1
\end{array}$$

$$\begin{array}{l} 1\overset{1}{e_7} \otimes \overset{3,3}{e_6} \otimes \overset{2,2}{e_6} 123 \\ 1\overset{1}{e_7} \otimes \overset{3,3}{e_6} \otimes \overset{2,2}{e_6} 13151 \\ 31\overset{1}{e_6} \otimes \overset{2,2}{e_6} \otimes \overset{3,3}{e_7} I^{\oplus 2} \end{array}$$

$$\begin{aligned}
& 1\epsilon_7^1 \otimes \epsilon_6^{4,3} \otimes \epsilon_6^{2,2} \otimes, \\
& 31\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_7^{3,4} 1, \\
& 1\epsilon_6^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_6^{3,3} I^{\oplus 2} \\
& 5131513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \mathfrak{so}1, \\
& \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \mathfrak{so}13, \\
& 1513221 I^{\oplus 2} \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \mathfrak{so}1, \\
& 1513221\epsilon_8^1 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \mathfrak{so}13, \\
& I^{\oplus 3} \epsilon_8^{I^{\oplus 3}} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \mathfrak{so}13 \\
& 1\epsilon_7' \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,2}, \\
& 321\epsilon_7'^1 \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,2} 1, \\
& I^{\oplus 2} \epsilon_7'^1 \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,2}, \\
& I^{\oplus 2} \epsilon_7'^1 \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,2} I^{\oplus 3}, \\
& 1\epsilon_6^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,2}, \\
& 1\epsilon_6^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,2} I^{\oplus 3}, \\
& I^{\oplus 2} \epsilon_7'^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_6^{2,2}, \\
& 1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2}, \\
& 1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} 132, \\
& 1\epsilon_7'^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{2,2}, \\
& 1\epsilon_6^1 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_7'^1, \\
& 1\epsilon_7'^1 \otimes \epsilon_6^{3,3} \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{2,2}, \\
& 31\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^1, \\
& \otimes \epsilon_7^{2,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2} \mathfrak{so}1, \\
& 1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_7^{3,4} I^{\oplus 2}, \\
& 1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_7^{3,5} I^{\oplus 2}, \\
& 1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_7^{3,5} I^{\oplus 5}, \\
& I^{\oplus 2} \epsilon_8^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \mathfrak{so}13, \\
& 1\epsilon_7^1 \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3}, \\
& 1\epsilon_7^1 \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} I^{\oplus 3}, \\
& I^{\oplus 2} \epsilon_7'^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3}, \\
& 1\epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7'^{4,4} \otimes \epsilon_7'^1 \\
& 1\epsilon_6^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3}, \\
& 1\epsilon_7^1 \otimes \epsilon_7'^{3,3} \otimes \epsilon_6^{3,3} 13, \\
& I^{\oplus 2} \epsilon_7'^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_6^{3,3} I^{\oplus 2}, \\
& I^{\oplus 2} \epsilon_7^1 \otimes \epsilon_7'^{3,3} \otimes \epsilon_7'^{3,3} \otimes \epsilon_6^1 \\
& 1\epsilon_7'^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_6^{3,3} 1315 \\
& 1\epsilon_7'^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_6^{4,3} \otimes \epsilon_6^1 \\
& 1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4}, \\
& 1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} I^{\oplus 2} \\
& 1\epsilon_7'^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,3} \otimes \epsilon_6^{5,3} I^{\oplus 2}, \\
& 1\epsilon_7'^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_6^1 \\
& 1\epsilon_7'^1 \otimes \epsilon_6^{3,3} \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{2,2} I^{\oplus 2}, \\
& 1\epsilon_6^1 \otimes \epsilon_6^{2,2} \otimes \epsilon_7'^{3,4} \otimes \epsilon_7'^1 \\
& I^{\oplus 3} \epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2} \mathfrak{so}1, \\
& I^{\oplus 3} \epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2} \mathfrak{so}1, \\
& I^{\oplus 3} \epsilon_7^1 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2} \mathfrak{so}1
\end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_7^{I^{\oplus 2} \begin{smallmatrix} 3,3 \\ 3,2 \end{smallmatrix}} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}1$$

$$\begin{smallmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{smallmatrix} \begin{smallmatrix} 1 \\ \otimes \mathfrak{e}_6 \end{smallmatrix} \begin{smallmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{smallmatrix} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_6^{1 \begin{smallmatrix} 2,2 \\ 2,2 \end{smallmatrix}} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}1,$$

$$31 \mathfrak{e}_6^{31 \begin{smallmatrix} 2,2 \\ 2,2 \end{smallmatrix}} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \begin{smallmatrix} 3,2 \\ 1,1 \end{smallmatrix}} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1,$$

$$1 \mathfrak{e}_7^{1 \begin{smallmatrix} 3,2 \\ 1,1 \end{smallmatrix}} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}13,$$

$$1 \mathfrak{e}_7^{1 \begin{smallmatrix} 4,2 \\ 1,1 \end{smallmatrix}} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$321 \mathfrak{e}_7^{1 \begin{smallmatrix} 4,5 \\ 4,2 \end{smallmatrix}} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_7^{1 \begin{smallmatrix} 4,5 \\ 4,2 \end{smallmatrix}} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13,$$

$$1 \mathfrak{e}_7^{1 \begin{smallmatrix} 5,5 \\ 4,2 \end{smallmatrix}} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13$$

$$1 \mathfrak{e}_7'^{1 \begin{smallmatrix} 4,5 \\ 5,4 \end{smallmatrix}} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7' I^{\oplus 2},$$

$$1 \mathfrak{e}_7'^{1 \begin{smallmatrix} 4,5 \\ 5,5 \end{smallmatrix}} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{e}_6^{1 \begin{smallmatrix} 3,5 \\ 4,2 \end{smallmatrix}} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13,$$

$$31 \mathfrak{e}_6^{1 \begin{smallmatrix} 3,5 \\ 4,2 \end{smallmatrix}} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_7'^{1 \begin{smallmatrix} 3,3 \\ 3,2 \end{smallmatrix}} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}1$$

$$1 \mathfrak{e}_7'^{1 \begin{smallmatrix} 4,5 \\ 4,2 \end{smallmatrix}} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}13$$

$$1 \mathfrak{e}_7'^{1 \begin{smallmatrix} 3,3 \\ 3,3 \end{smallmatrix}} \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7'^{1 \begin{smallmatrix} 3,3 \\ 3,3 \end{smallmatrix}} \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_7'^{1 \begin{smallmatrix} 3,2 \\ 1,1 \end{smallmatrix}} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1 \mathfrak{e}_6^{1 \begin{smallmatrix} 3,3 \\ 3,2 \end{smallmatrix}} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}1$$

$$1 \mathfrak{e}_6^{1 \begin{smallmatrix} 2,2 \\ 1,1 \end{smallmatrix}} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

2.4.9 SSgLgLIgS(165)

[illegible]

$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1\ 5,3},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1\ 5,4},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1\ 5,5}$
$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{122313},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8 I^{\oplus 5},$	$\epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 2}} 122315$
$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{12231512},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{1223151},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 2,4,2}}$
$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 2}} I^{\oplus 4},$	$I^{\oplus 2} \epsilon_7^{1\ 4,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 3}} I^{\oplus 3},$	$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1\ 5,3}$
$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1\ 5,4},$	$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1\ 5,5},$	$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{122313}$
$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 5}},$	$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{12231512},$	$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{1223151}$
$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 2,4,2}},$	$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 2}} I^{\oplus 4},$	$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8^{I^{\oplus 3}}$
$\epsilon_7^{1\ 5,5} \otimes \epsilon_8^{5,5\ 1} \epsilon_8,$		

$$I^{\oplus 2} \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2} 1,$$

$$31 \mathbf{e}_6^{31} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} \mathbf{e}_6^1 \otimes,$$

$$31 \mathbf{e}_6^{31} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2} 1$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_8^{4,5} \mathbf{e}_8^1 \otimes,$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_8^{4,5} \mathbf{e}_8^1 I^{\oplus 4},$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_8^{4,5} I^{\oplus 3} I^{\oplus 2}$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_7^{5,4} I^{\oplus 2} 1,$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_7^{5,5} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1 \otimes,$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1 \otimes,$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 3} 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} I^{\oplus 2}$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} 1,$$

$$31 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,3} \mathbf{e}_7^1,$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,4} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{3,3} \mathbf{e}_6^1$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} 1,$$

$$I^{\oplus 2} \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} \otimes \mathbf{e}_8^{5,4} I^{\oplus 2} 1,$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} \otimes \mathbf{e}_8^{5,5} \mathbf{e}_7^1$$

$$1 \mathbf{50} \otimes \mathbf{50}^{1,1} \otimes \mathbf{50}^{1,1} \mathbf{1}$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_8^{4,5} \otimes \mathbf{e}_8^{5,4} \mathbf{e}_7^1$$

$$1 \mathbf{e}_7' \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} \otimes \mathbf{e}_8^{5,4} \mathbf{e}_7^1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} I^{\oplus 2} 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6^{3,3} \mathbf{e}_7^1$$

2.4.10 SSgLIgIg(25)

$$1513221 \mathbf{e}_8 \overset{1}{\otimes} \overset{5,5}{\mathbf{e}_8} \overset{1}{\otimes} \overset{5,4}{\mathbf{e}_8} \otimes \mathbf{e}'_7,$$

$$I^{\oplus 4} \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_7',$$

$$I^{\oplus 2} \begin{smallmatrix} 3 & 2 & 2 & 1 \\ \mathbf{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5, 5 \\ \mathbf{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 1 & 5, 4 \\ \mathbf{e}_8 \otimes \mathbf{e}'_7 \end{smallmatrix},$$

$$I^{\oplus 3} I^{\oplus 2}_{\mathfrak{e}_8} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}'_7$$

$$\begin{array}{ccc} 2,4 & 1 & 5,5 \\ \otimes & \otimes & \otimes \\ \mathbf{e}_8 & \mathbf{e}_8 & \mathbf{e}'_7 \end{array}$$

$$1513221 \mathbf{e}_8^{1\ 5,5} \otimes \mathbf{e}_8^{1\ 5,3} \otimes \mathbf{e}_6,$$

$$I^{\oplus 4} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_6,$$

$$I^{\oplus 2} \begin{smallmatrix} 3221 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6,$$

$$I^{\oplus 3} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$\begin{array}{ccccc} 2,4 & 1 & 5,5 & 1 & 5,4 \\ \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{e}_6 \end{array}$$

$$\begin{array}{ccccc} 2,4 & 1 & 5,5 & I^{\oplus 2} & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{so}, \end{array}$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{5,5}{\otimes} I^{\oplus 2} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}$$

$$I^{\oplus 4} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,5 \\ \mathfrak{e}_8 \end{smallmatrix} I^{\oplus 2} \begin{smallmatrix} 4,2 \\ \mathfrak{e}_8 \end{smallmatrix} \otimes \mathfrak{so},$$

$$513221 \begin{matrix} I^{\oplus 2} & 5,5 & 1 & 4,2 \\ \mathfrak{e}_8 & \otimes & \mathfrak{e}_8 & \otimes & \mathfrak{so} \end{matrix}$$

$$31513221 \mathbf{e}_8^{1 \ 5,5} \otimes \mathbf{e}_8^{1 \ 5,4} \otimes \mathbf{e}_7,$$

$$1513221 \overset{1}{e}_8 \overset{5,5}{\otimes} \overset{1}{e}_8 \overset{5,5}{\otimes} e_7,$$

$$I^{\oplus 2} \begin{smallmatrix} 3221 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 1 \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} \\ \mathfrak{e}_7 \end{smallmatrix}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{1,3,2} \otimes \mathfrak{so}$$

$$321 \mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_7} \otimes \mathbf{e}_6$$

$$2321 \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7'$$

$$1_{\mathfrak{e}_7} \otimes I^{\oplus 2} \otimes 1_{\mathfrak{e}_8} \otimes 1_{\mathfrak{so}}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{1}{\otimes} \overset{5,5}{e'_7}$$

$$1 \overset{1}{\mathbf{e}}_7 \overset{4,5}{\otimes} \overset{1}{\mathbf{e}}_8 \overset{5,4}{\otimes} \mathbf{e}_6$$

$$1e_7' \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} 1e_7 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$1 \overset{1}{\mathfrak{e}_6} \overset{2,2}{\otimes} 1 \overset{2,2}{\otimes} \mathfrak{so}$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'{}^{1,3,3},$$

$$1 \mathfrak{e}_7'{}^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} I^{\oplus 3},$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'{}^{1,3,3} I^{\oplus 2}$$

$$\begin{matrix} 2,4 \\ \otimes \mathfrak{e}_8 \end{matrix} \begin{matrix} 1 \\ \otimes \mathfrak{e}_8 \end{matrix} \begin{matrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{matrix} \begin{matrix} 1 \\ \otimes \mathfrak{e}_6 \end{matrix} \begin{matrix} 5,3 \\ \otimes \mathfrak{e}_6 \end{matrix} 1,$$

$$I^{\oplus 4} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 1,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 1$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 13,$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2}} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 13$$

$$\begin{matrix} 2,4 \\ \otimes \mathfrak{e}_8 \end{matrix} \begin{matrix} 1 \\ \otimes \mathfrak{e}_8 \end{matrix} \begin{matrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{matrix} \begin{matrix} 1 \\ \otimes \mathfrak{e}_8 \end{matrix} \begin{matrix} 5,4 \\ \otimes \mathfrak{e}_7' \end{matrix} 1,$$

$$I^{\oplus 4} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7' 1,$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2},$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2},$$

$$31 \mathfrak{e}_6^1 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6 1$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1,$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6 1,$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6 13$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7 I^{\oplus 2},$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_7 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,4} \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,3} \otimes \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_7'{}^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1$$

$$1 \mathfrak{e}_7'{}^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7'{}^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6 1$$

$$1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so} 1$$

2.4.12 SSgLIgLIgS(22)

$$\begin{array}{lll}
1\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 122315132, & \overset{2,4}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{4,2}{\otimes}, & I^{\oplus 4}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{4,2}{\otimes} \\
3221\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 122315, & I^{\oplus 2}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 12231513, & I^{\oplus 3}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 1223151 \\
I^{\oplus 4}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 I^{\oplus 4}, & \overset{2,4}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 I^{\oplus 3} I^{\oplus 2}, & I^{\oplus 3}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 I^{\oplus 2} 1223 \\
1513221\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 I^{\oplus 2}, & I^{\oplus 4}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 I^{\oplus 3} I^{\oplus 2}, & I^{\oplus 2} I^{\oplus 2}\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 I^{\oplus 2} 1223 \\
I^{\oplus 3}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathbf{e}}_8 I^{\oplus 2} & &
\end{array}$$

$$1e_7^{1,4,5} \otimes e_8^{1,5,5} \otimes e_8^{1,4,2}, \quad 1e_7^{1,4,5} \otimes e_8^{1,5,5} \otimes e_8^{1,4} I^{\oplus 4}, \quad 1e_7^{1,4,5} \otimes e_8^{1,5,5} \otimes e_8^{1,4} I^{\oplus 3} I^{\oplus 2}$$

$$1\mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^1 123, \quad I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1,3,3} I^{\oplus 2} \mathbf{e}_7^1$$

$$1 \overset{1}{\mathbf{e}}_7 \overset{4,5}{\otimes} \overset{1}{\mathbf{e}}_8 \overset{5,4}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1 \overset{1}{\mathbf{e}}_7' \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} I^{\oplus 2} \overset{1}{\mathbf{e}}_7 1$$

$$1 \overset{1}{e}'_7 \overset{3,3}{\otimes} \overset{1}{e}_7 \overset{3,3}{\otimes} \overset{1}{e}'_7 1$$

$$1 \overset{1}{\mathbf{e}_6} \overset{2,2}{\otimes} \overset{1}{\mathbf{e}_6} \overset{2,2}{\otimes} \overset{1}{\mathbf{e}_6} 1$$

2.5 4 node(1730)

2.5.1 gLgLgLg(16)

$$\mathfrak{so}(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{3,3}{\otimes} \mathfrak{e}_6, \quad \mathfrak{so}(\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{3,3}{\bigcirc} \mathfrak{e}_6$$

$$50 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_6$$

$$50 \overset{3,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,3}{\otimes} 50$$

$$\mathfrak{so}\left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7\right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$50 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_7$$

$$50 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} e_7'$$

$$50 \overset{2,3}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{4,3}{\otimes} e_6$$

$$50 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_6$$

$$\mathfrak{so}\left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6\right)^2 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_7$$

$$\mathfrak{so}\left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6\right)^2 \begin{smallmatrix} 3,4 \\ \otimes \end{smallmatrix} \mathfrak{e}'_7$$

$$50 \overset{1,1}{\otimes} 50 \overset{2,3}{\otimes} e_7 \overset{3,2}{\otimes} 50$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} \otimes \mathfrak{e}_6 \otimes \mathfrak{so}^{2,2}$$

$$\mathbf{e}_6 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^2 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$e_6 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e'_7$$

$$\mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_6$$

2.5.2 gLIgLgLg(11)

$$e_7 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7'$$

$$e_7 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,4}{\otimes} e_6$$

$$e'_7 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 & 4,2 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$e'_7 \overset{4,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e'_7$$

$$\mathbf{e}_6 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 1,4,2 \\ \otimes \end{smallmatrix} \mathbf{so} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathbf{so}$$

$$e_6 \overset{3,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,4}{\otimes} e_6$$

$$e_6 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} 50$$

2.5.3 gLIgLIgLg(3)

$$50 \quad \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \quad \begin{smallmatrix} 1 \\ e_8 \end{smallmatrix} \quad \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \quad \begin{smallmatrix} 1 \\ e_8 \end{smallmatrix} \quad \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \quad e_7$$

$$50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e'_7$$

$$50 \begin{matrix} 2,4 \\ \otimes \end{matrix} e_8 \begin{matrix} 5,5 \\ \otimes \end{matrix} e_8 \begin{matrix} 5,3 \\ \otimes \end{matrix} e_6$$

2.5.4 SgLgLgLg(175)

$$\begin{array}{cccc}
3,5 & 5,5 & 5,4 & 3,3 \\
\otimes \epsilon_8 & \otimes \epsilon_8 & \otimes \epsilon_7 & \otimes \epsilon_6, \\
313221\epsilon_8 & \overset{5,5}{\otimes} \epsilon_8 & \overset{5,4}{\otimes} \epsilon_7 & \overset{3,3}{\otimes} \epsilon_6, \\
1513221\epsilon_8 & \overset{5,5}{\otimes} \epsilon_8 & \overset{5,4}{\otimes} \epsilon_7 & \overset{3,3}{\otimes} \epsilon_6, \\
I^{\oplus 3} \epsilon_7 & \overset{4,5}{\otimes} \epsilon_8 & \overset{5,5}{\otimes} \epsilon_8 & \overset{5,5}{\otimes} \epsilon_6,
\end{array}$$

$$\begin{array}{cccc}
4,5 & 5,5 & 5,4 & 3,3 \\
\otimes e_8 & \otimes e_8 & \otimes e_7 & \otimes e_6, \\
I^{+5} e_8 & \otimes e_8 & \otimes e_7 & \otimes e_6, \\
2,3 & 4,5 & 5,5 & 5,5 \\
\otimes e_7 & \otimes e_8 & \otimes e_8 & \otimes e_6, \\
I^{+2} e_7 (\otimes e_8)^2 & \otimes e_6
\end{array}$$

$$\begin{array}{cccc} 5,5 & 5,5 & 5,4 & 3,3 \\ \otimes e_8 & \otimes e_8 & \otimes e_7 & \otimes e_6 \\ 21513221e_8 & \otimes e_8 & \otimes e_7 & \otimes e_6 \\ 3,3 & 4,5 & 5,5 & 5,5 \\ \otimes e_7 & \otimes e_8 & \otimes e_8 & \otimes e_6 \end{array}$$

$$1e_7 \overset{4,4}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{3,2}{\otimes} so,$$

$$\begin{matrix} 2,4 \\ \otimes \end{matrix} \mathfrak{e}_7 \left(\begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{e}_7 \right)^2 \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{so},$$

$${}^{3,4} \otimes e_7 ({}^{3,3} \otimes e_7)^2 {}^{3,3} \otimes 50$$

$$\begin{aligned} & \overset{4,4}{\otimes} \mathfrak{e}_7 (\overset{3,3}{\otimes} \mathfrak{e}_7)^2 \overset{3,3}{\otimes} \mathfrak{so}, \\ & 31321 \mathfrak{e}_7 (\overset{3,3}{\otimes} \mathfrak{e}_7)^2 \overset{3,3}{\otimes} \mathfrak{so}, \\ & 1512321 \mathfrak{e}_7 (\overset{3,3}{\otimes} \mathfrak{e}_7)^2 \overset{3,3}{\otimes} \mathfrak{so}, \end{aligned}$$

$$\begin{aligned} & \otimes \mathfrak{e}_7 \left(\otimes \mathfrak{e}_7 \right)^2 \otimes \mathfrak{so}, \\ 2151321 \mathfrak{e}_7 \left(\otimes \mathfrak{e}_7 \right)^2 \otimes \mathfrak{so}, \\ 151321 \mathfrak{e}_7 \left(\otimes \mathfrak{e}_7 \right)^2 \otimes \mathfrak{so} \end{aligned}$$

$$I^{\oplus 4} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{so}$$

$$\begin{aligned} &2321\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^2\overset{3,3}{\otimes}\epsilon_7, \\ &321\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^2\overset{3,4}{\otimes}\epsilon_7, \\ &I^{\oplus 2}\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^2\overset{3,5}{\otimes}\epsilon_7, \end{aligned}$$

$$\begin{aligned} & 2321\mathbf{e}_7 \otimes \mathbf{e}_6 (\otimes \mathbf{e}_6)^2, \\ & 15131\mathbf{e}_6 (\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7, \\ & 31\mathbf{e}_6 \otimes \mathbf{e}_6 \otimes \mathbf{e}_7 \otimes \mathbf{e}_6, \end{aligned}$$

$$\begin{array}{c} 315131\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^2\overset{3,3}{\otimes}\epsilon_7 \\ \overset{2,2}{\otimes}\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^2\overset{3,5}{\otimes}\epsilon_7 \\ 31\epsilon_6\overset{3,3}{\otimes}\epsilon_7\overset{3,3}{\otimes}\epsilon_6\overset{2,2}{\otimes}\epsilon_6 \end{array}$$

$$\begin{array}{cccc} 2,3 & 3,3 & 3,3 & 3,3 \\ \otimes & \otimes & \otimes & \otimes \\ e_7 & e_7 & e'_7 & e_6, \\ I^{\oplus 3} e_7 & \otimes & \otimes & \otimes \\ & 3,3 & 3,3 & 3,3 \\ & e_7 & e'_7 & e_6, \end{array}$$

$$\begin{array}{cccc} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 & \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 & \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7' & \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_6, \\ I^{\oplus 2} e_7' & (\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7)^2 & \begin{smallmatrix} 4,3 \\ \otimes \end{smallmatrix} e_6, \end{array}$$

$$\begin{array}{c} 321e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_7 \otimes^{3,3} e_6 \\ 1e_6 (\otimes^{3,3} e_7)^2 \otimes^{4,4} e_7' \end{array}$$

$$\begin{aligned} & 3221e_7 \otimes e_7 (\otimes e_7')^2, \\ & 512321e_7 \otimes e_7 (\otimes e_7')^2, \end{aligned}$$

$$\begin{aligned} & 2321e_7' \otimes^{3,3} e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_7, \\ & 151321e_7 \otimes^{3,3} e_7 (\otimes^{3,3} e_7')^2, \end{aligned}$$

$$2321e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_7 \otimes^{3,3} e_7'$$

$$I^{\oplus 2} e_7' (\otimes^{3,3} e_7)^2 \otimes^{4,4} e_7'$$

$$\begin{aligned} & 3221e_7(\otimes e_7)^2 \otimes e_7, \\ & \otimes e_7(\otimes e_7)^2 \otimes e_7, \end{aligned}$$

$$512321e_7(\otimes e_7)^2 \otimes e_7, \\ \otimes e_7(\otimes e_7)^2 \otimes e_7,$$

$$151321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{4,4}{\otimes}\mathbf{e}_7$$

$$I^{\oplus 3}\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{4,5}{\otimes}\mathbf{e}_7$$

$$\begin{aligned} & 3151321e_7^{3,3} \otimes e_7^{3,3} \otimes e_7^{'3,3} \otimes e_7, \\ & 321e_7^{'3,3} (\otimes e_7)^2 \otimes e_7^{4,4}, \end{aligned}$$

$$\begin{aligned} & \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^2 \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} e'_7, \\ & I^{\oplus 3} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^2 \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} e'_7, \end{aligned}$$

$$I^{\oplus 2} e'_7 \left(\bigotimes e_7 \right)^2 \bigotimes e_7$$

$$2313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_6,$$

$$51231513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_6,$$

$$I^{\oplus 6} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$31513221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes^{5,5}\epsilon_8)^2\otimes^{5,5}\epsilon_6 \\ 15131513221\epsilon_8(\otimes^{5,5}\epsilon_8)^2\otimes^{5,5}\epsilon_6 \end{array}$$

$$\begin{aligned} &2313221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,3}{\otimes}\mathfrak{so}, \\ &51231513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,3}{\otimes}\mathfrak{so}, \end{aligned}$$

$$I^{\oplus 6} \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{(5,3)},$$

$$31513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{(5,3)},$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \mathfrak{so} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \mathfrak{so} \end{array}$$

$$\begin{aligned} & \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^2 \begin{smallmatrix} 4,3 \\ \otimes \end{smallmatrix} e_6, \\ & 31 e_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^2 \begin{smallmatrix} 4,4 \\ \otimes \end{smallmatrix} e_7, \end{aligned}$$

$$\begin{array}{l} \begin{array}{l} 3,3 \\ \otimes \end{array} e_7 \left(\begin{array}{l} 3,3 \\ \otimes \end{array} e_7 \right)^2 \begin{array}{l} 4,3 \\ \otimes \end{array} e_6, \\ 1e_6 \left(\begin{array}{l} 3,3 \\ \otimes \end{array} e_7 \right)^2 \begin{array}{l} 4,5 \\ \otimes \end{array} e_7 \end{array}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes^{(3,3)} \mathbf{e}_7 \right)^2 \bigotimes^{(4,3)} \mathbf{e}_6$$

$$31513221\mathbf{e}_8 \otimes^{5,3} \mathbf{e}_6 (\otimes^{2,2} \mathbf{e}_6)^2,$$

$$22315131\mathbf{e}_6\left(\overset{2,2}{\otimes}\mathbf{e}_6\right)^2\overset{3,5}{\otimes}\mathbf{e}_8,$$

$$31\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6$$

$$31\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$1\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_6$$

$$231513221\mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7(\overset{3,3}{\otimes} \mathbf{e}_7)^2,$$

$$223151321\mathbf{e}_7(\otimes^{3,3}\mathbf{e}_7)^2\otimes^{4,5}\mathbf{e}_8,$$

$$2321e_7 \otimes^{3,3} e_7 \otimes^{4,5} e_8 \otimes^{5,4} e_7$$

$$2321e_7 \overset{4,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes} e_7,$$

$$321e_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7$$

$$1513221e_8 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes} e_7 \overset{3,3}{\otimes} e_6,$$

$$321e_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,3}{\otimes} e_6,$$

$$I^{\oplus 2} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_6$$

$$I^{\oplus 2} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6,$$

$$1e_7 \overset{5,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes} e_6$$

$$\begin{matrix} 2,3 & 4,5 & 5,5 & 5,3 \\ \otimes e_7 & \otimes e_8 & \otimes e_8 & \otimes 50, \end{matrix}$$

$$\begin{matrix} 3,3 & 4,5 & 5,5 & 5,3 \\ \otimes e_7 & \otimes e_8 & \otimes e_8 & \otimes 50, \end{matrix}$$

$$I^{\oplus 3} \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \oplus 50$$

$$I^{\oplus 2} \mathfrak{e}_7 \left(\bigotimes^{5,5} \mathfrak{e}_8 \right)^2 \bigotimes^{5,3} \mathfrak{so}$$

$$\begin{matrix} 2,3 & 3,3 & 3,2 & 1,1 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{so} & \otimes \mathfrak{so}, \end{matrix}$$

$$\begin{matrix} 3,3 & 3,3 & 3,2 & 1,1 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{so} & \otimes \mathfrak{so}, \end{matrix}$$

$$I^{\oplus 3} \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1_{\mathfrak{so}} \otimes^{2,3} e_7 \otimes^{3,3} e_7 \otimes^{3,3} \mathfrak{so}$$

$$\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6,$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\bigotimes^{2,2} \mathbf{e}_6 \right)^2 \bigotimes^{3,3} \mathbf{e}_6,$$

$$\overset{2,2}{\otimes} \mathfrak{e}_6 (\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{3,3}{\bigcirc} \mathfrak{e}_6$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\overset{2,2}{\otimes} \mathbf{e}_6 \right)^2 \overset{3,3}{\bigcirc} \mathbf{e}_6$$

$$I^{\oplus 2} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{3,2}{\otimes} \mathfrak{so},$$

$$\begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\begin{matrix} 3,3 \\ \otimes \end{matrix} e'_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} e_7)^2 \begin{matrix} 3,3 \\ \otimes \end{matrix} so$$

$$I^{\oplus 3} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{so}$$

$$5131513221\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$\begin{matrix} 2,2 & 3,5 & 5,5 & 5,5 \\ \otimes & \otimes & \otimes & \otimes \\ \mathfrak{e}_6 & \mathfrak{e}_8 & \mathfrak{e}_8 & \mathfrak{e}_6, \end{matrix}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$3221\mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6,$$

$$512321\mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6,$$

$$151321\mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6$$

$$1\mathbf{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \begin{smallmatrix} 4,3 \\ \otimes \end{smallmatrix} \mathbf{e}_6$$

$$321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}'_7,$$

$$15131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}'_7,$$

$$\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array} \left(\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array} \right)^2 \begin{array}{c} 3,4 \\ \otimes \mathbf{e}_7' \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\bigotimes^{2,2} \mathbf{e}_6 \right)^2 \bigotimes^{3,4} \mathbf{e}_7'$$

$$I^{\oplus 2} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$321\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$51321\mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{4,2}{\otimes} \mathbf{50}$$

$$5131\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}$$

$$1e_6 \overset{2,2}{\otimes} e_6 \overset{3,3}{\otimes} e_7 \overset{3,2}{\otimes} 50$$

$$150 \overset{2,4}{\otimes} e_8 \overset{5,3}{\otimes} e_6 \overset{2,2}{\otimes} 50$$

$$1\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_6$$

$$31\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{2,4}{\otimes}\mathfrak{e}_7$$

$$150 \left(\begin{smallmatrix} 1,1 \\ \otimes 50 \end{smallmatrix} \right)^2 \begin{smallmatrix} 2,3 \\ \otimes e_6 \end{smallmatrix}$$

$$1e'_7 \overset{4,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes} e_6$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\bigotimes \mathfrak{e}_7 \right)^2 \bigotimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \oplus \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1\mathbf{e}'_7(\overset{5,5}{\otimes}\mathbf{e}_8)^2\overset{5,5}{\otimes}\mathbf{e}_6$$

2.5.5 SgLgLgLgS(500)

$$I^{\oplus 2} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 I^{\oplus 2}$$

$$\begin{array}{ccccccccccc} 2,2 & 3,5 & 5,5 & 5,5 & & & & & & & \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & & & & & & & \\ & & & & 122315123 & & & & & & \end{array}$$

$$\begin{array}{cccccccc} 2,2 & 3,5 & 5,5 & 5,5 \\ \otimes \mathfrak{e}_6 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & 12231513151 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 122315123$$

$$I^{\oplus 2} \mathbf{e}_6^{3,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} 12231513151$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 122315123$$

$$1e_6 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231513151$$

$$1e'_7 \otimes^{3,3} e_7 \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,5}$$

$$1e_7' \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231512$$

$$I^{\oplus 2} \mathbf{e}'_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix}$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes}$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_7 I^{\oplus 2}$$

$$1\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes}$$

$$1e_6 \overset{2,2}{\otimes} e_6 \overset{3,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 12231512$$

$$31\mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 I^{\oplus 4}$$

$$I^{\oplus 2} \mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{c} 2,2 \\ \otimes \mathfrak{so} \end{array} \left(\begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \right)^2 \begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \begin{array}{c} 2,4 \\ \otimes \end{array}$$

$$\begin{array}{c} 3,2 \\ \otimes \mathfrak{so} \end{array} \left(\begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \right)^2 \begin{array}{c} 1,1 \\ \otimes \mathfrak{so} \end{array} \begin{array}{c} 2,3 \\ \otimes \end{array}$$

$$I^{\oplus 2} \mathfrak{so} \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \right)^2 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix}$$

$$I^{\oplus 2} \mathfrak{so} \left(\begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \right)^2 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix}$$

$$\begin{matrix} 2,2 & (2,2) & 3,5 & 4,2 \\ \otimes \mathbf{e}_6 & (\otimes \mathbf{e}_6)^2 & \otimes \mathbf{e}_8 & \otimes \end{matrix}$$

$$\begin{aligned} &150(\otimes \epsilon_6)^2 \otimes \epsilon_6 \otimes, \\ &150(\otimes \epsilon_6)^2 \otimes \epsilon_6 \otimes, \\ &150(\otimes \epsilon_6)^2 \otimes \epsilon_6 1313, \end{aligned}$$

$$\begin{aligned} & 150 \left(\overset{2,2}{\otimes} \epsilon_6 \right)^2 \overset{2,2}{\otimes} \epsilon_6 \overset{3,3}{\otimes}, \\ & 150 \left(\overset{2,2}{\otimes} \epsilon_6 \right)^2 \overset{2,2}{\otimes} \epsilon_6 \bigcirc, \\ & 150 \left(\overset{2,2}{\otimes} \epsilon_6 \right)^2 \overset{2,2}{\otimes} \epsilon_6 131512, \end{aligned}$$

$$\begin{array}{l} 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \right)^2 \otimes \begin{smallmatrix} 2,2 \\ \epsilon_6 \end{smallmatrix} \otimes \begin{smallmatrix} 3,4 \\ \epsilon_6 \end{smallmatrix} \\ 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \right)^2 \otimes \begin{smallmatrix} 2,2 \\ \epsilon_6 \end{smallmatrix} I^{\oplus 3} \\ 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \right)^2 \otimes \begin{smallmatrix} 2,2 \\ \epsilon_6 \end{smallmatrix} 13151^1 \end{array}$$

$$\begin{aligned} & 1\epsilon_7'(\overset{3,3}{\otimes}\epsilon_7)^2\overset{4,5}{\otimes}\epsilon_8 1223151, \\ & 2321\epsilon_7'(\overset{3,3}{\otimes}\epsilon_7)^2\overset{4,5}{\otimes}\epsilon_8 I^{\oplus 2}, \\ & 1\epsilon_7'\overset{4,5}{\otimes}\epsilon_8\overset{5,4}{\otimes}\epsilon_7\overset{3,3}{\otimes}\epsilon_7 I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & I^{\oplus 2} e_7' \binom{3,3}{\otimes} e_7 \binom{4,5}{\otimes} e_8 \otimes, \\ & I^{\oplus 2} e_7' \binom{3,3}{\otimes} e_7 \binom{4,5}{\otimes} e_8 I^{\oplus 4}, \\ & 1 e_7' \binom{3,3}{\otimes} e_7 \binom{4,5}{\otimes} e_8 \binom{5,5}{\otimes} e_7 1 \end{aligned}$$

$$\begin{array}{l} 321e'_7(\overset{3,3}{\otimes} e_7)^2 \overset{4,5}{\otimes} e_8 I^{\oplus 3} \\ 1e'_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,4}{\otimes} e_7 I^{\oplus 2} \end{array}$$

$$\begin{aligned} & 1\mathbf{e}_6 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes} \mathbf{e}_7 I^{\oplus 2}, \\ & \overset{2,2}{\otimes} \mathbf{e}_6 (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes}, \\ & I^{\oplus 2} \mathbf{e}_6 (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes}, \end{aligned}$$

$$\begin{aligned} & 31e_6^{3,3} \otimes e_7^{3,3} \otimes e_7'^{3,3} \otimes e_7 1, \\ & \otimes e_6^{2,2} (\otimes e_7^{3,3})^2 \otimes e_7'^{3,3} I^{\oplus 3}, \\ & I^{\oplus 2} e_6^{3,3} (\otimes e_7^{3,3})^2 \otimes e_7'^{3,3} I^{\oplus 3} \end{aligned}$$

$$\begin{array}{c} \begin{array}{ccccccc} 2,2 & & 3,3 & & 3,3 & & 3,2 \\ \otimes & \mathbf{e}_6 & (\otimes \mathbf{e}_7)^2 & \otimes & \mathbf{e}'_7 & \otimes & \\ I^{\oplus 2} & \mathbf{e}_6 & (\otimes \mathbf{e}_7)^2 & \otimes & \mathbf{e}'_7 & \otimes & \end{array} \end{array}$$

$$\begin{aligned} & 1\mathbf{e}_6(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 123, \\ & \otimes \mathbf{e}_6(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 I^{\oplus 2}, \\ & I^{\oplus 2} \mathbf{e}_6(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 1, \end{aligned}$$

$$\begin{array}{c} 321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}_71, \\ I^{\oplus 2}\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}_7I^{\oplus 2}, \\ 1\mathbf{e}_6\overset{2,2}{\otimes}\mathbf{e}_6\overset{3,3}{\otimes}\mathbf{e}_7\overset{3,3}{\otimes}\mathbf{e}_61 \end{array}$$

$$\begin{array}{l} 15131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,3}{\otimes}\mathbf{e}_71 \\ \overset{2,2}{\otimes}\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^2\overset{3,4}{\otimes}\mathbf{e}_71 \end{array}$$

$$\begin{array}{l} 1\overset{1,1}{s}\overset{2,4}{o}\overset{5,5}{\otimes}\overset{2,4}{e}\overset{5,5}{\otimes}\overset{5,5}{e}1223132, \\ 1\overset{1,1}{s}\overset{2,4}{o}\overset{5,5}{\otimes}\overset{2,4}{e}\overset{5,5}{\otimes}\overset{5,5}{e}12231513215, \\ 31\overset{2,4}{s}\overset{5,5}{\otimes}\overset{5,5}{e}\overset{4,2}{\otimes}\overset{4,2}{e}\overset{4,2}{s}013 \end{array}$$

$$150^{1,1} \otimes 50^{2,4} \otimes e_8^{5,5} \otimes e_8 I^{\oplus 6},$$

$$150^{1,1} \otimes 50^{2,4} \otimes e_8^{5,5} \otimes e_8 12231^1 513,$$

$$\begin{array}{l} 1\overset{1,1}{50} \otimes 5\overset{2,4}{0} \otimes \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 122315123 \\ 1\overset{1,1}{50} \otimes 5\overset{2,4}{0} \otimes \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513151 \end{array}$$

$$\begin{aligned} & 1\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,2}, \\ & 31\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7 I^{\oplus 2}, \\ & I^{\oplus 2} \epsilon_6^{(3,3)} (\otimes \epsilon_7)^2 \otimes \epsilon_6^{3,3} I^{\oplus 2} \end{aligned}$$

$$1\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3},$$

$$\otimes \epsilon_6^{2,2} (\otimes \epsilon_7^{3,3})^2 \otimes \epsilon_6^{3,3} \otimes,$$

$$1\epsilon_6^{2,2} \otimes \epsilon_6^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7 I^{\oplus 3}$$

$$I^{\oplus 2} \epsilon_6^{3,3} (\otimes \epsilon_7)^2 \otimes \epsilon_6^{2,2}$$

$$\begin{array}{cccc} 2,2 & 3,5 & 5,5 & 5,4 \\ \otimes e_6 & \otimes e_8 & \otimes e_8 & \otimes e'_7 I^{\oplus 2}, \\ I^{\oplus 2} e_6 & \otimes e_8 & \otimes e_8 & \otimes e'_7 1, \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}'_7 I^{\oplus 2},$$

$$\begin{array}{cccc} 2,2 & 3,5 & 5,5 & 5,5 \\ \otimes e_6 & \otimes e_8 & \otimes e_8 & \otimes e'_7 1 \\ 1e_6 & 4,5 & 5,5 & 5,5 \\ & \otimes e_8 & \otimes e_8 & \otimes e'_7 1 \end{array}$$

$$3150 \overset{2,4}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 1223132,$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 6},$$

$$3150 \overset{2,4}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 122315123$$

$$3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513215, \quad 3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513, \quad 3150 \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 2,4 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} \otimes_{\mathfrak{e}_8}^{\begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}} 12231513151$$

$$\begin{array}{lll} 150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} 12315, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} \otimes_{\mathfrak{e}_7}^{3,2}, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{5,4} \otimes_{\mathfrak{e}_7}^{3,3} \\ 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7}^{5,4} I^{\oplus 3}, & 3150 \otimes_{\mathfrak{e}_8}^{2,4} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_7} I^{\oplus 2} & \end{array}$$

$$\begin{array}{lll} 1\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} \otimes \epsilon_8^{(4,2)}, & 1\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} I^{\oplus 4}, & 31\epsilon_6^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{(4,5)} I^{\oplus 3} \\ 1\epsilon_6^{(3,5)} \otimes \epsilon_8^{(5,4)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7 I^{\oplus 2}, & 31\epsilon_6^{(3,5)} \otimes \epsilon_8^{(5,4)} \otimes \epsilon_7^{(3,3)} \otimes \epsilon_7 1 & \end{array}$$

$$\begin{array}{lll}
150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 1315, & 3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 I^{\oplus 2} \\
3150^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_6 1 & &
\end{array}$$

$$\begin{array}{lll} 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,2}, & 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3}, & 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7 I^{\oplus 3} \\ 1\mathfrak{s}\mathfrak{o}^{1,1} \otimes \mathfrak{s}\mathfrak{o}^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7 I^{\oplus 2} & & \end{array}$$

$$I^{\oplus 2} e_7' \otimes_{e_8}^{4,5} \otimes_{e_8}^{5,5} \otimes_{e_8}^{5,4} e_7' I^{\oplus 2}, \quad I^{\oplus 2} e_7' \otimes_{e_8}^{4,5} \otimes_{e_8}^{5,5} \otimes_{e_8}^{5,5} e_7' 1, \quad 1 e_7' (\otimes_{e_8}^{5,5})^2 \otimes_{e_7}^{5,5} e_7' 1$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix}, \quad 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix}, \quad 150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} I^{\oplus 3}$$

$$150 \otimes_{e_7}^{2,3} \otimes_{e_7}^{3,3} \otimes_{e_7}^{3,3} \otimes_{e_7'}^{3,2}, \quad 150 \otimes_{e_7}^{2,3} \otimes_{e_7}^{3,3} \otimes_{e_7'}^{3,3} \otimes_{e_7'}^{3,3}, \quad 150 \otimes_{e_7}^{2,3} \otimes_{e_7}^{3,3} \otimes_{e_7'}^{3,3} \otimes_{e_7'}^{3,3} I^{\oplus 3}$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6 \otimes, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,3}{\otimes} \mathfrak{e}_6 I^{\oplus 2}, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_6 1$$

$$150(\overset{1,1}{\otimes} \mathfrak{so})^2 \overset{2,3}{\otimes} \mathfrak{e}_7 I^{\oplus 2}, \quad 3150(\overset{1,1}{\otimes} \mathfrak{so})^2 \overset{2,3}{\otimes} \mathfrak{e}_7 1, \quad 150(\overset{1,1}{\otimes} \mathfrak{so})^2 \overset{2,4}{\otimes} \mathfrak{e}_7 1$$

$$1\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 I^{\oplus 2}, \quad 31\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,4} \otimes \epsilon_7 1, \quad 1\epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_7 1$$

$$\begin{array}{cc} \begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array} (\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array})^2 \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7 \end{array} 1, & I^{\oplus 2} \mathbf{e}_6 (\begin{array}{c} 2,2 \\ \otimes \mathbf{e}_6 \end{array})^2 \begin{array}{c} 3,3 \\ \otimes \mathbf{e}'_7 \end{array} 1 \end{array}$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,4 \\ e_8 \end{smallmatrix} \otimes e_7' I^{\oplus 2}, \quad 3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes \begin{smallmatrix} 5,5 \\ e_8 \end{smallmatrix} \otimes e_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,5} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2},$$

$$31\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{so}^{3,2} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{so}^{2,2} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so}^{1,1})^2 \otimes \mathfrak{e}_7^{2,3} I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so}^{1,1})^2 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,4} I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2}$$

2.5.6 SgLgLIgLg(57)

$$\begin{array}{lll}
5131513221\epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{1}{\otimes}} \otimes \epsilon_7^{\overset{5,4}{\otimes}}, & 3,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\
\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_7, & \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_7, & 4,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,5 \\
\otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \epsilon_7, & 313221\epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{1}{\otimes}} \otimes \epsilon_7^{\overset{5,5}{\otimes}}, & I^{\oplus 5} \epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{1}{\otimes}} \otimes \epsilon_7^{\overset{5,5}{\otimes}} \\
21513221\epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{1}{\otimes}} \otimes \epsilon_7^{\overset{5,5}{\otimes}}, & 1513221\epsilon_8^{\overset{1}{\otimes}} \otimes \epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_8^{\overset{5,5}{\otimes}} \otimes \epsilon_7^{\overset{1}{\otimes}} &
\end{array}$$

$$\begin{array}{lll} \overset{3,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{1}{\otimes} \overset{5,4}{\mathbf{e}}_7', & \overset{4,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{1}{\otimes} \overset{5,4}{\mathbf{e}}_7', & \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{1}{\otimes} \overset{5,4}{\mathbf{e}}_7' \\ 313221 \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{1}{\otimes} \overset{5,4}{\mathbf{e}}_7', & I^{\oplus 5} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{1}{\otimes} \overset{5,4}{\mathbf{e}}_7', & 21513221 \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{1}{\otimes} \overset{5,4}{\mathbf{e}}_7' \\ {}^1_{15} 13221 \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{5,5}{\otimes} \mathbf{e}_8 \otimes \overset{1}{\otimes} \overset{5,4}{\mathbf{e}}_7' & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 3,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} & \begin{array}{l} 4,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ I^{\oplus 5}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \\ \\ \end{array} & \begin{array}{l} 5,5 \quad 5,5 \quad 5,5 \quad 1 \quad 5,3 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \\ \\ 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \end{array}$$

$$I^{\oplus 2} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}, \quad 31 \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4}, \quad 1 \epsilon_6^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,5}$$

$$1 \epsilon_7^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$\begin{array}{lll} 3221\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{1,3,3}} \otimes \epsilon_7', & 2321\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{1,3,3}} \otimes \epsilon_7, & 512321\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{1,3,3}} \otimes \epsilon_7' \\ 151321\epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{3,3}} \otimes \epsilon_7^{\overline{1,3,3}} \otimes \epsilon_7' & & \end{array}$$

$$\begin{array}{ccc} \begin{array}{ccccc} 2,3 & 3,3 & 3,3 & 1 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, & \end{array} & \begin{array}{ccccc} 3,3 & 3,3 & 3,3 & 1 & 3,3 \\ \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_7 & \otimes \mathfrak{e}_6, & \end{array} & I^{\oplus 3} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_7'^{5,4}, \quad 1 \mathbf{e}_7'^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_7^{5,5}, \quad 1 \mathbf{e}_7^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \otimes \mathbf{e}_7'^{5,4}$$

$$321e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,4}, \quad I^{\oplus 2}e_7^{4,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,5}, \quad 1e_7^{5,5} \otimes e_8^{5,5} \otimes e_8^1 \otimes e_7^{5,5}$$

$$I^{\oplus 2} \mathfrak{e}'_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^{1,3,3} \mathfrak{e}_6, \quad 31 \mathfrak{e}_6 \otimes^{3,3} \mathfrak{e}_7 \otimes^{3,3} \mathfrak{e}_7 \otimes^{1,3,3} \mathfrak{e}'_7$$

$$321\epsilon_6^{2,2} \otimes \epsilon_6^{2,2} \otimes \epsilon_6^1 \otimes \epsilon_6^{2,2}, \quad 15131\epsilon_6^{2,2} \otimes \epsilon_6^{2,2} \otimes \epsilon_6^1 \otimes \epsilon_6^{2,2}$$

$$1\epsilon_6 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4},$$

$$1\epsilon_7' \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$I^{\oplus 2} \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

$$321\epsilon_7' \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_7^{3,3}$$

$$3151321\epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_7^{3,3}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 2}} \otimes \mathfrak{so}^{4,2}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,5}$$

$$1\mathfrak{so}^{2,4} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,4}$$

$$1\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^1 \otimes \mathfrak{so}^{1,1}$$

$$1\epsilon_7' \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_7^{5,4}$$

$$1\epsilon_7' \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

$$1\epsilon_6 \otimes \epsilon_8^{3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \epsilon_6^{5,3}$$

$$1\epsilon_6 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3} \otimes \epsilon_7^1 \otimes \epsilon_6^{3,3}$$

$$1\epsilon_6 \otimes \epsilon_6^{2,2} \otimes \epsilon_6^{3,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}^{4,2}$$

2.5.7 SgLIgLG(28)

$$\begin{aligned}
& \begin{array}{l} \begin{array}{c} 3,5 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 4,5 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 5,5 \quad 5,5 \quad 1 \quad 4,2 \quad 1,1 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \begin{array}{c} 313221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} I^{\oplus 5} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \end{array} \quad \begin{array}{c} 2,4 \quad 5,5 \quad 1 \quad 5,5 \quad 5,5 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7, \end{array} \quad \begin{array}{c} I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{array} \\ \\ \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{array} \quad \begin{array}{c} 2,4 \quad 5,5 \quad 1 \quad 5,5 \quad 5,4 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6, \end{array} \quad \begin{array}{c} I^{\oplus 4} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}, \end{array} \quad \begin{array}{c} 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 31513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7, \end{array} \quad \begin{array}{c} 1513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \end{array} \\ \\ \\ \\ \begin{array}{c} I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} 2321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}'_7 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}'_7 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}'_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} \end{array} \\ \\ \\ \\ \begin{array}{c} 1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{so} \end{array} \end{aligned}$$

2.5.8 SgLIgLgLgS(93)

$$321\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'1, \quad 1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 3}, \quad I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1, \quad I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1, \quad I^{\oplus 3}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 13$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' I^{\oplus 2}, \quad 1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes, \quad 1\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 I^{\oplus 2}, \quad 31\mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 1$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1, \quad I^{\oplus 4}\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$I^{\oplus 2}\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1, \quad 1\mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 13$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 I^{\oplus 2}, \quad 1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_6 1$$

2.5.9 SgLIgLIgLg(3)

$$I^{\oplus 4} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{1}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}$$

$$1e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{4,2}{\otimes} 50$$

2.5.10 SgLIgLIgLgS(1)

$$150 \overset{2,4}{\otimes} \overset{1}{\mathfrak{e}_8} \overset{5,5}{\otimes} \overset{1}{\mathfrak{e}_8} \overset{4,2}{\otimes} 501$$

2.5.11 SSgLgLgLg(76)

$$\begin{array}{c} 5,5 \\ \otimes \end{array} \mathbf{e}_8 \begin{array}{c} (5,5) \\ \otimes \end{array} \mathbf{e}_8)^2 \begin{array}{c} 5,5 \\ \otimes \end{array} \mathbf{e}_6$$

$$21513221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^2\otimes^{5,5}\mathbf{e}_6$$

$$I^{\oplus 4} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \\ \mathfrak{e}_6 \end{smallmatrix}$$

$$\begin{matrix} 5,5 & 1 & (5,5) & 5,3 \\ \otimes & \mathfrak{e}_8 & (\otimes \mathfrak{e}_8)^2 & \otimes \mathfrak{so} \end{matrix}$$

$$21513221\mathfrak{e}_8^1(\otimes^{5,5}\mathfrak{e}_8)^2\otimes^{5,3}\mathfrak{so}$$

$$I^{\oplus 4} I^{\oplus 2} \begin{smallmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{smallmatrix} \begin{smallmatrix} 5,3 \\ \otimes \mathfrak{so} \end{smallmatrix}$$

$$I^{\oplus 3} \mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_6$$

$$1 \mathbf{e}_7 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^2 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes_{\mathbf{e}_7}^{3,3} \right)^2 \bigotimes_{\mathbf{e}_7}^{3,3} \mathbf{so}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,4}{\otimes} \overset{3,3}{e_7} \otimes e_6$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,4}{\otimes} \overset{3,3}{e_7} \otimes e_6$$

$$1 \overset{1}{e}'_7 (\overset{3,3}{\otimes} e_7)^2 \overset{4,5}{\otimes} e_7$$

$$1 \overset{1}{e}'_7 (\overset{3,3}{\otimes} e_7)^2 \overset{4,5}{\otimes} e_7$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, \quad I^{\oplus 2} 3221 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6, \quad 1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,5}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_8^{I^{\oplus 2} 5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_6^{4,3}$$

$$321 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^2,$$

$$1 \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,4}$$

$$321 \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,4},$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{4,5}$$

$$31 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,4},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,5}$$

$$31 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,3},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_7^{3,4}$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7^{1'} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{so}^{3,3}$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \otimes \mathfrak{e}_6^{3,3},$$

$$1 \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^2 \bigcirc \mathfrak{e}_6^{3,3}$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1'},$$

$$I^{\oplus 2} 3221 \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1'}$$

$$1513221 \mathfrak{e}_8^1 \otimes \mathfrak{so}^{4,2} (\otimes \mathfrak{so})^{1,1},$$

$$I^{\oplus 2} 3221 \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6^{4,3}$$

$$513221 I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$321 \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6$$

$$31513221 \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_8^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,5}$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_8^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}^{5,3}$$

$$1 \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7'$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$1 \mathfrak{e}_7' \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{so}^{3,2} \otimes \mathfrak{so}^{1,1}$$

$$1 \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}$$

$$1 \mathfrak{e}_6^{1,3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so}^{3,3}$$

$$1 \mathfrak{e}_6^{1,2,2} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

2.5.12 SSgLgLgLgS(471)

[illegible]

$$31\epsilon_6^{31}(\otimes \epsilon_6^2)^{2,2} \otimes \epsilon_6^{2,2} I^{\oplus 3}$$

$$31\epsilon_6^{31}(\otimes \epsilon_6^2)^{2,2} \otimes \epsilon_6^{2,2} 13151^1$$

$$\begin{aligned}
& 1\epsilon_7^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} 122315123 \\
& 1\epsilon_7^1 \otimes \epsilon_8^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8 12231513151 \\
& \otimes \epsilon_8^{5,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,4} I^{\oplus 2} \\
& 21513221\epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,4} I^{\oplus 2} \\
& I^{\oplus 4} \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,4} I^{\oplus 2} \\
& \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,5} \epsilon_7^1 \\
& I^{\oplus 5} \epsilon_8^1 (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,5} \epsilon_7^1 \\
& I^{\oplus 2} \epsilon_8^{2,4} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_7^{5,5} \epsilon_7^1
\end{aligned}$$

$$\begin{aligned}
& 1\overset{1}{\mathbf{e}}_7\overset{3,3}{\otimes}\overset{4,5}{\mathbf{e}}_7\overset{5,5}{\otimes}\overset{5,5}{\mathbf{e}}_8\overset{5,5}{\otimes}\mathbf{e}_8 \\
& 1\overset{1}{\mathbf{e}}_7\overset{3,3}{\otimes}\overset{4,5}{\mathbf{e}}_7\overset{5,5}{\otimes}\mathbf{e}_8\overset{5,5}{\otimes}\mathbf{e}_8 12231512 \\
& 2,4\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\mathbf{e}_8\overset{5,4}{\mathbf{e}}_8\overset{3,3}{\otimes}\mathbf{e}_7\overset{3,3}{\otimes}\mathbf{e}_7 I^{\oplus 2} \\
& I^{\oplus 2}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\mathbf{e}_8\overset{5,4}{\otimes}\mathbf{e}_7\overset{3,3}{\otimes}\mathbf{e}_7 1232 \\
& I^{\oplus 3}\overset{1}{\mathbf{e}}_8\overset{5,5}{\otimes}\mathbf{e}_8\overset{5,4}{\otimes}\mathbf{e}_7\overset{3,3}{\otimes}\mathbf{e}_7 I^{\oplus 2} \\
& I^{\oplus 2}\overset{1}{\mathbf{e}}_7\overset{4,5}{\otimes}\mathbf{e}_8\overset{5,5}{\otimes}\mathbf{e}_8\overset{5,4}{\otimes}\mathbf{e}_7 I^{\oplus 3} \\
& 1\overset{1}{\mathbf{e}}_7(\overset{5,5}{\mathbf{e}}_8)^2\overset{5,4}{\otimes}\mathbf{e}_7 I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
I^{\oplus 3} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,2} \\
I^{\oplus 2} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1223} \\
2321 & \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8 I^{\oplus 2} \\
I^{\oplus 2} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{123151} \\
I^{\oplus 2} & \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \\
1\overline{e}7 & \mathbf{e}_8^{1,4,5} \otimes \mathbf{e}_7^{5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 \otimes^{4,4} \\
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 12313 \\
& 1\mathbf{e}_6^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_7 1232151 \\
& \otimes^{3,3} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6^{2,2} \\
& \otimes^{3,3} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6 I^{\oplus 2} \\
& I^{\oplus 2} \mathbf{e}_7^{1,3,3}(\otimes \mathbf{e}_7)^2 \otimes^{3,3} \mathbf{e}_6 I^{\oplus 2}
\end{aligned}$$

$$\begin{array}{l} 1_{\mathfrak{e}_8}^{1, 5, 3} \otimes \mathfrak{e}_6^{2, 2} \otimes \mathfrak{e}_6^{2, 2} 1232 \\ I^{\oplus 3} \mathfrak{e}_8^{1, 5, 3} \otimes \mathfrak{e}_6^{2, 2} \otimes \mathfrak{e}_6^{2, 2} \otimes \mathfrak{e}_6^{2, 2} \\ I^{\oplus 2} \mathfrak{e}_8^{1, 5, 3} \otimes \mathfrak{e}_6^{2, 2} \otimes \mathfrak{e}_6^{2, 2} \mathfrak{e}_6 13151 \\ I^{\oplus 2} \mathfrak{e}_8^{1, 5, 3} \otimes \mathfrak{e}_6^{2, 2} \otimes \mathfrak{e}_6^{2, 2} \otimes \mathfrak{e}_6 I^{\oplus 2} \end{array}$$

$$\begin{array}{l} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 5} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 2} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{4,2} \mathfrak{so}13 \end{array}$$

$$\begin{array}{c} I^{\oplus 3} \mathbf{e}_8^{1 \oplus 2, 5, 5} \otimes \mathbf{e}_8^{5, 4} \otimes \mathbf{e}_7^{3, 3} \otimes \mathbf{e}'_7 1 \\ 1 \mathbf{e}'_7^{1, 4, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 4} \otimes \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 2} \mathbf{e}_7^{1, 4, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}_8^{5, 5} \otimes \mathbf{e}'_7 1 \end{array}$$

$$\begin{array}{cccc}
5,5 & 1 & 5,5 & 4,2 & 1,1 \\
\otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & 1 \\
21513221 & \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & 1 \\
I^{\oplus 4} & \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & \otimes \mathfrak{so} & 1
\end{array}$$

$$1e_6 \overset{1}{\otimes} \overset{3,5}{e_8} \overset{5,5}{\otimes} \overset{5,4}{e_8} \otimes e_7 I^{\oplus 3}$$

$$\begin{aligned} & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{2,2}, \\ & 1 \mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_7 I^{\oplus 2}, \\ & 1 \mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_6^{5,4} 1 \end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 I^{\oplus 2},$$

$$1 \mathfrak{e}_7^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2},$$

$$1e_7^{1\ 5,5}(\otimes e_8)^2\ 5,3\ \otimes e_6\ 2,2$$

$$I^{\oplus 2}e_7^{1\ 4,5}\ \otimes e_8\ 5,5\ \otimes e_8\ 5,4\ \otimes e_6 1$$

$$\begin{aligned} & 1\mathfrak{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^2 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix}, \\ & 1\mathfrak{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^2 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 I^{\oplus 4}, \\ & I^{\oplus 2} \mathfrak{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}'_7 I^{\oplus 2}, \end{aligned}$$

$$\begin{aligned} & 3221^1_{\mathbf{e}_8} \otimes^{5,4} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}'_7 1, \\ & I^{\oplus 2} \mathbf{e}_8^1 \otimes^{5,4} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}'_7 123, \\ & 1 \mathbf{e}_7^1 \otimes^{3,3} \mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 \otimes^{5,4} \mathbf{e}'_7 1, \end{aligned}$$

$$\begin{array}{cccc} 1 & 5,4 & 3,3 & 3,3 \\ 1e_8 \otimes e_7 \otimes e_7 \otimes e_7' & 1232 \\ I^{\oplus 3} & 1 & 5,4 & 3,3 & 3,3 \\ e_8 \otimes e_7 \otimes e_7 \otimes e_7' & I^{\oplus 2} \\ 1 & 4,5 & 5,4 & 3,3 \\ 1e_7 \otimes e_8 \otimes e_7 \otimes e_7' & 1 \end{array}$$

$$\begin{array}{l} 2,4 \quad 1 \quad 5,5 \quad 5,3 \quad 2,2 \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6 1, \\ I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2} 13, \\ 1 \mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2}, \end{array}$$

$$\begin{aligned} & I^{\oplus 4} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6 1, \\ & I^{\oplus 2} I^{\oplus 2} \mathbf{e}_8^{1,5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6 13, \\ & 1 \mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_6 1 \end{aligned}$$

$$I^{\oplus 3} \begin{array}{ccccc} \mathbf{e}_8^{5,5} & \otimes & \mathbf{e}_8^{5,3} & \otimes & \mathbf{e}_6^{2,2} \\ \mathbf{e}_6^{1,3,5} & \otimes & \mathbf{e}_8^{5,5} & \otimes & \mathbf{e}_8^{5,3} & \otimes & \mathbf{e}_6^{2,2} \end{array}$$

$$\begin{aligned} & \otimes_{\mathfrak{e}8}^{2,4,1} \otimes_{\mathfrak{so}}^{4,2} \otimes_{\mathfrak{so}}^{1,1} \otimes_{\mathfrak{so}1}^{1,1}, \\ & I^{\oplus 3} \otimes_{\mathfrak{e}8}^{I^{\oplus 2} 4,2} \otimes_{\mathfrak{so}}^{1,1} \otimes_{\mathfrak{so}}^{1,1} \otimes_{\mathfrak{so}1}, \\ & I^{\oplus 2} \otimes_{\mathfrak{e}8}^{I^{\oplus 2} 4,2} \otimes_{\mathfrak{so}}^{1,1} \otimes_{\mathfrak{so}}^{1,1} \otimes_{\mathfrak{so}13} \end{aligned}$$

$$\begin{aligned} I^{\oplus 4} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^1, \\ I^{\oplus 2} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^1 32, \end{aligned}$$

$$I^{\oplus 3} \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1} 1322$$

$$1e_7' \otimes^{1,3,3} e_7 \otimes^{3,3} e_7' \otimes^{3,3} e_7 I^{\oplus 2},$$

$$I^{\oplus 2} e_7' \otimes^{1,3,3} (e_7)^2 \otimes^{3,3} e_7' \otimes^{3,2},$$

$$1e_7^{1,3,3} \otimes e_7^{3,3} \otimes e_7^{3,3} \otimes e_7' I^{\oplus 2},$$

$$I^{\oplus 2} e_7' (\otimes e_7)^{2,3,3} \otimes e_7^{3,3} \otimes e_7',$$

$$\begin{array}{l} I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7'^{3,3} \otimes \mathbf{e}_7' 1 \\ I^{\oplus 2} \mathbf{e}_7' (\otimes \mathbf{e}_7)^{2,3,3} \otimes \mathbf{e}_7' I^{\oplus 3} \end{array}$$

$$1e_7^{1,3,3} \otimes e_7'^{3,3} \otimes e_7^{3,3} \otimes e_6 1, \\ 1e_6^{1,3,3} (\otimes e_7)^2 \otimes e_7'^{3,3} I^{\oplus 3},$$

$$1\mathbf{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix},$$

$$I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix},$$

$$1e_6^{1(3,3)}(\otimes e_7)^{2(3,3)} \otimes e_7'^{3(3,3)} \otimes I^{\oplus 2} e_7'^{1(3,3)}(\otimes e_7)^{2(3,3)} \otimes e_6^{3(3,3)} I^{\oplus 2}$$

$$\begin{aligned} & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^2 \otimes^{1,1} \mathfrak{so} \otimes^{2,2}, \\ & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^2 \otimes^{1,1} \mathfrak{so} 132, \end{aligned}$$

$$\begin{aligned} & 1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2 \overset{1,1}{\otimes} \mathfrak{so} \overset{2,3}{\otimes}, \\ & 1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2 \overset{1,1}{\otimes} \mathfrak{so} I^{\oplus 2} \end{aligned}$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^2 \otimes^{1,1} \mathfrak{so} \otimes^{2,4}$$

$$\begin{array}{l} 1e_7' \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,3}{\otimes} e_6 \overset{2,2}{\otimes}, \\ 1e_6 \overset{3,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_7' 1, \end{array}$$

$$\begin{array}{cccc} 1 & 3,5 & 5,5 & 5,4 \\ 1e_6 & \otimes e_8 & \otimes e_8 & \otimes e_7' I^{\oplus 2}, \\ \\ 1 & 4,5 & 5,5 & 5,4 \\ 1e_7' & \otimes e_8 & \otimes e_8 & \otimes e_6 1 \end{array}$$

$$1e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,3} e_6 I^{\oplus 2}$$

$$1\mathbf{e}_7 \overset{1}{\otimes} \overset{3,3}{\mathbf{e}_6} \overset{2,2}{\otimes} \overset{2,2}{\mathbf{e}_6} \overset{2,2}{\otimes} \overset{2,2}{\mathbf{e}_6},$$

$$1\mathbf{e}_6^{(2,2)} \otimes \mathbf{e}_6^{(3,3)} \otimes \mathbf{e}_7 I^{\oplus 2},$$

$$1e_7^{1,3,3} \otimes e_6^{2,2} \otimes e_6^{2,2} \otimes e_6 I^{\oplus 2}$$

$$31\mathbf{e}_6^1(\otimes^{2,2}\mathbf{e}_6)^2\otimes^{3,3}\mathbf{e}_71,$$

$$1\mathbf{e}_6^{(2,2)}(\otimes \mathbf{e}_6)^2 \otimes \mathbf{e}_7 1$$

$$\otimes^{2,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^2 \otimes^{3,2} \mathfrak{so}1,$$

$$\otimes^{3,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^2 \otimes^{3,2} \mathfrak{so}1,$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^2 \bigotimes \mathfrak{so} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^{\oplus 2} \left(\bigotimes_{\mathfrak{e}_7}^{3,3} \right)^2 \bigotimes_{\mathfrak{e}_7}^{3,2} \mathfrak{so}1$$

$$I^{\oplus 3} \mathbf{e}_8^1 \otimes^{5,4} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}_7 \otimes^{3,3} \mathbf{e}_6 1,$$

$$I^{\oplus 2} I_8^{\oplus 2} \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_6 1,$$

$$I^{\oplus 2} \mathbf{e}_8^{1,5,4} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{13}$$

$$1e_7^1 \otimes e_7^{3,3} \otimes e_7^{4,5} \otimes e_8^{5,3} \otimes e_6^1$$

$$I^{\oplus 2} \mathbf{e}_7^{1,3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_6^{2,2} \otimes \mathbf{e}_6 1,$$

$$1e_7 \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{2,2}{e_6} \overset{2,2}{\otimes} e_6 13,$$

$$1 \overset{1}{\mathbf{e}}_6 (\overset{3,3}{\otimes} \overset{2}{\mathbf{e}}_7) \overset{3,3}{\otimes} \overset{2,2}{\mathbf{e}}_6 \otimes$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$${}^{2,2} \otimes \mathfrak{e}_6 ({}^{2,2} \otimes \mathfrak{e}_6)^2 {}^{2,2} \otimes \mathfrak{so}1,$$

$$I^{\oplus 2} \mathfrak{e}_6 \binom{2,2}{\otimes} \mathfrak{e}_6)^2 \binom{2,2}{\otimes} \mathfrak{so}1,$$

$$31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6)^{2,2} \otimes \mathfrak{so}1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \oplus \mathfrak{so}13,$$

$$1e_7^{1(5,5)}(\otimes e_8)^2 \otimes so_{13}^{4,2}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1,$$

$$1e_7 \overset{1,5,5}{\otimes} e_8 \overset{4,2}{\otimes} so \overset{1,1}{\otimes} so1$$

$$1e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{5,4} e_7' I^{\oplus 2},$$

$$1 \overset{1}{e}'_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e'_7 1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{4,2} \oplus \mathfrak{so}1$$

$$I^{\oplus 2} \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^2 {}^{3,2} \otimes \mathfrak{so}1$$

$$1e_7' \otimes^{4,5} e_8 \otimes^{5,5} e_8 \otimes^{4,2} so_{13}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,3}{\otimes} \overset{2,2}{e_6} \otimes e_6 1$$

$$1e'_7 \begin{smallmatrix} 1 \\ 4,5 \end{smallmatrix} \otimes e_8 \begin{smallmatrix} 4,2 \\ \end{smallmatrix} \otimes so \begin{smallmatrix} 1,1 \\ \end{smallmatrix} \otimes so1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_7' \otimes \epsilon_7'1$$

$$1\epsilon_7' \otimes \epsilon_7 \otimes \epsilon_6 \otimes \epsilon_61$$

$$1\epsilon_7 \otimes \mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \epsilon_8 \otimes \mathfrak{so}13$$

$$1\epsilon_6 \otimes \epsilon_8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\epsilon_6(\otimes \epsilon_7)^2 \otimes \mathfrak{so}1$$

$$1\epsilon_6 \otimes \epsilon_6 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$1\epsilon_6(\otimes \epsilon_6)^2 \otimes \epsilon_7'1$$

2.5.13 SSgLgIgS(149)

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8122313,$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8122313,$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 I^{\oplus 5},$$

$$21513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512,$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_8 \otimes$$

$$\otimes \epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_81223151$$

$$313221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$I^{\oplus 5}\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

$$1513221\epsilon_8(\otimes \epsilon_8)^2 \otimes \epsilon_812231512$$

[illegible]

$$\begin{array}{lll}
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 122313, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 5}, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 12231512 \\
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 1223151, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 24,2}, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 2} \\
1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} I^{\oplus 3} & & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 I^{\oplus 4}
\end{array}$$

$$\begin{array}{lll}
1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2}, & 1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2} I^{\oplus 4}, & 1_{\mathbf{e}_7}^{1,3,3} \otimes \mathbf{e}_7^{4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1,4,2} I^{\oplus 2} \\
I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,4} \otimes \mathbf{e}_7^{1,4,2} I^{\oplus 2}, & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_7^{1,4,2}, & 1_{\mathbf{e}_7}^{1,5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7^{5,5} \otimes \mathbf{e}_7^{1,4,2}
\end{array}$$

$$\begin{array}{lll} 1\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} 1223, & 2321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} 1, & 321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 2} \\ I^{\oplus 2}\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 3} 1, & I^{\oplus 2}\epsilon_7^{(3,3)}(\otimes \epsilon_7)^2 \otimes \epsilon_8^{4,5} I^{\oplus 2}, & 1\epsilon_7^{(3,3)} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,4} 1 \end{array}$$

[illegible]

$$\begin{array}{lll}
1 \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} \overset{3,3}{\mathbf{e}}'_7 \overset{3,3}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \overset{I^{\oplus 2}}{\mathbf{e}}_7 1, & I^{\oplus 2} \overset{1}{\mathbf{e}}'_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,2}{\otimes}, & I^{\oplus 2} \overset{1}{\mathbf{e}}'_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} \\
I^{\oplus 2} \overset{1}{\mathbf{e}}'_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \overset{I^{\oplus 3}}{\mathbf{e}}_7 1, & I^{\oplus 2} \overset{1}{\mathbf{e}}'_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \overset{I^{\oplus 2}}{\mathbf{e}}_7 I^{\oplus 2} &
\end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{3,3}{\otimes}\mathbf{e}_7\overset{1,3,2}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{3,3}{\otimes}\mathbf{e}_7\overset{1,3,3}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{3,3}{\otimes}\mathbf{e}_7\overset{1\oplus 3}{\otimes} \\ 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^2\overset{3,3}{\otimes}\mathbf{e}_7\overset{1\oplus 2}{\otimes} & & \end{array}$$

$$1\epsilon_6^{(2,2)}(\otimes \epsilon_6)^2 \otimes \epsilon_8 1, \quad 1\epsilon_6^{(2,2)}(\otimes \epsilon_6)^2 \otimes \epsilon_8 I^{\oplus 2}, \quad 31\epsilon_6^{(2,2)}(\otimes \epsilon_6)^2 \otimes \epsilon_8 1$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes e_7^{\oplus 2}, \quad 1e_7' \otimes e_8 \otimes e_8 \otimes e_7^{\oplus 1}$$

$$1\mathbf{e}_7' \otimes \mathbf{e}_7 \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' \otimes \mathbf{e}_7' 1, \quad I^{\oplus 2} \mathbf{e}_7' (\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7' 1$$

$$1e_7'(\otimes^{3,3} e_7)^2 \otimes^{4,5} e_8 I^{\oplus 3}, \quad 1e_7'(\otimes^{3,3} e_7)^2 \otimes^{4,5} e_8 I^{\oplus 2}$$

$$1 \overset{1}{e}_6 \overset{3,5}{\otimes} \overset{5,5}{e}_8 \overset{5,4}{\otimes} I^{\oplus 2} \overset{5,5}{e}_8 \overset{5,5}{\otimes} \overset{1}{e}_7 1,$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_71$$

$$1\mathfrak{e}_6\overset{1}{\otimes}\overset{3,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_71$$

$$1\mathfrak{e}_6\overset{1}{\otimes}\overset{3,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,3}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_61$$

$$1\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}'_7I^{\oplus 2}1$$

$$1\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_61$$

$$1\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_71$$

2.5.14 SSgLgLIg(21)

$$1513221\mathfrak{e}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\otimes}\mathfrak{e}_7,$$

$$I^{\oplus 2}3221\mathfrak{e}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_7,$$

$$\overset{2,4}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_7$$

$$I^{\oplus 4}\mathfrak{e}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\mathfrak{e}_7,$$

$$I^{\oplus 3}I^{\oplus 2}\mathfrak{e}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_7$$

$$\overset{2,4}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_7,$$

$$I^{\oplus 4}\mathfrak{e}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,4}{\otimes}\mathfrak{e}'_7,$$

$$I^{\oplus 3}I^{\oplus 2}\mathfrak{e}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,4}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}'_7$$

$$\overset{2,4}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,3}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_6,$$

$$I^{\oplus 4}\mathfrak{e}_8\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{5,5}{\otimes}\overset{1}{\mathfrak{e}}_8\overset{5,3}{\otimes}\mathfrak{e}_6,$$

$$I^{\oplus 3}I^{\oplus 2}\mathfrak{e}_8\overset{5,5}{\otimes}\overset{5,5}{\mathfrak{e}}_8\overset{1}{\otimes}\overset{5,3}{\mathfrak{e}}_8\overset{1}{\otimes}\mathfrak{e}_6$$

$$I^{\oplus 2}\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$321\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}}_7\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}}_7\overset{3,3}{\otimes}\mathfrak{e}'_7$$

$$31\mathfrak{e}_6\overset{1}{\otimes}\overset{2,2}{\mathfrak{e}}_6\overset{2,2}{\otimes}\overset{1}{\mathfrak{e}}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$I^{\oplus 3} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 1$$

$$321e_7^1 \otimes^{3,3} e_7^{3,3} \otimes^{1,3,3} e_7^1 \otimes^{3,3} e_7^1,$$

$$1e_7 \overset{1}{\otimes} \overset{5,5}{e_8} \overset{5,5}{\otimes} \overset{1}{e_8} \overset{4,2}{\otimes} so1$$

$$I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7' 1$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} \overset{1}{e_8} \overset{5,4}{\otimes} e_7 1$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes sol$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} e_7' 1$$

$$1e_6 \otimes^{1,3,5} e_8 \otimes^{5,5} e_8 \otimes^{1,4,2} \text{sol}$$

$$1 \overset{1}{\mathbf{e}}_6 \overset{2,2}{\otimes} \overset{2,2}{\mathbf{e}}_6 \overset{1}{\otimes} \overset{2,2}{\mathbf{e}}_6 1$$

2.5.16 SSgLgLIgLIgS(15)

$$\begin{array}{lll}
31513221 \begin{smallmatrix} 1 & 5,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 & 1 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 & 1 \\ \otimes & \end{smallmatrix} 1, & \begin{smallmatrix} 2,4 & 1 & 5,5 & 5,5 & 1 & 5,5 & 1 \\ \otimes & \otimes & \otimes & \otimes & \otimes & \otimes & \end{smallmatrix} I^{\oplus 3} 1, & I^{\oplus 3} \begin{smallmatrix} 1 & 5,5 & 5,5 & 1 & 5,5 & 1 \\ \otimes & \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} 1223 \\
1513221 \begin{smallmatrix} 1 & 5,5 & 5,5 & 1 & 5,5 \\ \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} I^{\oplus 2} 1, & I^{\oplus 4} \begin{smallmatrix} 1 & 5,5 & 5,5 & 1 & 5,5 \\ \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} I^{\oplus 3} 1, & \begin{smallmatrix} 2,4 & 1 & 5,5 & 5,5 & 1 & 5,5 \\ \otimes & \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} I^{\oplus 2} \\
I^{\oplus 2} \begin{smallmatrix} 1 & 5,5 & 5,5 & 1 & 5,5 & 1 \\ \otimes & \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} 1223, & I^{\oplus 2} \begin{smallmatrix} 3221 & 5,5 & 5,5 & 1 & 5,5 \\ \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} I^{\oplus 2} 1, & I^{\oplus 4} \begin{smallmatrix} 1 & 5,5 & 5,5 & 1 & 5,5 \\ \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} I^{\oplus 2} \\
I^{\oplus 3} \begin{smallmatrix} 1 & 5,5 & 5,5 & 1 & 5,5 \\ \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} I^{\oplus 3} 1, & I^{\oplus 3} \begin{smallmatrix} 1 & 5,5 & 5,5 & 1 & 5,5 \\ \otimes & \otimes & \otimes & \otimes & \otimes \end{smallmatrix} I^{\oplus 2} 1, &
\end{array}$$

$$1e_7 \otimes^{1,4,5} e_8 \otimes^{5,5} e_8 \otimes^{1,5,5} e_8 \otimes^{I^{\oplus 2}} e_8 \otimes^{I^{\oplus 2}}$$

$$I^{\oplus 2} \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{1 \ 3,3} \otimes \mathbf{e}_7^1$$

$$1e_7' \overset{1}{\otimes} \overset{3,3}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} \overset{3,3}{\otimes} \overset{1}{e_7} 1$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 1, \quad I^{\oplus 2} I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 1, \quad I^{\oplus 2} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6 13$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,2}, \quad 1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3}, \quad 1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 I^{\oplus 3}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7' 1, \quad I^{\oplus 2} I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so} 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6 1$$

2.5.19 SSgLIgLgLIgS(15)

$$\begin{aligned} 1 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 12231513, & \quad I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{e}_8, \\ I^{\oplus 2} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 1223151, & \quad I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 I^{\oplus 4}, \\ I^{\oplus 2} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{I^{\oplus 2}} 1223, & \quad 3221 \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{I^{\oplus 2}} I^{\oplus 2}, \\ I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2}, & \quad I^{\oplus 2} I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{I^{\oplus 3}} I^{\oplus 2} \end{aligned}$$

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^1 1, \quad I^{\oplus 2} I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^1 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 I^{\oplus 2} 1$$

$$1 \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7' 1$$

2.5.20 SSgLIgLIg(2)

$$I^{\oplus 3} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}, \quad I^{\oplus 2} I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} \otimes \mathfrak{e}_8^{1,4,2} \otimes \mathfrak{so}$$

2.6 5 node(1332)

2.6.1 gLgLGgLG(17)

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7')^2$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2}$$

$$\mathfrak{so}^{2,3} \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7^{4,4}$$

$$\mathfrak{so}^{2,2} (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,4}$$

$$\mathfrak{so}^{2,2} (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,3}$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_6^{3,3}$$

$$\mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2 \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_7^{5,5}$$

$$\mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2 \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_6^{5,4}$$

$$\mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7'$$

$$\mathfrak{e}_6^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'$$

$$\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7'$$

$$\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_6$$

2.6.2 gLgLIgLg(4)

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7$$

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7'$$

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6$$

$$\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

2.6.3 gLIgLgLg(10)

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7'$$

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6$$

$$\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2$$

$$\mathfrak{e}_7^{5,5} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$\mathfrak{e}_7'^{4,5} \otimes \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$\mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7'^{5,5}$$

$$\mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,4}$$

$$e'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 2 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} 50$$

$$\mathfrak{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} \mathfrak{so} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^1 \left(\mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6 \right)^2 \otimes \mathfrak{so}^{2,2}$$

2.6.4 gLIgLgLIgLg(1)

$$50 \otimes_{\mathbb{F}_8}^{2,4} \mathbb{F}_8 \otimes_{\mathbb{F}_8}^{5,5} \mathbb{F}_8 \otimes_{\mathbb{F}_8}^{5,5} \mathbb{F}_8 \otimes_{\mathbb{F}_8}^{1,5,4} \mathbb{F}_7$$

2.6.5 SgLgLgLgLg(133)

$$\begin{aligned} & 3,5 \otimes \mathfrak{e}_8 \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,3 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \otimes \begin{pmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{pmatrix}, \\ & 313221 \mathfrak{e}_8 \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,3 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \otimes \begin{pmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{pmatrix}, \\ & \begin{pmatrix} 1 \\ 1513221 \mathfrak{e}_8 \end{pmatrix} \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,3 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \otimes \begin{pmatrix} 2,2 \\ \otimes \mathfrak{e}_6 \end{pmatrix}, \\ & 1 \mathfrak{e}_6 \otimes \begin{pmatrix} 4,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix} \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^2 \otimes \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_6 \end{pmatrix} \end{aligned}$$

$$\begin{array}{l} 4,5 \quad 5,5 \quad 5,3 \quad 2,2 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6, \\ I^{\oplus 5} \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6, \\ 2,2 \quad 3,5 \quad 5,5 \quad 5,5 \\ \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6, \end{array}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \\ \otimes \end{array} \mathfrak{e}_8 \begin{array}{c} 5,5 \\ (\otimes \mathfrak{e}_8)^2 \end{array} \otimes \begin{array}{c} 5,3 \\ \mathfrak{e}_6 \end{array} \otimes \begin{array}{c} 2,2 \\ \mathfrak{e}_6 \end{array} \\ 21513221 \mathfrak{e}_8 \begin{array}{c} 5,5 \\ (\otimes \mathfrak{e}_8)^2 \end{array} \otimes \begin{array}{c} 5,3 \\ \mathfrak{e}_6 \end{array} \otimes \begin{array}{c} 2,2 \\ \mathfrak{e}_6 \end{array} \\ I^{\oplus 2} \mathfrak{e}_6 \begin{array}{c} 3,5 \\ \otimes \end{array} \mathfrak{e}_8 \begin{array}{c} 5,5 \\ (\otimes \mathfrak{e}_8)^2 \end{array} \otimes \begin{array}{c} 5,5 \\ \mathfrak{e}_6 \end{array} \end{array}$$

$$\begin{aligned} & \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \epsilon_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \epsilon_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}, \\ & \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \epsilon_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \epsilon_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}, \\ & 2151321 \epsilon_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \epsilon_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}, \\ & \begin{smallmatrix} 1 \\ 15 \end{smallmatrix} 1321 \epsilon_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \epsilon_7^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so} \end{aligned}$$

$$\begin{aligned} & 3,4 \quad (3,3) \quad 3,3 \\ & \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}, \\ & I^{\oplus 4} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}, \\ & 2321 \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}, \end{aligned}$$

$$\begin{array}{l} 4,4 \quad (3,3) \quad 3,3 \\ \otimes e_7 (\otimes e_7)^3 \otimes \mathfrak{so} \\ 31321 e_7 (\otimes e_7)^3 \otimes \mathfrak{so} \\ 1512321 e_7 (\otimes e_7)^3 \otimes \mathfrak{so} \end{array}$$

$$\begin{aligned} & 3,5 \quad 5,5 \quad 4,2 \quad (1,1) \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2, \\ & 313221 \mathfrak{e}_8 \quad 5,5 \quad 4,2 \quad (1,1) \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2, \\ & 1513221 \mathfrak{e}_8 \quad 5,5 \quad 4,2 \quad (1,1) \\ & \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2, \end{aligned}$$

$$\begin{array}{ccccccc}
4,5 & 5,5 & 4,2 & (1,1) \\
\otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & (\otimes \mathfrak{so})^2, \\
I^{\oplus 5} \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} & (\otimes \mathfrak{so})^2, \\
1,1 & 2,4 & 5,5 & 5,3 \\
\mathfrak{so} & \otimes \mathfrak{so} & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 \otimes \mathfrak{so}
\end{array}$$

$$\begin{array}{c} 5,5 \quad 5,5 \quad 4,2 \quad (1,1) \\ \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \\ 21513221\mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \end{array}$$

$$\begin{array}{l} 3,5 \\ \otimes \mathbf{e}_8 \end{array} \begin{array}{l} (5,5) \\ \otimes \mathbf{e}_8 \end{array}^2 \begin{array}{l} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7' \end{array}, \\ 313221 \mathbf{e}_8 \begin{array}{l} (5,5) \\ \otimes \mathbf{e}_8 \end{array}^2 \begin{array}{l} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7' \end{array}, \\ 1513221 \mathbf{e}_8 \begin{array}{l} (5,5) \\ \otimes \mathbf{e}_8 \end{array}^2 \begin{array}{l} 5,4 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7' \end{array}$$

$$\begin{aligned} & \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \otimes \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7, \\ & I^{\oplus 5} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \otimes \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7, \end{aligned}$$

$$\begin{array}{c} \begin{array}{ccccc} 5,5 & (5,5) & 5,4 & 3,3 & \\ \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^2 & \otimes \mathbf{e}_7 & \otimes \mathbf{e}'_7 & \end{array} \\ 21513221 \mathbf{e}_8 (5,5) \otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_7 \otimes \mathbf{e}'_7 \end{array}$$

$$2313221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^3\overset{5,5}{\otimes}\mathbf{e}_6,$$

$$I^{\oplus 6} \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^3 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$321513221\mathbf{e}_8\left(\bigotimes^{5,5}\mathbf{e}_8\right)^3\bigotimes^{5,5}\mathbf{e}_6$$

$$51231513221\epsilon_8^{(5,5)}(\otimes\epsilon_8)^3\otimes\epsilon_6, \quad 31513221\epsilon_8^{(5,5)}(\otimes\epsilon_8)^3\otimes\epsilon_6, \quad 15131513221\epsilon_8^{(5,5)}(\otimes\epsilon_8)^3\otimes\epsilon_6$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & I^{+6}\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & 321513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so} \\ 51231513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & {}^131513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so}, & 15131513221\epsilon_8(\otimes\epsilon_8)^3\otimes\mathfrak{so} \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathbf{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathbf{e}_8 \end{array} (\otimes \mathbf{e}_8)^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_6 \end{array}, & \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathbf{e}_8 \end{array} (\otimes \mathbf{e}_8)^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_6 \end{array}, & I^{\oplus 3} \mathbf{e}_7 \otimes \begin{array}{l} 4,5 \\ \otimes \mathbf{e}_8 \end{array} (\otimes \mathbf{e}_8)^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_6 \end{array} \\ I^{\oplus 2} \mathbf{e}_7 (\otimes \mathbf{e}_8)^3 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_6 \end{array} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array}, & \begin{array}{l} 3,3 \\ \otimes \mathfrak{e}_7 \end{array} \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array}, & I^{\oplus 3} \mathfrak{e}_7 \otimes \begin{array}{l} 4,5 \\ \otimes \mathfrak{e}_8 \end{array} (\otimes \mathfrak{e}_8)^2 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array} \\ I^{\oplus 2} \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^3 \otimes \begin{array}{l} 5,3 \\ \otimes \mathfrak{so} \end{array} & & \end{array}$$

$$\begin{array}{lll} \begin{array}{l} 2,3 \\ \otimes \mathbf{e}_7(\otimes \mathbf{e}_7)^2(\otimes \mathbf{e}_7')^2, \end{array} & \begin{array}{l} 3,3 \\ \otimes \mathbf{e}_7(\otimes \mathbf{e}_7)^2(\otimes \mathbf{e}_7')^2, \end{array} & 321 \mathbf{e}_7'(\otimes \mathbf{e}_7)^2 \otimes \mathbf{e}_7' \otimes \mathbf{e}_7 \\ I^{\oplus 3} \mathbf{e}_7(\otimes \mathbf{e}_7)^2(\otimes \mathbf{e}_7')^2 & & \end{array}$$

$$3221\epsilon_7(\otimes \epsilon_7)^2 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3}, \quad 512321\epsilon_7(\otimes \epsilon_7)^2 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3}, \quad 151321\epsilon_7(\otimes \epsilon_7)^2 \otimes \epsilon_7^{3,3} \otimes \epsilon_7^{3,3}$$

$$I^{\oplus 2}\epsilon_7'(\otimes \epsilon_7)^3 \otimes \epsilon_7^{4,4}$$

$$321\mathfrak{e}_6\binom{2,2}{\otimes\mathfrak{e}_6}^3\otimes\mathfrak{e}_7, \quad 15131\mathfrak{e}_6\binom{2,2}{\otimes\mathfrak{e}_6}^3\otimes\mathfrak{e}_7, \quad \binom{2,2}{\otimes\mathfrak{e}_6}\binom{2,2}{\otimes\mathfrak{e}_6}^3\otimes\mathfrak{e}_7$$

$$\begin{array}{lll}
31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,5}, & \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3}, & I^{\oplus 2} \mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3} \\
1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{5,3} & &
\end{array}$$

$$\begin{array}{lll}
1e_7^{3,3} \otimes e_7'({}^{3,3}(\otimes e_7)^2 \otimes \mathfrak{so}, & {}^{2,3} \otimes e_7'({}^{3,3}(\otimes e_7)^3 \otimes \mathfrak{so}, & {}^{3,3} \otimes e_7'({}^{3,3}(\otimes e_7)^3 \otimes \mathfrak{so} \\
I^{\oplus 3} e_7'({}^{3,3}(\otimes e_7)^3 \otimes \mathfrak{so} & &
\end{array}$$

$$\begin{array}{ccc} \begin{array}{c} 2,3 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7) \end{array}^3 \begin{array}{c} 4,4 \\ \otimes \mathbf{e}_7 \end{array}, & \begin{array}{c} 3,3 \\ \otimes \mathbf{e}_7 \end{array} \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7) \end{array}^3 \begin{array}{c} 4,4 \\ \otimes \mathbf{e}_7 \end{array}, & I^{\oplus 3} \mathbf{e}_7 \begin{array}{c} 3,3 \\ (\otimes \mathbf{e}_7) \end{array}^3 \begin{array}{c} 4,4 \\ \otimes \mathbf{e}_7 \end{array} \end{array}$$

$$\begin{array}{ccc} \overset{2,3}{\otimes} e_7 (\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_6 \overset{2,2}{\otimes} e_6, & \overset{3,3}{\otimes} e_7 (\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_6 \overset{2,2}{\otimes} e_6, & I^{\oplus 3} e_7 (\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_6 \overset{2,2}{\otimes} e_6 \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7', \quad 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7', \quad 1 \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7'$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}$$

$$\otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7',$$

$$I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7'$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{so},$$

$$I^{\oplus 2} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_6 (\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6,$$

$$1 \mathfrak{e}_7' (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so},$$

$$1 \mathfrak{e}_7' (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} (\otimes \mathfrak{so})^2,$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} (\otimes \mathfrak{so})^2$$

$$31 \mathfrak{so} (\otimes \mathfrak{so})^3 \otimes \mathfrak{e}_7,$$

$$1 \mathfrak{so} (\otimes \mathfrak{so})^3 \otimes \mathfrak{e}_7$$

$$31 \mathfrak{e}_6 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7,$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_6$$

$$I^{\oplus 2} \mathfrak{e}_7' \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

$$I^{\oplus 2} \mathfrak{e}_7' (\otimes \mathfrak{e}_7)^2 (\otimes \mathfrak{e}_7')^2$$

$$I^{\oplus 2} \mathfrak{e}_7' (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 \otimes \mathfrak{e}_6$$

$$5131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$3221\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$321\mathfrak{e}_7\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_7'\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$23151321\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^3\overset{4,5}{\otimes}\mathfrak{e}_8$$

$$2315131\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^3\overset{3,5}{\otimes}\mathfrak{e}_8$$

$$1\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{so}\overset{2,3}{\otimes}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{so}$$

$$1\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^3\overset{2,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^3\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{e}_7'\overset{4,5}{\otimes}\mathfrak{e}_8\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\epsilon'_7 \otimes^{4,5} \epsilon_8 \otimes^{4,2} \mathfrak{so}(\otimes^{1,1} \mathfrak{so})^2$$

$$1e'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50$$

$$1e'_7(\overset{3,3}{\otimes} e_7)^2 \overset{4,5}{\otimes} e_8 \overset{4,2}{\otimes} 50$$

$$1\mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_6 (\overset{2,2}{\otimes} \mathfrak{e}_6)^2 \overset{2,2}{\otimes} \mathfrak{so}$$

$$1\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \overset{3,3}{\otimes} \mathbf{e}'_7$$

$$1\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 \overset{5,5}{\otimes} \mathbf{e}_8 \overset{5,3}{\otimes} \mathbf{e}_6 \overset{2,2}{\otimes} \mathbf{e}_6$$

$$1\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so}(\overset{1,1}{\otimes} \mathfrak{so})^2$$

$$1\mathbf{e}_6 \binom{3,3}{\otimes} \mathbf{e}_7)^3 \binom{4,4}{\otimes} \mathbf{e}_7$$

$$1e_6(\overset{3,3}{\otimes} e_7)^2(\overset{3,3}{\otimes} e_7')^2$$

$$1\mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7)^2 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_6 \begin{pmatrix} 2,2 \\ \otimes \end{pmatrix} \mathbf{e}_6$$

$$1\mathfrak{e}_6 \overset{2,2}{\otimes} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so}$$

$$1513221\epsilon_8 \otimes^{4,2} \mathfrak{so}(\otimes^{1,1} \mathfrak{so})^3$$

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$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix},$$

$${}^{2,4} \otimes \mathfrak{e}_7 ({}^{3,3} \otimes \mathfrak{e}_7)^3 {}^{3,3} \otimes \mathfrak{e}_7 \otimes {}^{4,5},$$

$$\otimes^{2,4} \mathbf{e}_7 (\otimes^{3,3} \mathbf{e}_7)^3 \otimes^{3,3} \mathbf{e}_7 1231512,$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 4,3 \\ \otimes \end{smallmatrix},$$

$$I^{\oplus 4} \mathfrak{e}_7 \left(\overset{(3,3)}{\otimes} \mathfrak{e}_7 \right)^3 \overset{3,3}{\otimes} \mathfrak{e}_7 \overset{4,2}{\otimes},$$

$$\begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 1232,$$

$$\begin{array}{ccccccc} 2,4 & & 3,3 & & 3,3 & & 4,4 \\ \otimes & \mathbf{e}_7 & (\otimes \mathbf{e}_7)^3 & \otimes & \mathbf{e}_7 & \otimes & \end{array}$$

$$\overset{2,4}{\otimes} \mathbf{e}_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^3 \overset{3,3}{\otimes} \mathbf{e}_7 12313$$

$$\otimes^{2,4} \mathbf{e}_7 (\otimes^{3,3} \mathbf{e}_7)^3 \otimes^{3,3} \mathbf{e}_7 1232151$$

$$31513221\epsilon_8(\otimes^{5,5}\epsilon_8)^3\otimes^{5,5}\epsilon_812231513151, \quad 15131513221\epsilon_8(\otimes^{5,5}\epsilon_8)^3\otimes^{5,5}\epsilon_812231513151$$

$$\begin{array}{l}
\begin{array}{ccccc}
2,2 & 3,3 & & 3,3 & 4,4 \\
\otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_7)^3 & \otimes & \mathfrak{e}_7 & \otimes
\end{array} \\
\begin{array}{ccccc}
2,2 & 3,3 & & 3,3 & \\
\otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_7)^3 & \otimes & \mathfrak{e}_7 & 12313
\end{array} \\
\begin{array}{ccccc}
2,2 & 3,3 & & 3,3 & \\
\otimes \mathfrak{e}_6 & (\otimes \mathfrak{e}_7)^3 & \otimes & \mathfrak{e}_7 & 1232151
\end{array} \\
I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \begin{array}{c} 4,3 \\ \otimes \end{array} \\
I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 I^{\oplus 4} \\
I^{\oplus 2} \mathfrak{e}_6 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \begin{array}{c} 1 \\ 1232 \end{array}
\end{array}$$

$$\begin{array}{l}
\begin{array}{l}
2,2 \\ \otimes \mathbf{e}_6
\end{array} \otimes \begin{array}{l} 3,5 \\ \otimes \mathbf{e}_8 \end{array} (\begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array})^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} 122315123 \\
\begin{array}{l}
2,2 \\ \otimes \mathbf{e}_6
\end{array} \otimes \begin{array}{l} 3,5 \\ \otimes \mathbf{e}_8 \end{array} (\begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array})^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} 12231513151 \\
I^{\oplus 2} \begin{array}{l} 3,5 \\ \otimes \mathbf{e}_6 \end{array} \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} (\begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array})^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} 122315123 \\
I^{\oplus 2} \begin{array}{l} 3,5 \\ \otimes \mathbf{e}_6 \end{array} \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} (\begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array})^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} 12231513151 \\
1 \begin{array}{l} 4,5 \\ \otimes \mathbf{e}_6 \end{array} \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} (\begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array})^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} 122315123 \\
1 \begin{array}{l} 4,5 \\ \otimes \mathbf{e}_6 \end{array} \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} (\begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array})^2 \otimes \begin{array}{l} 5,5 \\ \otimes \mathbf{e}_8 \end{array} 12231513151
\end{array}$$

$$\begin{array}{l}
1\epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \\
1\epsilon_7^{3,3} \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{12231512} \\
2,3 \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\
I^{\oplus 3} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,4} \otimes \epsilon_7^{3,3} \\
3,3 \otimes \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_7^{5,5} I^{\oplus 2}
\end{array}$$

$$\begin{array}{l} \begin{array}{ccccc} 2,2 & (1,1) & & 1,1 & 2,4 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes \end{array} \\ \begin{array}{ccccc} 3,2 & (1,1) & & 1,1 & 2,3 \\ \otimes \mathfrak{so} & (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & & 1,1 & & 2,3 \\ (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes & \end{array} \\ I^{\oplus 2} \mathfrak{so} \begin{array}{ccccc} (1,1) & & 1,1 & & 2,4 \\ (\otimes \mathfrak{so})^3 & \otimes & \mathfrak{so} & \otimes & \end{array} \end{array}$$

$$\begin{array}{l} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 2} \mathbf{e}_6 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ 1 \mathbf{e}_6 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \\ 1 \mathbf{e}_6 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 (\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8)^2 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 2} \end{array}$$

$$\begin{aligned}
& I^{\oplus 2} \epsilon_7^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} 12 \\
& I^{\oplus 2} \epsilon_7'^{4,5} \otimes \epsilon_8^{5,5} (\otimes \epsilon_8)^2 \otimes \epsilon_8^{5,5} 12 \\
& 1 \epsilon_7^{5,5} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5} 12231512 \\
& 1 \epsilon_7'^{5,5} (\otimes \epsilon_8)^3 \otimes \epsilon_8^{5,5} 12231513.
\end{aligned}$$

$$\begin{array}{l} 1\mathfrak{so}^{\begin{smallmatrix} 2,3 \\ \end{smallmatrix}} \otimes \mathfrak{e}_7^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}} (\otimes \mathfrak{e}_7^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}})^2 \otimes \mathfrak{e}_7^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}} \otimes^{\begin{smallmatrix} 4,4 \\ \end{smallmatrix}} \\ 1\mathfrak{so}^{\begin{smallmatrix} 2,3 \\ \end{smallmatrix}} \otimes \mathfrak{e}_7^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}} (\otimes \mathfrak{e}_7^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}})^2 \otimes^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}} \mathfrak{e}_7 12313 \\ 1\mathfrak{so}^{\begin{smallmatrix} 2,3 \\ \end{smallmatrix}} \otimes \mathfrak{e}_7^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}} (\otimes \mathfrak{e}_7^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}})^2 \otimes^{\begin{smallmatrix} 3,3 \\ \end{smallmatrix}} \mathfrak{e}_7 1232151 \end{array}$$

$$\begin{array}{l} 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \right)^3 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \begin{smallmatrix} 3,4 \\ \otimes \end{smallmatrix} \\ 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \right)^3 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 I^{\oplus 3} \\ 150 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 \right)^3 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \epsilon_6 13151^1 \end{array}$$

$$I^{\oplus 2} \epsilon_7'{}^{4,5} \otimes \epsilon_8({}^{5,5} \epsilon_8)^2 \otimes \epsilon_7 I^{\oplus 3}$$

$$1 \epsilon_7'({}^{5,5} \otimes \epsilon_8)^3 \otimes \epsilon_7 I^{\oplus 3}$$

$$\begin{array}{l} \begin{array}{c} 3,3 \\ \otimes \end{array} \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7^3 \begin{array}{c} 4,5 \\ \otimes \end{array} \mathfrak{e}_8 I^{\oplus 3} \\ 512321 \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7^3 \begin{array}{c} 4,5 \\ \otimes \end{array} \mathfrak{e}_8 I^{\oplus 2} \end{array}$$

$$\begin{array}{c} 2,3 \\ \otimes \end{array} e'_7 \left(\begin{array}{c} 3,3 \\ \otimes \end{array} e_7 \right)^3 \begin{array}{c} 3,3 \\ \otimes \end{array} e'_7 \begin{array}{c} 3,3 \\ \otimes \end{array}$$

$$I^{\oplus 3} \mathbf{e}'_7 (\overset{(3,3)}{\otimes} \mathbf{e}_7)^3 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes}$$

$$\begin{array}{ccccccc} 1,1 & 2,4 & 5,5 & 5,5 & & & \\ 1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 & 122315123 & & & & & \\ 1,1 & 2,4 & 5,5 & 5,5 & & & \\ 1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 & 12231513151 & & & & & \end{array}$$

$$\begin{array}{c} 2231\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^3\overset{2,4}{\otimes}\mathfrak{e}_81 \\ 1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2\overset{2,4}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}1 \end{array}$$

$$\begin{array}{c} \begin{array}{cccc} 2,2 & 3,3 & 3,3 & 3,3 \\ \otimes & \otimes & \otimes & \otimes \\ \mathbf{e}_6 & (\mathbf{e}_7)^3 & \mathbf{e}_7' & \end{array} \\ I^{\oplus 2} \mathbf{e}_6 (\otimes \mathbf{e}_7)^3 \otimes \mathbf{e}_7' \end{array}$$

$$\begin{array}{l} 315131\mathfrak{e}_6(\bigotimes\mathfrak{e}_6)^3\bigotimes\mathfrak{e}_81 \\ I^{\oplus 2}\mathfrak{e}_6(\bigotimes\mathfrak{e}_6)^3\bigotimes\mathfrak{e}_8I^{\oplus 3} \end{array}$$

$$\begin{array}{ccccccc} 2,2 & 3,5 & (5,5) & 5,5 \\ \otimes e_6 & \otimes e_8 & (\otimes e_8)^2 & \otimes e'_7 1 \\ 1e_6 & \otimes e_8 & (\otimes e_8)^2 & \otimes e'_7 1 \end{array}$$

$$I^{\oplus 2} \mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,3}{\otimes} \mathbf{e}_6 I^{\oplus 2}$$

$$1 \mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{l} 31\mathfrak{s}\mathfrak{o} \otimes \mathfrak{e}_8 \otimes (\mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 122315123 \\ 31\mathfrak{s}\mathfrak{o} \otimes \mathfrak{e}_8 \otimes (\mathfrak{e}_8)^2 \otimes \mathfrak{e}_8 12231513151 \end{array}$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^2 \otimes^{5,4} e_7 I^{\oplus 3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7' 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} (\otimes \mathfrak{so})^{1,1} \otimes \mathfrak{e}_7^{2,3} 1$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7 1$$

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$$\otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$\otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$\otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$313221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$I^{\oplus 5} \mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4},$$

$$21513221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$1513221\mathfrak{e}_8 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_7^{1,5,4}$$

$$\otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$\otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$321\mathfrak{e}_7' (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$I^{\oplus 3} \mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$3221\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$512321\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3},$$

$$151321\mathfrak{e}_7 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3}$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \right)^2 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6$$

$$1e_7(\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{5,4}{\otimes} e_7$$

$$I^{\oplus 2} e'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^2 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7$$

$$31e_6(\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} \overset{3,3}{\otimes} e_7$$

$$1_{50} \overset{2,4}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_7$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e'_7$$

$$150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} e_6$$

$$1e'_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,4}{\otimes} e_7$$

$$1e_7 \overset{3,3}{\otimes} e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{4,2}{\otimes} 50$$

$$1e_6 \overset{3,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,4}{\otimes} e_7$$

$$1e_6(\overset{3,3}{\otimes} e_7)^2 \overset{3,3}{\otimes} e_7 \overset{1}{\otimes} \overset{3,3}{e'_7}$$

2.6.8 SgLgLIgLgLg(4)

$$\begin{array}{cccccc} 2,4 & 5,5 & 5,5 & 1 & 5,5 & 4,2 \\ \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so}, \end{array} \quad \begin{array}{cccccc} I^{\oplus 4} \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{e}_8 & \otimes \mathfrak{so} \end{array}$$

$$150 \otimes_{\mathbb{Z}}^{2,4} \mathfrak{e}_8 \otimes_{\mathbb{Z}}^{5,5} \mathfrak{e}_8 \otimes_{\mathbb{Z}}^{1,4,2} 50 \otimes_{\mathbb{Z}}^{1,1} 50$$

$$1e_7 \overset{4,5}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} e_8 \overset{5,5}{\otimes} e_8 \overset{4,2}{\otimes} 50$$

$$\begin{aligned}
& 3221 \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1, & 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1232, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 123 \\
& I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 2}, & I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{e}_7 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 1, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \\
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 3}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 \otimes, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 I^{\oplus 2}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 1
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 \otimes, & 1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7 I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 1, & I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6 13
\end{aligned}$$

$$\begin{aligned}
& 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' I^{\oplus 2}, & 1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' 1
\end{aligned}$$

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7' 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 13$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7' I^{\oplus 2}$$

$$1 \mathfrak{e}_7 \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6 1$$

2.6.12 SgLIgLgLIg(1)

$$I^{\oplus 3} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

2.6.13 SgLIgLgLIgS(1)

$$1 \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} 1$$

$$31\mathfrak{e}_6(\overset{1}{\otimes}\overset{2,2}{\mathfrak{e}_6})^3\overset{3,3}{\otimes}\mathfrak{e}_7,$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{2,2}{\mathfrak{e}_6})^3\overset{3,4}{\otimes}\mathfrak{e}_7$$

$$1513221\mathfrak{e}_8(\overset{1}{\otimes}\overset{5,5}{\mathfrak{e}_8})^2\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7,$$

$$I^{\oplus 2}{}^{3221}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7$$

$$I^{\oplus 2}\mathfrak{e}_8\overset{1}{\otimes}\overset{5,3}{\mathfrak{e}_6}(\overset{2,2}{\otimes}\mathfrak{e}_6)^2\overset{2,2}{\otimes}\mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^3\overset{4,4}{\otimes}\mathfrak{e}_7$$

$$I^{\oplus 2}\mathfrak{e}_7'(\overset{3,3}{\otimes}\mathfrak{e}_7)^3\overset{3,3}{\otimes}\mathfrak{so}$$

$$I^{\oplus 2}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2(\overset{3,3}{\otimes}\mathfrak{e}_7')^2$$

$$I^{\oplus 2}\mathfrak{e}_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^2\overset{3,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$3221\mathfrak{e}_8\overset{1}{\otimes}\overset{4,2}{\mathfrak{so}}(\overset{1,1}{\otimes}\mathfrak{so})^3$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,4}{\otimes}\mathfrak{e}_7\overset{3,3}{\otimes}\mathfrak{e}_7'$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{5,5}{\otimes}\mathfrak{e}_8\overset{5,3}{\otimes}\mathfrak{e}_6\overset{2,2}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,5}{\otimes}\mathfrak{e}_6$$

$$1\mathfrak{e}_7'\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}(\overset{5,5}{\otimes}\mathfrak{e}_8)^2\overset{5,3}{\otimes}\mathfrak{so}$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}_8}\overset{4,2}{\otimes}\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^2$$

$$1\mathfrak{e}_7\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7}\overset{4,5}{\otimes}\mathfrak{e}_8\overset{4,2}{\otimes}\mathfrak{so}\overset{1,1}{\otimes}\mathfrak{so}$$

$$\begin{aligned} I^{\oplus 4} \mathbf{e}_8^{\otimes 2} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{12231513} \\ I^{\oplus 3} \mathbf{e}_8^{\otimes 3} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{122315123} \\ I^{\oplus 3} \mathbf{e}_8^{\otimes 3} \left(\bigotimes \mathbf{e}_8 \right)^3 \bigotimes \mathbf{e}_8^{12231513151} \end{aligned}$$

[illegible]

$$\begin{array}{l}
2,3 \quad 1 \quad 3,3 \quad 3,3 \quad 4,4 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 12313 \\
2,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 1232151 \\
I^{\oplus 3} \quad 1 \quad 3,3 \quad 3,3 \quad 4,2 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
3,3 \quad 1 \quad 3,3 \quad 3,3 \quad 4,5 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes \\
3,3 \quad 1 \quad 3,3 \quad 3,3 \\
\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 1231512
\end{array}$$

$$\begin{array}{c} 31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_6^{2,2} I^{\oplus 3} \\ 31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^3 \otimes \epsilon_6^{2,2} 13151^1 \end{array}$$

$$\begin{aligned}
& 1 \epsilon_7' \otimes \epsilon_8 (\otimes \epsilon_8)^2 \otimes \epsilon_8 122315123 \\
& 1 \epsilon_7' \otimes \epsilon_8 (\otimes \epsilon_8)^2 \otimes \epsilon_8 12231513151 \\
& \otimes \epsilon_8 (\otimes \epsilon_8)^3 \epsilon_7' I^{\oplus 2} \\
& 21513221 \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' I^{\oplus 2} \\
& I^{\oplus 4} \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' I^{\oplus 2} \\
& \otimes \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' 1 \\
& I^{\oplus 5} \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' 1 \\
& I^{\oplus 2} \epsilon_8 (\otimes \epsilon_8)^3 \otimes \epsilon_7' 1
\end{aligned}$$

$$\begin{aligned}
& \otimes_{\mathbf{e}_7}^{2,3,1} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3} \\
& \otimes_{\mathbf{e}_7}^{2,3,1} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3} I^{\oplus 3} \\
& I^{\oplus 3} \otimes_{\mathbf{e}_7}^{1, (3,3)} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,3} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^{1, (3,3)} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{4,4} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^{1, (3,3)} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{12313} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7'}^{1, (3,3)} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7}^{3,3} \otimes_{\mathbf{e}_7}^{1232151} \\
& I^{\oplus 2} \otimes_{\mathbf{e}_7}^{1, \oplus 2} (\otimes_{\mathbf{e}_7}^{3,3})^3 \otimes_{\mathbf{e}_7'}^{3,3} \otimes_{\mathbf{e}_7'}^{3,2}
\end{aligned}$$

$$\begin{aligned}
& 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 \otimes^{4,4} \\
& 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 12313 \\
& 1\epsilon_6^{1,3,3}(\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_7 1232151 \\
& \otimes^{3,3} \epsilon_7 (\otimes^{3,3} \epsilon_7)^3 \otimes^{3,3} \epsilon_6 \otimes^{2,2} \\
& \otimes^{3,3} \epsilon_7 (\otimes^{3,3} \epsilon_7)^3 \otimes^{3,3} \epsilon_6 I^{\oplus 2} \\
& I^{\oplus 2} \epsilon_7^{1,3,3} (\otimes \epsilon_7)^3 \otimes^{3,3} \epsilon_6 I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned} & I^{\oplus 3} \mathbf{e}_8^{1 \oplus 2} (\otimes \mathbf{e}_8)^{2 \oplus 5, 5} \otimes \mathbf{e}_7^{5, 4} \otimes \mathbf{e}_7^{3, 3} 1 \\ & I^{\oplus 2} \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4, 5} (\otimes \mathbf{e}_8)^{5, 5} \otimes \mathbf{e}_7^{5, 4} I^{\oplus 3} \\ & 1 \mathbf{e}_7^1 (\otimes \mathbf{e}_8)^{5, 5} \otimes \mathbf{e}_7^{5, 4} I^{\oplus 3} \end{aligned}$$

$$\begin{array}{l} \otimes^{5,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \\ 21513221 \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \\ I^{\oplus 4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{4,2} \mathfrak{so}13 \end{array}$$

$$\begin{array}{l} \begin{array}{c} 5,5 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \end{array} \\ 21513221 \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \\ I^{\oplus 4} \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} \end{array}$$

$$\begin{aligned} & 2321^1_{\mathbf{e}_7} \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_8 1 \\ & I^{\oplus 2} \mathbf{e}_8^1 \otimes \mathbf{e}_7^{5,4} \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^2 \otimes \mathbf{e}_7^{3,3} \otimes \mathbf{e}_7^{3,2} \\ & I^{\oplus 2} \mathbf{e}_7^1 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_8^{4,5} I^{\oplus 3} \end{aligned}$$

$$\begin{aligned} & 1_{\mathbf{e}_6}^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_7^{5,4} I^{\oplus 3} \\ & 1_{\mathbf{e}_7}^{1,5,5} \otimes \mathbf{e}_8^{5,3} \otimes \mathbf{e}_6^{2,2} \\ & I^{\oplus 2} \mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{5,4} \end{aligned}$$

$$1_{\mathbf{e}'_7} \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^2 \otimes^{5,4} \mathbf{e}_7 I^{\oplus 3}$$

$$I^{\oplus 2} \mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^2 \otimes^{5,5} \mathbf{e}'_7 1$$

$$\begin{aligned}
& 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} 123, & 1\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_8^{3,5} I^{\oplus 3}, & 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} 13151 \\
& I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2}, & I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^2 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}, & 31\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_8^{3,5} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2}, & 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,3}, & 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \\
& 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,2} 132, & 1\mathfrak{so}(\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} I^{\oplus 2} &
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7^{5,4} I^{\oplus 2}, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2} \\
& 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7^{5,5} \otimes \mathfrak{e}'_7 1, & 1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,4} 1 &
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,2}, & 1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,3}, & 1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} I^{\oplus 3} \\
& I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2}, & I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2} &
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1, & \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1, & I^{\oplus 3} \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1 \\
& I^{\oplus 2} \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2} 1 & &
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 3} 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 1, & 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 132, & I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 1 \\
& I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^2 \otimes \mathfrak{so}^{1,1} 13 & &
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}_7^{3,3} 1, & I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,2}, & I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} \otimes \mathfrak{e}'_7^{3,3} \\
& I^{\oplus 2} \mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}'_7^{3,3} I^{\oplus 3} & &
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{so}^{2,2} 1, & I^{\oplus 2} 1\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{so}^{2,2} 1, & 31\mathfrak{e}_6(\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{so}^{2,2} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}'_7^{3,3} 123, & 1\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 3}, & I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}'_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}, & 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2}, & 1\mathfrak{e}_6 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_6^{5,4} 1
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6^{3,3} 1, & 1\mathfrak{e}_8 \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_6^{3,3} 13 &
\end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1,$$

$$1\mathfrak{e}_7^{1,5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13,$$

$$1\mathfrak{e}_7^{1,5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7^{5,5}1$$

$$1\mathfrak{e}_6^{1,3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{e}_6^{1,3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$I^{\oplus 2} \mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2}1$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{5,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_71$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_7^{1,4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_7^{1,3,3} \otimes \mathfrak{e}_7^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_6^{1,3,3} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so}^{3,2}1$$

$$1\mathfrak{e}_6^{1,2,2} (\otimes \mathfrak{e}_6)^3 \otimes \mathfrak{e}_7^{3,3}1$$

$$\begin{array}{lll} \begin{array}{l} 3,3 \ 1 \\ \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 \otimes, \end{array} & \begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^1 \otimes, \end{array} & \begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^1 I^{\oplus 3} \end{array} \\ \begin{array}{l} 2,3 \ 1 \\ \otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7 I^{\oplus 2}, \end{array} & \begin{array}{l} \otimes \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^1 I^{\oplus 2}, \end{array} & \begin{array}{l} I^{\oplus 3} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^1 I^{\oplus 2} \end{array} \\ I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_7^1 I^{\oplus 2} & & \end{array}$$

$$\begin{array}{lll}
1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,3}, & 1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,4}, & 1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5} \\
1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,122313}, & 1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1, I^{\oplus 5}}, & 1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,12231512} \\
1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,1223151}, & 1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 2,4,2}}, & 1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 2}} \\
1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 3}}, & & 1\epsilon_7' \otimes \epsilon_8 \otimes \epsilon_8^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 4}}
\end{array}$$

$$\begin{array}{lll}
1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,3}, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,4}, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5} \\
1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{122313}, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1\oplus 5}, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{12231512} \\
1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1223151}, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 2,4,2}}, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 2}} \\
1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^2 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{I^{\oplus 3}}, & &
\end{array}$$

[illegible]

$$\begin{array}{lll}
I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_7 \begin{smallmatrix} 1,3,2 \\ \otimes \end{smallmatrix}, & I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_7 \begin{smallmatrix} 1,3,3 \\ \otimes \end{smallmatrix}, & I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_7 \begin{smallmatrix} 1 \\ \oplus \end{smallmatrix} \\
I^{\oplus 2} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \otimes \mathbf{e}_7 \begin{smallmatrix} 1 \\ \oplus \end{smallmatrix} I^{\oplus 2} & &
\end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{1,3,2}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{1,3,3}{\otimes}, & 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{I^{\oplus 3}}{1} \\ 1\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^3\overset{3,3}{\otimes}\mathbf{e}_7\overset{I^{\oplus 2}}{I^{\oplus 2}} & & \end{array}$$

$$I_7^{\oplus 2} \overset{1}{e_7} \overset{4,5}{\otimes} \overset{5,5}{e_8} (\overset{5,5}{\otimes} e_8)^2 \overset{5,4}{\otimes} I_7^{\oplus 2} 1, \quad I_7^{\oplus 2} \overset{1}{e_7} \overset{4,5}{\otimes} \overset{5,5}{e_8} (\overset{5,5}{\otimes} e_8)^2 \overset{5,5}{\otimes} \overset{1}{e_7} 1, \quad 1 \overset{1}{e_7} (\overset{5,5}{\otimes} e_8)^3 \overset{5,5}{\otimes} \overset{1}{e_7} 1$$

$$321 \mathfrak{e}_7^{1(3,3)} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_8^{4,5} 1, \quad I^{\oplus 2} \mathfrak{e}_7^{1(3,3)} (\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 2} 1$$

$$1e_7' \otimes e_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes e_8 \end{smallmatrix}^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} I^{\oplus 2} 1,$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,5} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_8^{3,5} I^{\oplus 2},$$

$$31\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_8^{3,5} \mathfrak{e}_8^1$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_7'^{3,3} I^{\oplus 2}$$

$$1\mathfrak{so}^1 (\otimes \mathfrak{so}^{1,1})^3 \otimes \mathfrak{so}^{1,1} \mathfrak{so}^1$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7'^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 2}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7'^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_7'^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

2.6.17 SSgLgLgLIgLg(9)

$$\otimes \mathfrak{e}_8^{2,4} \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4},$$

$$I^{\oplus 4} \mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4},$$

$$I^{\oplus 3} \mathfrak{e}_8^{I^{\oplus 2}} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7'^{3,3}$$

$$321\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^2 \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7^1 \otimes \mathfrak{e}_7^{3,3}$$

$$231513221\mathfrak{e}_8^1 (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_8^{5,5} \mathfrak{e}_8^1 \otimes \mathfrak{e}_8^{5,5}$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} \overset{5,5}{\otimes} \overset{5,5}{e_8} \overset{1}{\otimes} \overset{5,4}{e_7}$$

$$1 \overset{1}{e}'_7 (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \overset{1}{e}_7 \overset{3,3}{\otimes} \overset{1}{e}'_7$$

$$1\mathbf{e}_6^{1(2,2)}(\otimes \mathbf{e}_6)^{2(2,2)} \otimes \mathbf{e}_6^{1(2,2)} \otimes \mathbf{e}_6^{2,2}$$

2.6.18 SSgLgLgLIgLgS(25)

[illegible]

$$I^{\oplus 2} \mathbf{e}_7^{1\ 4,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 4,2} \oplus 1 \mathbf{e}_7^{1\ 5,5} (\otimes \mathbf{e}_8)^2 \otimes \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^{1\ 4,2} \oplus$$

$$I^{\oplus 2} \mathbf{e}_7^1 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^2 \overset{3,3}{\otimes} \mathbf{e}_7^1 \overset{3,3}{\otimes} \mathbf{e}_7^1$$

$$1e_7^{1,4,5} \otimes e_8^{5,5} \otimes e_8^{5,5} \otimes e_8^{1,5,5} \otimes e_8 I^{\oplus 3}$$

$$1e_7' \otimes e_8 \otimes e_8 \otimes e_8 \otimes e_8 \otimes so_1$$

$$1 \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^2 \overset{3,3}{\otimes} \overset{1}{\mathbf{e}}_7 \overset{3,3}{\otimes} \mathbf{e}_7 1$$

$$1e_6^{1,3,5} \otimes e_8^{5,5} \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes so1$$

2.6.19 SSgLgLgLIgLIgS(6)

2.6.23 SSgLIgLgLIgS(6)

$$1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \begin{smallmatrix} 1 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^2 \begin{smallmatrix} 5,5 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_8 1223151, \quad I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \begin{smallmatrix} 1 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^2 \begin{smallmatrix} 5,5 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 4,2 \\ 1 \end{smallmatrix}, \quad 1 \mathfrak{e}_8 \otimes \mathfrak{e}_8 \begin{smallmatrix} 1 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^2 \begin{smallmatrix} 5,5 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_8 I^{\oplus 2} 1223$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \begin{smallmatrix} 1 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^2 \begin{smallmatrix} 5,5 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_8 I^{\oplus 4}, \quad I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \begin{smallmatrix} 1 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^2 \begin{smallmatrix} 5,5 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_8 I^{\oplus 3} I^{\oplus 2}$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \begin{smallmatrix} 1 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^2 \begin{smallmatrix} 5,4 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_7 1$$

2.6.24 SSgLIgLgLIg(1)

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \begin{smallmatrix} 1 \\ 5,5 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ 1 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 4,2 \\ 1 \end{smallmatrix} \otimes \mathfrak{so}$$

2.7 6 node(1127)

2.7.1 g6(12)

$$\mathfrak{so} \begin{smallmatrix} 1,1 \\ (\otimes \mathfrak{so})^3 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \\ 4,2 \end{smallmatrix} \otimes \mathfrak{so}, \quad \mathfrak{so} \begin{smallmatrix} 1,1 \\ (\otimes \mathfrak{so})^2 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \\ 4,2 \end{smallmatrix} \otimes \mathfrak{so} \otimes \mathfrak{so} \begin{smallmatrix} 1,1 \\ 1,1 \end{smallmatrix}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \begin{smallmatrix} 3,5 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^3 \begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix} \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \begin{smallmatrix} 3,5 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^3 \begin{smallmatrix} 5,3 \\ 5,3 \end{smallmatrix} \otimes \mathfrak{so}$$

$$\mathfrak{so} \begin{smallmatrix} 3,3 \\ (\otimes \mathfrak{e}_7)^4 \end{smallmatrix} \otimes \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ 3,3 \end{smallmatrix}$$

$$\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ 3,3 \end{smallmatrix} (\otimes \mathfrak{e}_7)^2 \otimes \mathfrak{e}_7' \begin{smallmatrix} 3,3 \\ 3,3 \end{smallmatrix} \otimes \mathfrak{e}_7$$

$$\mathfrak{so} \begin{smallmatrix} 2,2 \\ (\otimes \mathfrak{e}_6)^4 \end{smallmatrix} \otimes \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ 3,3 \end{smallmatrix}$$

$$\mathfrak{so} \otimes \mathfrak{so} \begin{smallmatrix} 1,1 \\ 2,4 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ 5,4 \end{smallmatrix} \otimes \mathfrak{e}_7 \begin{smallmatrix} 3,3 \\ 3,3 \end{smallmatrix} \otimes \mathfrak{e}_7'$$

$$\mathfrak{so} \otimes \mathfrak{so} \begin{smallmatrix} 1,1 \\ 2,4 \end{smallmatrix} \otimes \mathfrak{e}_8 \begin{smallmatrix} 5,5 \\ 5,3 \end{smallmatrix} \otimes \mathfrak{e}_6 \begin{smallmatrix} 2,2 \\ 2,2 \end{smallmatrix} \otimes \mathfrak{e}_6$$

$$\mathfrak{e}_7' \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8)^3 \end{smallmatrix} \otimes \mathfrak{e}_7 \begin{smallmatrix} 5,4 \\ 3,3 \end{smallmatrix} \otimes \mathfrak{e}_7$$

$$\mathfrak{e}_6 \begin{smallmatrix} 5,5 \\ (\otimes \mathfrak{e}_8)^4 \end{smallmatrix} \otimes \mathfrak{e}_6 \begin{smallmatrix} 5,5 \\ 5,5 \end{smallmatrix}$$

$$\mathfrak{e}_6 \otimes \mathfrak{e}_8 \begin{smallmatrix} 4,5 \\ 5,5 \end{smallmatrix} (\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \begin{smallmatrix} 5,4 \\ 3,3 \end{smallmatrix} \otimes \mathfrak{e}_7$$

2.7.2 Sg6(81)

$$\begin{aligned} & \otimes \epsilon_7^{2,4} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3}, \\ & \otimes \epsilon_7^{5,4} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3}, \\ & 2151321 \epsilon_7^{(3,3)} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3}, \\ & 151321 \epsilon_7^{(3,3)} (\otimes \epsilon_7^{3,3})^4 \otimes \mathfrak{so}^{3,3} \end{aligned}$$

$$\begin{array}{l} 3,4 \\ \otimes \mathbf{e}_7(\otimes \mathbf{e}_7)^4 \otimes \mathbf{50}, \\ I^{\oplus 4} \mathbf{e}_7(\otimes \mathbf{e}_7)^4 \otimes \mathbf{50}, \\ 2321 \mathbf{e}_7(\otimes \mathbf{e}_7)^4 \otimes \mathbf{50}, \end{array}$$

$$\begin{array}{l} 4,4 \quad 3,3 \quad 3,3 \\ \otimes e_7 (\otimes e_7)^4 \otimes 50 \\ 31321 e_7 (\otimes e_7)^4 \otimes 50 \\ 1512321 e_7 (\otimes e_7)^4 \otimes 50 \end{array}$$

$$\begin{aligned} & \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7, \\ & 313221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7, \\ & \overset{1}{15} 13221 \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7 \end{aligned}$$

$$\begin{array}{c} 4,5 \quad 5,5 \quad 5,4 \quad 3,3 \\ \otimes \mathbf{e}_8 (\otimes \mathbf{e}_8)^3 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7, \\ I^{\oplus 5} \mathbf{e}_8 (\otimes \mathbf{e}_8)^3 \otimes \mathbf{e}_7 \otimes \mathbf{e}_7, \end{array}$$

$$\begin{array}{c} \begin{array}{cccc} 5,5 & 5,5 & 5,4 & 3,3 \\ \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 \end{array} \\ 21513221\mathbf{e}_8 \begin{array}{cccc} 5,5 & & 5,4 & 3,3 \\ \otimes & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}_7 & \otimes \mathbf{e}_7 \end{array} \end{array}$$

$$2313221\mathbf{e}_8(\otimes\mathbf{e}_8)^4\otimes\mathbf{e}_6,$$

$$51231513221\mathbf{e}_8(\otimes\mathbf{e}_8)^4\otimes\mathbf{e}_6,$$

$$I^{\oplus 6} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$31513221 \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes^{5,5}\epsilon_8)^4\otimes^{5,5}\epsilon_6 \\ 15131513221\epsilon_8(\otimes^{5,5}\epsilon_8)^4\otimes^{5,5}\epsilon_6 \end{array}$$

$$\begin{aligned} &2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}, \\ &51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}, \end{aligned}$$

$$I^{\oplus 6} \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,3}{\otimes} \mathfrak{so},$$

$$31513221 \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,3}{\otimes} \mathfrak{so},$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^4 \otimes \mathfrak{so}^{5,5} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^4 \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{c} 2,3 \quad 4,5 \quad (5,5) \quad 5,5 \\ \otimes \mathbf{e}_7 \quad \otimes \mathbf{e}_8 \quad (\otimes \mathbf{e}_8)^3 \quad \otimes \mathbf{e}_6, \\ I^{\oplus 2} \mathbf{e}_7 (\otimes \mathbf{e}_8)^4 \quad \otimes \mathbf{e}_6 \end{array}$$

$$\begin{matrix} 3,3 \\ \otimes \end{matrix} \mathbf{e}_7 \begin{matrix} 4,5 \\ \otimes \end{matrix} \mathbf{e}_8 \left(\begin{matrix} 5,5 \\ \otimes \end{matrix} \mathbf{e}_8 \right)^3 \begin{matrix} 5,5 \\ \otimes \end{matrix} \mathbf{e}_6,$$

$$I^{\oplus 3} \mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$\begin{array}{c} 2,3 \quad 4,5 \quad (5,5) \quad 5,3 \\ \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_8 (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}, \\ I^{\oplus 2} \mathfrak{e}_7 (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so} \end{array}$$

$$\begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{e}_7 \begin{matrix} 4,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \left(\begin{matrix} 5,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \right)^3 \begin{matrix} 5,3 \\ \otimes \end{matrix} \mathfrak{so},$$

$$I^{\oplus 3} \mathfrak{e}_7 \otimes^{4,5} \mathfrak{e}_8 \left(\otimes^{5,5} \mathfrak{e}_8 \right)^3 \otimes^{5,3} \mathfrak{so}$$

$$\begin{array}{l} 3150 \otimes e_8(5,5)^3 \otimes e_6, \\ 1e_6 \otimes e_8(5,5)^3 \otimes 50 \end{array}$$

$$\begin{matrix} 2,2 & 3,5 & 5,5 & 5,3 \\ \otimes & \otimes & \otimes & \otimes \\ \mathfrak{e}_6 & \mathfrak{e}_8 & \mathfrak{e}_8 & \mathfrak{so}, \end{matrix}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 \left(\overset{5,5}{\otimes} \mathfrak{e}_8 \right)^3 \overset{5,3}{\otimes} \mathfrak{so}$$

$$\begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7' \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}'_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{so},$$

$$I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^4 \bigotimes^{3,3} \mathfrak{so}$$

$$\begin{matrix} 2,3 \\ \otimes \end{matrix} e_7 \left(\begin{matrix} 3,3 \\ \otimes \end{matrix} e_7 \right)^3 \begin{matrix} 3,3 \\ \otimes \end{matrix} e'_7 \begin{matrix} 3,3 \\ \otimes \end{matrix} e_7,$$

$$\begin{matrix} 3,3 \\ \otimes \end{matrix} e_7 \left(\begin{matrix} 3,3 \\ \otimes \end{matrix} e_7 \right)^3 \begin{matrix} 3,3 \\ \otimes \end{matrix} e'_7 \begin{matrix} 3,3 \\ \otimes \end{matrix} e_7,$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^3 \bigotimes^{3,3} \mathbf{e}'_7 \bigotimes^{3,3} \mathbf{e}_7$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} (\otimes \mathfrak{so})^2$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3} \mathfrak{e}_7'$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} (\otimes \mathfrak{so})^{1,1} \otimes \mathfrak{e}_7^{2,3}$$

$$1\mathfrak{e}_7' \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1\mathfrak{e}_7 \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_8^{4,5} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}$$

$$1\mathfrak{e}_6 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{e}_7^{5,4} \otimes \mathfrak{e}_7^{3,3}$$

$$1\mathfrak{e}_6 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7' \otimes \mathfrak{e}_7^{3,3}$$

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$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1231512},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{123151},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,5},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$I^{\oplus 4} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,2},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1232},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$I^{\oplus 4} \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3},$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4}$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{12313}$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{2,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1232151}$$

$$\otimes \mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,4} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,4}$$

$$31321\mathfrak{e}_7 (\otimes \mathfrak{e}_7)^{3,3} \otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{4,3}$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} \\ I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 3} \\ I^{\oplus 3} \mathbf{e}_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_7 I^{\oplus 2} \end{array}$$

[illegible]

$$\begin{array}{l} 150 \binom{2,2}{\otimes \epsilon_6}^4 \otimes \epsilon_6 \otimes \epsilon_6 \binom{3,4}{\otimes} \\ 150 \binom{2,2}{\otimes \epsilon_6}^4 \otimes \epsilon_6 \binom{2,2}{\otimes} I^{\oplus 3} \\ 150 \binom{2,2}{\otimes \epsilon_6}^4 \otimes \epsilon_6 \binom{2,2}{\otimes} 13151^1 \end{array}$$

$$I^{\oplus 2} \mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^3 \otimes^{5,4} \mathbf{e}_7 I^{\oplus 3}$$

$$1 \mathbf{e}'_7 (\otimes^{5,5} \mathbf{e}_8)^4 \otimes^{5,4} \mathbf{e}_7 I^{\oplus 3}$$

$$1s^0 \otimes^{1,1} s^0 \otimes^{2,4} e_8 (\otimes e_8)^2 \otimes^{5,5} e_8 122315123$$

$$\begin{array}{l} I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}'_7 \bigotimes \\ I^{\oplus 3} \mathbf{e}'_7 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{e}'_7 I^{\oplus 3} \end{array}$$

$$\begin{array}{ccccccc} 2,2 & 3,5 & (5,5) & (5,5) & & & \\ \otimes \mathbf{e}_6 & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}'_7 1 & & & \\ 1\mathbf{e}_6 & (4,5) & (5,5) & (5,5) & & & \\ & \otimes \mathbf{e}_8 & (\otimes \mathbf{e}_8)^3 & \otimes \mathbf{e}'_7 1 & & & \end{array}$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,3}{\otimes} \mathfrak{e}_6 I^{\oplus 2}$$

$$\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \begin{matrix} 3,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \left(\begin{matrix} 5,5 \\ \otimes \end{matrix} \mathfrak{e}_8 \right)^3 \begin{matrix} 5,4 \\ \otimes \end{matrix} \mathfrak{e}_6 1,$$

$$I^{\oplus 2} \mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^3 \overset{5,4}{\otimes} \mathfrak{e}_6 1,$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$\begin{array}{c} 2,2 \\ \otimes \end{array} \mathbf{e}_6 \left(\begin{array}{c} 3,3 \\ \otimes \end{array} \mathbf{e}_7 \right)^4 \begin{array}{c} 3,3 \\ \otimes \end{array} \mathbf{e}'_7 \begin{array}{c} 3,2 \\ \otimes \end{array},$$

$$\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7' \otimes \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix},$$

$$\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7)^4 \begin{matrix} 3,3 \\ \otimes \end{matrix} \mathfrak{e}'_7 I^{\oplus 3}$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^4 \bigotimes^{3,3} \mathbf{e}_7' \bigotimes^{3,2},$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^4 \overset{3,3}{\otimes} \mathbf{e}'_7 \overset{3,3}{\otimes},$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^4 \overset{3,3}{\otimes} \mathbf{e}_7' I^{\oplus 3}$$

$$3221e_7(\overset{3,3}{\otimes} e_7)^4 \overset{4,5}{\otimes} e_8 1,$$

$$512321\mathbf{e}_7\left(\overset{3,3}{\otimes}\mathbf{e}_7\right)^4\overset{4,5}{\otimes}\mathbf{e}_81,$$

$$151321\mathbf{e}_7\binom{3,3}{\otimes\mathbf{e}_7}^4\binom{4,5}{\otimes\mathbf{e}_8}1$$

$$\bigotimes^{2,3} \mathfrak{e}_7 \left(\bigotimes^{3,3} \mathfrak{e}_7 \right)^4 \bigotimes^{4,5} \mathfrak{e}_8 I^{\oplus 2},$$

$$\otimes^{3,3} \mathfrak{e}_7 (\otimes^{3,3} \mathfrak{e}_7)^4 \otimes^{4,5} \mathfrak{e}_8 I^{\oplus 2},$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes^{3,3} \mathbf{e}_7 \right)^4 \bigotimes^{4,5} \mathbf{e}_8 I^{\oplus 2}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_8 1223132,$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_8 I^{\oplus 6},$$

$$3150 \otimes^{2,4} e_8(\otimes^{5,5} e_8)^3 \otimes^{5,5} e_8 122315123$$

$$3150 \otimes^{2,4} e_8(\otimes^{5,5} e_8)^3 \otimes^{5,5} e_8 12231513215,$$

$$3150 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^3 \overset{5,5}{\otimes} e_8 12231 \overset{1}{5} 13,$$

$$3150 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^3 \overset{5,5}{\otimes} e_8 12231513151$$

$$321\mathfrak{e}_6\left(\overset{2,2}{\otimes}\mathfrak{e}_6\right)^4\overset{3,5}{\otimes}\mathfrak{e}_81,$$

$$15131\mathfrak{e}_6\binom{2,2}{\otimes\mathfrak{e}_6}^4\binom{3,5}{\otimes\mathfrak{e}_8}1,$$

$$\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \left(\begin{matrix} 2,2 \\ \otimes \end{matrix} \mathfrak{e}_6 \right)^4 \begin{matrix} 3,5 \\ \otimes \end{matrix} \mathfrak{e}_8 I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^4 \begin{smallmatrix} 3,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 I^{\oplus 2}$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,4} e_7 \otimes^{3,2},$$

$$3150 \otimes^{2,4} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,4} e_7 \otimes^{3,3},$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,4} \mathfrak{e}_7 I^{\oplus 3}$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_7 I^{\oplus 2}$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} e_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^2 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix},$$

$$1_{50} \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^2 \overset{5,4}{\otimes} e_7 \overset{3,3}{\otimes},$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^2 \overset{5,4}{\otimes} e_7 I^{\oplus 3}$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{5,5} \mathfrak{e}_7 I^{\oplus 2}$$

$$\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix},$$

$$I^{\oplus 2} \mathfrak{e}_6 \left(\overset{3,3}{\otimes} \mathfrak{e}_7 \right)^4 \overset{3,3}{\otimes} \mathfrak{e}_6 \overset{2,2}{\otimes},$$

$$I^{\oplus 2} \mathfrak{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathfrak{e}_6 I^{\oplus 2}$$

$$I^{\oplus 2} \mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^3 \otimes^{5,4} \mathbf{e}'_7 I^{\oplus 2},$$

$$I^{\oplus 2} \mathbf{e}'_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^3 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^4 \overset{5,5}{\otimes} e'_7 1$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,3} \mathfrak{e}_6 \otimes^{2,2},$$

$$3150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,3} \mathfrak{e}_6 I^{\oplus 2},$$

$$3150 \begin{smallmatrix} 2,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 \right)^3 \begin{smallmatrix} 5,4 \\ \otimes \end{smallmatrix} \mathfrak{e}_6 1$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix},$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix},$$

$$150 \otimes^{2,3} e_7 (\otimes^{3,3} e_7)^3 \otimes^{3,3} e'_7 I^{\oplus 3}$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^2 \otimes^{5,3} \mathfrak{e}_6 \otimes^{2,2},$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^2 \overset{5,3}{\otimes} e_6 I^{\oplus 2},$$

$$1\mathfrak{so}^{1,1} \otimes \mathfrak{so}^{2,4} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^2 \otimes \mathfrak{e}_6 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 I^{\oplus 3},$$

$$231\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 1,$$

$$31\mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{e}_8 I^{\oplus 2}$$

$$321\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 1,$$

$$I^{\oplus 2}\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 I^{\oplus 2}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6 \otimes,$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{e}_6 I^{\oplus 2}$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{e}_6(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 I^{\oplus 2},$$

$$31\mathfrak{e}_6(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{e}_7 \otimes \mathfrak{e}_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^3 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{e}_6)^4 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^2 \otimes \mathfrak{so} 13$$

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$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$\otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 5}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$21513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$1513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 2}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6,$$

$$I^{\oplus 4}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$I^{\oplus 3}\mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6$$

$$\begin{array}{c} 5,5 \quad 1 \\ \otimes \mathfrak{e}_8 \end{array} \left(\begin{array}{c} 5,5 \\ \otimes \mathfrak{e}_8 \end{array} \right)^4 \begin{array}{c} 5,3 \\ \otimes \mathfrak{so} \end{array}$$

$$21513221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^4\otimes^{5,3}\mathbf{so}$$

$$I^{\oplus 4} I^{\oplus 2} \begin{pmatrix} 5,5 \\ \otimes \mathfrak{e}_8 \end{pmatrix}^4 \otimes \mathfrak{so}_{50}$$

$$I^{\oplus 3} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^4 \bigotimes \mathbf{so}$$

$$I^{\oplus 3} I^{\oplus 2} \begin{pmatrix} 5,5 \\ \otimes \mathbf{e}_8 \end{pmatrix}^3 \begin{pmatrix} 5,4 \\ \otimes \mathbf{e}_7 \end{pmatrix} \begin{pmatrix} 3,3 \\ \otimes \mathbf{e}_7 \end{pmatrix}$$

$$I^{\oplus 4} \mathbf{e}_8^1 \left(\bigotimes^{5,5} \mathbf{e}_8 \right)^3 \bigotimes^{5,4} \mathbf{e}_7 \bigotimes^{3,3} \mathbf{e}_7,$$

$$1e_7^{(1)}(\otimes_{5,5} e_8)^4 \otimes_{5,5} e_6$$

$$1e_7^{1(5,5)}(\otimes e_8)^{4(5,3)} \otimes so$$

$$I^{\oplus 2} \mathbf{e}_7' \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^4 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so}$$

$$I^{\oplus 2} \mathbf{e}_7^1 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7 \right)^3 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7' \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathbf{e}_7$$

$$2321\mathbf{e}_7({}^1\otimes\mathbf{e}_7)^4\otimes\mathbf{e}_8$$

$$1e_8 \otimes^{5,4} e_7 (\otimes^{3,3} e_7)^3 \otimes^{3,2} so$$

$$1_{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,3}{\mathfrak{e}_6} \left(\overset{2,2}{\otimes} \mathfrak{e}_6 \right)^3 \overset{2,2}{\otimes} \mathfrak{so}$$

$$1e_7' \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^3 \overset{5,5}{\otimes} e_6$$

$$1e_7' \otimes^{1,4,5} e_8 (\otimes^{5,5} e_8)^3 \otimes^{5,3} \mathfrak{so}$$

$$1\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 3},$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 2}$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7'^{5,4} I^{\oplus 2},$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7'^{5,5} 1,$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_6^{5,4} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_6^{3,2},$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_6^{3,3},$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7'^{3,3} I^{\oplus 3}$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_6^{3,3} \otimes \mathfrak{e}_6^{2,2},$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$\otimes \mathfrak{e}_7^{2,3} \otimes \mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{so}^{3,2} 1,$$

$$\otimes \mathfrak{e}_7^{3,3} \otimes \mathfrak{e}_7^{1,3,3} (\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{so}^{3,2} 1,$$

$$I^{\oplus 3} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{so}^{3,2} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_8^1 \otimes \mathfrak{e}_6^{5,3} (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_8^{3,5} I^{\oplus 2},$$

$$1\mathfrak{e}_8^1 \otimes \mathfrak{e}_6^{5,3} (\otimes \mathfrak{e}_6^{2,2})^3 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2}$$

$$31\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_8^{3,5} 1$$

$$\otimes \mathfrak{e}_6^{2,2} \otimes \mathfrak{e}_6^{1,2,2} (\otimes \mathfrak{e}_6)^4 \otimes \mathfrak{so}^{2,2} 1,$$

$$I^{\oplus 2} \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{so}^{2,2} 1,$$

$$31\mathfrak{e}_6^{31} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{so}^{2,2} 1$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_7'^{3,2},$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7'^{3,3} \otimes \mathfrak{e}_7'^{3,3},$$

$$I^{\oplus 2} \mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7'^{3,3} I^{\oplus 3}$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_6^{5,3} \otimes \mathfrak{e}_6^{2,2},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_6^{5,3} I^{\oplus 2},$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_6^{5,4} 1$$

$$I^{\oplus 2} \mathfrak{e}_8^1 \otimes \mathfrak{so}^{4,2} (\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} 1,$$

$$1\mathfrak{e}_8^1 \otimes \mathfrak{so}^{4,2} (\otimes \mathfrak{so})^3 \otimes \mathfrak{so}^{1,1} 13$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{so}^{4,2} 13,$$

$$1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}^{4,2} 13$$

$$I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^2 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1,$$

$$1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7'^{5,4} I^{\oplus 2},$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7'^{5,5} 1$$

$$1\mathfrak{e}_7'^1 \otimes \mathfrak{e}_7^{3,3} (\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_8^{4,5} I^{\oplus 2},$$

$$1\mathfrak{e}_8^1 \otimes \mathfrak{e}_7^{5,4} (\otimes \mathfrak{e}_7^{3,3})^3 \otimes \mathfrak{e}_7'^{3,3} I^{\oplus 2}$$

$$1\mathfrak{e}_6^{1(3,3)}(\otimes \mathfrak{e}_7)^4 \otimes \mathfrak{e}_6^{3,3} I^{\oplus 2}$$

$$1 \overset{1}{\mathbf{e}}_8 \overset{5,4}{\otimes} \mathbf{e}_7 \left(\overset{3,3}{\otimes} \mathbf{e}_7 \right)^3 \overset{3,3}{\otimes} \mathbf{e}_6 1$$

$$1\mathbf{e}_7' \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^2 \overset{4,2}{\otimes} \mathbf{so} \overset{1,1}{\otimes} \mathbf{so} 1$$

$$1\mathfrak{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{4,2}{\otimes} \mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so} 1$$

2.7.6 SSg6IS(135)

$${}^{3,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^4 {}^{5,5}_1 \otimes \mathfrak{e}_8 \otimes$$

$${}^{3,5}{}^1_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^4 {}^{5,5}{}^1_1 \otimes \mathfrak{e}_8 12231512$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^4 {}^{5,5}_1 \otimes \mathfrak{e}_8 \otimes$$

$${}^{4,5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 ({}^{5,5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8)^4 {}^{5,5} \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \mathfrak{e}_8 12231512$$

$$313221 \mathbf{e}_8^1 (\otimes^{5,5} \mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_8 \otimes^{1,5,5}$$

$${}^{5,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5}_1 \otimes \mathfrak{e}_8)^4 {}^{5,5}_1 \otimes \mathfrak{e}_8 1223151$$

$$313221\mathbf{e}_8^1(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8^{5,5} 12231512$$

$$I^{\oplus 5} \mathbf{e}_8^1 (\bigotimes \mathbf{e}_8^{5,5})^4 \bigotimes \mathbf{e}_8^{5,5} \mathbf{e}_8^1 12231512$$

$$1513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8^{(5,5)} 12231512$$

$${}^{4,5}_1 \otimes \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^4 {}^{5,5} I^{\oplus 2} {}^{4,2} \otimes$$

$$I^{\oplus 5} \mathbf{e}_8 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \right)^4 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} I^{\oplus 24,2} \begin{smallmatrix} \otimes \\ \mathbf{e}_8 \end{smallmatrix} \otimes$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} 1223151, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 2} 4,2, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 2} \\
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_8^{5,5} I^{\oplus 3} I^{\oplus 3}
\end{aligned}$$

$$\begin{aligned}
& \otimes \mathfrak{e}_6^{2,2} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & I^{\oplus 2} \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & I^{\oplus 2} \mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2} \\
& 31\mathfrak{e}_6^{3,1} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} \otimes, & 31\mathfrak{e}_6^{3,1} (\otimes \mathfrak{e}_6^{2,2})^4 \otimes \mathfrak{e}_6^{2,2} I^{\oplus 2} 1
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 3} \\
& I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} \otimes, & 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 3} \\
& 1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2}
\end{aligned}$$

$$\begin{aligned}
& I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & I^{\oplus 2} \mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1, & 1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & 1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$\begin{aligned}
& 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} I^{\oplus 2} 1, & 1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,5} 1
\end{aligned}$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_8^{4,5} 1$$

$$I^{\oplus 2} \mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} 1$$

$$1\mathfrak{so}^1 (\otimes \mathfrak{so}^{1,1})^4 \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{e}_7^{5,4} 1$$

$$1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{e}_8^{4,5} 1$$

$$1\mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^3 \otimes \mathbf{e}_7^{5,4} \mathbf{e}_7'^1 1$$

$$1 \mathfrak{e}_6^{1,3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_6^{5,3} 1$$

$$1 \mathbf{e}_6 ({}^{3,3} \otimes \mathbf{e}_7)^4 {}^{3,3} I^{\oplus 2} \otimes \mathbf{e}_7' 1$$

$$1 \mathbf{e}_6^{1(3,3)} (\otimes \mathbf{e}_7)^4 \mathbf{e}_6^{3,3} 1$$

$$1\mathbf{e}_6^{(1)}(\otimes^{(2,2)}\mathbf{e}_6)^4\otimes^{(3,5)}\mathbf{e}_81$$

2.7.7 gLIg5(3)

$$50 \otimes^{2,4} e_8 \otimes^{1,5,5} (e_8 \otimes e_8)^2 \otimes^{5,4} e_7 \otimes^{3,3} e_7$$

$$e_7 \begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix} e_8 \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8)^2 \begin{smallmatrix} 4,2 \\ \otimes \end{smallmatrix} so \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} so$$

$$e_7 \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 1 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix} 50$$

2.7.8 SgLIg5(5)

$$I^{\oplus 3} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8^{4,2} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{so}^{1,1})^2 \otimes \mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{1,5,5} (\otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}^{5,3})^3 \otimes \mathfrak{so}^{5,3}$$

$$I^{\oplus 2} \mathbf{e}_8^{5,5} \otimes \mathbf{e}_8^1 \left(\otimes \mathbf{e}_8^{5,5} \right)^3 \otimes \mathbf{e}_7^{5,5}$$

$$I^{\oplus 2} \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^1 \left(\otimes \mathfrak{e}_8^{5,5} \right)^3 \otimes \mathfrak{e}_6^{5,4}$$

$$150 \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^3 \otimes^{5,5} \mathfrak{e}_6$$

2.7.9 SgLIg3LIgLg(1)

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^2 \otimes \mathfrak{e}_8 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{e}'_7 I^{\oplus 2},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{e}'_7 1$$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{so} 13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^2 \otimes \mathfrak{so} \otimes \mathfrak{so} 1$$

2.8 n≥7 node(10,217)

2.8.1 g(n)(47)

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^6 \otimes \mathfrak{e}_6$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^6 \otimes \mathfrak{so}$$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{e}_6$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{14} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{13} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{12} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{11} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{10} \otimes \mathfrak{e}_6^{5,5}$$

$$\mathfrak{so}^{3,5} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^{10} \otimes \mathfrak{so}^{5,3}$$

$$\mathfrak{so}^{3,3} \left(\otimes \mathfrak{e}_7 \right)^9 \otimes \mathfrak{so}^{3,3}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^7 \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^6 \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^5 \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{15} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{14} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{13} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{12} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{11} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so}(\overset{3,3}{\otimes} \mathfrak{e}_7)^{10} \overset{3,3}{\otimes} \mathfrak{so}$$

$$\mathfrak{so} \overset{2,3}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} (\overset{3,3}{\otimes} \mathfrak{e}_7)^5 \overset{4,5}{\otimes} \mathfrak{e}_8$$

$$\mathfrak{so}(\overset{2,2}{\otimes} \mathfrak{e}_6)^6 \overset{3,5}{\otimes} \mathfrak{e}_8$$

$$\mathfrak{so} \overset{1,1}{\otimes} \mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^2 \overset{5,4}{\otimes} \mathfrak{e}_7 \overset{3,3}{\otimes} \mathfrak{e}_7$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^8 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^6 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^5 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{15} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{13} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{12} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{11} \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$\mathfrak{e}_6(\overset{5,5}{\otimes} \mathfrak{e}_8)^{10} \overset{5,5}{\otimes} \mathfrak{e}_6$$

2.8.2 Sg(n)(549)

$$13221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$1223151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$2151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$\overset{1}{15}1321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$13151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$2221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$\overset{1}{23}21\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$123151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$31321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$1512321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^9 \overset{3,3}{\otimes} \mathfrak{so}$$

$$13221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$1223151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$13151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$2221\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so},$$

$$123151321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so}$$

$$31321\mathfrak{e}_7(\overset{3,3}{\otimes} \mathfrak{e}_7)^8 \overset{3,3}{\otimes} \mathfrak{so}$$

$$1512321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^8\overset{3,3}{\otimes}\mathfrak{so}$$

$$\begin{array}{l} 123151321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^7\overset{3,3}{\otimes}\mathfrak{so} \\ 31321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^7\overset{3,3}{\otimes}\mathfrak{so} \\ 1512321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^7\overset{3,3}{\otimes}\mathfrak{so} \end{array}$$

$$\begin{array}{l} 123151321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^6\overset{3,3}{\otimes}\mathbf{50} \\ 31321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^6\overset{3,3}{\otimes}\mathbf{50} \\ 1512321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^6\overset{3,3}{\otimes}\mathbf{50} \end{array}$$

$$\begin{array}{l} 123151321\mathbf{e}_7(\otimes\mathbf{e}_7)^5\otimes\mathbf{so} \\ 31321\mathbf{e}_7(\otimes\mathbf{e}_7)^5\otimes\mathbf{so} \\ 1512321\mathbf{e}_7(\otimes\mathbf{e}_7)^5\otimes\mathbf{so} \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\otimes\epsilon_7)^{15}\otimes\mathfrak{so}^{3,3} \\ 31321\epsilon_7(\otimes\epsilon_7)^{15}\otimes\mathfrak{so}^{3,3} \\ 1512321\epsilon_7(\otimes\epsilon_7)^{15}\otimes\mathfrak{so}^{3,3} \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\otimes\epsilon_7)^{14}\otimes\mathfrak{so}^{3,3} \\ 31321\epsilon_7(\otimes\epsilon_7)^{14}\otimes\mathfrak{so}^{3,3} \\ 1512321\epsilon_7(\otimes\epsilon_7)^{14}\otimes\mathfrak{so}^{3,3} \end{array}$$

$$123151321e_7(\overset{3,3}{\otimes} e_7)^{13} \overset{3,3}{\otimes} 50$$

$$\begin{array}{l} 31321e_7(\otimes e_7)^{13} \otimes so \\ 1512321e_7(\otimes e_7)^{13} \otimes so \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\otimes \epsilon_7)^{12} \otimes \mathfrak{so}^{3,3} \\ 31321\epsilon_7(\otimes \epsilon_7)^{12} \otimes \mathfrak{so}^{3,3} \\ 1512321\epsilon_7(\otimes \epsilon_7)^{12} \otimes \mathfrak{so}^{3,3} \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\otimes \epsilon_7)^{11} \otimes \mathfrak{so} \\ 31321\epsilon_7(\otimes \epsilon_7)^{11} \otimes \mathfrak{so} \\ 1512321\epsilon_7(\otimes \epsilon_7)^{11} \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{l} 123151321\epsilon_7(\otimes \epsilon_7)^{10} \otimes \mathfrak{so}^{3,3} \\ 31321\epsilon_7(\otimes \epsilon_7)^{10} \otimes \mathfrak{so}^{3,3} \\ 1512321\epsilon_7(\otimes \epsilon_7)^{10} \otimes \mathfrak{so}^{3,3} \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes^{5,5}\epsilon_8)^9\otimes^{5,5}\epsilon_6 \\ 15131513221\epsilon_8(\otimes^{5,5}\epsilon_8)^9\otimes^{5,5}\epsilon_6 \end{array}$$

$$\begin{array}{l} 321513221\epsilon_8(\otimes \epsilon_8)^9 \otimes \mathfrak{so} \\ 15131513221\epsilon_8(\otimes \epsilon_8)^9 \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{l} 321513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,5}{\otimes}\mathfrak{e}_6 \\ 15131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,5}{\otimes}\mathfrak{e}_6 \end{array}$$

$$321513221\mathfrak{e}_8(\bigotimes\mathfrak{e}_8)^8\bigotimes\mathfrak{so}_{5,3}$$

$$\begin{array}{lll} 2313221\mathfrak{c}_8(\overset{5,5}{\otimes}\mathfrak{c}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so}, & 222221\mathfrak{c}_8(\overset{5,5}{\otimes}\mathfrak{c}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so}, & 321513221\mathfrak{c}_8(\overset{5,5}{\otimes}\mathfrak{c}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so} \\ 51231513221\mathfrak{c}_8(\overset{5,5}{\otimes}\mathfrak{c}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so}, & \overset{1}{3}1513221\mathfrak{c}_8(\overset{5,5}{\otimes}\mathfrak{c}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so}, & 15131513221\mathfrak{c}_8(\overset{5,5}{\otimes}\mathfrak{c}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so} \end{array}$$

$$\begin{array}{lll} 2313221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_6, & 222221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_6, & 321513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_6 \\ 51231513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_6, & \overset{1}{3}1513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_6, & 15131513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,5}{\otimes}\mathfrak{e}_6 \end{array}$$

$$\begin{array}{lll} 2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 321513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so} \\ 51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 31513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}, & 15131513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{lll} 2313221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 321513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6 \\ 51231513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 31513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6, & 15131513221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6 \end{array}$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes \epsilon_8)^{11} \otimes \mathfrak{so}, & 222221\epsilon_8(\otimes \epsilon_8)^{11} \otimes \mathfrak{so}, & 321513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes \mathfrak{so} \\ 51231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes \mathfrak{so}, & 31513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes \mathfrak{so}, & 15131513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes \mathfrak{so} \end{array}$$

$$\begin{array}{lll} 2313221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^{10}\overset{5,5}{\otimes}\epsilon_6, & 222221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^{10}\overset{5,5}{\otimes}\epsilon_6, & 321513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^{10}\overset{5,5}{\otimes}\epsilon_6 \\ 51231513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^{10}\overset{5,5}{\otimes}\epsilon_6, & \overset{1}{3}1513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^{10}\overset{5,5}{\otimes}\epsilon_6, & 15131513221\epsilon_8(\overset{5,5}{\otimes}\epsilon_8)^{10}\overset{5,5}{\otimes}\epsilon_6 \end{array}$$

$$\begin{array}{lll} 2313221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 222221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 321513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so} \\ 51231513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 31513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so}, & 15131513221\epsilon_8(\otimes \epsilon_8)^{5,5} \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{array}{lll} 31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6^{5,5}, & 131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so}^{5,3}, & 21\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so}^{5,3} \\ 1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so}^{5,3} & & \end{array}$$

$$\begin{array}{lll}
31\mathfrak{so}^{2,4} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_6, & 131\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so}, & 21\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so} \\
1\mathfrak{e}_6^{4,5} \otimes \mathfrak{e}_8^{(5,5)} \otimes \mathfrak{e}_8^{5,3} \otimes \mathfrak{so} & &
\end{array}$$

$$3150 \otimes_{\mathfrak{e}_8}^{2,4} (\otimes_{\mathfrak{e}_8}^{5,5})^7 \otimes_{\mathfrak{e}_6}^{5,5}, \quad 131\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} (\otimes_{\mathfrak{e}_8}^{5,5})^7 \otimes_{\mathfrak{so}}^{5,3}, \quad 21\mathfrak{e}_6 \otimes_{\mathfrak{e}_8}^{3,5} (\otimes_{\mathfrak{e}_8}^{5,5})^7 \otimes_{\mathfrak{so}}^{5,3}$$

$$221e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^9 \otimes^{5,3} 50$$

$$221\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^8 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,3} 50$$

$$221\mathfrak{e}_7 \overset{4,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_6$$

$$221e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^7 \otimes^{5,3} 50$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^6 \otimes^{5,5} e_6$$

$$221e_7 \otimes^{4,5} e_8(\otimes^{5,5} e_8)^6 \otimes^{5,3} \mathfrak{so}$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,5} e_6$$

$$221e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^5 \otimes^{5,3} 50$$

$$221e_7 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^4 \overset{5,5}{\otimes} e_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes} \mathbf{e}_8)^5 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^4 \otimes^{5,3} so,$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,3} \mathfrak{so},$$

$$221e_7 \otimes^{4,5} e_8 \left(\otimes^{5,5} e_8 \right)^4 \otimes^{5,3} so$$

$$21e_7(\overset{5,5}{\otimes} e_8)^5 \overset{5,3}{\otimes} so$$

$$1321\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{13} \otimes^{5,5} \mathbf{e}_6,$$

$$12321e_7 \overset{4,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{13} \overset{5,5}{\otimes} e_6,$$

$$221e_7 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^{13} \overset{5,5}{\otimes} e_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{14}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,3} \mathfrak{so},$$

$$221e_7 \overset{4,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{13} \overset{5,3}{\otimes} so$$

$$21e_7(\otimes^{5,5} e_8)^{14} \otimes^{5,3} so$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,5} e_6,$$

$$12321e_7 \overset{4,5}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^{12} \overset{5,5}{\otimes} e_6,$$

$$221e_7 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^{12} \overset{5,5}{\otimes} e_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{13}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,3} \mathfrak{so},$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,3} so$$

$$21e_7(\overset{5,5}{\otimes} e_8)^{13} \overset{5,3}{\otimes} 50$$

$$1321\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{11} \otimes^{5,5} \mathbf{e}_6,$$

$$12321e_7^{\begin{smallmatrix} 4,5 \\ \otimes \end{smallmatrix}} e_8^{\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}} (\otimes e_8)^{11} \otimes e_6^{\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix}},$$

$$221\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{12}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{11} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{11} \otimes^{5,3} \mathfrak{so},$$

$$221e_7^{4,5} \otimes e_8^{(5,5)} \otimes e_8^{11} \otimes so^{5,3}$$

$$21e_7(\overset{5,5}{\otimes} e_8)^{12} \overset{5,3}{\otimes} 50$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,5} e_6,$$

$$12321\mathbf{e}_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{10} \otimes^{5,5} \mathbf{e}_6,$$

$$221\mathbf{e}_7 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{10} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$21\mathbf{e}_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{11}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$1321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,3} \mathfrak{so},$$

$$12321e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,3} \mathfrak{so},$$

$$221e_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,3} so$$

$$21e_7(\overset{5,5}{\otimes} e_8)^{11} \overset{5,3}{\otimes} 50$$

$$3221\mathbf{e}_7\left(\bigotimes^{3,3}\mathbf{e}_7\right)^5\bigotimes^{4,5}\mathbf{e}_8,$$

$$512321e_7(\overset{3,3}{\otimes} e_7)^5 \overset{4,5}{\otimes} e_8,$$

$$151321\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^5\overset{4,5}{\otimes}\mathbf{e}_8$$

$$1321e'_7(\overset{3,3}{\otimes} e_7)^9 \overset{3,3}{\otimes} \mathfrak{so},$$

$$12321\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^9\overset{3,3}{\otimes}\mathbf{so},$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^9 \overset{3,3}{\otimes} 50$$

$$1321\mathfrak{e}'_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^8\overset{3,3}{\otimes}\mathfrak{so},$$

$$12321e'_7(\otimes^{3,3} e_7)^8 \otimes^{3,3} \mathfrak{so},$$

$$221e'_7(\otimes^{3,3} e_7)^8 \otimes^{3,3} \mathfrak{so}$$

$$1321\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^7\overset{3,3}{\otimes}\mathfrak{so},$$

$$12321e'_7(\otimes^{3,3} e_7)^7 \otimes^{3,3} \mathfrak{so},$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^7 \overset{3,3}{\otimes} 50$$

$$1321\mathbf{e}_7\left(\overset{3,3}{\otimes}\mathbf{e}_7\right)^6\overset{4,5}{\otimes}\mathbf{e}_8,$$

$$12321e_7(\overset{3,3}{\otimes} e_7)^6 \overset{4,5}{\otimes} e_8,$$

$$221\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^6\overset{4,5}{\otimes}\mathbf{e}_8$$

$$1321e'_7(\overset{3,3}{\otimes} e_7)^6 \overset{3,3}{\otimes} \mathfrak{so},$$

$$12321e'_7(\overset{3,3}{\otimes} e_7)^6 \overset{3,3}{\otimes} \mathfrak{so},$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^6 \overset{3,3}{\otimes} \mathfrak{so}$$

$$1321e'_7(\overset{3,3}{\otimes} e_7)^5 \overset{3,3}{\otimes} \mathfrak{so},$$

$$12321\mathbf{e}'_7\left(\bigotimes^{3,3}\mathbf{e}_7\right)^5\bigotimes^{3,3}\mathfrak{so},$$

$$221e'_7(\otimes^{3,3} e_7)^5 \otimes^{3,3} \mathfrak{so}$$

$$1321\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{15}\overset{3,3}{\otimes}\mathbf{so},$$

$$12321e'_7(\otimes^{3,3} e_7)^{15} \otimes^{3,3} \mathfrak{so},$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{15}\overset{3,3}{\otimes}\mathbf{so}$$

$$1321\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{14}\overset{3,3}{\otimes}\mathbf{so},$$

$$12321e'_7(\overset{3,3}{\otimes} e_7)^{14} \overset{3,3}{\otimes} \mathfrak{so},$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{14}\overset{3,3}{\otimes}\mathbf{so}$$

$$1321e'_7(\overset{3,3}{\otimes} e_7)^{13} \overset{3,3}{\otimes} \mathfrak{so},$$

$$12321e'_7 \binom{3,3}{\otimes e_7}^{13} \otimes \binom{3,3}{\otimes 50},$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{13}\overset{3,3}{\otimes}\mathbf{so}$$

$$1321e'_7(\overset{(3,3)}{\otimes} e_7)^{12} \overset{(3,3)}{\otimes} \mathfrak{so},$$

$$12321e'_7(\overset{(3,3)}{\otimes} e_7)^{12} \overset{3,3}{\otimes} 50,$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^{12} \overset{3,3}{\otimes} \mathfrak{so}$$

$$1321e'_7(\overset{(3,3)}{\otimes} e_7)^{11} \overset{3,3}{\otimes} \mathfrak{so},$$

$$12321e'_7(\overset{(3,3)}{\otimes} e_7)^{11} \overset{3,3}{\otimes} 50,$$

$$221\mathbf{e}'_7(\overset{3,3}{\otimes}\mathbf{e}_7)^{11}\overset{3,3}{\otimes}\mathbf{so}$$

$$1321e'_7(\overset{3,3}{\otimes} e_7)^{10} \overset{3,3}{\otimes} \mathfrak{so},$$

$$12321e'_7(\otimes e_7)^{10} \otimes^{3,3} 50,$$

$$221e'_7(\overset{3,3}{\otimes} e_7)^{10} \overset{3,3}{\otimes} \mathfrak{so}$$

$$131\mathbf{e}_6 \overset{3,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \mathbf{e}_6,$$

$$21\mathfrak{e}_6 \overset{3,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^9 \overset{5,5}{\otimes} \mathfrak{e}_6,$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \mathbf{e}_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,3} 50,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^9 \overset{5,3}{\otimes} \mathfrak{so}$$

$$21\mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^7 \otimes^{5,5} \mathbf{e}_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^8 \overset{5,5}{\otimes} e_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^8 \overset{5,3}{\otimes} \mathfrak{so}$$

$$21\mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^6 \otimes^{5,5} \mathbf{e}_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^7 \overset{5,5}{\otimes} e_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^6 \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^7 \overset{5,3}{\otimes} so$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,5} e_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^6 \overset{5,5}{\otimes} e_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,3} 50,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^6 \overset{5,3}{\otimes} \mathfrak{so}$$

$$21\mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^4 \otimes^{5,5} \mathbf{e}_6,$$

$$1\mathbf{e}'_7(\overset{5,5}{\otimes}\mathbf{e}_8)^5\overset{5,5}{\otimes}\mathbf{e}_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^5 \overset{5,3}{\otimes} so$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{14} \otimes^{5,3} \mathfrak{so},$$

$$1e'_7 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^{15} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} 50$$

$$21e'_7 \overset{4,5}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^{13} \overset{5,5}{\otimes} e_6,$$

$$1\mathbf{e}'_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{14}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13} \otimes^{5,3} so,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{14} \overset{5,3}{\otimes} so$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,5} e_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{13} \overset{5,5}{\otimes} e_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12} \otimes^{5,3} \mathfrak{so},$$

$$1e'_7 \left(\begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} e_8 \right)^{13} \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} 50$$

$$21\mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{11} \otimes^{5,5} \mathbf{e}_6,$$

$$1\mathbf{e}'_7(\overset{5,5}{\otimes}\mathbf{e}_8)^{12}\overset{5,5}{\otimes}\mathbf{e}_6$$

$$21\mathfrak{e}'_7 \otimes^{4,5} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{11} \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{12} \overset{5,3}{\otimes} 50$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,5} e_6,$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{11} \overset{5,5}{\otimes} e_6$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10} \otimes^{5,3} \mathfrak{so},$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^{11} \overset{5,3}{\otimes} 50$$

$$21\mathfrak{e}_6^{3,5} \otimes \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{e}_6^{5,5},$$

$$1\mathbf{e}_6 \overset{4,5}{\otimes} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{14} \overset{5,5}{\otimes} \mathbf{e}_6$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^9\overset{3,3}{\otimes}\mathfrak{so},$$

$$21e_6(\overset{3,3}{\otimes} e_7)^9 \overset{3,3}{\otimes} 50$$

$$131e_6(\otimes^{3,3} e_7)^8 \otimes^{3,3} \mathfrak{so},$$

$$21e_6(\overset{3,3}{\otimes} e_7)^8 \overset{3,3}{\otimes} 50$$

$$131e_6(\overset{3,3}{\otimes} e_7)^7 \overset{3,3}{\otimes} 50,$$

$$21e_6(\overset{3,3}{\otimes} e_7)^7 \overset{3,3}{\otimes} 50$$

$$131e_6(\overset{3,3}{\otimes} e_7)^6 \overset{3,3}{\otimes} so,$$

$$21e_6(\otimes^{3,3} e_7)^6 \otimes^{3,3} so$$

$$131e_6(\overset{3,3}{\otimes} e_7)^5 \overset{3,3}{\otimes} so,$$

$$21e_6(\overset{3,3}{\otimes} e_7)^5 \overset{3,3}{\otimes} 50$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{15}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{15}\overset{3,3}{\otimes}\mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{14}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{14}\overset{3,3}{\otimes}\mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{13}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21e_6(\overset{3,3}{\otimes} e_7)^{13} \overset{3,3}{\otimes} \mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{12}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{12}\overset{3,3}{\otimes}\mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{11}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{11}\overset{3,3}{\otimes}\mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{10}\overset{3,3}{\otimes}\mathfrak{so},$$

$$21\mathfrak{e}_6(\overset{3,3}{\otimes}\mathfrak{e}_7)^{10}\overset{3,3}{\otimes}\mathfrak{so}$$

$$131\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^6\overset{3,5}{\otimes}\mathfrak{e}_8,$$

$$21\mathfrak{e}_6(\overset{2,2}{\otimes}\mathfrak{e}_6)^6\overset{3,5}{\otimes}\mathfrak{e}_8$$

$$321\mathfrak{e}'_7(\overset{3,3}{\otimes}\mathfrak{e}_7)^5\overset{4,5}{\otimes}\mathfrak{e}_8$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^9\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^7\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^6\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^5\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^4\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{14}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{13}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so}\overset{2,4}{\otimes}\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{11}\overset{5,3}{\otimes}\mathfrak{so}$$

$$31\mathfrak{so} \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix}^{5,5} 10 \otimes \mathfrak{so} \begin{smallmatrix} 5,3 \end{smallmatrix}$$

$$31\mathfrak{so} \begin{smallmatrix} 1,1 \\ (\otimes \mathfrak{so}) \end{smallmatrix}^6 \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \end{smallmatrix}$$

$$31\mathfrak{e}_6 \begin{smallmatrix} 3,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^5 \otimes \mathfrak{e}_8 \begin{smallmatrix} 4,5 \end{smallmatrix}$$

$$231\mathfrak{so} \begin{smallmatrix} 1,1 \\ (\otimes \mathfrak{so}) \end{smallmatrix}^5 \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \end{smallmatrix}$$

$$21\mathfrak{e}_7' \begin{smallmatrix} 3,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^6 \otimes \mathfrak{e}_8 \begin{smallmatrix} 4,5 \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \begin{smallmatrix} 2,4 \\ (\otimes \mathfrak{e}_8) \end{smallmatrix}^3 \otimes \mathfrak{e}_7 \begin{smallmatrix} 5,4 \\ \otimes \mathfrak{e}_7 \end{smallmatrix} \otimes \mathfrak{e}_7 \begin{smallmatrix} 3,3 \end{smallmatrix}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^9$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^8$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^7$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^5$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^4$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^{14}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7 \begin{smallmatrix} 2,3 \\ (\otimes \mathfrak{e}_7) \end{smallmatrix}^3 \otimes \mathfrak{so} \begin{smallmatrix} 3,3 \end{smallmatrix}^{13}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^9 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{so}^{1,1} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{e}_6^{5,5}$$

2.8.3 Sg(n)S(3795)

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\begin{array}{c} 3131\epsilon_6(\otimes^{2,2}\epsilon_6)^{14}\otimes^{2,2}\epsilon_613151 \\ 15131\epsilon_6(\otimes^{2,2}\epsilon_6)^{14}\otimes^{2,2}\epsilon_613151 \end{array}$$

$$\begin{aligned}
& 1321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_613151321 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_61231 \\
& 1321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131\overset{1}{5}1 \\
& 12321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131513221 \\
& 12321\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_61313 \\
& 12315131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_613151321 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_613151321 \\
& 12315131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131\overset{1}{5}1 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131513221 \\
& 122315131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131\overset{1}{5}1 \\
& 3131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_61315131 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6122 \\
& 221\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131\overset{1}{5}1 \\
& 3131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131\overset{1}{5}1 \\
& \overset{1}{1}5131\mathbf{e}_6(\overset{2,2}{\otimes}\mathbf{e}_6)^{13}\overset{2,2}{\otimes}\mathbf{e}_6131\overset{1}{5}1
\end{aligned}$$

$$\begin{aligned}
& 1321\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_613151321 \\
& 221\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_61231 \\
& 1321\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_61315^11 \\
& 12321\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_6131513221 \\
& 12321\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_61313 \\
& 12315131\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_613151321 \\
& 221\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_613151321 \\
& 12315131\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_61315^11 \\
& 221\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_6131513221 \\
& 122315131\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_61315^11 \\
& 3131\epsilon_6(\overset{2,2}{\otimes}\epsilon_6)^{12}\overset{2,2}{\otimes}\epsilon_61315131
\end{aligned}$$

[illegible]

[illegible]

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$$\begin{aligned} & 221\mathbf{e}_7 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^7 \otimes \mathbf{e}_8^{5,5} 122315123 \\ & 221\mathbf{e}_7 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^7 \otimes \mathbf{e}_8^{5,5} 12231513151 \\ & 21\mathbf{e}_7 (\otimes \mathbf{e}_8^{5,5})^8 \otimes \mathbf{e}_8^{5,5} 122315123 \\ & 21\mathbf{e}_7 (\otimes \mathbf{e}_8^{5,5})^8 \otimes \mathbf{e}_8^{5,5} 12231513151 \end{aligned}$$

$$\begin{aligned}
& 1321\epsilon_7^{\overline{4,5}} \otimes \epsilon_8(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^6 \otimes \epsilon_8 122315123 \\
& 1321\epsilon_7^{\overline{4,5}} \otimes \epsilon_8(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^6 \otimes \epsilon_8 12231513151 \\
& 12321\epsilon_7^{\overline{4,5}} \otimes \epsilon_8(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^6 \otimes \epsilon_8 122315123 \\
& 12321\epsilon_7^{\overline{4,5}} \otimes \epsilon_8(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^6 \otimes \epsilon_8 12231513151 \\
& 221\epsilon_7^{\overline{4,5}} \otimes \epsilon_8(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^6 \otimes \epsilon_8 122315123 \\
& 221\epsilon_7^{\overline{4,5}} \otimes \epsilon_8(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^6 \otimes \epsilon_8 12231513151 \\
& 21\epsilon_7^{\overline{5,5}}(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^7 \otimes \epsilon_8 122315123 \\
& 21\epsilon_7^{\overline{5,5}}(\overline{\otimes} \epsilon_8)^{\overline{5,5}}{}^7 \otimes \epsilon_8 12231513151
\end{aligned}$$

$$\begin{aligned}
& 1321\epsilon_7^{\otimes 4,5} \otimes \epsilon_8^{\otimes 5,5} \otimes \epsilon_8^{\otimes 5,5} 122315123 \\
& 1321\epsilon_7^{\otimes 4,5} \otimes \epsilon_8^{\otimes 5,5} \otimes \epsilon_8^{\otimes 5,5} 12231513151 \\
& 12321\epsilon_7^{\otimes 4,5} \otimes \epsilon_8^{\otimes 5,5} \otimes \epsilon_8^{\otimes 5,5} 122315123 \\
& 12321\epsilon_7^{\otimes 4,5} \otimes \epsilon_8^{\otimes 5,5} \otimes \epsilon_8^{\otimes 5,5} 12231513151 \\
& 221\epsilon_7^{\otimes 4,5} \otimes \epsilon_8^{\otimes 5,5} \otimes \epsilon_8^{\otimes 5,5} 122315123 \\
& 221\epsilon_7^{\otimes 4,5} \otimes \epsilon_8^{\otimes 5,5} \otimes \epsilon_8^{\otimes 5,5} 12231513151 \\
& 21\epsilon_7^{\otimes 5,5} \otimes \epsilon_8^{\otimes 6,5} \otimes \epsilon_8^{\otimes 5,5} 122315123 \\
& 21\epsilon_7^{\otimes 5,5} \otimes \epsilon_8^{\otimes 6,5} \otimes \epsilon_8^{\otimes 5,5} 12231513151
\end{aligned}$$

$$\begin{aligned}
& 1321\mathbf{e}_7 \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 122315123 \\
& 1321\mathbf{e}_7 \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513151 \\
& 12321\mathbf{e}_7 \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 122315123 \\
& 12321\mathbf{e}_7 \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513151 \\
& 221\mathbf{e}_7 \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 122315123 \\
& 221\mathbf{e}_7 \otimes \mathbf{e}_8(\otimes \mathbf{e}_8)^4 \otimes \mathbf{e}_8 12231513151 \\
& 21\mathbf{e}_7(\otimes \mathbf{e}_8)^5 \otimes \mathbf{e}_8 122315123
\end{aligned}$$

[illegible]

[illegible]

[illegible]

$$\begin{array}{l} 150 \left(\otimes \epsilon_6 \right)^{12} \otimes^{2,2} \epsilon_6 122 \\ 150 \left(\otimes \epsilon_6 \right)^{12} \otimes^{2,2} \epsilon_6 13151 \end{array}$$

$$\begin{array}{l} 150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 13151321 \\ 150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 122 \\ 150(\otimes \mathfrak{e}_6)^{11} \otimes \mathfrak{e}_6 13151 \end{array}$$

$$\begin{aligned} &150(\otimes \epsilon_6)^{10} \otimes \epsilon_6 13151321 \\ &150(\otimes \epsilon_6)^{10} \otimes \epsilon_6 122 \\ &150(\otimes \epsilon_6)^{10} \otimes \epsilon_6 13151 \end{aligned}$$

$$\begin{array}{l} 21\epsilon_7' \otimes^{4,5} \epsilon_8 (\otimes^{5,5} \epsilon_8)^9 \otimes^{5,4} \epsilon_7 122 \\ 1\epsilon_7' (\otimes^{5,5} \epsilon_8)^{10} \otimes^{5,4} \epsilon_7 122 \end{array}$$

$$\begin{array}{l} 21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,4} e_7 122 \\ 1e'_7 (\otimes^{5,5} e_8)^9 \otimes^{5,4} e_7 122 \end{array}$$

$$\begin{array}{l} 21\mathbf{e}'_7 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^7 \otimes \mathbf{e}_7^{5,4} 122 \\ 1\mathbf{e}'_7 (\otimes \mathbf{e}_8^{5,5})^8 \otimes \mathbf{e}_7^{5,4} 122 \end{array}$$

$$\begin{array}{l} 21\epsilon'_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_7^{5,4} 122 \\ 1\epsilon'_7 (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_7^{5,4} 122 \end{array}$$

$$\begin{array}{l} 21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^5 \otimes^{5,4} e_7 122 \\ 1e'_7 (\otimes^{5,5} e_8)^6 \otimes^{5,4} e_7 122 \end{array}$$

$$1e'_7(\overset{5,5}{\otimes} e_8)^6 \overset{5,5}{\otimes} e_7 12$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^4 \otimes^{5,5} e_8$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{13}$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{12}$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{11}$$

$$21e'_7 \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^{10}$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^9 \otimes^{5,5}$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^8 \otimes^{5,5}$$

[illegible]

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{14} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{13} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{12} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{11} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \begin{smallmatrix} 2,3 \\ \otimes \end{smallmatrix} e_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{10} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e'_7 122$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^9 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^8 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^7 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^6 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 (\overset{5,5}{\otimes} e_8)^5 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^4 \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8 \left(\overset{5,5}{\otimes} e_8 \right)^3 \overset{5,4}{\otimes} e_6 1$$

$$1_{\mathfrak{so}} \overset{1,1}{\otimes} \mathfrak{so} \overset{2,4}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^{13} \overset{5,4}{\otimes} \mathfrak{e}_6 1$$

$$1_{\mathfrak{so}} \otimes^{1,1} \mathfrak{so} \otimes^{2,4} \mathfrak{e}_8 (\otimes^{5,5} \mathfrak{e}_8)^{12} \otimes^{5,4} \mathfrak{e}_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^{11} \overset{5,3}{\otimes} e_6 131, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^{11} \overset{5,3}{\otimes} e_6 12, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} e_8(\overset{5,5}{\otimes} e_8)^{11} \overset{5,4}{\otimes} e_6 1$$

$$150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8(\overset{5,5}{\otimes} 8)^{10} \overset{5,3}{\otimes} 6131, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8(\overset{5,5}{\otimes} 8)^{10} \overset{5,3}{\otimes} 612, \quad 150 \overset{1,1}{\otimes} 50 \overset{2,4}{\otimes} 8(\overset{5,5}{\otimes} 8)^{10} \overset{5,4}{\otimes} 61$$

$$1321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^5\overset{4,5}{\otimes}\epsilon_81, \quad 12321\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^5\overset{4,5}{\otimes}\epsilon_81, \quad 221\epsilon_7(\overset{3,3}{\otimes}\epsilon_7)^5\overset{4,5}{\otimes}\epsilon_81$$

$$131\epsilon_6^{(3,3)}(\otimes \epsilon_7)^9 \otimes \epsilon_6 131, \quad 21\epsilon_6^{(3,3)}(\otimes \epsilon_7)^9 \otimes \epsilon_6 131, \quad 21\epsilon_6^{(3,3)}(\otimes \epsilon_7)^9 \otimes \epsilon_6 12$$

$$131\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^8\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^8\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^8\overset{3,3}{\otimes}\mathbf{e}_612$$

$$131\epsilon_6(\otimes \epsilon_7)^7 \otimes \epsilon_6 131, \quad 21\epsilon_6(\otimes \epsilon_7)^7 \otimes \epsilon_6 131, \quad 21\epsilon_6(\otimes \epsilon_7)^7 \otimes \epsilon_6 12$$

$$131\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^6 \otimes \mathbf{e}_6 131, \quad 21\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^6 \otimes \mathbf{e}_6 131, \quad 21\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^6 \otimes \mathbf{e}_6 12$$

$$131\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6 131, \quad 21\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6 131, \quad 21\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6 12$$

$$131\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{15} \otimes \mathbf{e}_6 131, \quad 21\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{15} \otimes \mathbf{e}_6 131, \quad 21\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{15} \otimes \mathbf{e}_6 12$$

$$131\mathbf{e}_6\overset{(3,3)}{(\otimes\mathbf{e}_7)}^{14}\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6\overset{(3,3)}{(\otimes\mathbf{e}_7)}^{14}\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6\overset{(3,3)}{(\otimes\mathbf{e}_7)}^{14}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$131\epsilon_6^{(3,3)}(\otimes \epsilon_7)^{13} \otimes \epsilon_6 131, \quad 21\epsilon_6^{(3,3)}(\otimes \epsilon_7)^{13} \otimes \epsilon_6 131, \quad 21\epsilon_6^{(3,3)}(\otimes \epsilon_7)^{13} \otimes \epsilon_6 12$$

$$131\mathbf{e}_6\overset{(3,3)}{(\otimes\mathbf{e}_7)}^{12}\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6\overset{(3,3)}{(\otimes\mathbf{e}_7)}^{12}\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6\overset{(3,3)}{(\otimes\mathbf{e}_7)}^{12}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$131\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^{11}\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^{11}\overset{3,3}{\otimes}\mathbf{e}_6131, \quad 21\mathbf{e}_6(\overset{3,3}{\otimes}\mathbf{e}_7)^{11}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$131\epsilon_6^{(3,3)}(\otimes \epsilon_7)^{10} \otimes \epsilon_6 131, \quad 21\epsilon_6^{(3,3)}(\otimes \epsilon_7)^{10} \otimes \epsilon_6 131, \quad 21\epsilon_6^{(3,3)}(\otimes \epsilon_7)^{10} \otimes \epsilon_6 12$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}'_7 12,$$

$$1\mathfrak{so} \otimes \mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}'_7 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^5 \otimes \mathfrak{e}_8 12,$$

$$31\mathfrak{so}(\otimes \mathfrak{so})^5 \otimes \mathfrak{e}_8 1$$

$$131\mathfrak{e}_6(\otimes \mathfrak{e}_6)^5 \otimes \mathfrak{e}_8 1,$$

$$21\mathfrak{e}_6(\otimes \mathfrak{e}_6)^5 \otimes \mathfrak{e}_8 1$$

$$222221\mathfrak{e}_8(\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{e}_8 122222$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{e}_8 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^9 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^8 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^7 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^6 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7(\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^4 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{14} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{13} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_7^{2,3} (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^9 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^8 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^7 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^6 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^5 \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^{15} \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^{14} \otimes \mathfrak{so}^{2,2} 1$$

$$1\mathfrak{so} (\otimes \mathfrak{e}_6^{2,2})^{13} \otimes \mathfrak{so}^{2,2} 1$$

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$$\begin{aligned} & 131513221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^9 \otimes \epsilon_6, \\ & 313221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^9 \otimes \epsilon_6, \\ & 1513221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^9 \otimes \epsilon_6, \\ & 221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^9 \otimes \epsilon_6 \end{aligned}$$

$$\begin{aligned} &1231513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6, \\ &22221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6, \\ &21^{13221} \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}_6, \end{aligned}$$

$$\begin{array}{l} 12231513221\mathbf{e}_8(\otimes\mathbf{e}_8)^9\otimes\mathbf{e}_6^{5,5} \\ 21513221\mathbf{e}_8(\otimes\mathbf{e}_8)^9\otimes\mathbf{e}_6^{5,5} \\ 2221\mathbf{e}_8^{21}(\otimes\mathbf{e}_8)^9\otimes\mathbf{e}_6^{5,5} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{c}_8^{(5,5)} (\otimes \mathbf{c}_8)^9 \otimes \mathbf{50}, \\ &313221 \mathbf{c}_8^{(5,5)} (\otimes \mathbf{c}_8)^9 \otimes \mathbf{50}, \\ &1513221 \mathbf{c}_8^{(5,5)} (\otimes \mathbf{c}_8)^9 \otimes \mathbf{50}, \\ &221 \mathbf{c}_8^{(5,5)} (\otimes \mathbf{c}_8)^9 \otimes \mathbf{50} \end{aligned}$$

$$\begin{aligned} &1231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^9 \otimes \mathbf{so}, \\ &22221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^9 \otimes \mathbf{so}, \\ &21^{13}221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^9 \otimes \mathbf{so}, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \\ 21513221 \mathfrak{e}_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \\ 2221 \mathfrak{e}_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{aligned} &131513221^1 \mathfrak{e}_8^{\left(\otimes \mathfrak{e}_8\right)^{5,5}} \otimes \mathfrak{e}_6^{5,5}, \\ &313221^1 \mathfrak{e}_8^{\left(\otimes \mathfrak{e}_8\right)^{5,5}} \otimes \mathfrak{e}_6^{5,5}, \\ &1513221^1 \mathfrak{e}_8^{\left(\otimes \mathfrak{e}_8\right)^{5,5}} \otimes \mathfrak{e}_6^{5,5}, \\ &221^1 \mathfrak{e}_8^{\left(\otimes \mathfrak{e}_8\right)^{5,5}} \otimes \mathfrak{e}_6^{5,5} \end{aligned}$$

$$\begin{aligned} &1231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}_6, \\ &22221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}_6, \\ &21^{13221} \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{e}_6, \end{aligned}$$

$$\begin{array}{l} 12231513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,5}{\otimes}\mathfrak{e}_6 \\ 21513221\mathfrak{e}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,5}{\otimes}\mathfrak{e}_6 \\ 2221\mathfrak{e}_8^{21}(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,5}{\otimes}\mathfrak{e}_6 \end{array}$$

$$\begin{aligned} &131513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \mathbf{c}_8^{5,3} \mathbf{so}, \\ &313221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \mathbf{c}_8^{5,3} \mathbf{so}, \\ &1513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \mathbf{c}_8^{5,3} \mathbf{so}, \\ &221 \mathbf{c}_8^{221} (\otimes \mathbf{c}_8)^{5,5} \mathbf{c}_8^{5,3} \mathbf{so} \end{aligned}$$

$$\begin{aligned} &1231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{so}, \\ &22221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{so}, \\ &21^{13221} \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^8 \otimes \mathbf{so}, \end{aligned}$$

$$\begin{array}{l} 12231513221 \epsilon_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \\ 21513221 \epsilon_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \\ 2221^{21} \epsilon_8^{(5,5)} \otimes^{5,3} \mathfrak{so} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} 7^{5,5} \otimes \mathbf{c}_6, \\ &313221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} 7^{5,5} \otimes \mathbf{c}_6, \\ &1513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} 7^{5,5} \otimes \mathbf{c}_6, \\ &221 \mathbf{c}_8^{221} (\otimes \mathbf{c}_8)^{5,5} 7^{5,5} \otimes \mathbf{c}_6 \end{aligned}$$

$$\begin{aligned} &1231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6, \\ &22221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6, \\ &21^{13221} \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{e}_6, \end{aligned}$$

$$\begin{array}{l} 12231513221\mathfrak{e}_8^1(\otimes \mathfrak{e}_8)^{5,5}{}^7 \otimes \mathfrak{e}_6^{5,5} \\ 21513221\mathfrak{e}_8^1(\otimes \mathfrak{e}_8)^{5,5}{}^7 \otimes \mathfrak{e}_6^{5,5} \\ 2221\mathfrak{e}_8^{21}(\otimes \mathfrak{e}_8)^{5,5}{}^7 \otimes \mathfrak{e}_6^{5,5} \end{array}$$

$$\begin{aligned} &131513221\mathbf{e}_8(\otimes\mathbf{e}_8)^7\otimes\mathbf{50}, \\ &313221\mathbf{e}_8(\otimes\mathbf{e}_8)^7\otimes\mathbf{50}, \\ &1513221\mathbf{e}_8(\otimes\mathbf{e}_8)^7\otimes\mathbf{50}, \end{aligned}$$

$$\begin{aligned} &1231513221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{so}, \\ &22221 \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{so}, \\ &21^{13221} \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^7 \otimes \mathbf{so}, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathbf{e}_8^{(5,5)}(\otimes \mathbf{e}_8)^7 \otimes \mathfrak{so}^{5,3} \\ 21513221 \mathbf{e}_8^{(5,5)}(\otimes \mathbf{e}_8)^7 \otimes \mathfrak{so}^{5,3} \\ 2221 \mathbf{e}_8^{(5,5)}(\otimes \mathbf{e}_8)^7 \otimes \mathfrak{so}^{5,3} \end{array}$$

$$2221\mathfrak{e}_8^{21}(\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{so}^{5,3}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}_6^{5,5} \\ 21513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}_6^{5,5} \\ 2221 \mathfrak{e}_8^{21(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}_6^{5,5} \end{array}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3} \\ 21513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3} \\ 2221 \mathfrak{e}_8^{21(5,5)} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} 12231513221 \epsilon_8^1 (\otimes \epsilon_8)^{5,5} {}^{12} \otimes \epsilon_6^{5,5} \\ 21513221 \epsilon_8^1 (\otimes \epsilon_8)^{5,5} {}^{12} \otimes \epsilon_6^{5,5} \\ 2221 \epsilon_8^{21} (\otimes \epsilon_8)^{5,5} {}^{12} \otimes \epsilon_6^{5,5} \end{array}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \\ 21513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \\ 2221 \mathfrak{e}_8^{21(5,5)} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} 12231513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{(5,5)} \\ 21513221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{(5,5)} \\ 2221 \mathbf{e}_8^{(5,5)} \otimes \mathbf{e}_6^{(5,5)} \end{array}$$

$$12231513221\mathbf{e}_8(\bigotimes^{5,5}\mathbf{e}_8)^{11}\bigotimes^{5,3}\mathbf{so}$$

$$\begin{array}{l} 313221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}^{5,3}, \\ 1513221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}^{5,3}, \\ 221 \epsilon_8^{221} (\otimes \epsilon_8)^{5,5} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} 22221 \epsilon_8^{(1)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}_{5,3}, \\ 21^{13} 221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}_{5,3}, \end{array}$$

$$\begin{array}{l} 21513221 \epsilon_8^{(1)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}^{5,3} \\ 2221 \epsilon_8^{(21)} (\otimes \epsilon_8)^{11} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{e}_8^1 (\mathbf{e}_8^{5,5})^{10} \mathbf{e}_6^{5,5}, \\ &313221 \mathbf{e}_8^1 (\mathbf{e}_8^{5,5})^{10} \mathbf{e}_6^{5,5}, \\ &1513221 \mathbf{e}_8^1 (\mathbf{e}_8^{5,5})^{10} \mathbf{e}_6^{5,5}, \\ &221 \mathbf{e}_8^{221} (\mathbf{e}_8^{5,5})^{10} \mathbf{e}_6^{5,5} \end{aligned}$$

$$\begin{aligned} &1231513221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{10} \otimes \epsilon_6, \\ &22221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{10} \otimes \epsilon_6, \\ &21^{13} 221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{10} \otimes \epsilon_6, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6^{5,5} \\ 21513221 \mathfrak{e}_8^{(5,5)} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{e}_6^{5,5} \\ 2221 \mathfrak{e}_8^{21} (\otimes \mathfrak{e}_8)^{5,5} \otimes \mathfrak{e}_6^{5,5} \end{array}$$

$$\begin{aligned} &131513221 \mathbf{e}_8^{1,5,5} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{50}, \\ &313221 \mathbf{e}_8^{1,5,5} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{50}, \\ &1513221 \mathbf{e}_8^{1,5,5} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{50}, \\ &221 \mathbf{e}_8^{221,5,5} (\otimes \mathbf{e}_8)^{10} \otimes \mathbf{50} \end{aligned}$$

$$\begin{aligned} &1231513221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{10} \otimes^{5,3} \mathfrak{so}, \\ &22221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{10} \otimes^{5,3} \mathfrak{so}, \\ &21^{13} 221 \epsilon_8^{(5,5)} (\otimes \epsilon_8)^{10} \otimes^{5,3} \mathfrak{so}, \end{aligned}$$

$$\begin{array}{l} 12231513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so}^{5,3} \\ 21513221 \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so}^{5,3} \\ 2221 \mathfrak{e}_8^{21(5,5)} (\otimes \mathfrak{e}_8)^{10} \otimes \mathfrak{so}^{5,3} \end{array}$$

$$\begin{array}{l} 1321^1_7(\otimes \mathbf{e}_7)^9 \otimes^{3,3} \mathbf{so}, \\ 21^{21}_7(\otimes \mathbf{e}_7)^9 \otimes^{3,3} \mathbf{so} \end{array}$$

$$12321\mathfrak{e}_7^1(\otimes^{3,3}\mathfrak{e}_7)^9\otimes^{3,3}\mathfrak{so},$$

$$221e_7^1(\otimes^{3,3} e_7)^9 \otimes^{3,3} so$$

$$\begin{array}{l} 1321^1_7(\otimes e_7)^8 \otimes^{3,3} \mathfrak{so}, \\ 21^{21}_7(\otimes e_7)^8 \otimes^{3,3} \mathfrak{so} \end{array}$$

$$12321 e_7^1 (\otimes^{3,3} e_7)^8 \otimes^{3,3} \mathfrak{so},$$

$$221e_7^1(\otimes^{3,3} e_7)^8 \otimes^{3,3} so$$

$$\begin{array}{l} 1321^1_7(\otimes^{3,3} e_7)^7 \otimes^{3,3} \mathfrak{so}, \\ 21^{21}_7(\otimes^{3,3} e_7)^7 \otimes^{3,3} \mathfrak{so} \end{array}$$

$$12321e_7^1(\otimes_{e_7}^{3,3})^7 \otimes_{e_7}^{3,3} \mathfrak{so},$$

$$221e_7^1(\otimes e_7)^{3,3} \otimes so$$

$$\begin{array}{l} 1321\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^6 \otimes \mathbf{so}, \\ 21\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^6 \otimes \mathbf{so} \end{array}$$

$$12321 e_7^1 (\otimes e_7)^{3,3} (\otimes e_7)^{3,3} \otimes \mathfrak{so},$$

$$221e_7({}^{3,3}\otimes e_7)^6 {}^{3,3}\otimes so$$

$$\begin{array}{l} 1321e_7^{(3,3)} \otimes e_7^{(3,3)} \otimes \mathfrak{so}, \\ 21e_7^{(3,3)} \otimes e_7^{(3,3)} \otimes \mathfrak{so} \end{array}$$

$$12321\mathbf{e}_7^1(\otimes^{3,3}\mathbf{e}_7)^5\otimes^{3,3}\mathfrak{so},$$

$$221e_7^1(\otimes^{3,3} e_7)^5 \otimes^{3,3} so$$

$$1321e_7^1(\overset{3,3}{\otimes} e_7)^{15} \overset{3,3}{\otimes} \mathfrak{so},$$

$$12321 \mathbf{e}_7^1 (\otimes \mathbf{e}_7^{3,3})^{15} \otimes \mathbf{so}^{3,3},$$

$$221e_7(\overset{1}{\otimes} \overset{3,3}{e_7})^{15} \overset{3,3}{\otimes} so$$

$$21^{21}_{\mathfrak{e}_7}(\otimes^{3,3}_{\mathfrak{e}_7})^{15} \otimes^{3,3}_{\mathfrak{so}}$$

$$1321^1_7 \left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} e_7 \right)^{14} \begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix} \mathfrak{so},$$

$$12321\mathfrak{e}_7\left(\bigotimes^{3,3}\mathfrak{e}_7\right)^{14}\bigotimes^{3,3}\mathfrak{so},$$

$$221e_7(\overset{3,3}{\otimes} e_7)^{14} \overset{3,3}{\otimes} \mathfrak{so}$$

$$21^{21} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{14} \bigotimes \mathbf{so}$$

$$1321 \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{13} \bigotimes \mathfrak{so},$$

$$12321e_7^1(\otimes^{3,3}e_7)^{13}\otimes^{3,3}\mathfrak{so},$$

$$221e_7^1(\otimes^{3,3} e_7)^{13} \otimes^{3,3} \mathfrak{so}$$

$$21^{21} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{13} \bigotimes \mathbf{so}$$

$$1321 \mathbf{e}_7^1 \left(\bigotimes_{\mathbf{e}_7}^{3,3} \right)^{12} \bigotimes_{\mathbf{e}_7}^{3,3} \mathbf{so},$$

$$12321\mathfrak{e}_7\left(\bigotimes^{3,3}\mathfrak{e}_7\right)^{12}\bigotimes^{3,3}\mathfrak{so},$$

$$221e_7(\overset{3,3}{\otimes} e_7)^{12} \overset{3,3}{\otimes} \mathfrak{so}$$

$$21^{21} \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{12} \bigotimes \mathbf{so}$$

$$1321 \mathbf{e}_7^1 (\otimes \mathbf{e}_7^{3,3})^{11} \otimes \mathbf{so}^{3,3},$$

$$12321 e_7^1 \left(\bigotimes^{3,3} e_7 \right)^{11} \bigotimes^{3,3} \mathfrak{so},$$

$$221e_7^1(\otimes^{3,3}e_7)^{11}\otimes^{3,3}so$$

$$21^{21}_{\mathfrak{e}_7} ({}^{3,3} \otimes \mathfrak{e}_7)^{11} {}^{3,3} \otimes \mathfrak{so}$$

$$1321 \mathbf{e}_7 \left(\bigotimes \mathbf{e}_7 \right)^{10} \bigotimes \mathfrak{so},$$

$$12321\mathfrak{e}_7\left(\bigotimes^{3,3}\mathfrak{e}_7\right)^{10}\bigotimes^{3,3}\mathfrak{so},$$

$$221e_7^1(\otimes^{3,3}e_7)^{10}\otimes^{3,3}so$$

$$21_{\mathbf{e}_7}^{21}(\otimes_{\mathbf{e}_7}^{3,3})^{10} \otimes_{\mathbf{so}}^{3,3}$$

$$21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^9 \otimes \mathbf{e}_6^{5,5},$$

$$1e_7^{(5,5)}(\otimes e_8)^{10} \otimes e_6^{(5,5)}$$

$$21e_7^{1,4,5} \otimes e_8^{(5,5)} \otimes e_8^9 \otimes so^{5,3},$$

$$1e_7^{(5,5)} \otimes e_8^{(5,3)} \otimes 50$$

$$21\mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^8 \otimes \mathbf{e}_6^{5,5},$$

$$1e_7^{(1)}(\otimes_{5,5} e_8)^9 \otimes_{5,5} e_6$$

$$21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^8 \otimes so^{5,3}.$$

$$1e_7(\overset{5,5}{\otimes} e_8)^9 \overset{5,3}{\otimes} 50$$

$$21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^7 \otimes e_6^{5,5},$$

$$1e_7^{(1)}(\otimes_{5,5} e_8)^8 \otimes_{5,5} e_6$$

$$21\mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^7 \otimes \mathbf{so}^{5,3}.$$

$$1e_7^{(5,5)}(\otimes e_8)^8 \otimes^{5,3} \mathfrak{so}$$

$$21\mathfrak{e}_7^1 \otimes \mathfrak{e}_8^{4,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{5,3},$$

$$1\mathfrak{e}_7^1 (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$321\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^5 \otimes \mathfrak{e}_8^{4,5}$$

$$31\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^5 \otimes \mathfrak{e}_8^{3,5}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{e}_8^{4,5}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^5 \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{15} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{14} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{13} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}_7'^1 (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,3}$$

$$21\mathfrak{e}'_7(\otimes^{3,3}\mathfrak{e}_7)^{10}\otimes^{3,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^9\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^9\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^8\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^8\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^7\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^7\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^6\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^6\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^5\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^5\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^4\otimes^{5,5}\mathfrak{e}_6$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^4\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}'_7\otimes^{4,5}\mathfrak{e}_8(\otimes^{5,5}\mathfrak{e}_8)^{14}\otimes^{5,3}\mathfrak{so}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{12} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_6^{1\ 3,5} \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_6^{5,5}$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \mathfrak{so}^{5,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^9 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^8 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^7 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^6 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^5 \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{15} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{14} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{13} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{12} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{11} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_7^{3,3})^{10} \otimes \mathfrak{so}^{3,3}$$

$$1\epsilon_6^1 (\otimes \epsilon_6^{2,2})^6 \otimes \epsilon_8^{3,5}$$

$$\begin{array}{l} 2221^{21} \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513 \\ 221^{221} \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_8 122315123 \\ 221^{221} \mathfrak{e}_8(\overset{5,5}{\otimes} \mathfrak{e}_8)^7 \overset{5,5}{\otimes} \mathfrak{e}_8 12231513151 \end{array}$$

$$\begin{aligned}
& 131513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 131513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 1231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 1231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 12231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 12231513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 21513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 1223132 \\
& 313221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 313221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 22221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513215 \\
& 21513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122222 \\
& 1513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 21513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 1513221 \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 21^{13221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 21^{13221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151 \\
& 2221^{21} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122222 \\
& 2221^{21} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513 \\
& 221^{221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 122315123 \\
& 221^{221} \mathbf{c}_8^1 (\otimes \mathbf{c}_8)^{5,5} \otimes \mathbf{c}_8^{5,5} 12231513151
\end{aligned}$$

$$\begin{aligned} &131513221 \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8 122315123 \\ &131513221 \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8 12231513151 \\ &1231513221 \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8 122315123 \\ &1231513221 \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8 12231513151 \\ &12231513221 \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8^{(5,5)} \otimes \mathbf{c}_8 122315123 \end{aligned}$$

[illegible]

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_8^{5,5} 1223132, \quad 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_8^{5,5} 122222, \quad 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_8^{5,5} 122315123$$

$$\begin{array}{lll} 131\mathbf{e}_6^{1(2,2)}\otimes\mathbf{e}_6^{2,2}\otimes\mathbf{e}_6^{2,2}1231, & 131\mathbf{e}_6^{1(2,2)}\otimes\mathbf{e}_6^{2,2}\otimes\mathbf{e}_6^{2,2}12321, & 131\mathbf{e}_6^{1(2,2)}\otimes\mathbf{e}_6^{2,2}\otimes\mathbf{e}_6^{2,2}13151321 \\ 131\mathbf{e}_6^{1(2,2)}\otimes\mathbf{e}_6^{2,2}\otimes\mathbf{e}_6^{2,2}131513221, & 131\mathbf{e}_6^{1(2,2)}\otimes\mathbf{e}_6^{2,2}\otimes\mathbf{e}_6^{2,2}1315131, & 131\mathbf{e}_6^{1(2,2)}\otimes\mathbf{e}_6^{2,2}\otimes\mathbf{e}_6^{2,2}122 \end{array}$$

$$21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^5 \otimes \mathbf{e}_6^{5,4} 1$$

$$\begin{aligned} & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{14} \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

$$\begin{array}{l} 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{12} \otimes \epsilon_6^{5,3} 131 \\ 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{array}$$

$$\begin{aligned} & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 122 \\ & 1\epsilon_7^{1,5,5} (\otimes \epsilon_8)^{11} \otimes \epsilon_6^{5,3} 131 \\ & 21\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 \end{aligned}$$

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[illegible]

[illegible]

$$21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^8 \otimes \mathbf{e}_7'^1$$

$$\begin{array}{l} 1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^7 \otimes e_7^{5,4} 122 \\ 21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^7 \otimes e_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^6 \otimes e_7^{5,4} 122 \\ 21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^6 \otimes e_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^5 \otimes e_7^{5,4} 122 \\ 21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^5 \otimes e_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^4 \otimes e_7^{5,4} 122 \\ 21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^4 \otimes e_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1\epsilon_7' \otimes \epsilon_8(\overset{1}{\otimes} \overset{4,5}{\epsilon_8})^{13} \overset{5,4}{\otimes} \epsilon_7 122 \\ 21\epsilon_7 \otimes \epsilon_8(\overset{1}{\otimes} \overset{4,5}{\epsilon_8})^{13} \overset{5,5}{\otimes} \epsilon_7' 1 \end{array}$$

$$\begin{array}{l} 1^1_{e'_7} \otimes e_8(\overset{5,5}{\otimes} e_8)^{12} \otimes e_7^{5,4} 122 \\ 21^1_{e_7} \otimes e_8(\overset{5,5}{\otimes} e_8)^{12} \otimes e'^{5,5}_7 1 \end{array}$$

$$1e_7' \otimes e_8 \left(\otimes e_8 \right)^{11} \otimes e_7 122$$

$$\begin{aligned} & 21\mathbf{e}_7^{1,4,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^{11} \otimes \mathbf{e}'_7{}^{5,4}{}_{12}, \\ & 1\mathbf{e}_7^{1,5,5} (\otimes \mathbf{e}_8)^{12} \otimes \mathbf{e}'_7{}^{5,4}{}_{12}, \end{aligned}$$

$$\begin{array}{l} 1e_7' \otimes e_8 ({}^{5,5} \otimes e_8)^{11} \otimes e_7 12, \\ 1e_7 ({}^{5,5} \otimes e_8)^{12} \otimes e_7' 1 \end{array}$$

$$21 \mathbf{e}_7^1 \otimes \mathbf{e}_8^{4,5} (\otimes \mathbf{e}_8^{5,5})^{11} \otimes \mathbf{e}_7'^{5,5} 1$$

$$\begin{aligned} & 1\mathbf{e}'_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_8^{10} \otimes \mathbf{e}'_7 1231, \\ & 21\mathbf{e}'_7 \otimes \mathbf{e}_8 \otimes \mathbf{e}_8^{10} \otimes \mathbf{e}'_7 12, \\ & 1\mathbf{e}'_7 \otimes \mathbf{e}_8^{11} \otimes \mathbf{e}'_7 12, \end{aligned}$$

$$\begin{aligned} & 1\mathbf{e}_7^1 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{10} \otimes^{5,4} \mathbf{e}_7 12321, \\ & 1\mathbf{e}_7^1 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^{10} \otimes^{5,5} \mathbf{e}_7 12, \\ & 1\mathbf{e}_7 (\otimes^{5,5} \mathbf{e}_8)^{11} \otimes^{5,5} \mathbf{e}_7' 1 \end{aligned}$$

$$\begin{array}{l} 1\epsilon_7' \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_7^{5,4} 122 \\ 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_7^{5,5} 1 \end{array}$$

$$\begin{array}{l} 1_{\mathfrak{so}}^{1,1}(\otimes \mathfrak{so})^9 \otimes^{1,1} \mathfrak{so} 131, \\ 1_{\mathfrak{so}}^{1,1}(\otimes \mathfrak{so})^9 \otimes^{1,1} \mathfrak{so} 132, \end{array}$$

$$\begin{array}{l} 1\mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^9 \otimes^{1,1} \mathfrak{so}1321, \\ 1\mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^9 \otimes^{1,1} \mathfrak{so}12 \end{array}$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^9 \otimes \mathfrak{so}^{1,1}13221$$

$$\begin{array}{l} 1_{\mathfrak{so}}^{1,1}(\otimes \mathfrak{so})^8 \otimes^{1,1} \mathfrak{so} 131, \\ 1_{\mathfrak{so}}^{1,1}(\otimes \mathfrak{so})^8 \otimes^{1,1} \mathfrak{so} 132, \end{array}$$

$$\begin{array}{l} 1\mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^8 \otimes^{1,1} \mathfrak{so}1321, \\ 1\mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^8 \otimes^{1,1} \mathfrak{so}12 \end{array}$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^8 \otimes \mathfrak{so}13221$$

$$\begin{aligned} & 1_{\mathfrak{so}}^{1,1}(\otimes \mathfrak{so})^7 \otimes^{1,1} \mathfrak{so} 131, \\ & 1_{\mathfrak{so}}^{1,1}(\otimes \mathfrak{so})^7 \otimes^{1,1} \mathfrak{so} 132, \end{aligned}$$

$$\begin{array}{l} 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^7 \otimes^{1,1} \mathfrak{so}1321, \\ 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^7 \otimes^{1,1} \mathfrak{so}12 \end{array}$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^7 \otimes \mathfrak{so}^{1,1}13221$$

$$\begin{aligned} & 1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^6 \otimes {}^{1,1}\mathfrak{so}131, \\ & 1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^6 \otimes {}^{1,1}\mathfrak{so}132, \end{aligned}$$

$$\begin{array}{l} 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^6 \otimes^{1,1} \mathfrak{so}1321, \\ 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^6 \otimes^{1,1} \mathfrak{so}12 \end{array}$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^6 \otimes \mathfrak{so}^{1,1}13221$$

$$\begin{aligned} & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^5 \otimes^{1,1} \mathfrak{so}131, \\ & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^5 \otimes^{1,1} \mathfrak{so}132, \end{aligned}$$

$$\begin{array}{l} 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^5 \otimes^{1,1} \mathfrak{so}1321, \\ 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^5 \otimes^{1,1} \mathfrak{so}12 \end{array}$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^5 \otimes^{1,1} \mathfrak{so}13221$$

$$\begin{aligned} & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{15} \otimes^{1,1} \mathfrak{so} 131, \\ & 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{15} \otimes^{1,1} \mathfrak{so} 132, \end{aligned}$$

$$\begin{array}{l} 1\mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so}^{1,1} 1321, \\ 1\mathfrak{so}^{(1,1)}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so}^{1,1} 12 \end{array}$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so}^{1,1}13221$$

$$\begin{array}{l} 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{14} \otimes \mathfrak{so}^{1,1} 131, \\ 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{14} \otimes \mathfrak{so}^{1,1} 2 132, \end{array}$$

$$\begin{array}{l} 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{14} \otimes^{1,1} \mathfrak{so}1321, \\ 1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{14} \otimes^{1,1} \mathfrak{so}12 \end{array}$$

$$1\mathfrak{so}(\overset{1}{\otimes}\overset{1,1}{\mathfrak{so}})^{14}\overset{1,1}{\otimes}\mathfrak{so}13221$$

$$1\mathfrak{so}^1(\overset{1,1}{\otimes}\mathfrak{so})^{13}\overset{1,1}{\otimes}\mathfrak{so}131,$$

$$1\mathfrak{so}^1(\otimes^{1,1}\mathfrak{so})^{13}\otimes^{1,1}\mathfrak{so}1321.$$

$$1\mathfrak{so}^1(\otimes^{1,1}\mathfrak{so})^{13}\otimes^{1,1}\mathfrak{so}13221$$

$$1_{50}^{1,1}(\otimes 50)^{13} \otimes 50^{1,1} 132,$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^{13} {}^{1,1}\otimes \mathfrak{so}12$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^{12} \otimes \mathfrak{so}131,$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{12}\overset{1,1}{\otimes}\mathfrak{so}1321,$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{12} \otimes \mathfrak{so}^{1,1}13221$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^{12} \otimes \mathfrak{so}^{132},$$

$$1_{\mathfrak{so}}({}^{1,1}\otimes \mathfrak{so})^{12} \quad {}^{1,1}\otimes \mathfrak{so}12$$

$$150 \binom{1,1}{\otimes 50}^{11} \otimes 131,$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{11} \otimes^{1,1} \mathfrak{so} 1321,$$

$$1\mathfrak{so}(\overset{1}{\otimes}\overset{1,1}{\mathfrak{so}})^{11}\overset{1,1}{\otimes}\mathfrak{so}13221$$

$$150 \binom{1,1}{\otimes so}^{11} \binom{1,1}{\otimes so}^2 132,$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{11} \otimes^{1,1} \mathfrak{so}^{12}$$

$$150 \binom{1,1}{\otimes 50}^{10} \binom{1,1}{\otimes 50}^{131},$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{10}\overset{1,1}{\otimes}\mathfrak{so}1321,$$

$$1\mathfrak{so}^1(\overset{1,1}{\otimes}\mathfrak{so})^{10}\overset{1,1}{\otimes}\mathfrak{so}13221$$

$$150 \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50^{10} \begin{smallmatrix} 1,1 \\ \otimes \end{smallmatrix} 50^{132},$$

$$1\mathfrak{so}^{1,1}(\otimes \mathfrak{so})^{10} \otimes \mathfrak{so}^{12}$$

$$1\overset{1}{\mathbf{e}}_7' \otimes_{4,5} \mathbf{e}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \otimes_{5,3} \mathbf{e}_6 131,$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} \otimes e_7^{5,4} 12,$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathbf{e}_6 12$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} (\otimes e_8)^9 \otimes e_7'^{5,5} 1,$$

$$1\mathbf{e}'_7 \otimes^{4,5} \mathbf{e}_8 (\otimes^{5,5} \mathbf{e}_8)^9 \otimes^{5,4} \mathbf{e}_6 1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,3} e_6 131,$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} (\otimes e_8)^8 \otimes e_7'^{5,4} 12,$$

$$1e_7' \otimes e_8^{4,5} (\otimes e_8^{5,5})^8 \otimes e_6^{5,3} 12$$

$$1e_6 \otimes^{1,3,5} e_8 (\otimes^{5,5} e_8)^8 \otimes^{5,5} e_7' 1,$$

$$1\mathbf{e}'_7 \otimes_{4,5} \mathbf{e}_8 (\otimes_{5,5} \mathbf{e}_8)^8 \otimes_{5,4} \mathbf{e}_6 1$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,3} e_6 131,$$

$$1e_6 \otimes e_8^{3,5} (\otimes e_8^{5,5})^7 \otimes e_7^{5,4} 12,$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^7 \otimes^{5,3} e_6 12$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} (\otimes e_8)^7 \otimes e_7'^{5,5} 1,$$

$$1\mathbf{e}'_7 \otimes_{4,5} \mathbf{e}_8 (\otimes_{5,5} \mathbf{e}_8)^7 \otimes_{5,4} \mathbf{e}_6 1$$

$$1e_7' \otimes e_8 \binom{5,5}{\otimes e_8}^6 \otimes e_6 131,$$

$$1e_6 \otimes e_8 (e_8)^6 \otimes e_7' 12,$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes & \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,3 \\ \otimes \end{smallmatrix} \mathbf{e}_6 12$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,5}{\otimes} \mathbf{e}'_7 1,$$

$$1 \overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1\epsilon_7' \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^5 \otimes \epsilon_6 131,$$

$$1\mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^5 \otimes \mathbf{e}_7'^{5,4} 12,$$

$$1\mathbf{e}'_7 \otimes \mathbf{e}_8 \begin{pmatrix} 5,5 \\ \otimes \mathbf{e}_8 \end{pmatrix}^5 \otimes \mathbf{e}_6 12$$

$$1\mathbf{e}_6^{1,3,5} \otimes \mathbf{e}_8^{(5,5)} (\otimes \mathbf{e}_8)^5 \otimes \mathbf{e}_7'^{5,5} 1,$$

$$1 \overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^5 \overset{5,4}{\otimes} \mathbf{e}_6 1$$

$$1\epsilon_7' \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^4 \otimes \epsilon_6 131,$$

$$1e_6^1 \otimes e_8^{3,5} (\otimes e_8^{5,5})^4 \otimes e_7'^{5,4} 12,$$

$$1\overset{1}{\mathbf{e}}_7' \otimes \overset{4,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,3}{\otimes} \mathbf{e}_6 12$$

$$1e_6^{1,3,5} \otimes e_8^{(5,5)} (\otimes e_8)^4 \otimes e_7'^{5,5} 1,$$

$$1 \overset{1}{e}_7' \otimes \overset{4,5}{e}_8 (\overset{5,5}{\otimes} e_8)^4 \overset{5,4}{\otimes} e_6 1$$

$$\begin{array}{lll} 1\epsilon_7' \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7' \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,3} 12 \\ 1\epsilon_6^{1,3,5} \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7' \otimes \epsilon_8(\otimes \epsilon_8)^{13} \otimes \epsilon_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll}
1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_6^{5,3} 12 \\
1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_6^{5,4} 1 &
\end{array}$$

$$\begin{array}{lll} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_6^{5,3} 12 \\ 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{11} \otimes \epsilon_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,3} 131, & 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,4} 12, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,3} 12 \\ 1\epsilon_6^{1,3,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_7^{5,5} 1, & 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_6^{5,4} 1 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_7'^{3,3}1231, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_7'^{3,3}12321, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_7'^{3,3}122 \\ 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_6^{3,3}131, & 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^9 \otimes \epsilon_6^{3,3}12 & \end{array}$$

$$\begin{array}{lll} 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_7'^{3,3}1231, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_7'^{3,3}12321, & 1\epsilon_6^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_7'^{3,3}122 \\ 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_6^{3,3}131, & 21\epsilon_7'^{1\ 3,3}(\otimes \epsilon_7)^8 \otimes \epsilon_6^{3,3}12 & \end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7'1231, & 1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7'12321, & 1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_7'122 \\ 21\mathbf{e}_7'(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_6131, & 21\mathbf{e}_7'(\otimes \mathbf{e}_7)^7 \otimes \mathbf{e}_612 & \end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes \mathbf{e}_7'^{3,3} 1231, & 1\mathbf{e}_6^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes \mathbf{e}_7'^{3,3} 12321, & 1\mathbf{e}_6^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes \mathbf{e}_7'^{3,3} 122 \\ 21\mathbf{e}_7'^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes \mathbf{e}_6^{3,3} 131, & 21\mathbf{e}_7'^{1\begin{pmatrix} 3,3 \\ \otimes \end{pmatrix}6} \otimes \mathbf{e}_6^{3,3} 12 & \end{array}$$

$$\begin{array}{lll} 1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_7'^{3,3}1231, & 1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_7'^{3,3}12321, & 1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_7'^{3,3}122 \\ 21\mathbf{e}_7'^{1(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6^{3,3}131, & 21\mathbf{e}_7'^{1(3,3)}(\otimes \mathbf{e}_7)^5 \otimes \mathbf{e}_6^{3,3}12 & \end{array}$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{15}{}^{3,3} \mathbf{e}_7' 1231, \quad 1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{15} \otimes \mathbf{e}_7' 12321, \quad 1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{15} \otimes \mathbf{e}_7' 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}}_7)^{15}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{15}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1e_6^{(3,3)}(\otimes e_7)^{14} \otimes e_7' 1231,$$

$$1e_6^{(3,3)}(\otimes e_7)^{14} \otimes e_7' 12321,$$

$$1e_6^{(3,3)}(\otimes e_7)^{14} \otimes e_7' 122$$

$$21\mathbf{e}'_7 \binom{1}{(3,3)} (\otimes \mathbf{e}_7)^{14} \binom{3,3}{\otimes} \mathbf{e}_6 131,$$

$$21\overset{1}{\mathbf{e}}_7'(\overset{3,3}{\otimes}\mathbf{e}_7)^{14}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1\mathbf{e}_6\left(\overset{3,3}{\otimes}\mathbf{e}_7\right)^{13}\overset{3,3}{\otimes}\mathbf{e}_7'1231,$$

$$1e_6^{(3,3)}(\otimes e_7)^{13} \otimes e_7' 12321,$$

$$1e_6^{(3,3)}(\otimes e_7)^{13} \otimes e_7' 122$$

$$21\mathbf{e}'_7 \binom{1}{(3,3)} \binom{3,3}{\otimes \mathbf{e}_7}^{13} \otimes \mathbf{e}_6 131,$$

$$21\overset{1}{\mathbf{e}}_7'(\overset{3,3}{\otimes}\mathbf{e}_7)^{13}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{12} \otimes \mathbf{e}_7' 1231,$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{12} \otimes \mathbf{e}_7'^{3,3}12321,$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{12} \otimes \mathbf{e}_7' 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{12}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{12}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7^{3,3} 1231,$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7'^{3,3}12321,$$

$$1\mathbf{e}_6^{(3,3)}(\otimes \mathbf{e}_7)^{11} \otimes \mathbf{e}_7' 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{11}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{11}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1\mathbf{e}_6 \binom{3,3}{\otimes} \mathbf{e}_7^{10} \binom{3,3}{\otimes} \mathbf{e}_7' 1231,$$

$$1\mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7^{10} \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7' 12321,$$

$$1\mathbf{e}_6^{1(3,3)}(\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 122$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{10}\overset{3,3}{\otimes}\mathbf{e}_6131,$$

$$21\mathbf{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{10}\overset{3,3}{\otimes}\mathbf{e}_612$$

$$1321e_7^1(\otimes^{3,3} e_7)^9 \otimes^{3,2} so1,$$

$$12321\mathbf{e}_7^1(\otimes\mathbf{e}_7)^9\otimes\mathbf{so}1,$$

$$221\mathbf{e}_7({}^{(3,3)}\otimes\mathbf{e}_7)^9\otimes{}^{(3,2)}\mathfrak{so}1$$

$$21\mathbf{e}_7\left(\begin{smallmatrix} 3,3 \\ \otimes \end{smallmatrix}\mathbf{e}_7\right)^9\begin{smallmatrix} 3,2 \\ \otimes \end{smallmatrix}\mathfrak{so}1$$

$$1321e_7^{(3,3)}(\otimes e_7)^8 \otimes^{3,2} \mathfrak{so}1,$$

$$12321\mathfrak{e}_7^1(\otimes \mathfrak{e}_7)^{3,3}{}^8 \otimes \mathfrak{so}1,{}^{3,2}$$

$$221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^8 \otimes \mathbf{so}1$$

$$21\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^8\overset{3,2}{\otimes}\mathfrak{so}1$$

$$1321e_7^{(3,3)}(\otimes e_7)^7 \otimes so_1,$$

$$12321e_7(\otimes e_7)^7 \otimes so1,$$

$$221\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^7\otimes\mathbf{so}1$$

$$21\mathbf{e}_7(\overset{3,3}{\otimes}\mathbf{e}_7)^7\overset{3,2}{\otimes}\mathbf{so}1$$

$$1321e_7^{(3,3)}(\otimes e_7)^6 \otimes^{3,2} so1,$$

$$12321 \mathbf{e}_7^1 (\otimes \mathbf{e}_7)^{3,3} \otimes \mathbf{so}1, \quad \mathbf{e}_7^{3,2} \otimes \mathbf{so}1,$$

$$221\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^6{}^{3,2}\otimes\mathbf{so}1$$

$$21\mathfrak{e}_7^{21}(\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{so}^{3,2}1$$

$$\begin{array}{l} 1321^1_{\mathbf{e}_7}(\otimes \mathbf{e}_7)^5 \otimes^{3,2} \mathbf{sol}, \\ 21^{21}_{\mathbf{e}_7}(\otimes \mathbf{e}_7)^5 \otimes^{3,2} \mathbf{sol} \end{array}$$

$$12321\mathbf{e}_7^1(\otimes \mathbf{e}_7)^{3,3}{}^5 \otimes \mathbf{so}1,{}^{3,2}$$

$$221\mathbf{e}_7({}^1\otimes\mathbf{e}_7)^5\otimes\mathbf{sol}$$

$$\begin{array}{l} 1321e_7^{(3,3)}(\otimes e_7)^{15} \otimes so_1, \\ 21e_7^{(3,3)}(\otimes e_7)^{15} \otimes so_1 \end{array}$$

$$12321\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^{15}{}^{3,2}\otimes\mathbf{so}1,$$

$$221\mathbf{e}_7({}^1\otimes\mathbf{e}_7)^{15}{}^{3,2}\otimes\mathbf{so}1$$

$$\begin{array}{l} 1321e_7^{(3,3)}(\otimes e_7)^{14} \otimes so1, \\ 21e_7^{(3,3)}(\otimes e_7)^{14} \otimes so1 \end{array}$$

$$12321\mathbf{e}_7^1(\otimes \mathbf{e}_7)^{3,3}{}^{14} \otimes \mathbf{so}1,{}^{3,2}$$

$$221\mathbf{e}_7({}^1\otimes\mathbf{e}_7)^{14}{}^{3,2}\otimes\mathbf{so}1$$

$$\begin{array}{l} 1321e_7^{(3,3)}(\otimes e_7)^{13} \otimes \mathfrak{so}1, \\ 21e_7^{(3,3)}(\otimes e_7)^{13} \otimes \mathfrak{so}1 \end{array}$$

$$12321\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^{13}{}^{3,2}\otimes\mathfrak{so}1,$$

$$221\mathbf{e}_7({}^1\otimes\mathbf{e}_7)^{13}{}^{3,2}\otimes\mathbf{so}1$$

$$\begin{array}{l} 1321e_7^{(3,3)}(\otimes e_7)^{12} \otimes \mathfrak{so}1, \\ 21e_7^{(3,3)}(\otimes e_7)^{12} \otimes \mathfrak{so}1 \end{array}$$

$$12321\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^{12}{}^{3,2}\otimes\mathfrak{so}1,$$

$$221\mathbf{e}_7({}^1\otimes\mathbf{e}_7)^{12}{}^{3,2}\otimes\mathbf{so}1$$

$$\begin{array}{l} 1321\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{11} \otimes \mathfrak{so}1, \\ 21\epsilon_7^{(3,3)}(\otimes \epsilon_7)^{11} \otimes \mathfrak{so}1 \end{array}$$

$$12321\mathbf{e}_7({}^{3,3}\otimes\mathbf{e}_7)^{11}{}^{3,2}\otimes\mathbf{so}1,$$

$$221e_7({}^1\otimes e_7)^{11} {}^{3,2}\otimes so1$$

$$\begin{array}{l} 1321e_7^{(3,3)}(\otimes e_7)^{10} \otimes^{3,2} \mathfrak{so}1, \\ 21e_7^{(3,3)}(\otimes e_7)^{10} \otimes^{3,2} \mathfrak{so}1 \end{array}$$

$$12321e_7^1(\otimes e_7)^{10} \otimes so1,$$

$$221\mathbf{e}_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^{10}\overset{3,2}{\otimes}\mathbf{so}1$$

$$22221\mathbf{e}_8^1(\otimes^{5,5}\mathbf{e}_8)^{14}\otimes^{5,5}\mathbf{e}_8122222,$$

$$2221\mathbf{e}_8^{21}(\otimes^{5,5}\mathbf{e}_8)^{14}\otimes^{5,5}\mathbf{e}_8122222,$$

$$221 \overset{221}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^{14} \overset{5,5}{\otimes} \mathfrak{e}_8 122222$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^9\overset{3,3}{\otimes}\mathbf{e}'_71231,$$

$$21\overset{1}{\mathbf{e}}_7'(\overset{3,3}{\otimes}\mathbf{e}_7)^9\overset{3,3}{\otimes}\mathbf{e}_7'12321,$$

$$21 \overset{1}{\mathbf{e}}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^9 \overset{3,3}{\otimes} \mathbf{e}_7' 122$$

$$21\mathbf{e}'_7(\bigotimes^{3,3}\mathbf{e}_7)^8\bigotimes^{3,3}\mathbf{e}'_71231,$$

$$21\mathbf{e}'_7(\otimes^{3,3}\mathbf{e}_7)^8\otimes^{3,3}\mathbf{e}'_712321,$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^8\overset{3,3}{\otimes}\mathbf{e}'_7122$$

$$21\mathbf{e}'_7(\otimes_{\mathbf{e}_7}^{3,3})^7\otimes_{\mathbf{e}_7}^{3,3}\mathbf{e}'_71231,$$

$$21\mathbf{e}'_7(\otimes_{3,3}\mathbf{e}_7)^7\otimes_{3,3}\mathbf{e}'_712321,$$

$$21\mathbf{e}'_7(\overset{1}{\otimes}\overset{3,3}{\mathbf{e}_7})^7\overset{3,3}{\otimes}\mathbf{e}'_7122$$

$$31\mathfrak{e}_6^{31}(\otimes \mathfrak{e}_6)^{10} \otimes \mathfrak{so}1$$

$$1e_7^{1,5,5}(\otimes e_8)^{10} \otimes so^{4,2} \otimes so^{1,1}$$

$$1e_7^{1(5,5)}(\otimes e_8)^{10} \otimes so_{13}^{4,2}$$

$$1e_7^{1(5,5)}(\otimes e_8)^9 \otimes^{4,2} so \otimes^{1,1} so1$$

$$1e_7^{(5,5)}(\otimes e_8)^9 \otimes so_{13}$$

$$1e_7 \binom{5,5}{\otimes} e_8 \binom{4,2}{\otimes} so \binom{1,1}{\otimes} so1$$

$$1e_7^{(5,5)}(\otimes e_8)^8 \otimes so_{13}^{4,2}$$

$$1e_7^{1(5,5)}(\otimes e_8)^{7(4,2)} \otimes so^{1,1} \otimes so1$$

$$1e_7^{1(5,5)}(\otimes e_8)^7 \otimes so_{13}^{4,2}$$

$$1e_7 \binom{1}{5,5} (\otimes e_8)^6 \binom{4,2}{\otimes} \text{so} \binom{1,1}{\otimes} \text{so} 1$$

$$1e_7^{1(5,5)}(\otimes e_8)^6 \otimes^{4,2} so_{13}$$

$$1e_7(\overset{1}{\otimes} \overset{5,5}{e_8})^5 \overset{4,2}{\otimes} so \overset{1,1}{\otimes} so1$$

$$1e_7^{(1)}(\otimes e_8^{(5,5)})^5 \otimes so_{13}^{(4,2)}$$

$$1e_7(\otimes e_8)^4 \otimes^{4,2} so \otimes^{1,1} so1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\mathfrak{e}_6131,$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\mathfrak{e}_612$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\mathfrak{e}_6131,$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\mathfrak{e}_612$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\mathfrak{e}_6131,$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\mathfrak{e}_612$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^9\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^8\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^7\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^6\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{4,5}{\otimes}\mathfrak{e}_81$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{15}\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{14}\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{13}\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}_7'(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,2}{\otimes}\mathfrak{so}1$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{10} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}_8 \otimes \mathfrak{so}(\otimes \mathfrak{so})^4 \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{so}13$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so} \otimes \mathfrak{so}1$$

$$1\mathfrak{e}'_7 \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{so}13$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^3 \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{14} \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{13} \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{13} \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{12} \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{12} \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{11} \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{11} \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{10} \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon'_7 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^{10} \otimes \epsilon_8$$

$$1\epsilon'_7 \begin{pmatrix} 3,3 \\ \otimes \epsilon_7 \end{pmatrix}^5 \otimes \epsilon_8$$

$$1\epsilon_6 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^9 \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\epsilon_6 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^9 \otimes \epsilon_8$$

$$1\epsilon_6 \otimes \epsilon_8 \begin{pmatrix} 5,5 \\ \otimes \epsilon_8 \end{pmatrix}^8 \otimes \epsilon_8 \otimes \epsilon_8$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^8 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^7 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^6 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^5 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^4 \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{14} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{so}^{4,2} 13$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^5 \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{15} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{14} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{13} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{12} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{11} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^{10} \otimes \mathfrak{so}^{3,2} 1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_6^{2,2})^5 \otimes \mathfrak{e}_8^{3,5} 1$$

$131513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 122315131,$	$131513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151321,$	$131513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231513221$
$313221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 122315131,$	$22221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 122315131,$	$131513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231512$
$131513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151,$	$1231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151321,$	$1231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231513221$
$313221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151321,$	$22221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151321,$	$1231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231512$
$1231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151,$	$12231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231513221,$	$313221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231513221$
$22221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231513221,$	$12231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231512,$	$12231513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151$
$313221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 122313,$	$22221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 122313,$	$313221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231512$
$313221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151,$	$22221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12222,$	$22221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231512$
$22221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 1223151,$	$21513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231512,$	$1513221\epsilon_8(\otimes \epsilon_8)^{11} \otimes^{5,5}_8 \epsilon_8^1 12231512$

$$1\mathbf{e}_7 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} \mathbf{e}_8 \begin{pmatrix} 5,5 \\ \otimes \end{pmatrix} I^{\oplus 2} \mathbf{e}_8 12231$$

$$\begin{aligned}
& 21\mathbf{c}_7^1 \otimes \mathbf{c}_8^{4,5} (\otimes \mathbf{c}_8^{5,5})^5 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 12231513221 \\
& 21\mathbf{c}_7^1 \otimes \mathbf{c}_8^{4,5} (\otimes \mathbf{c}_8^{5,5})^5 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 12231512 \\
& 21\mathbf{c}_7^1 \otimes \mathbf{c}_8^{4,5} (\otimes \mathbf{c}_8^{5,5})^5 \otimes \mathbf{c}_8^{5,5} I^{\oplus 2} \mathbf{1} 222 \\
& 1\mathbf{c}_7^1 (\otimes \mathbf{c}_8^{5,5})^6 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 1223151321 \\
& 1\mathbf{c}_7^1 (\otimes \mathbf{c}_8^{5,5})^6 \otimes \mathbf{c}_8^{5,5} \mathbf{1} 12222 \\
& 1\mathbf{c}_7^1 (\otimes \mathbf{c}_8^{5,5})^6 \otimes \mathbf{c}_8^{5,5} I^{\oplus 2} \mathbf{1} 2231
\end{aligned}$$

$$\begin{aligned}
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} 12231513221 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} 1\epsilon_8 12231512 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 1222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} 1\epsilon_8 1223151321 \\
& 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} 1\epsilon_8 12222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 12231
\end{aligned}$$

$$\begin{aligned}
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 12231513221 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 1\epsilon_8 12231512 \\
& 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 1222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 1223151321 \\
& 1\epsilon_7^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} 12222 \\
& 1\epsilon_7^1 (\otimes \epsilon_8)^{5,5} \otimes \epsilon_8^{5,5} I^{\oplus 2} \epsilon_8 12231
\end{aligned}$$

$$\begin{aligned} & 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} 12231513221 \\ & 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} 12231512 \\ & 21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} I^{\oplus 2} 1222 \end{aligned}$$

[illegible]

$$\begin{array}{l} 221\mathbf{e}_7(\otimes \mathbf{e}_7)^{14} \otimes \mathbf{e}_7^{I \oplus 3} 1 \\ 221\mathbf{e}_7(\otimes \mathbf{e}_7)^{14} \otimes \mathbf{e}_7^{I \oplus 2} 12 \end{array}$$

$$\begin{array}{l} 221 \mathbf{e}_7 (\otimes \mathbf{e}_7)^{13} \otimes \mathbf{e}_7^{1231} \\ 221 \mathbf{e}_7 (\otimes \mathbf{e}_7)^{13} \otimes \mathbf{e}_7^{1 \oplus 3} \\ 221 \mathbf{e}_7 (\otimes \mathbf{e}_7)^{13} \otimes \mathbf{e}_7^{1 \oplus 2} \end{array}$$

$$\begin{array}{l} 221\epsilon_7^{(3,3)} \otimes \epsilon_7^{(3,3)} 1231 \\ 221\epsilon_7^{(3,3)} \otimes \epsilon_7^{(3,3)} I^{\oplus 3} 1 \\ 221\epsilon_7^{(3,3)} \otimes \epsilon_7^{(3,3)} I^{\oplus 2} 12 \end{array}$$

$$\begin{array}{l} 221 \mathbf{e}_7^{(3,3)} \otimes \mathbf{e}_7^{3,3} \mathbf{1} \\ 221 \mathbf{e}_7^{(3,3)} \otimes \mathbf{e}_7^{3,3} \mathbf{1}^{I \oplus 3} \\ 221 \mathbf{e}_7^{(3,3)} \otimes \mathbf{e}_7^{3,3} \mathbf{1}^{I \oplus 2} \end{array}$$

$$\begin{array}{l} 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 1231 \\ 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 1^{\oplus 3} \\ 221\mathbf{e}_7^{(3,3)}(\otimes \mathbf{e}_7)^{10} \otimes \mathbf{e}_7^{3,3} 12^{\oplus 2} \end{array}$$

$$\begin{array}{l} 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{1,22313} \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{1,22315} \\ 1\epsilon_7^{1,4,5} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{1,22} \end{array}$$

$$22221 \mathbf{e}_8 \left(\overset{5,5}{\otimes} \mathbf{e}_8 \right)^{14} \overset{5,5}{\otimes} \mathbf{e}_8 I^{\oplus 3} 122$$

$$2221 \mathfrak{e}_8^{21} (\otimes \mathfrak{e}_8^{5,5})^{14} \otimes \mathfrak{e}_8^{5,5} I^{\oplus 2} 1222,$$

$$2221\mathbf{e}_8(\overset{5,5}{\otimes}\mathbf{e}_8)^{14}\overset{5,5}{\otimes}I^{\oplus 3}\mathbf{e}_8122,$$

$$221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^{14} \bigotimes \mathbf{e}_8^{5,5} I^{\oplus 3} 122$$

$$\begin{aligned} & 131\epsilon_6^1(\otimes \epsilon_6)^{2,2} \otimes \epsilon_6^1 131, \\ & 31\epsilon_6^1(\otimes \epsilon_6)^{2,2} \otimes \epsilon_6^1 131, \end{aligned}$$

$$\begin{array}{l} 21^1_{\mathbf{e}_6} ({}^{2,2} \otimes \mathbf{e}_6)^9 {}^{2,2} \otimes \mathbf{e}_6 131, \\ 31^{31}_{\mathbf{e}_6} ({}^{2,2} \otimes \mathbf{e}_6)^9 {}^{2,2} I^{\oplus 2} \mathbf{e}_6 1 \end{array}$$

$$21 \mathbf{e}_6 ({}^{2,2} \otimes \mathbf{e}_6)^9 {}^{2,2} I^{\oplus 2} \otimes \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^{(2,2)}(\otimes \epsilon_6)^8 \otimes \epsilon_6^{(2,2)} 131, \\ &31\epsilon_6^{(2,2)}(\otimes \epsilon_6)^8 \otimes \epsilon_6^{(2,2)} 131, \end{aligned}$$

$$\begin{array}{l} 21^1 \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^8 \bigotimes \mathbf{e}_6^{2,2} 131, \\ 31^{31} \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^8 \bigotimes \mathbf{e}_6^{2,2} I^{\oplus 2} 1 \end{array}$$

$$21^1_6 ({}^{2,2} \otimes \mathbf{e}_6)^8 {}^{2,2} I^{\oplus 2} \otimes \mathbf{e}_6 1$$

$$\begin{aligned} & 131\epsilon_6^1(\otimes^{2,2}\epsilon_6)^7\otimes^{2,2}\epsilon_6^1131, \\ & 31\epsilon_6^{31}(\otimes^{2,2}\epsilon_6)^7\otimes^{2,2}\epsilon_6^1131, \end{aligned}$$

$$\begin{array}{l} 21^1 \epsilon_6^{(2,2)} \otimes \epsilon_6^{(2,2)} \otimes 131, \\ 31^{31} \epsilon_6^{(2,2)} \otimes \epsilon_6^{(2,2)} \otimes I^{\oplus 2} 1 \end{array}$$

$$21\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^7 \overset{2,2}{\otimes} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^{1(2,2)}(\otimes \epsilon_6)^6 \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31(2,2)}(\otimes \epsilon_6)^6 \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{array}{l} 21^1_{\mathbf{e}_6} (\otimes \mathbf{e}_6)^{2,2}_6 \otimes \mathbf{e}_6^{2,2}_6 131, \\ 31^{31}_{\mathbf{e}_6} (\otimes \mathbf{e}_6)^{2,2}_6 \otimes \mathbf{e}_6^{2,2}_6 I^{\oplus 2} 1 \end{array}$$

$$21 \mathbf{e}_6 ({}^{2,2} \otimes \mathbf{e}_6)^6 {}^{2,2} I^{\oplus 2} \otimes \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^{(2,2)}(\otimes \epsilon_6)^5 \otimes \epsilon_6^{(2,2)} 131, \\ &31\epsilon_6^{(2,2)}(\otimes \epsilon_6)^5 \otimes \epsilon_6^{(2,2)} 131, \end{aligned}$$

$$\begin{array}{l} 21^1_6 ({}^{2,2} \otimes \mathbf{e}_6)^5 {}^{2,2} \otimes \mathbf{e}_6 131, \\ 31^{31}_6 ({}^{2,2} \otimes \mathbf{e}_6)^5 {}^{2,2} I^{\oplus 2} \mathbf{e}_6 1 \end{array}$$

$$21 \mathbf{e}_6 \left(\bigotimes \mathbf{e}_6 \right)^5 \bigotimes \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^1(\otimes \epsilon_6)^{15} \otimes \epsilon_6^1 131, \\ &31\epsilon_6^{31}(\otimes \epsilon_6)^{15} \otimes \epsilon_6^1 131, \end{aligned}$$

$$\begin{array}{l} 21\epsilon_6^1(\otimes \epsilon_6)^{2,2} \otimes \epsilon_6^{2,2} 131, \\ 31\epsilon_6^{31}(\otimes \epsilon_6)^{2,2} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1 \end{array}$$

$$21 \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^{15} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^1(\otimes \epsilon_6^{2,2})^{14} \otimes \epsilon_6^1 131, \\ &31\epsilon_6^{31}(\otimes \epsilon_6^{2,2})^{14} \otimes \epsilon_6^1 131, \end{aligned}$$

$$\begin{array}{l} 21\epsilon_6^1(\otimes \epsilon_6)^{14} \otimes \epsilon_6^1 131, \\ 31\epsilon_6^{31}(\otimes \epsilon_6)^{14} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1 \end{array}$$

$$21\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^{14} \overset{2,2}{\otimes} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^1(\otimes \epsilon_6)^{13} \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31}(\otimes \epsilon_6)^{13} \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{array}{l} 21^1 \epsilon_6^{\left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \right)} \epsilon_6^{13} \otimes^{2,2} \epsilon_6^1 131, \\ 31^{31} \epsilon_6^{\left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \right)} \epsilon_6^{13} \otimes^{2,2} \epsilon_6^{I \oplus 2} 1 \end{array}$$

$$21\mathbf{e}_6(\overset{2,2}{\otimes} \mathbf{e}_6)^{13} \overset{2,2}{\otimes} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^1(\otimes \epsilon_6)^{12} \otimes \epsilon_6^1 131, \\ &31\epsilon_6^{31}(\otimes \epsilon_6)^{12} \otimes \epsilon_6^1 131, \end{aligned}$$

$$\begin{array}{l} 21 \mathbf{e}_6^1 (\otimes \mathbf{e}_6^{2,2})^{12} \otimes \mathbf{e}_6^{2,2} \mathbf{e}_6^1 131, \\ 31 \mathbf{e}_6^{31} (\otimes \mathbf{e}_6^{2,2})^{12} \otimes \mathbf{e}_6^{2,2} I^{\oplus 2} \mathbf{e}_6^1 1 \end{array}$$

$$21 \mathbf{e}_6 \left(\begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} \mathbf{e}_6 \right)^{12} \begin{smallmatrix} 2,2 \\ \otimes \end{smallmatrix} I^{\oplus 2} \mathbf{e}_6 1$$

$$\begin{aligned} &131\epsilon_6^{1(2,2)11} \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31(2,2)11} \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{aligned} &21\epsilon_6^{1(2,2)11} \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31(2,2)11} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1 \end{aligned}$$

$$21\epsilon_6^{1(2,2)11} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1$$

$$\begin{aligned} &131\epsilon_6^{1(2,2)10} \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31(2,2)10} \otimes \epsilon_6^{2,2} 131, \end{aligned}$$

$$\begin{aligned} &21\epsilon_6^{1(2,2)10} \otimes \epsilon_6^{2,2} 131, \\ &31\epsilon_6^{31(2,2)10} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1 \end{aligned}$$

$$21\epsilon_6^{1(2,2)10} \otimes \epsilon_6^{2,2} I^{\oplus 2} 1$$

$$\begin{aligned} &21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} 1231, \\ &21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)9} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} &21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} 1231, \\ &21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)8} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} &21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} 1231, \\ &21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)7} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} &21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} 1231, \\ &21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)6} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} &21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} 1231, \\ &21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)5} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} &21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} 1231, \\ &21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)15} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$\begin{aligned} &21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} 1231, \\ &21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} I^{\oplus 2} 12 \end{aligned}$$

$$21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} 12321,$$

$$21\epsilon_7^{1(3,3)14} \otimes \epsilon_7^{3,3} I^{\oplus 3} 1$$

$$21 \overset{1}{e}_7' (\overset{3,3}{\otimes} \mathbf{e}_7)^{13} \overset{3,3}{\otimes} I^{\oplus 3} \mathbf{e}_7 1$$

$$21 \mathbf{e}_7' ({}^{3,3} \otimes \mathbf{e}_7)^{12} {}^{3,3} I^{\oplus 3} \otimes \mathbf{e}_7 1$$

$$21_7^1 ({}^{3,3} \otimes \mathbf{e}_7)^{11} {}^{3,3} I^{\oplus 3} \otimes \mathbf{e}_7 1$$

$$21e_7'({}^{1,3,3} \otimes e_7)^{10} {}^{3,3} I^{\oplus 3} \otimes e_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1 \mathbf{e}_6 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} \mathbf{e}_7 \begin{pmatrix} 3,3 \\ \otimes \end{pmatrix} I^{\oplus 3} \mathbf{e}_7 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^9 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^8 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1 \mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^7 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^6 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^5 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1\mathbf{e}_6 \overset{1}{\otimes} \overset{3,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^4 \overset{5,5}{\otimes} \overset{1}{\mathbf{e}_7} 1$$

$$1 \overset{1}{\mathbf{e}}_6 \otimes \overset{3,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{13} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1 \mathbf{e}_6^1 \otimes \mathbf{e}_8^{3,5} (\otimes \mathbf{e}_8^{5,5})^{12} \otimes \mathbf{e}_7^1$$

$$1 \overset{1}{\mathbf{e}}_6 \otimes \overset{3,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{11} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$1 \overset{1}{\mathbf{e}}_6 \otimes \overset{3,5}{\mathbf{e}}_8 (\overset{5,5}{\otimes} \mathbf{e}_8)^{10} \overset{5,5}{\otimes} \overset{1}{\mathbf{e}}_7 1$$

$$21 \mathbf{e}'_7 ({}^{3,3} \otimes \mathbf{e}_7)^8 {}^{3,3} I^{\oplus 2} \otimes \mathbf{e}'_7 1$$

$$21 \mathbf{e}'_7 ({}^{3,3} \otimes \mathbf{e}_7)^6 {}^{3,3} I^{\oplus 2} \otimes \mathbf{e}'_7 1$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^5 \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{15} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{14} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{13} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{12} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{11} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$21\mathfrak{e}'_7(\otimes \mathfrak{e}_7)^{10} \otimes \mathfrak{e}'_7 I^{\oplus 2}$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^9 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^8 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^7 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^6 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^5 \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^{15} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\otimes \mathfrak{so})^{14} \otimes \mathfrak{so} 1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{13}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{12}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{11}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{so}(\overset{1,1}{\otimes}\mathfrak{so})^{10}\overset{1,1}{\otimes}\overset{1}{\mathfrak{so}}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^9\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^8\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^7\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^6\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^5\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^4\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{13}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{12}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{11}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\mathfrak{e}'_7\overset{1}{\otimes}\overset{4,5}{\mathfrak{e}}_8(\overset{5,5}{\otimes}\mathfrak{e}_8)^{10}\overset{5,4}{\otimes}\overset{1}{\mathfrak{e}'_7}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^9 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^8 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^8 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^7 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_7'^{5,4} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{14} \otimes \epsilon_6^{5,3} 1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_7'^{5,4} 1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{13} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{12} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{11} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{e}_7^{5,4} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 \otimes \mathfrak{e}_8^{3,5} (\otimes \mathfrak{e}_8^{5,5})^{10} \otimes \mathfrak{e}_6^{5,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^9 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^8 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^7 \otimes \mathfrak{e}_6^{3,3} \mathfrak{e}_6^1$$

$$1\mathfrak{e}_6^1 (\otimes \mathfrak{e}_7^{3,3})^6 \otimes \mathfrak{e}_7^{3,3} I^{\oplus 2} \mathfrak{e}_7^1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^6\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^5\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{15}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{15}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{14}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{14}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{13}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{13}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{12}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{11}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\overset{I^{\oplus 2}}{\mathfrak{e}_7'}1$$

$$1\mathfrak{e}_6(\overset{1}{\otimes}\overset{3,3}{\mathfrak{e}_7})^{10}\overset{3,3}{\otimes}\overset{1}{\mathfrak{e}_6}1$$

2.8.7 $\mathfrak{gLIg}(n-1)(3)$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{e}_7'$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{e}_6$$

$$\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so} \otimes \mathfrak{so}$$

2.8.8 $\mathfrak{gLIgLg}(n-4)\mathfrak{LIgLg}(1)$

$$\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{e}_8 \otimes \mathfrak{e}_8$$

2.8.9 $\mathfrak{SgLIg}(n-1)(26)$

$$I^{\oplus 2} \mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^3 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^4 \otimes \mathfrak{so} \otimes \mathfrak{so},$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^9 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^8 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^7 \otimes \mathfrak{so}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^6 \otimes \mathfrak{e}_6$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^6 \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^5 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{14} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{14} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{13} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{12} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{11} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{e}_6^{5,5}$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4} (\otimes \mathfrak{e}_8^{1,5,5})^{10} \otimes \mathfrak{so}^{5,3}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_7'^{5,5}$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8^{1,5,5})^4 \otimes \mathfrak{e}_6^{5,4}$$

2.8.10 **SgLIg(n-3)LIgLg(1)**

$$1\mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1(5,5)} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{e}_8^{1(4,2)} \otimes \mathfrak{so}$$

2.8.11 SgLIg(n-3)LIgLgS(11)

$$150 \otimes_{\mathfrak{e}_8}^{2,4} (\otimes_{\mathfrak{e}_8}^{5,5})^9 \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{1,4,2} 501$$

$$150 \otimes_{\mathfrak{e}_8}^{2,4} (\otimes_{\mathfrak{e}_8}^{5,5})^8 \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{1,4,2} 501$$

$$150 \otimes_{\mathfrak{e}_8}^{2,4} (\otimes_{\mathfrak{e}_8}^{5,5})^7 \otimes_{\mathfrak{e}_8}^{5,5} \otimes_{\mathfrak{e}_8}^{1,4,2} 501$$

$$150 \otimes_{\mathfrak{e}_8}^{2,4} \mathfrak{e}_8 \left(\otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 \right)^6 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 \otimes_{\mathfrak{e}_8}^{4,2} 501$$

$$150 \otimes^{2,4} \mathfrak{e}_8 \otimes^{1,5,5} (\otimes \mathfrak{e}_8)^5 \otimes^{5,5} \mathfrak{e}_8 \otimes^{1,4,2} \mathfrak{so}1$$

$$150 \otimes_{\mathfrak{e}_8}^{2,4} \mathfrak{e}_8 \left(\otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 \right)^4 \otimes_{\mathfrak{e}_8}^{5,5} \mathfrak{e}_8 \otimes_{\mathfrak{e}_8}^{4,2} 501$$

$$150 \otimes^{2,4} e_8 \otimes^{1,5,5} (e_8 \otimes e_8)^3 \otimes^{5,5} e_8 \otimes^{1,4,2} 501$$

$$150 \otimes^{2,4} e_8 \otimes^{5,5} (e_8)^{13} \otimes^{5,5} e_8 \otimes^{4,2} 501$$

$$150 \otimes^{2,4} e_8 \otimes^{5,5} (e_8 \otimes e_8)^{12} \otimes^{5,5} e_8 \otimes^{4,2} 501$$

$$1s0 \otimes^{2,4} e_8 \otimes^{5,5} (e_8)^{11} \otimes^{5,5} e_8 \otimes^{4,2} s01$$

$$150 \otimes^{2,4} e_8 \otimes^{5,5} (e_8)^{10} \otimes^{5,5} e_8 \otimes^{4,2} 501$$

2.8.12 SSgLIg(n-1)(1)

$$1_{\mathfrak{e}_8}^{1,5,5} \otimes \mathfrak{e}_8^{1,(5,5)} (\otimes \mathfrak{e}_8)^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}$$

2.8.13 SSgLIg(n-1)S(1)

$$1\mathfrak{e}_8 \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} \overset{1}{\otimes} \overset{5,5}{\mathfrak{e}_8} (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{4,2}{\otimes} \mathfrak{so}1$$

2.8.14 SSg(n-2)LIgLg(6)

$$\begin{array}{c} 2,4 \quad 1 \quad 5,5 \\ \otimes \mathfrak{e}_8 \left(\otimes \mathfrak{e}_8 \right)^5 \quad \otimes \mathfrak{e}_8 \quad \otimes \mathfrak{e}_8, \end{array}$$

$$I^{\oplus 4} \mathbf{e}_8^1 \left(\bigotimes^{5,5} \mathbf{e}_8 \right)^5 \bigotimes^{5,5} \mathbf{e}_8^1 \bigotimes^{5,5} \mathbf{e}_8,$$

$$I^{\oplus 3} I^{\oplus 2} \begin{pmatrix} 5,5 \\ \otimes \mathbf{e}_8 \end{pmatrix}^5 \begin{pmatrix} 5,5 \\ \otimes \mathbf{e}_8 \end{pmatrix}^1 \begin{pmatrix} 5,5 \\ \otimes \mathbf{e}_8 \end{pmatrix}$$

$$1513221\mathfrak{e}_8^1(\otimes^{5,5}\mathfrak{e}_8)^4\otimes^{5,5}\mathfrak{e}_8^1\otimes^{5,5}\mathfrak{e}_8,$$

$$I^{\oplus 2} {}^{3221} \mathfrak{e}_8 \left(\bigotimes \mathfrak{e}_8 \right)^4 \bigotimes \mathfrak{e}_8 \bigotimes \mathfrak{e}_8$$

$$1e_7 \overset{1}{\otimes} \overset{4,5}{e_8} (\overset{5,5}{\otimes} e_8)^4 \overset{5,5}{\otimes} e_8 \overset{1}{\otimes} \overset{5,5}{e_8}$$

2.8.15 SSg(n-2)LIgLgS(148)

$$131513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^9 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^9 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^9 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1$$

$$313221 \epsilon_8^{1, 5, 5} (\otimes \epsilon_8)^9 \otimes \epsilon_8^{5, 5} \otimes \epsilon_8^{1, 4, 2} \otimes \mathfrak{so}1, \quad 22221 \epsilon_8^{1, 5, 5} (\otimes \epsilon_8)^9 \otimes \epsilon_8^{5, 5} \otimes \epsilon_8^{1, 4, 2} \otimes \mathfrak{so}1, \quad 21513221 \epsilon_8^{1, 5, 5} (\otimes \epsilon_8)^9 \otimes \epsilon_8^{5, 5} \otimes \epsilon_8^{1, 4, 2} \otimes \mathfrak{so}1$$

$$1513221 \mathbf{c}_8 (\otimes \mathbf{c}_8)^9 \otimes \mathbf{c}_8 \otimes \mathbf{so}1, \quad 21^{13221} \mathbf{c}_8 (\otimes \mathbf{c}_8)^9 \otimes \mathbf{c}_8 \otimes \mathbf{so}1, \quad 2221^{21} \mathbf{c}_8 (\otimes \mathbf{c}_8)^9 \otimes \mathbf{c}_8 \otimes \mathbf{so}1$$

$$221^{221} \mathfrak{e}_8 \left(\bigotimes \mathfrak{e}_8 \right)^9 \bigotimes \mathfrak{e}_8 \bigotimes \mathfrak{so} 1$$

$$131513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^8 \epsilon_8^{5,5,1} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^8 \otimes \epsilon_8^{5,5,1} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^8 \epsilon_8^{5,5,1} \otimes \epsilon_8^{4,2} \otimes \mathfrak{so}1$$

$$313221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^8\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1, \quad 22221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^8\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1, \quad 21513221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^8\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1$$

$$1513221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^8 \bigotimes \mathbf{e}_8 \bigotimes \mathbf{so}1, \quad 21^{13221} \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^8 \bigotimes \mathbf{e}_8 \bigotimes \mathbf{so}1, \quad 2221 \mathbf{e}_8 \left(\bigotimes \mathbf{e}_8 \right)^8 \bigotimes \mathbf{e}_8 \bigotimes \mathbf{so}1$$

$$221 \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^8 {}^{5,5} \otimes \mathfrak{e}_8 \otimes \mathfrak{so}1$$

$$131513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^7 \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^7 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^7 \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$313221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^7\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1, \quad 22221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^7\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1, \quad 21513221\epsilon_8^{1\ 5,5}(\otimes\epsilon_8)^7\otimes\epsilon_8^{5,5}\otimes\epsilon_8^{1\ 4,2}\otimes\mathfrak{so}1$$

$$1513221 \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1, \quad 21^{13221} \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1, \quad 2221^{21} \mathfrak{e}_8^{5,5} (\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}_8^{5,5} \otimes \mathfrak{so}1$$

$$221 \mathfrak{e}_8 ({}^{5,5} \otimes \mathfrak{e}_8)^7 {}^{5,5} \otimes \mathfrak{e}_8 \otimes \mathfrak{so} 1$$

$$131513221 \epsilon_8^{(5,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(1,4,2)} \otimes \mathfrak{so}1, \quad 1231513221 \epsilon_8^{(5,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(1,4,2)} \otimes \mathfrak{so}1, \quad 12231513221 \epsilon_8^{(5,5)} \otimes \epsilon_8^{(5,5)} \otimes \epsilon_8^{(1,4,2)} \otimes \mathfrak{so}1$$

$$313221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(4,2)} \otimes \mathfrak{so}1, \quad 22221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(4,2)} \otimes \mathfrak{so}1, \quad 21513221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(4,2)} \otimes \mathfrak{so}1$$

$$1513221\epsilon_8(\otimes \epsilon_8)^6 \otimes \epsilon_8 \otimes \mathfrak{so}1, \quad 21^{13221}\epsilon_8(\otimes \epsilon_8)^6 \otimes \epsilon_8 \otimes \mathfrak{so}1, \quad 2221^{21}\epsilon_8(\otimes \epsilon_8)^6 \otimes \epsilon_8 \otimes \mathfrak{so}1$$

$$\begin{array}{lll} 1513221 \epsilon_8^{1,5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1, & 21^{13221} \epsilon_8^{5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1, & 2221 \epsilon_8^{21,5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1 \\ 221^{221} \epsilon_8^{5,5} (\otimes \epsilon_8)^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \mathfrak{so}1 & & \end{array}$$

$$13221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(5,5)} \otimes \epsilon_8 1, \quad 2221\epsilon_8^{1(5,5)}(\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(5,5)} \otimes \epsilon_8 1, \quad 221\epsilon_8^{21(5,5)}(\otimes \epsilon_8)^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1(5,5)} \otimes \epsilon_8 1$$

$$21e_7 \otimes e_8^{4,5} (\otimes e_8^{5,5})^9 \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes \mathfrak{so}l, \quad 1e_7 (\otimes e_8^{5,5})^{10} \otimes e_8^{5,5} \otimes e_8^{1,4,2} \otimes \mathfrak{so}l$$

$$21e_7^1 \otimes e_8^{4,5} (\otimes e_8^{5,5})^8 \otimes e_8^{5,5} \otimes e_8^1 \otimes so1^{4,2}, \quad 1e_7^1 (\otimes e_8^{5,5})^9 \otimes e_8^{5,5} \otimes e_8^1 \otimes so1^{4,2}$$

$$21e_7^{1,4,5} \otimes e_8^{(5,5)} \otimes e_8^{5,5,1,4,2} \otimes \mathfrak{so}l, \quad 1e_7^{(5,5)} \otimes e_8^{5,5,1,4,2} \otimes \mathfrak{so}l$$

$$21\mathbf{e}_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes \otimes \otimes \end{smallmatrix} \mathbf{sol}, \quad 1\mathbf{e}_7 \begin{smallmatrix} 1 & 5,5 \\ \otimes \end{smallmatrix} \mathbf{e}_8 \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes \otimes \otimes \end{smallmatrix} \mathbf{sol}$$

$$21\mathbf{e}_7 \otimes \mathbf{e}_8 \begin{pmatrix} 1 & 4,5 \\ & 5,5 \end{pmatrix} \otimes \mathbf{e}_8 \begin{pmatrix} 5,5 & 1 \\ & 4,2 \end{pmatrix} \otimes \mathbf{sol}, \quad 1\mathbf{e}_7 \begin{pmatrix} 1 & 5,5 \\ & 5,5 \end{pmatrix} \otimes \mathbf{e}_8 \begin{pmatrix} 5,5 & 1 \\ & 4,2 \end{pmatrix} \otimes \mathbf{sol}$$

$$21\mathbf{e}_7 \otimes \mathbf{e}_8 \begin{smallmatrix} 1 & 4,5 \\ \otimes \end{smallmatrix} \begin{smallmatrix} 5,5 \\ (\otimes \mathbf{e}_8)^4 \end{smallmatrix} \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes \otimes \otimes \end{smallmatrix} \mathbf{sol}, \quad 1\mathbf{e}_7 \begin{smallmatrix} 1 & 5,5 \\ (\otimes \mathbf{e}_8)^5 \end{smallmatrix} \begin{smallmatrix} 5,5 & 1 & 4,2 \\ \otimes \otimes \otimes \end{smallmatrix} \mathbf{sol}$$

$$21e_7 \otimes e_8 \otimes e_8^3 \otimes e_8 \otimes \text{sol}, \quad 1e_7 \otimes e_8^4 \otimes e_8 \otimes \text{sol}$$

$$21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}, \quad 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^{14} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}$$

$$21\mathbf{e}_7 \overset{1}{\otimes} \overset{4,5}{\mathbf{e}_8} (\overset{5,5}{\otimes} \mathbf{e}_8)^{12} \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \overset{4,2}{\mathbf{so}1}, \quad 1\mathbf{e}_7 (\overset{5,5}{\otimes} \mathbf{e}_8)^{13} \overset{5,5}{\otimes} \mathbf{e}_8 \overset{1}{\otimes} \overset{4,2}{\mathbf{so}1}$$

$$21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}, \quad 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}$$

$$21\epsilon_7^1 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}, \quad 1\epsilon_7^1 (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^1 \otimes \mathfrak{so}1^{4,2}$$

$$1e_7' \otimes^{4,5} e_8 (\otimes^{5,5} e_8)^9 \otimes^{5,5} e_8 \otimes^{4,2} so1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^6 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{13} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{12} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{11} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

$$1\epsilon_6^1 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,4,2} \otimes \mathfrak{so}1$$

2.8.16 Sg(n-2)LlgLg(15)

$$\otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$\otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$\otimes \epsilon_8^{5,5} (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$313221\epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$I^{\oplus 5} \epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$21513221\epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$1513221\epsilon_8 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$I^{\oplus 2} \epsilon_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5},$$

$$1\epsilon_7 (\otimes \epsilon_8^{5,5})^5 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$5131513221\epsilon_8 (\otimes \epsilon_8^{5,5})^4 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$321\epsilon_7 \otimes \epsilon_8^{4,5} (\otimes \epsilon_8^{5,5})^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$$31\epsilon_6 \otimes \epsilon_8^{3,5} (\otimes \epsilon_8^{5,5})^3 \otimes \epsilon_8^{5,5} \otimes \epsilon_8^{1,5,5}$$

$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^9 \otimes \mathfrak{e}'_7 1$
$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^8 \otimes \mathfrak{e}'_7 1$
$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^7 \otimes \mathfrak{e}'_7 1$
$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^6 \otimes \mathfrak{e}'_7 1$
$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^5 \otimes \mathfrak{e}'_7 1$
$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^4 \otimes \mathfrak{e}'_7 1$
$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{14} \otimes \mathfrak{e}'_7 1$
$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}'_7 12,$	$1\mathfrak{so} \otimes \mathfrak{e}_8(\otimes \mathfrak{e}_8)^{13} \otimes \mathfrak{e}'_7 1$

$$150 \otimes^{2,4} e_8 \otimes^{1(5,5)} (e_8 \otimes e_8)^{12} \otimes^{5,5} e_7' 1$$

$$150 \otimes e_8^{2,4} (\otimes e_8^{5,5})^{11} \otimes e_7^{5,5} 1$$

$$150 \otimes \epsilon_8^{2,4} (\otimes \epsilon_8^{5,5})^{10} \otimes \epsilon_7'^{5,5} 1$$

$$1\mathfrak{e}_8 \overset{5,5}{\otimes} \mathfrak{e}_8 (\overset{5,5}{\otimes} \mathfrak{e}_8)^4 \overset{5,5}{\otimes} \mathfrak{e}_7 1$$

$$150 \otimes^{2,4} e_8 \otimes^{1(5,5)} (e_8 \otimes e_8)^5 \otimes^{4,2} so13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^3 \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{14} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{13} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{13} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{12} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{12} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{11} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{11} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{10} \otimes \mathfrak{so}^{4,2} \otimes \mathfrak{so}^{1,1}1$$

$$1\mathfrak{so} \otimes \mathfrak{e}_8^{2,4}(\otimes \mathfrak{e}_8^1)^{5,5}{}^{10} \otimes \mathfrak{so}^{4,2}13$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{e}_7'^{5,4}1$$

$$1\mathfrak{e}_8 \otimes \mathfrak{e}_8^{5,5}(\otimes \mathfrak{e}_8^1)^{5,5}{}^4 \otimes \mathfrak{e}_6^{5,3}1$$